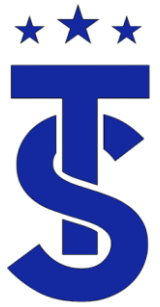




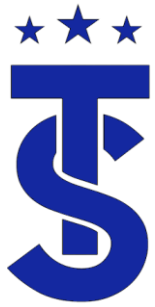
STS Training Series

DVM S & DVM S 2
Electrical Circuits



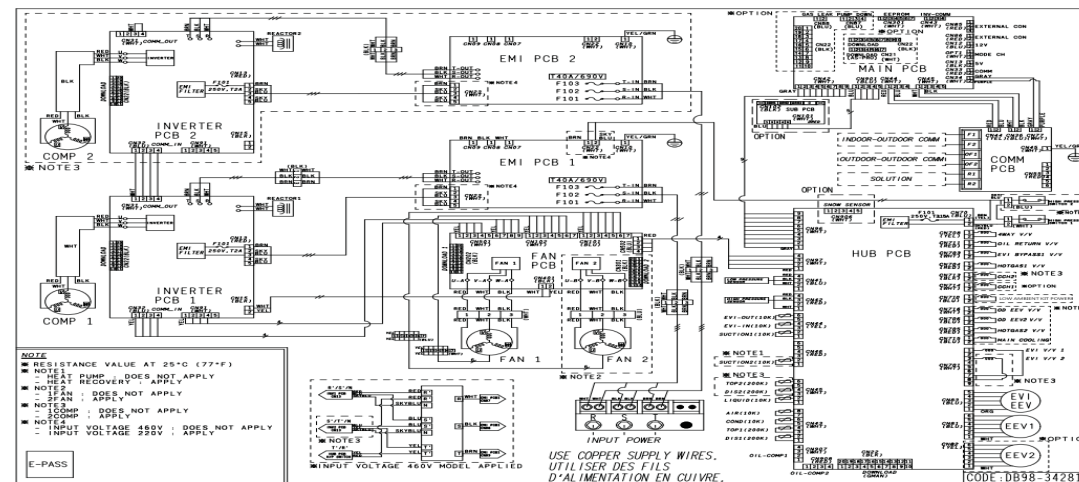
Troubleshooting Fundamentals

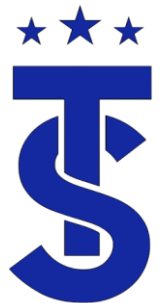
- All equipment operates with a logical sequence of operations.
- Trouble shooting is a logical process of elimination.
- In order to effectively troubleshoot a system, you must know:
 - The unit electrically
 - What is the unit supposed to be doing and how
 - The function and interrelationship of the components within the system.
 - How to read an electrical diagram to determine the flow of electricity through the system.



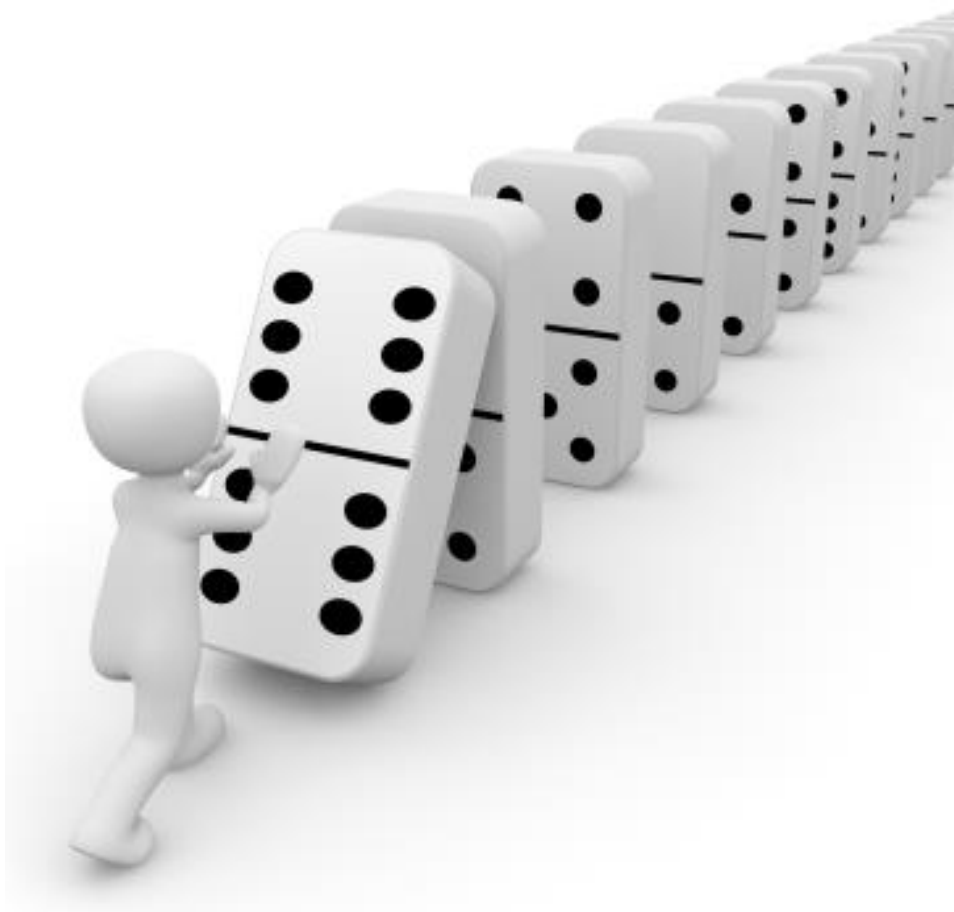
Electrical Diagrams

- Are a road map of electrical flow through a system.
- They show the components within the electrical system .
- They show the inter-relationship between components.
- Are a vital tool used in the troubleshooting process.
- Electrical Diagrams can be found in the service manual.





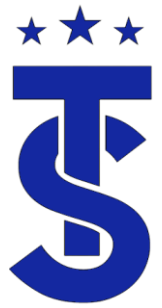
Sequence of Operation



**DVM S Control Training
Manual for North America
- Sequence of Operations -**

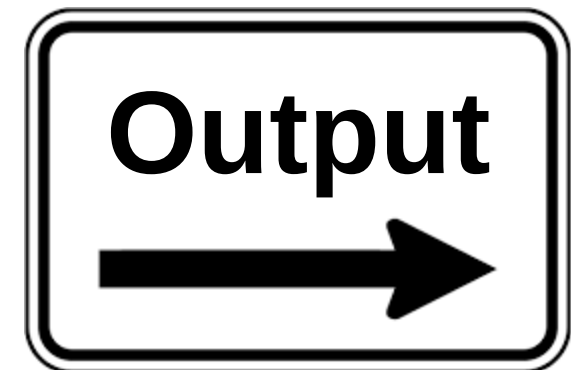
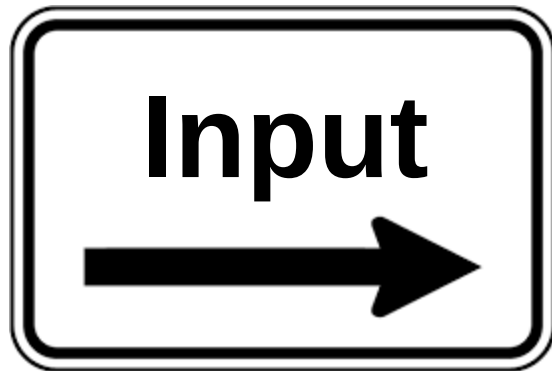
Version 12202018

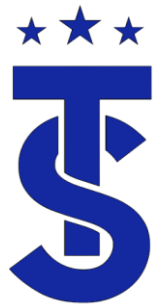
SAMSUNG



Troubleshooting 101

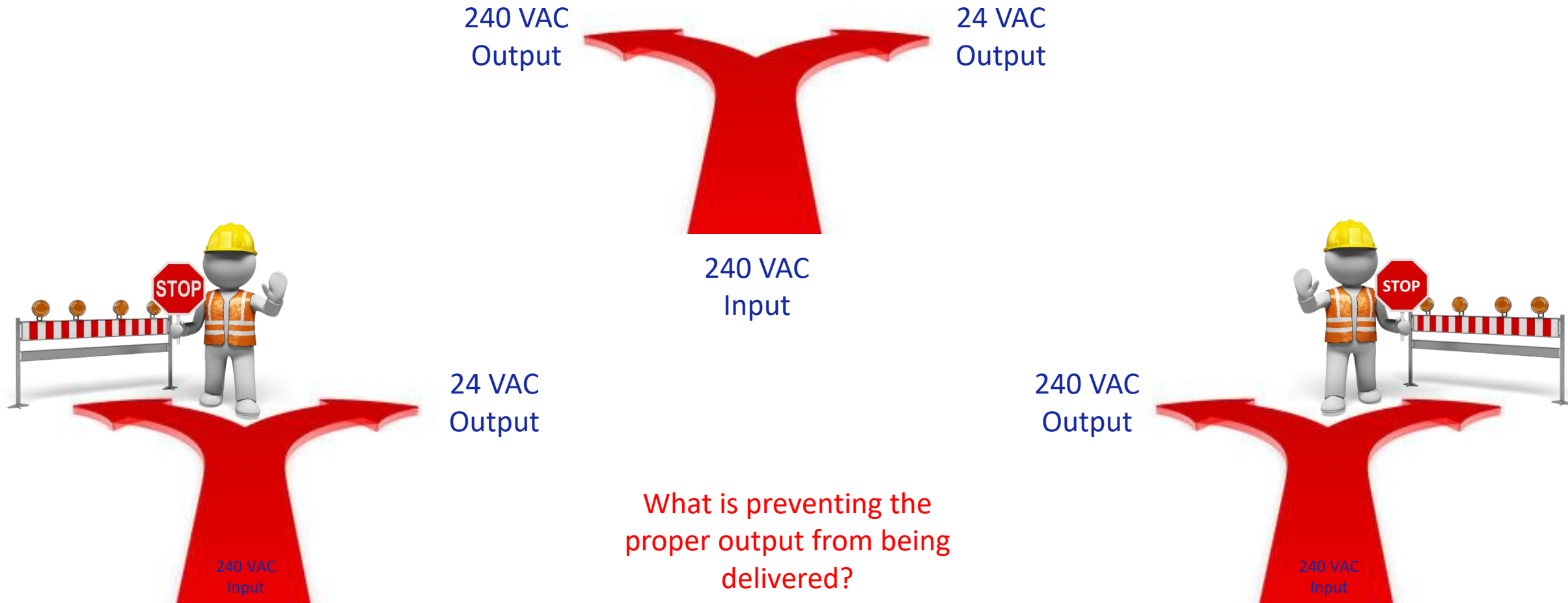
- Know the inputs and outputs of a device.
- Do you have the desired out put of a device?
- If not, do you have the proper input?

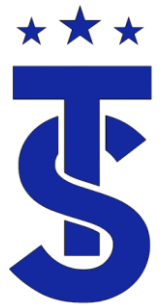




Circuit Board Troubleshooting

Power In - Power Out

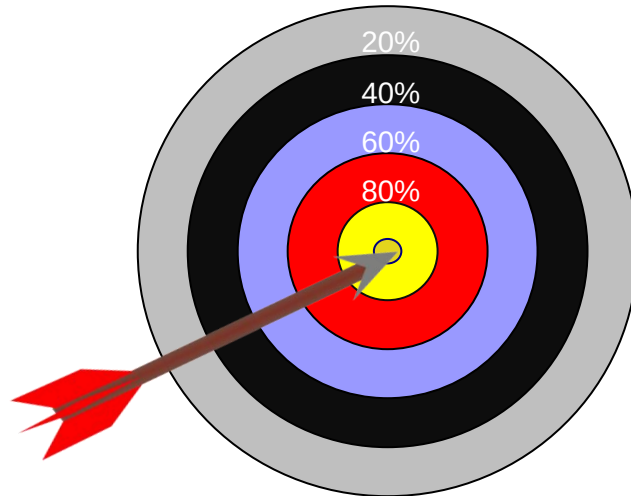




Troubleshooting Process

Process:

A series of actions or steps taken in order to achieve a particular end.

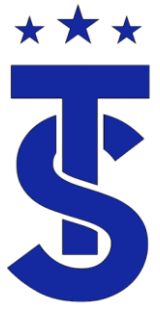


Outline

- This class will review the basic components and power flow for an DVM S outdoor unit.
- Using the electrical diagram, we will identify the electrical components that make up the system
- We will discuss basic function of each component.
- This class will focus on gaining a basic understanding of each component while identifying the basic input and output voltages of each component
- **Topics Covered:**
 - Parts of an inverter
 - Electrical system components
 - System electrical flow including line voltage and control voltage.
 - Component troubleshooting using power in power out

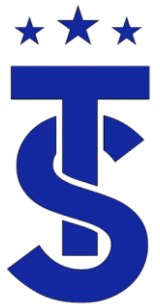


AM096FXVAJH2AA



Power Supplies

- This section we will review the power supplies within the VRF System
- line voltages
- Control Voltages
- Motor Supply Voltage



DVM S System Power Supplies & Control Circuits

- Line voltage

- Outdoor:

- Dedicated 3 Phase 208/230, 460 & 575Volt circuit
 - Dedicated 208/230vac 1 phase

- Indoor:

- Dedicated 208/230vac 1 phase circuit

- MCU:

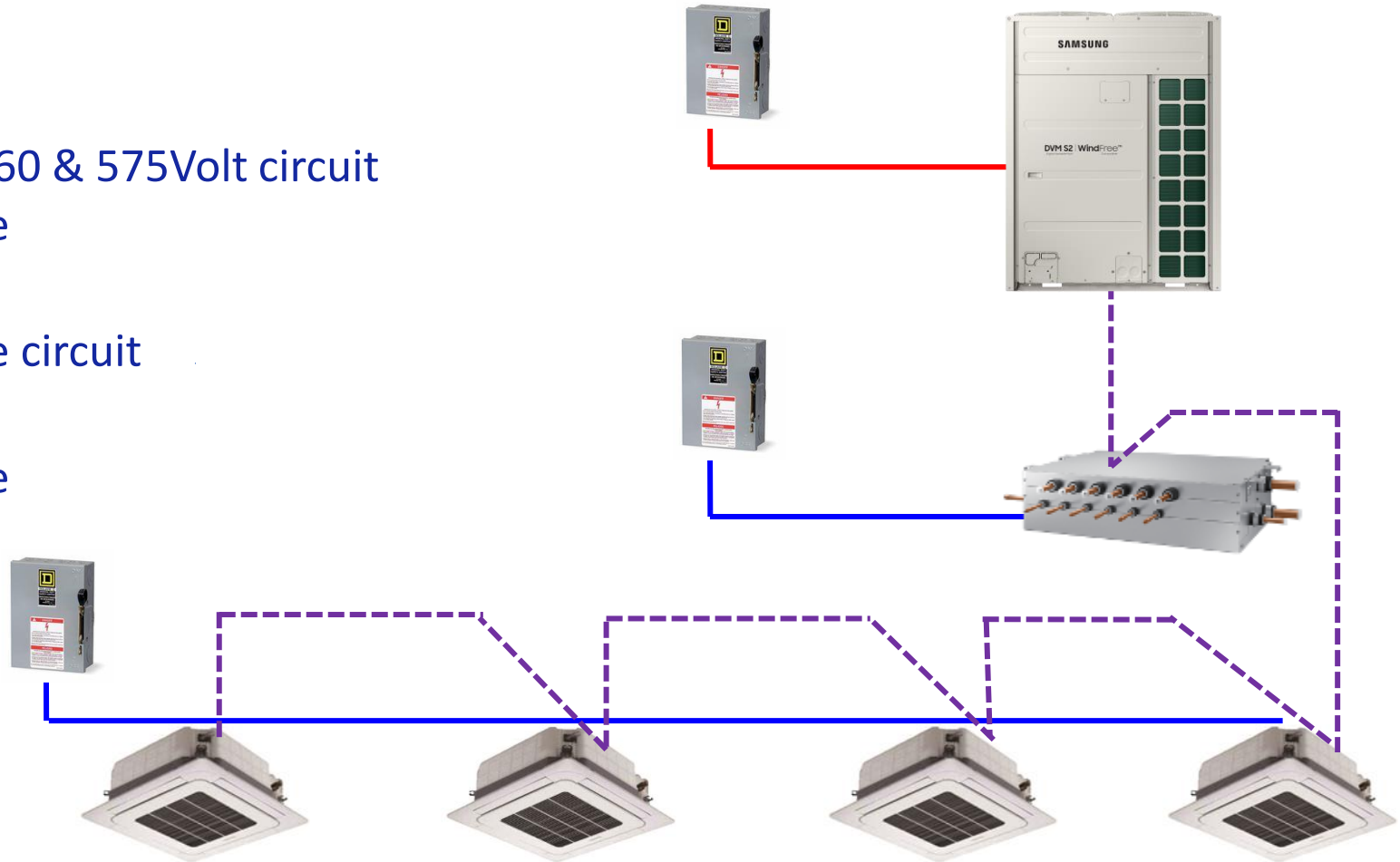
- Dedicated 208/230vac 1 phase

- Control Circuits

- Communication

- 2 conductor control wire

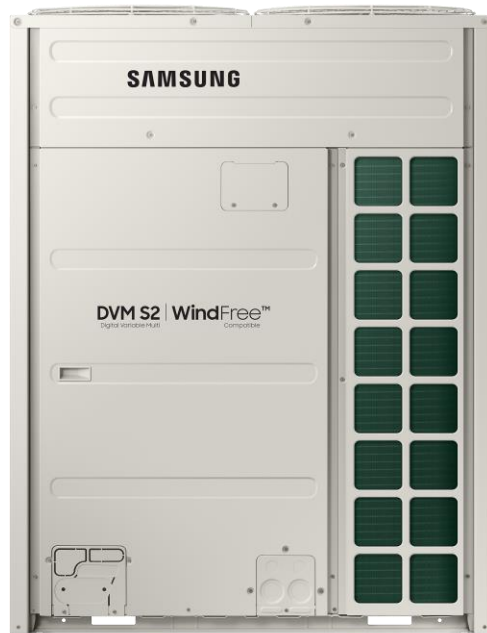
- 16/2 stranded with shield



DVM S Voltages



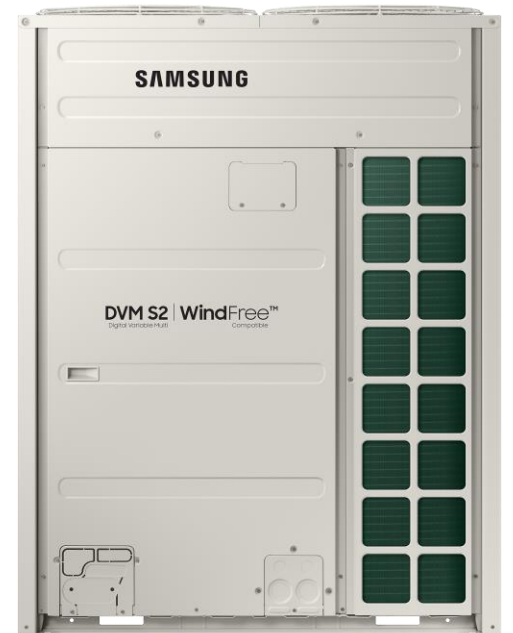
230 -240 VAC 1ph.



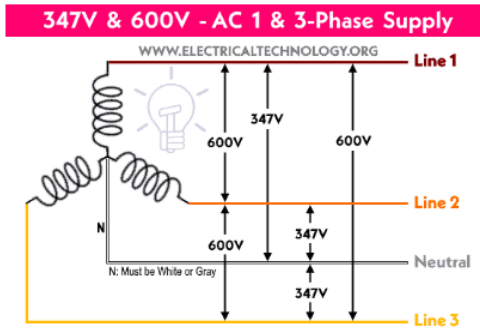
230 -240 VAC 3ph.



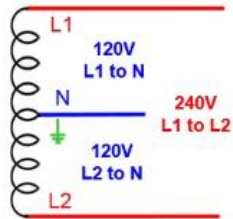
460 VAC 3ph.



575 VAC 3ph.

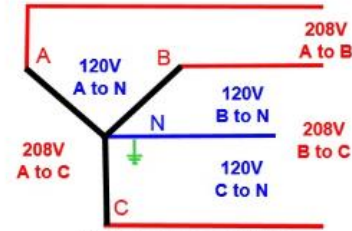


Common US and Canada Voltage Systems 600Vac and below



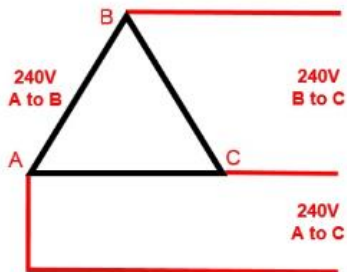
Single Phase

1Ø3W, 120/240Vac
This system is **grounded**
There is a **neutral**



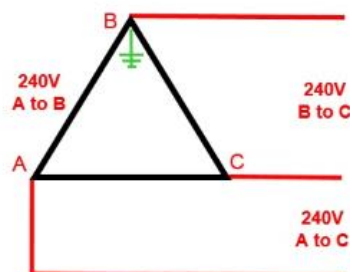
Three Phase Wye

3Ø4W, 208Y/120Vac Wye (shown)
3Ø4W, 480Y/277Vac Wye
3Ø4W, 600Y/347Vac Wye (Canada only)
This system is **grounded**
There is a **neutral**



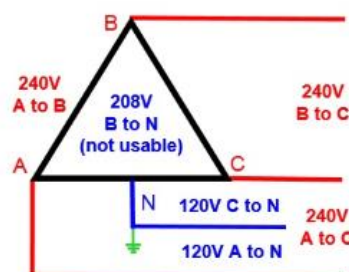
Three Phase Ungrounded Delta

3Ø3W, 240Vac Delta (shown)
3Ø3W, 480Vac Delta
3Ø3W, 600Vac Delta
This system is **not grounded**
There is **no neutral**



Three Phase Corner Grounded Delta

3Ø3W, 240Vac Grounded B phase (shown)
3Ø3W, 480Vac Grounded B phase
3Ø3W, 600Vac Grounded B phase
This system is **grounded**
There is **no neutral**



Three Phase High Leg Delta

3Ø4W, 240/120Vac High Leg Delta
This system is **grounded**
There is a **neutral**

Canada
North America

Capital
Ottawa

Population
37.7 Million
100% Access to Electricity

AC Voltages		
Single Phase Voltages: 120V - 2 Wire, 120V/240V - 3 Wire	Three Phase Voltages: 208V - 4 Wire, 240V - 3 Wire, 480V/600V - 3/4 Wire	Frequency: 60 Hz

Single Phase Plug & Socket Types

Type A	Type B	NEMA 5-15

sinalda
mains voltage control

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www.sinalda.com

United States
North America

Capital
Washington, D.C.

Population
331 Million
100% Access to Electricity

AC Voltages		
Single Phase Voltages: 120V - 2 Wire, 120V/240V - 3 Wire	Three Phase Voltages: 208V - 4 Wire, 240V - 3 Wire, 480V - 3/4 Wire	Frequency: 60 Hz

Single Phase Plug & Socket Types

Type A	Type B	NEMA 5-15

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mains voltage control

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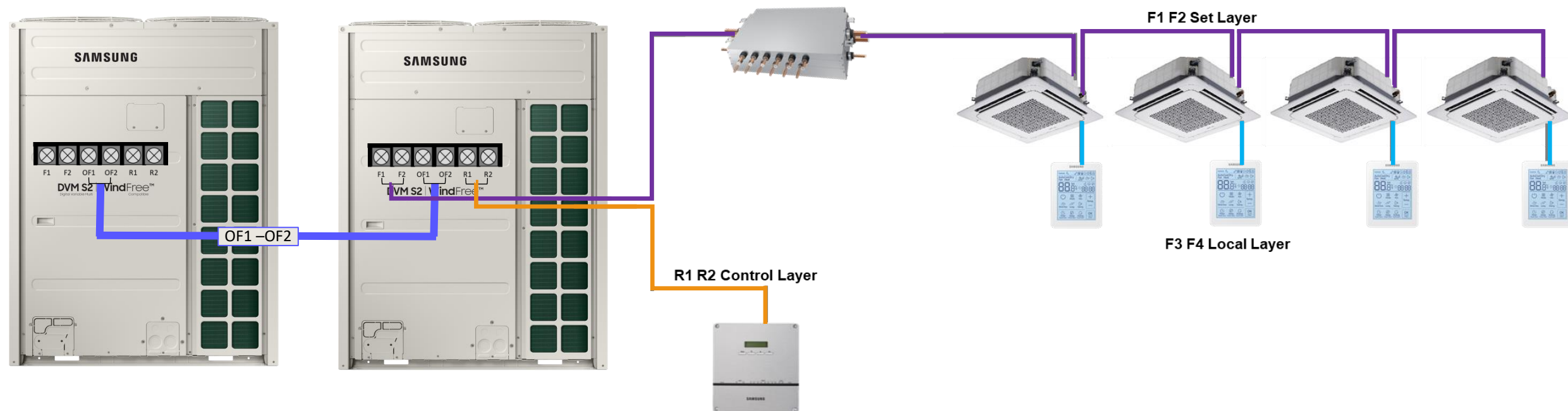
www.sinalda.com

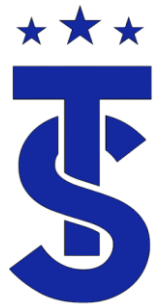


NASA Digital Control Layers



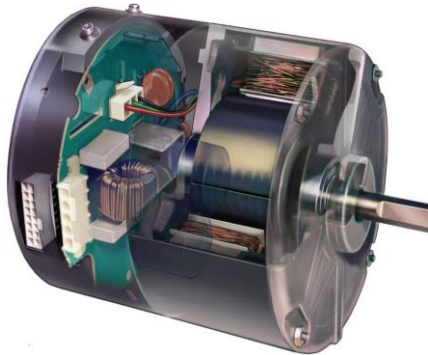
- **F1 F2 – Set Layer:** ODU to IDU's & MCU'S (incl. EEV Kit) daisy chain connections 0.5 VDC
- **F3 F4 – Local Layer:** Wired Remote Controller to indoor unit 12 VDC
- **R1 R2 – Control Layer:** Outdoor Unit to DMS Centralized Control 5 VDC
- **OF1-OF2 – ODU main & sub communication** 0.5 VDC
- **Control Wire:** 16/2 AWG stranded with shield
 - Never use solid core thermostat wire for digital communications





Inverter Driven Motors

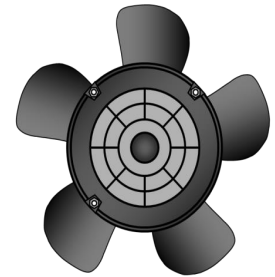
What do they all have in common



ECM Motors



BLDC Motors

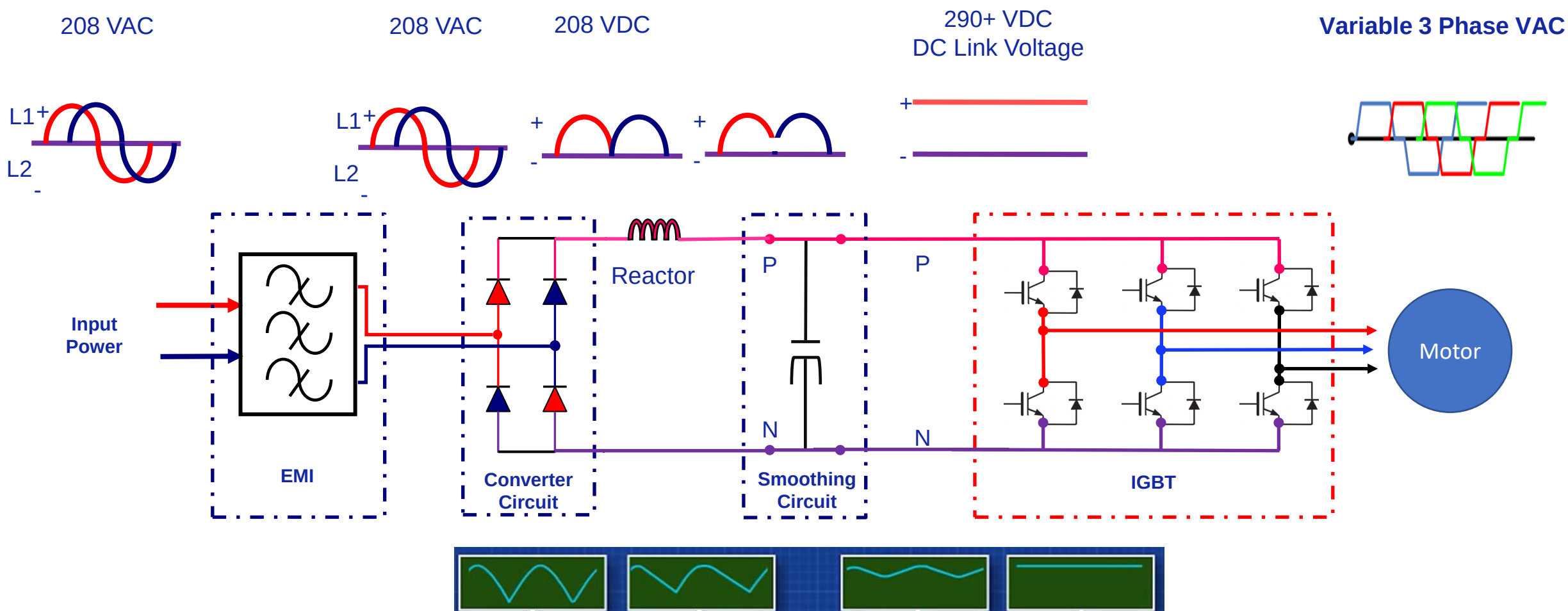


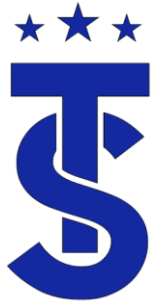
3 Phase Digitally
Commutated
AC Voltage





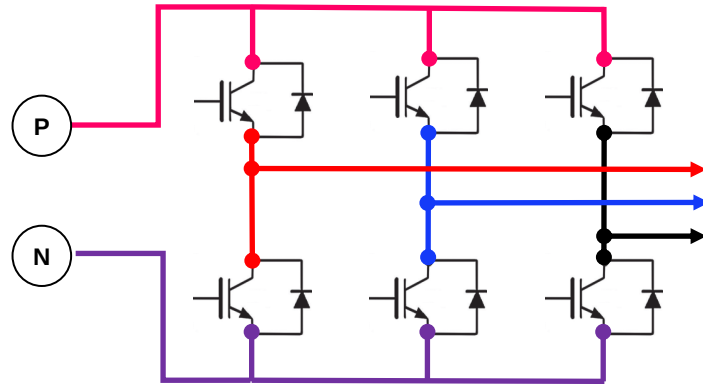
Basic Inverter Drive Power Flow





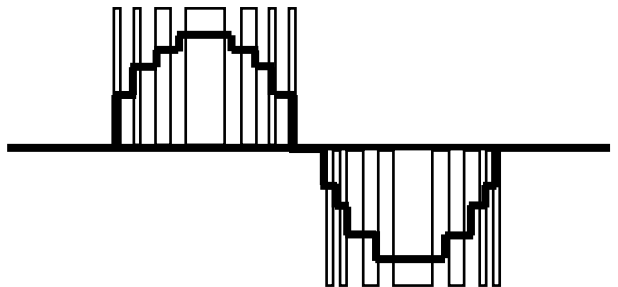
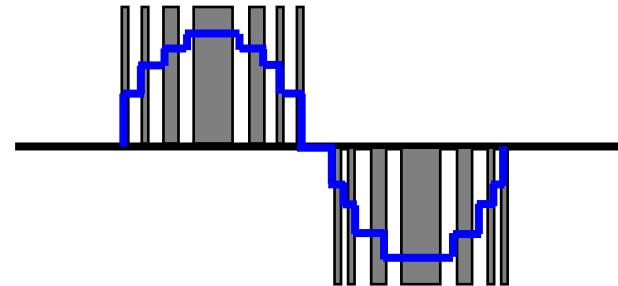
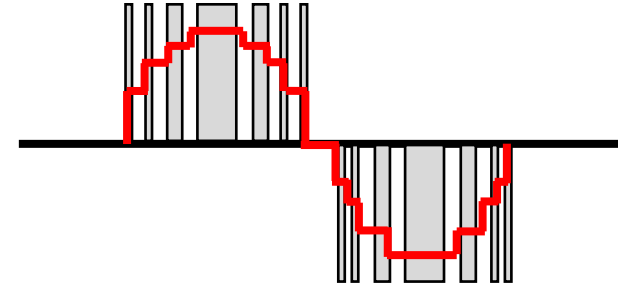
Digitally Commutated AC Voltage

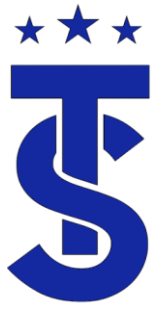
High DC Voltage
Input



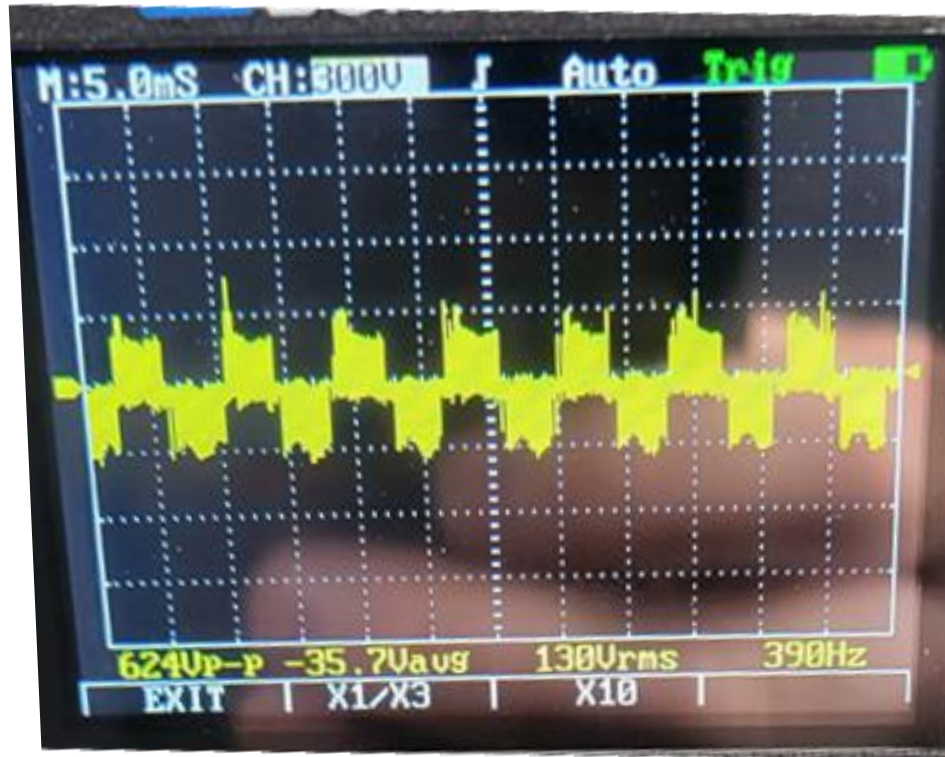
6 - IGBTs

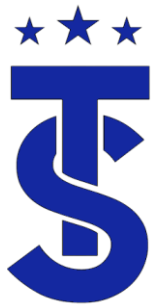
Rectifiers
DC to AC



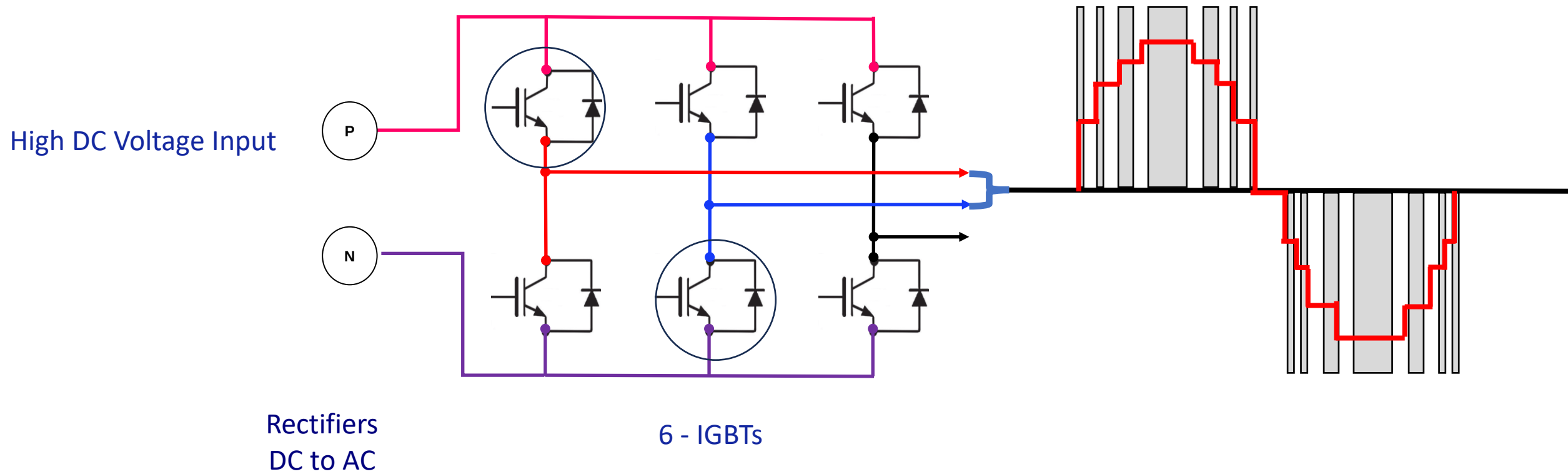


DC 3 phase AC Voltage





Phase generation

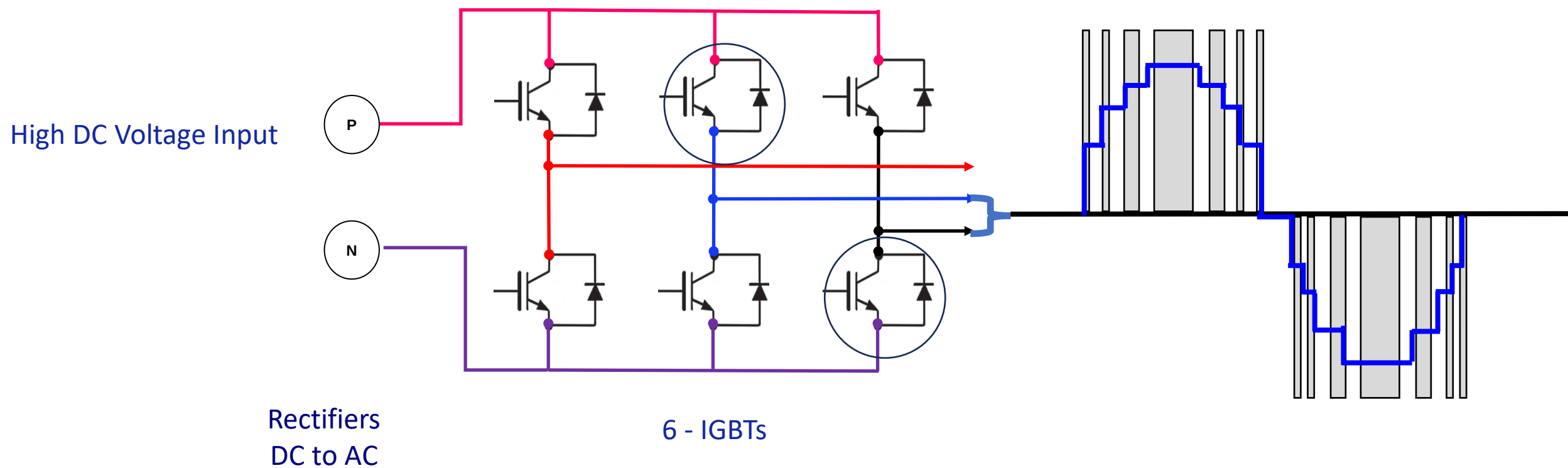


<https://www.youtube.com/watch?v=ilqhAX0I7II>

<https://www.youtube.com/watch?v=yEPe7RDtkgo&t=141s>



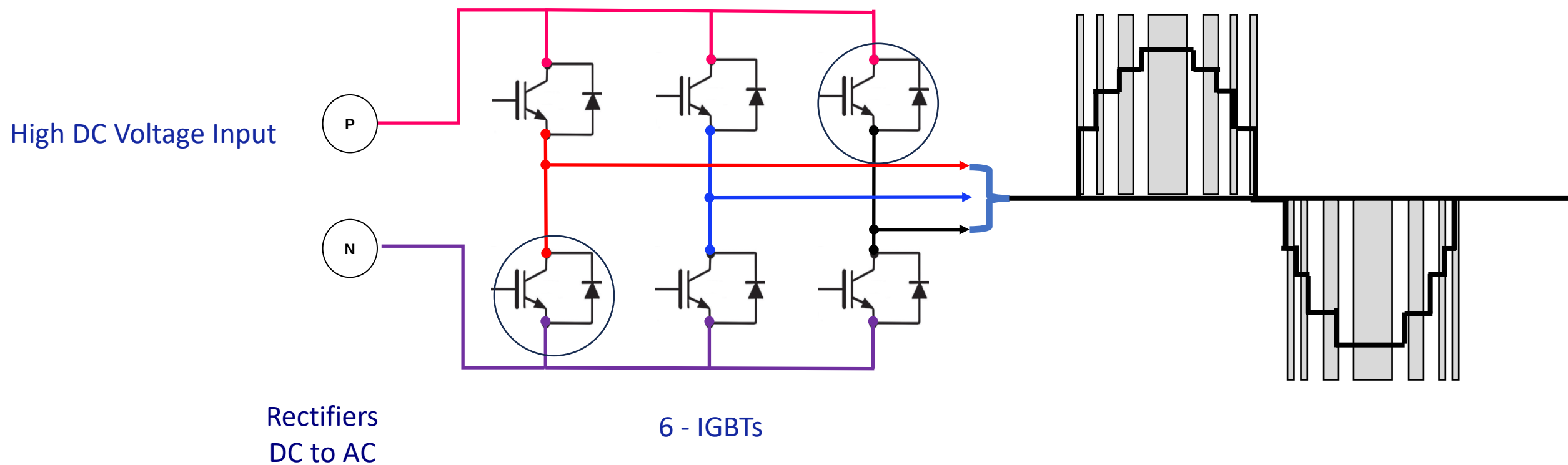
Phase generation



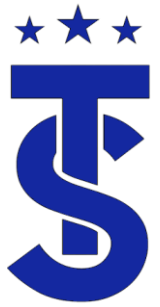
<https://www.youtube.com/watch?v=iIqhAX0I7II>



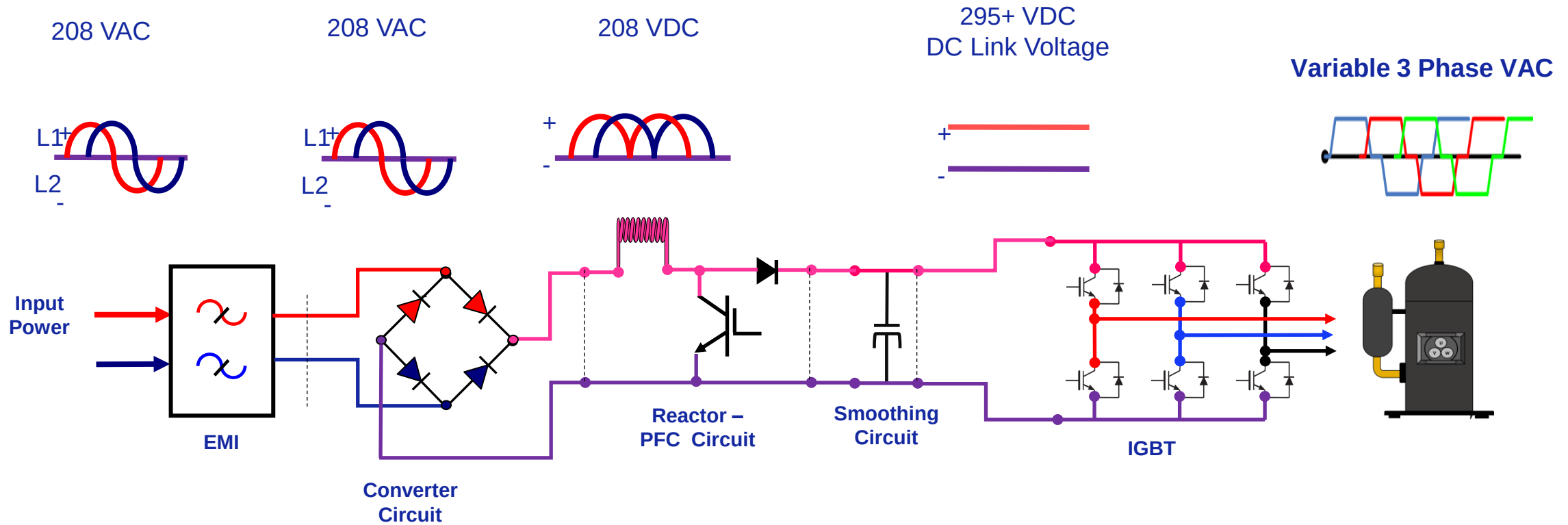
Phase generation



<https://www.youtube.com/watch?v=iIqhAX0I7II>

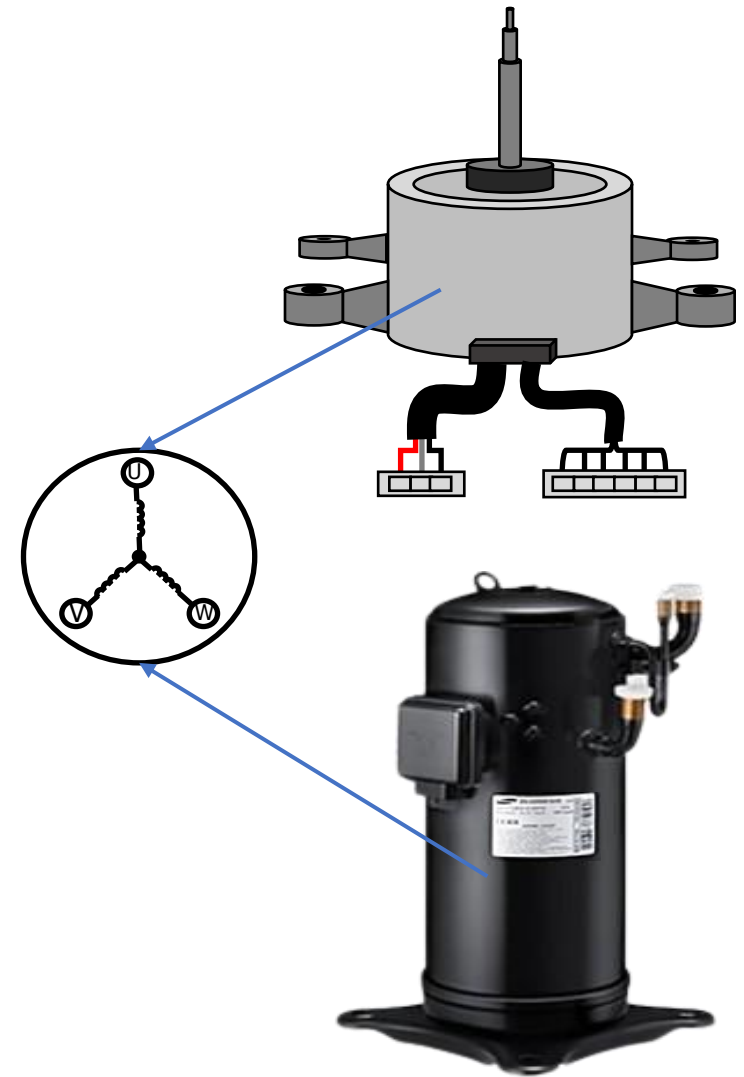


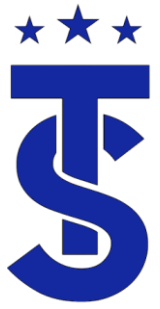
Samsung ODU Compressor



ODU Motors

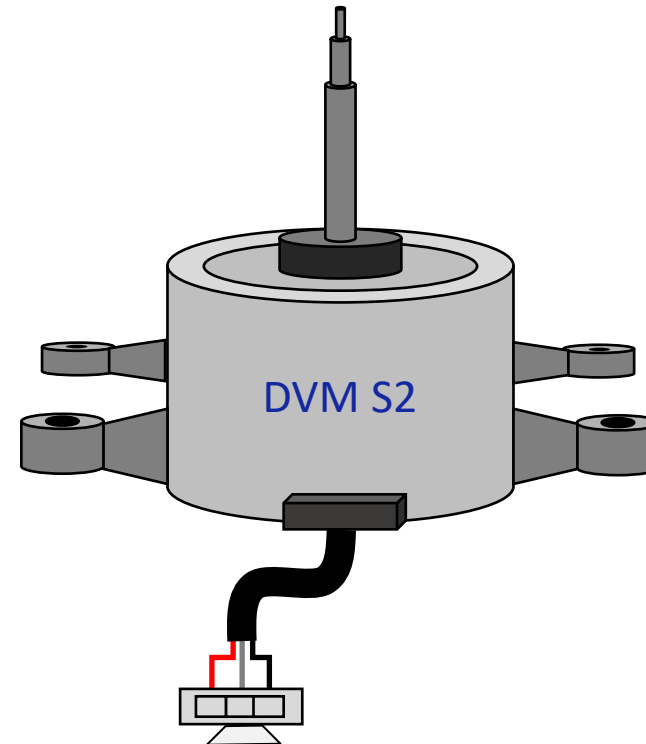
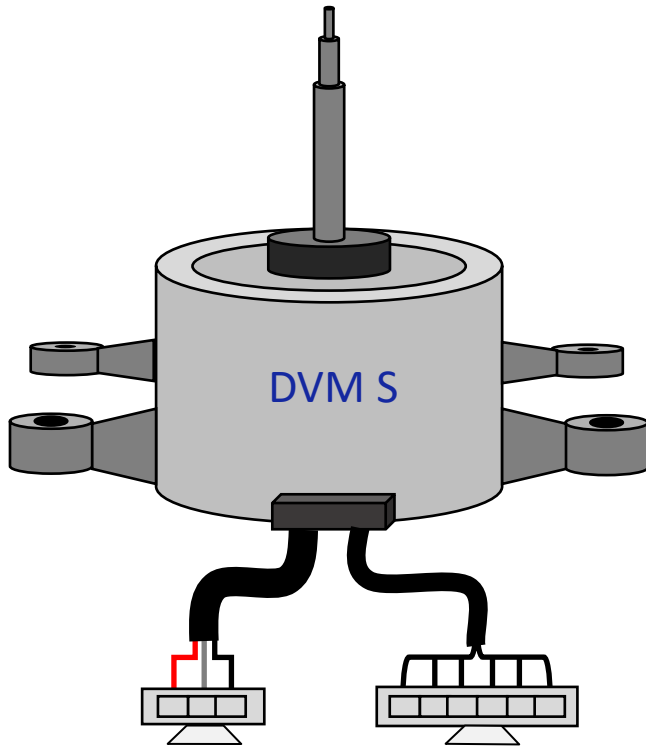
- The compressors and outdoor fans are inverter driven motors.
- They are supplied with a digitally commutated 3 phase AC voltage
- The voltage and frequency are varied to control the speed of the motor
- The resistance of all 3 winding must be equal

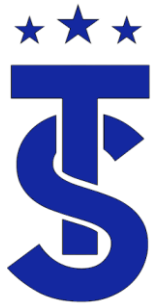




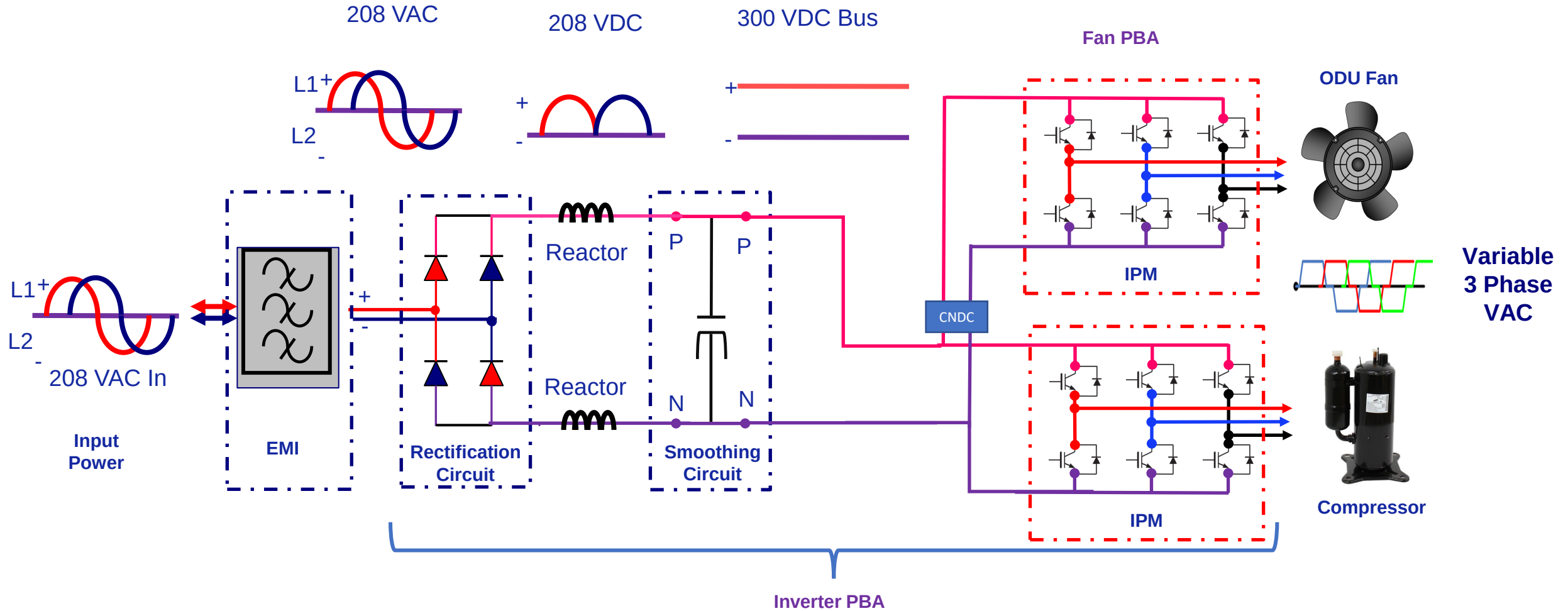
Outdoor Fan Motors

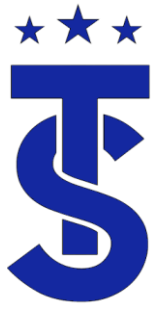
- There are 2 types of condenser fans used depending on the generation of equipment
- The motor will have either 1 or 2 plugs





ODU Motors Power Supply

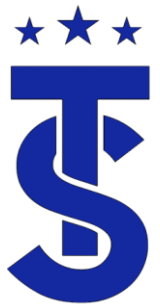




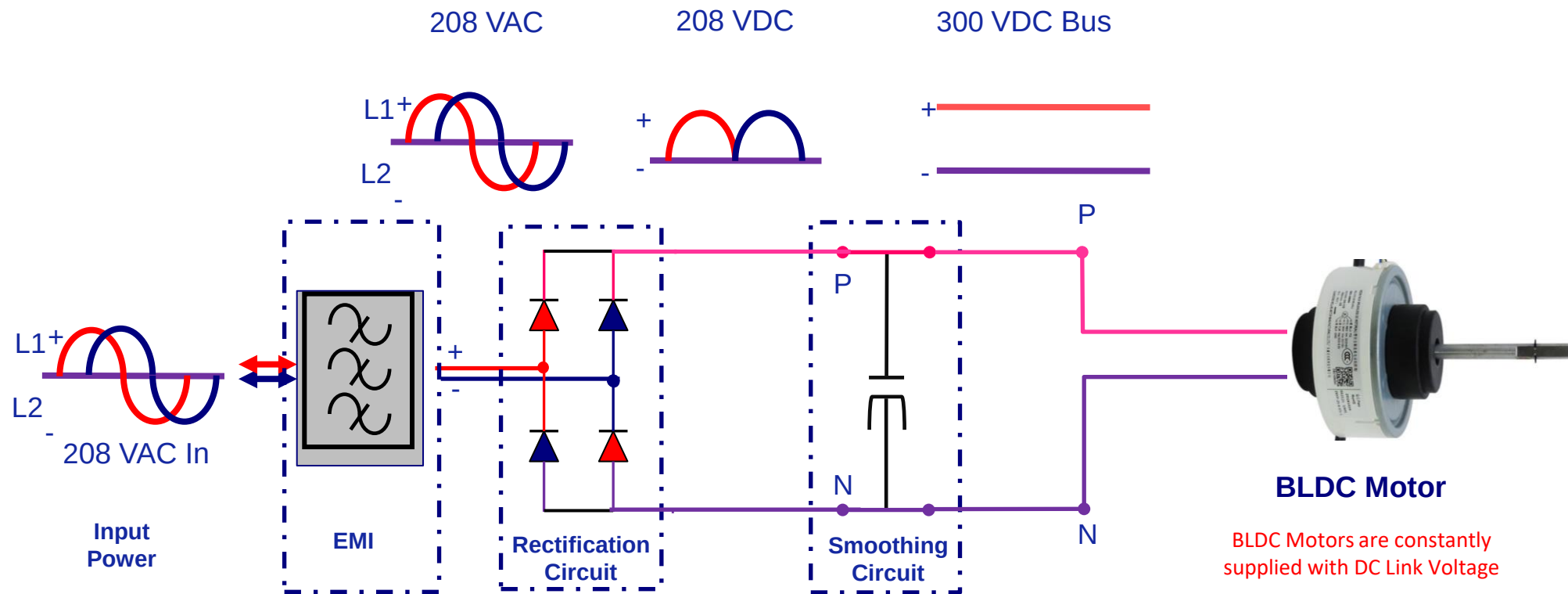
What do these have in common electrically

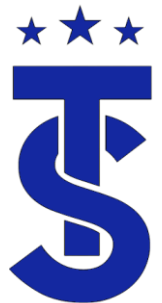


Everything that happens in the ODU happens in the IDU
The same goes for the RAC, CAC and FJM systems



Indoor Motor Power Supply



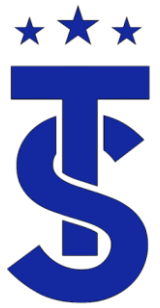


Where does the DC Bus Voltage Come From

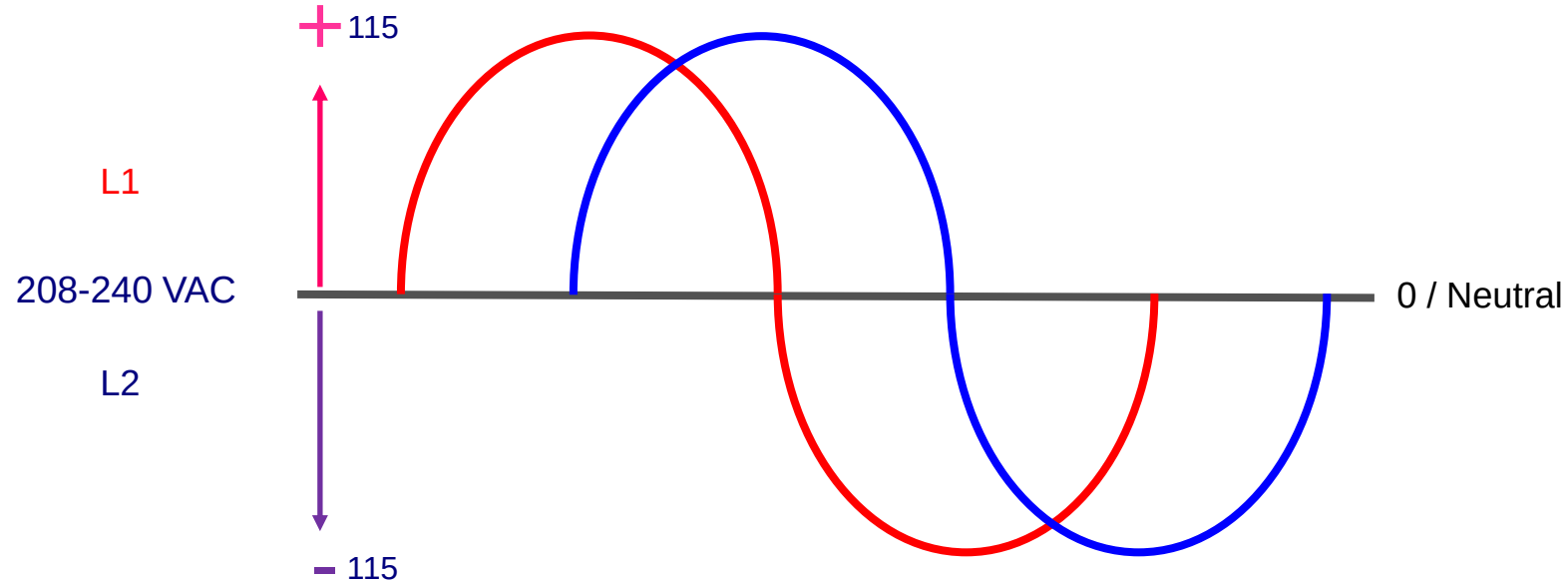
- One of the most common question I get is where does the DC Bus voltage come from?
- Is there a step up transformer in the system.
- Why is it a factor of the incoming power
- The next few slide will help explain.

Standard voltage range of DC and AC

AC Input Voltage	Standard AC Voltage(Vac)		Standard DC Voltage(Vdc)	
	Min.	Max.	Min.	Max.
208~230V	187	253	265	358
460V	414	506	585	715
380~415V	342	457	484	646
575V	518	633	732	894

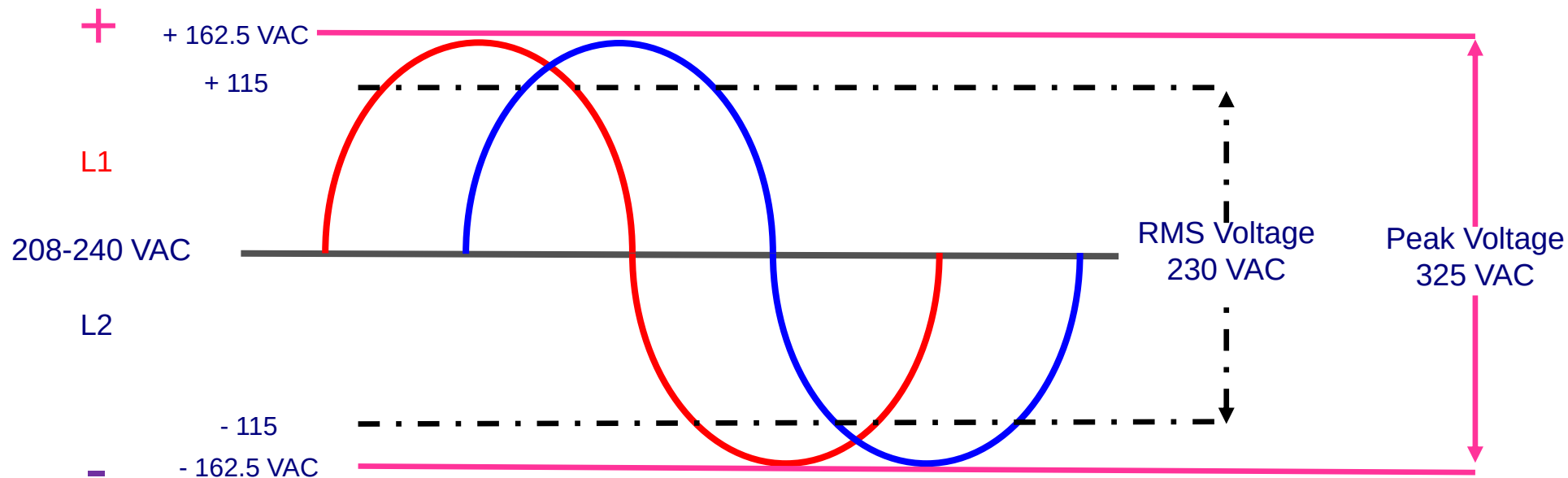


Parts of a Sine Wave

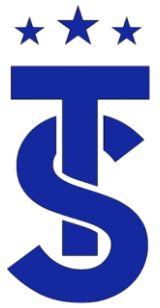




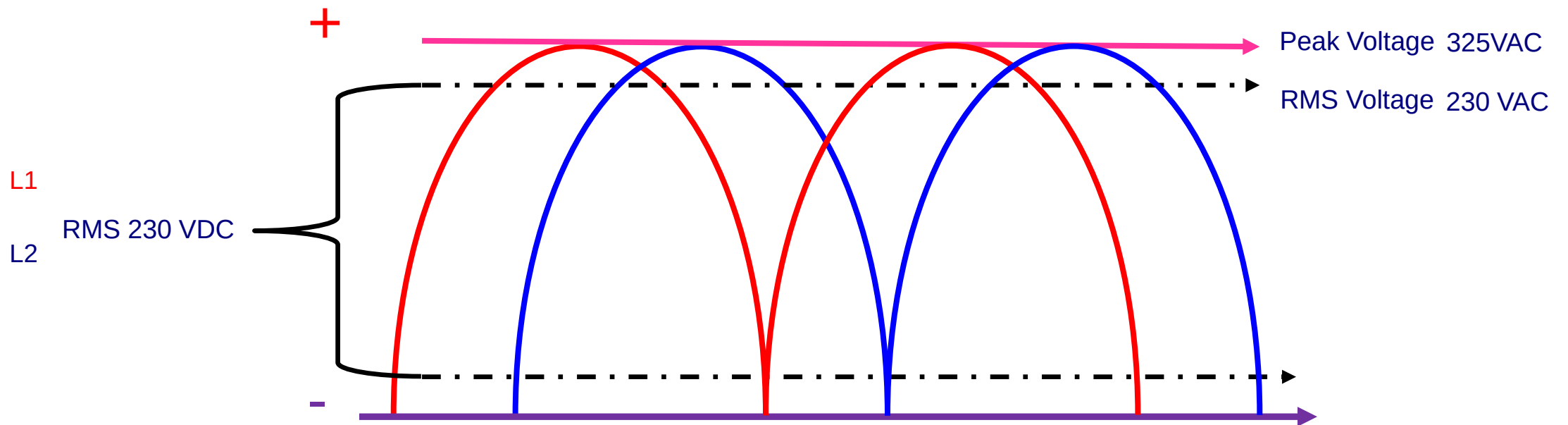
Peak Voltage Vs True RMS Voltage



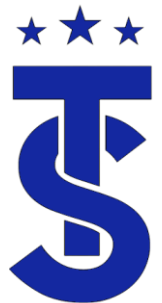
The rms value is 0.707 times the peak value, and the peak value is 1.41 times the value the voltmeter shows. The peak value for 230 V mains is 325 V



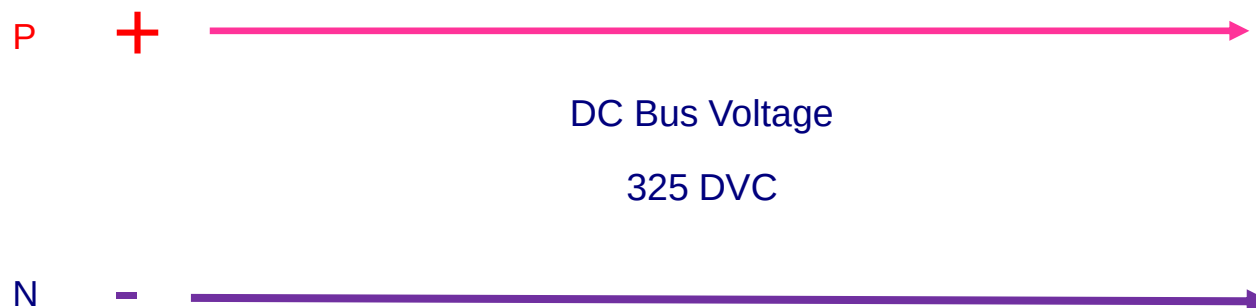
Rectified DC Sine Wave



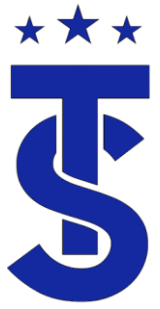
The voltage leaves the rectifier and a DC voltage. The signal is rigid and not stable



DC of a Sine Wave After The Smoothing Circuit.

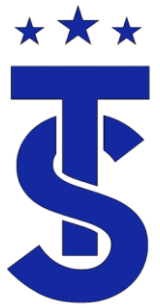


The rms value is 0.707 times the peak value, and the peak value is 1.41 times the value the voltmeter shows.
The peak value for 230 V mains is 325 V



DC Link Voltage

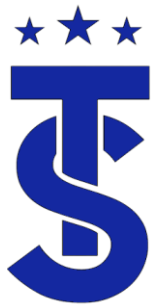
- The DC link voltage is used to power multiple components within the system.
- DC Link voltage is supplied to the following components:
 - The Compressor IPM
 - The SMPS in some units
 - The Condenser fan (BLDC Motors)
 - The condenser fan IPM (3 phase A/C motors)



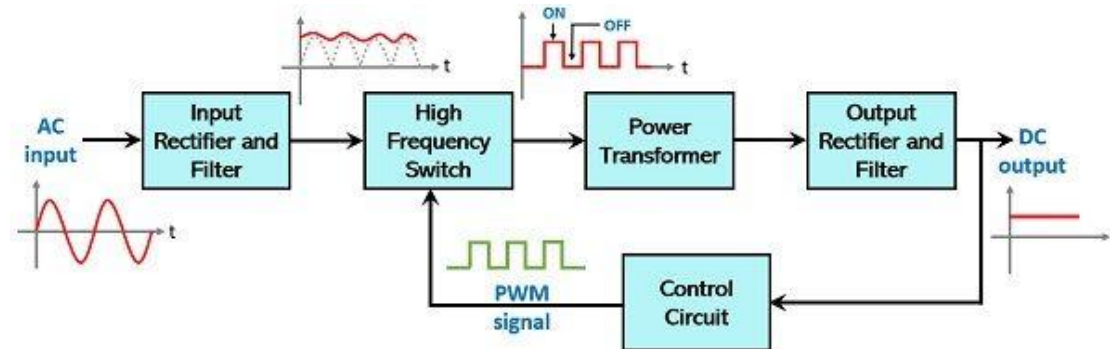
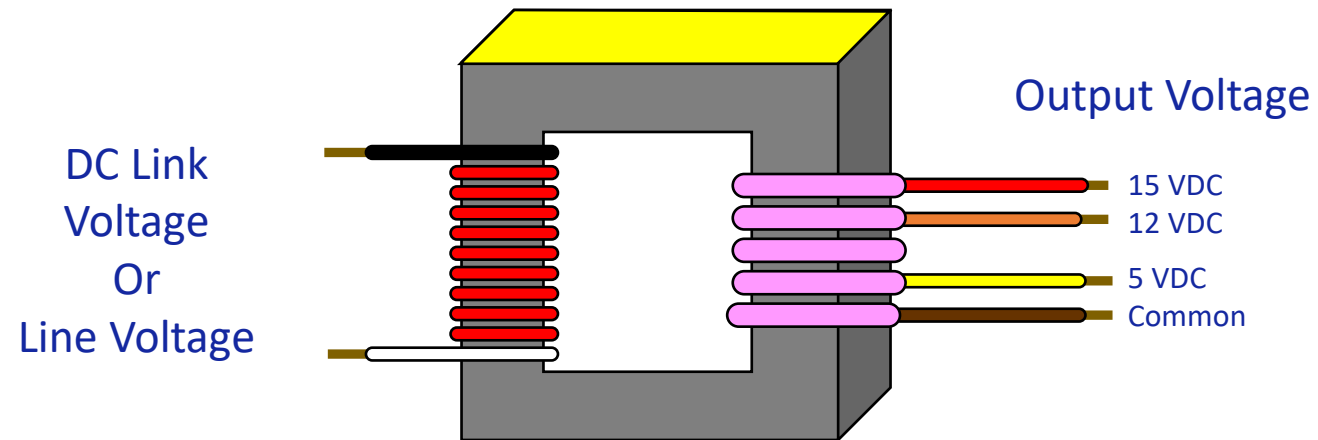
SMPS (Switch Mode Power Supply)

- The SMPS (Switch Mode Power Supply) supplies:
 - DC Bus Voltage (Older IDU's)
 - 15 VDC (BLDC Motor Control Voltage)
 - 12 DVC (EEV,s Communications)
 - 5 VCD (thermistors, Transducers communications)
- These voltages are critical to operation of the system and need to be verified
- There may be multiple SMPS's with in a system
- The SMPS can be supplied with either AC line voltage or DC Bus Voltage



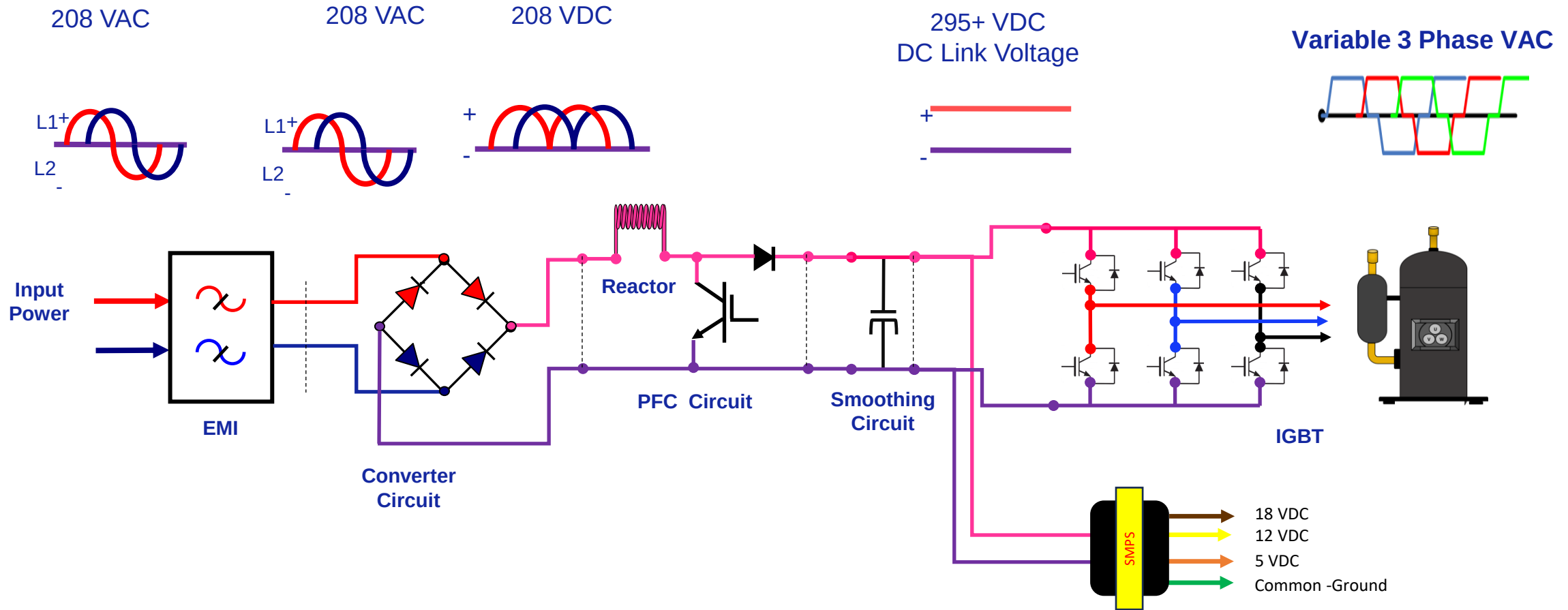


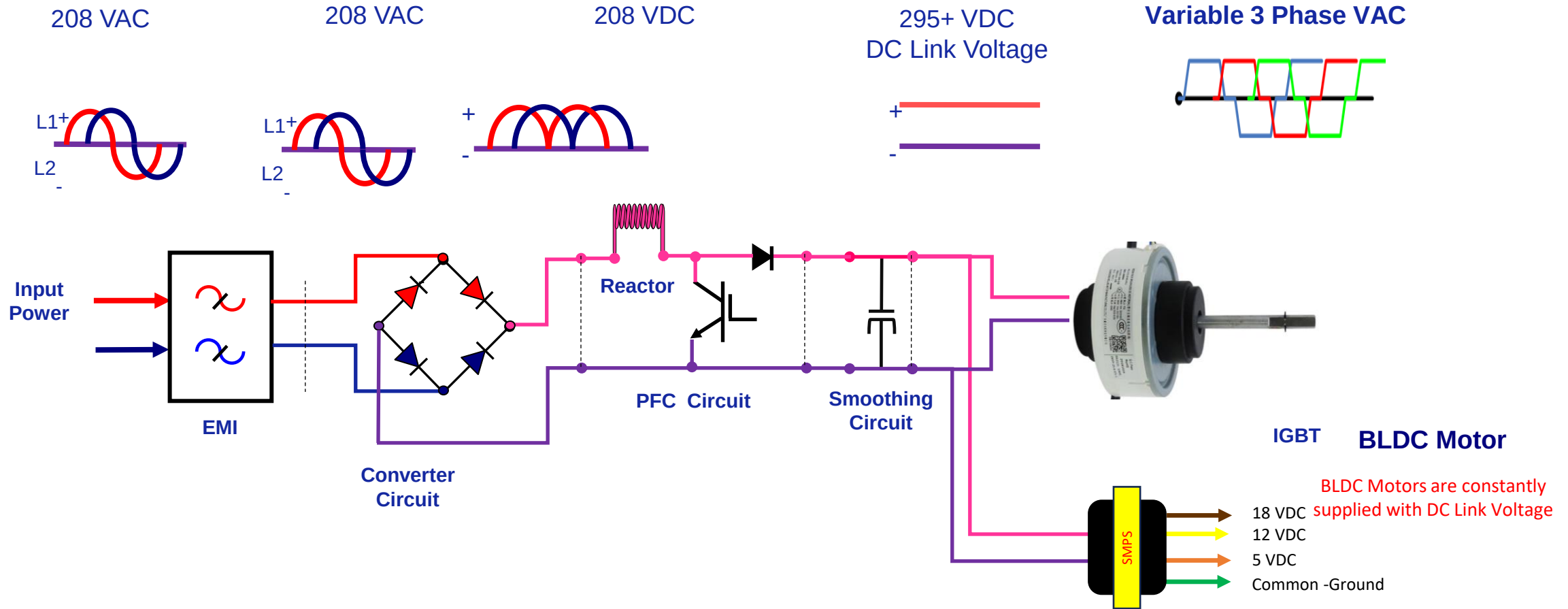
DC Transformer - SMPS



Block diagram of Switch Mode Power Supply

Samsung Inverter

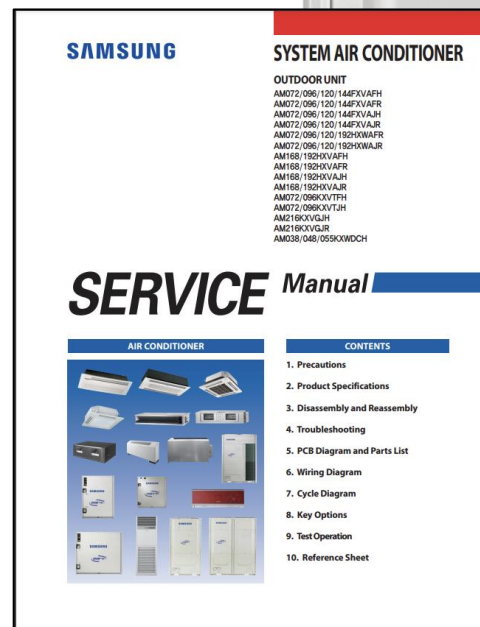


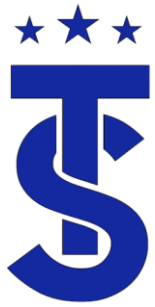




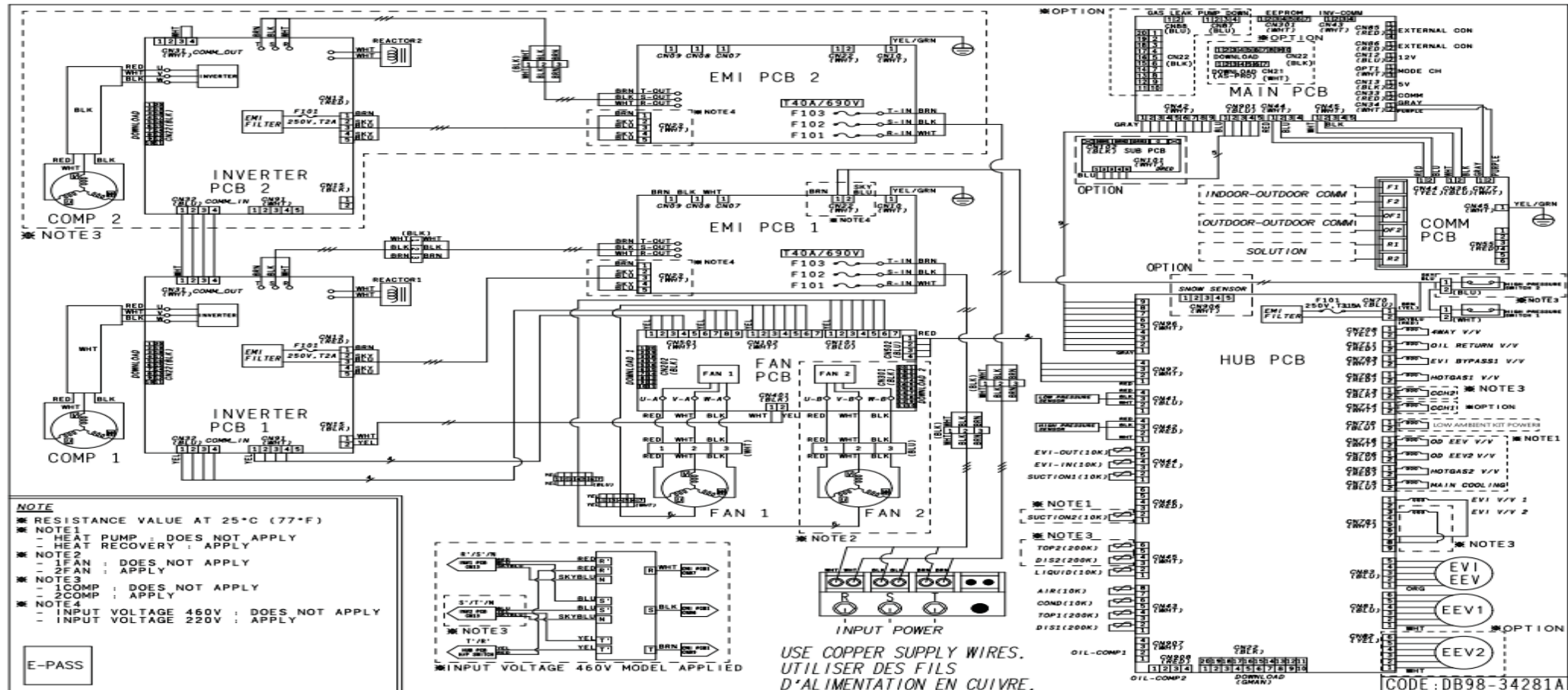
Electrical Components

- Wiring diagram is from AM072-144 HP-HR Unit
- Information was taken from the Service manual .
- All service manuals can be found on Samsung's web site.
- Information is subject to change depending on units.
- This presentation is meant to give the user a basic understanding of the system electrically.





Identify the components of the system





What's inside?

- Electrical Components

- Compressor
- Condenser fan
- Reactor
- High Pressure Switch
- Pressure Transducers
- EEV's
- Reversing Valve
- Solenoid Valves



- Circuit Boards

- Main PBA
- Communications PBA
- Hub PBA
- Inverter 1 & 2 PBA
- Fan PBA
- EMI 1 & 2

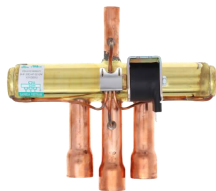


System Components

Component pictures are for demonstration purposes only.
Actual components may look slightly different



Solenoid Valves



Reversing Valve



EEV



High Pressure Sw.



Pressure Transducer



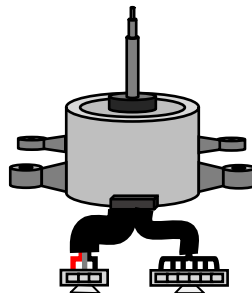
Thermistors



Compressor



Reactor



Condenser fan



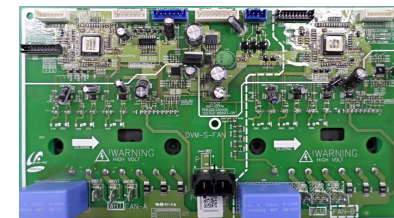
Main PBA



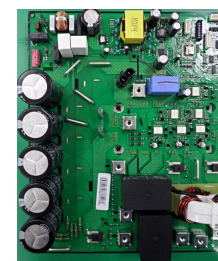
Communications PBA



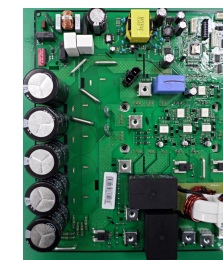
Hub PBA



Fan PBA



Inverter 1 PBA



Inverter 2 PBA



EMI 1



EMI 2

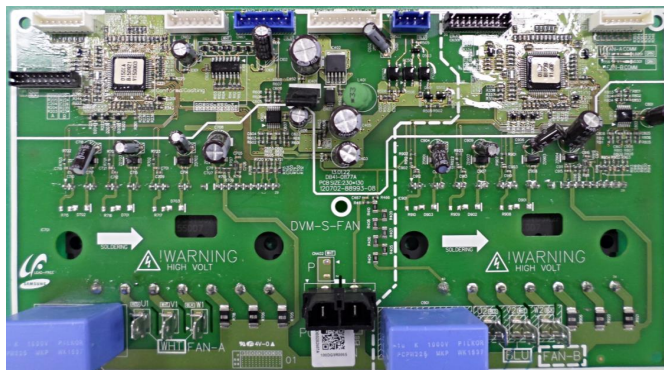


Circuit Boards Basic Overview

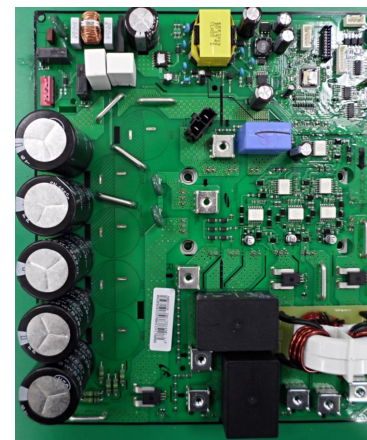
Main PBA



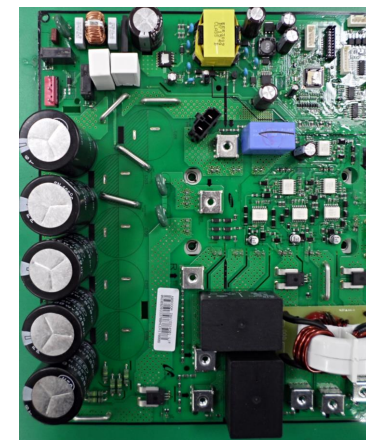
Fan PBA



Inverter # 1 PBA



Inverter # 2 PBA



Hub PBA



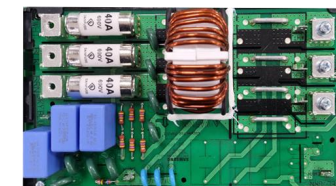
Comm. PBA



EMI # 1



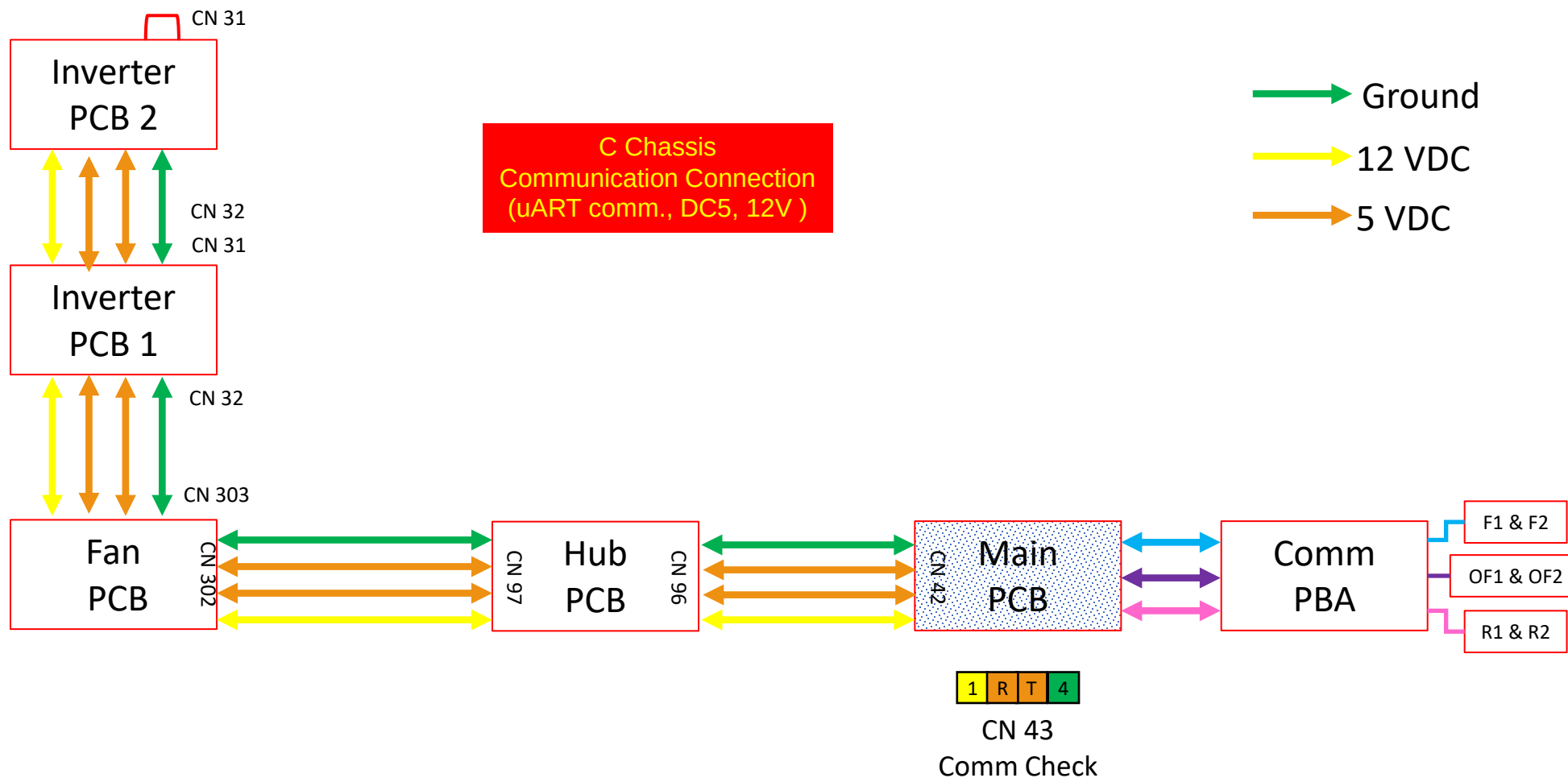
EMI # 2

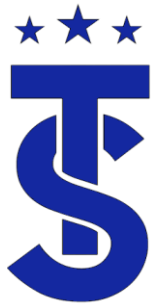




Communications Flow

DVM S





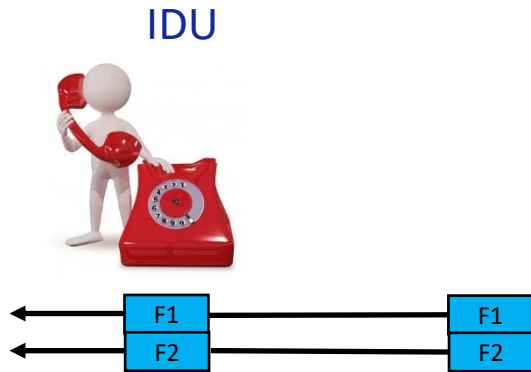
DVM S communications

C Chassis
Communication Connection
(uART comm., DC5, 12V)



CN 43
Comm Check

- The boards communicate using a transmit and receive network
- The main PCB is the source of communications signal, and the other boards communicate with it.
- The Tx and Rx voltages of the micom are DC 5V, and the voltage applied to the communication cable for each PBA is DC 12 V.
- These voltage are consistent through out the system



Mian PBA



Hub
PBA



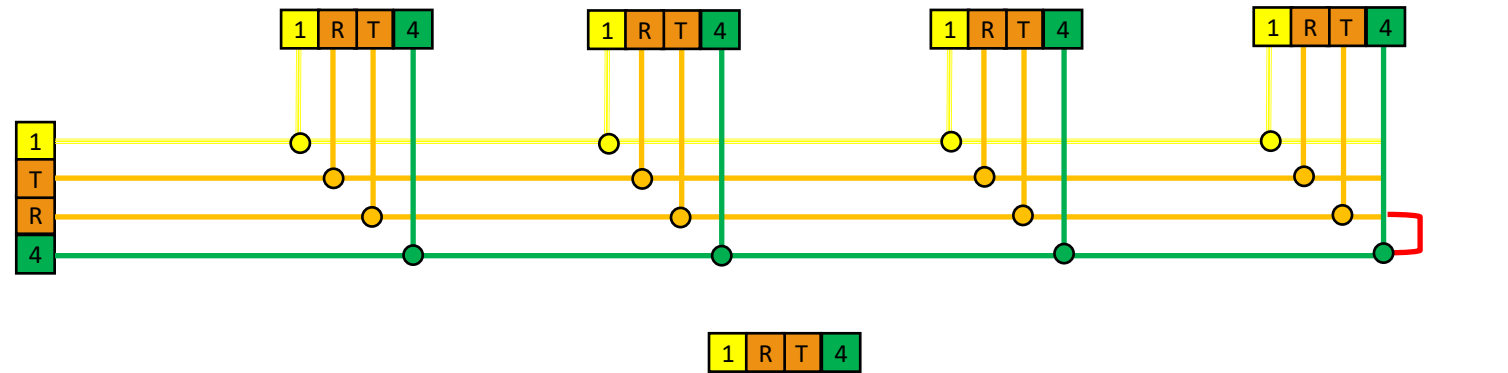
Fan
PBA



Inverter
PBA 1



Inverter
PBA 2

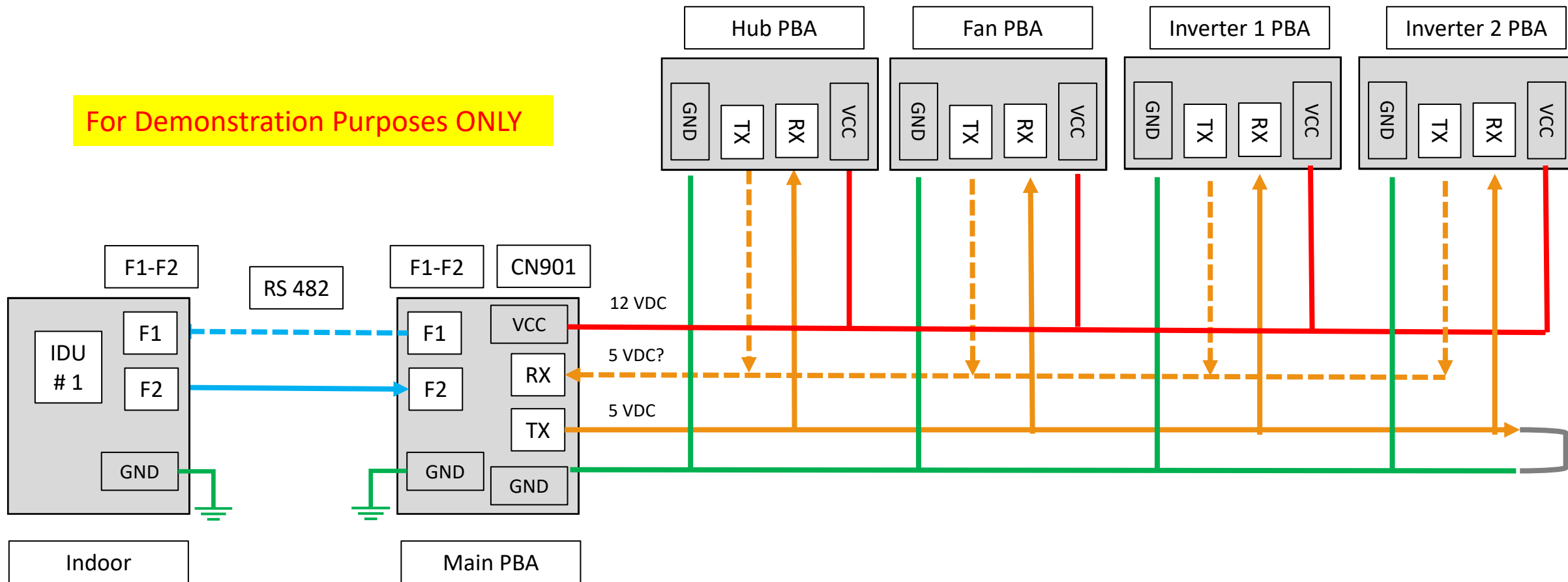


C Chassis
Communication Connection
(comm, DC5, 12V)



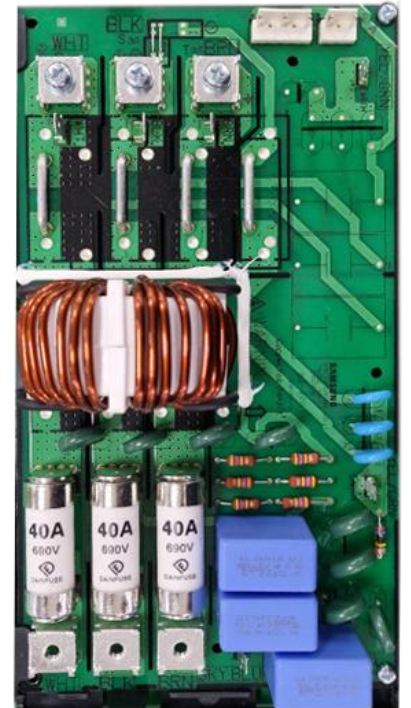
C Chassis Communication

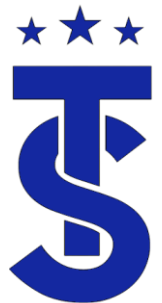
For Demonstration Purposes ONLY



EMI

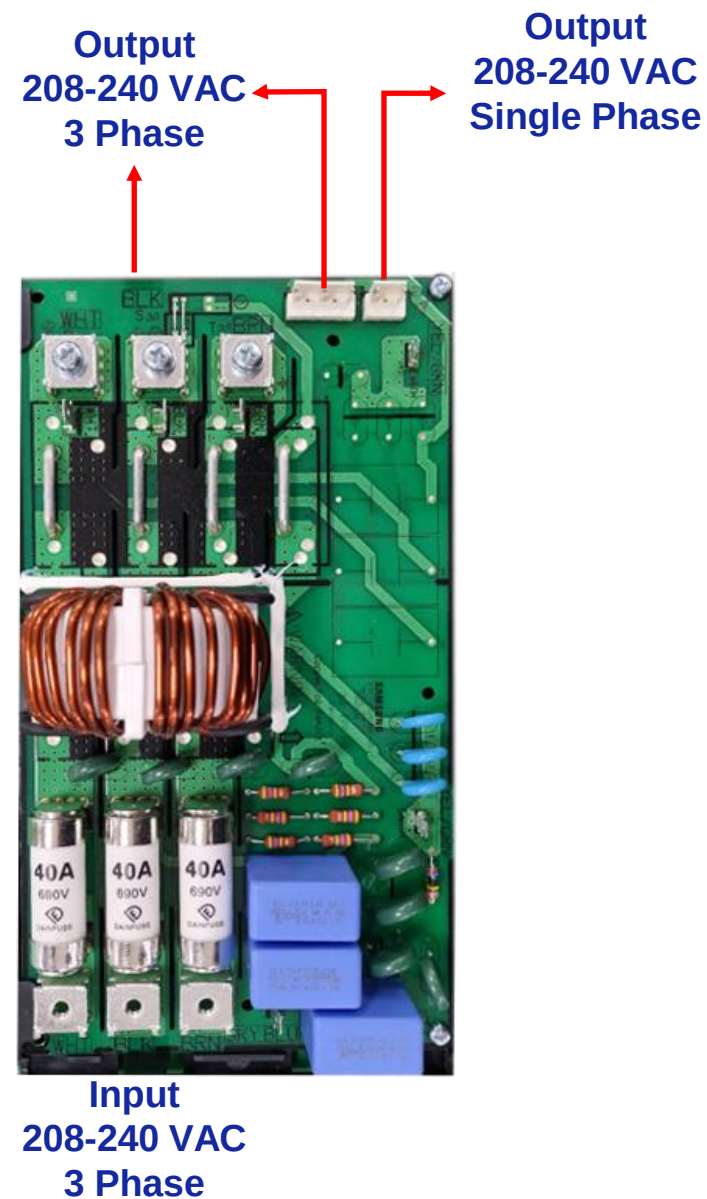
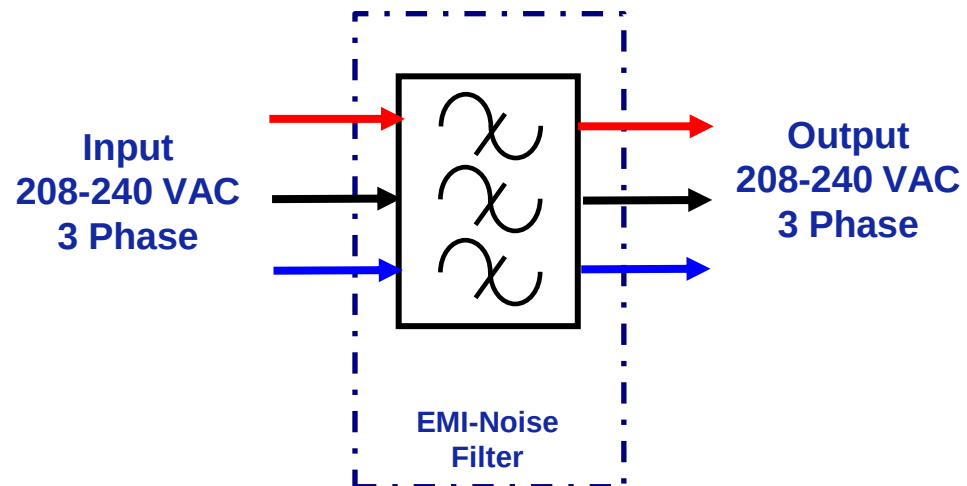
- Purpose:
 - Remove any harmonics and distortion from the power supply.
 - Provide clean power to the rest of the system.

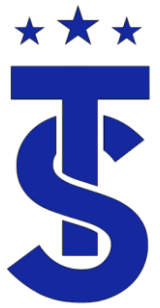




Inverter Circuit Components

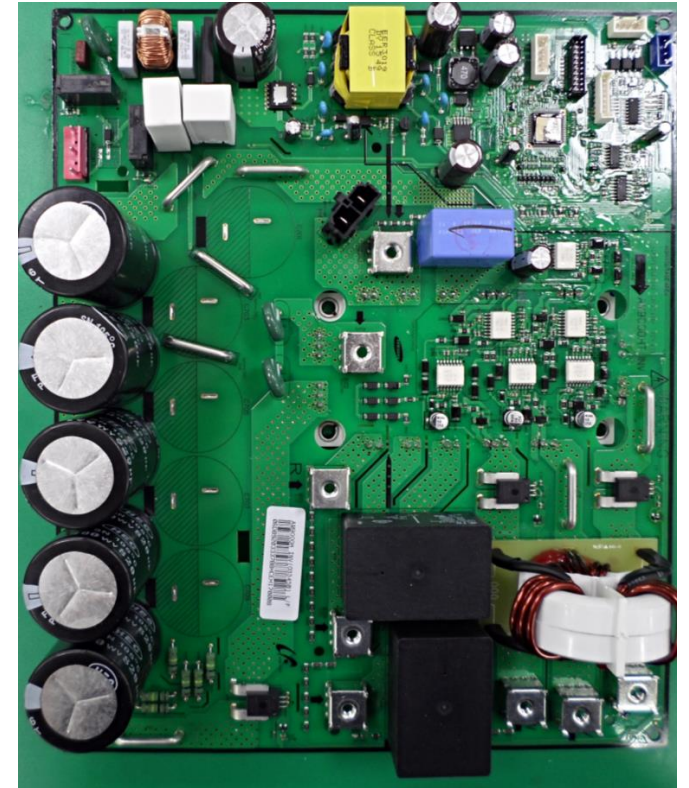
EMI

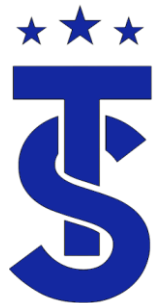




Inverter PCB

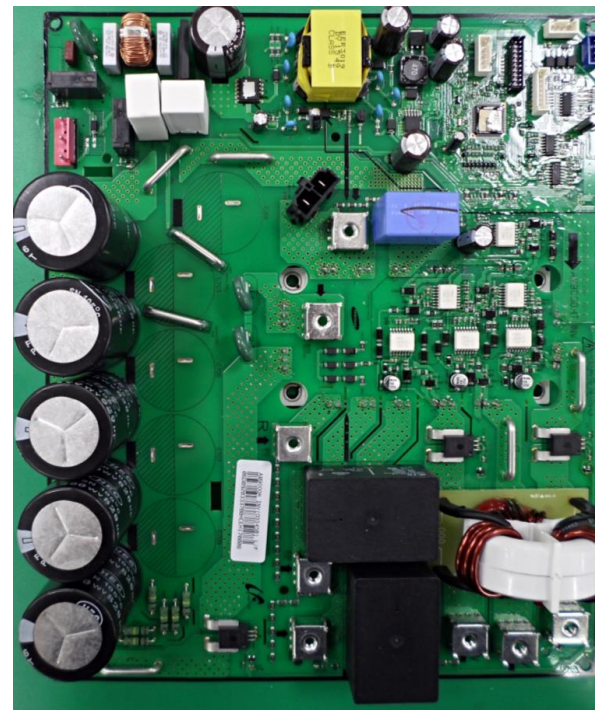
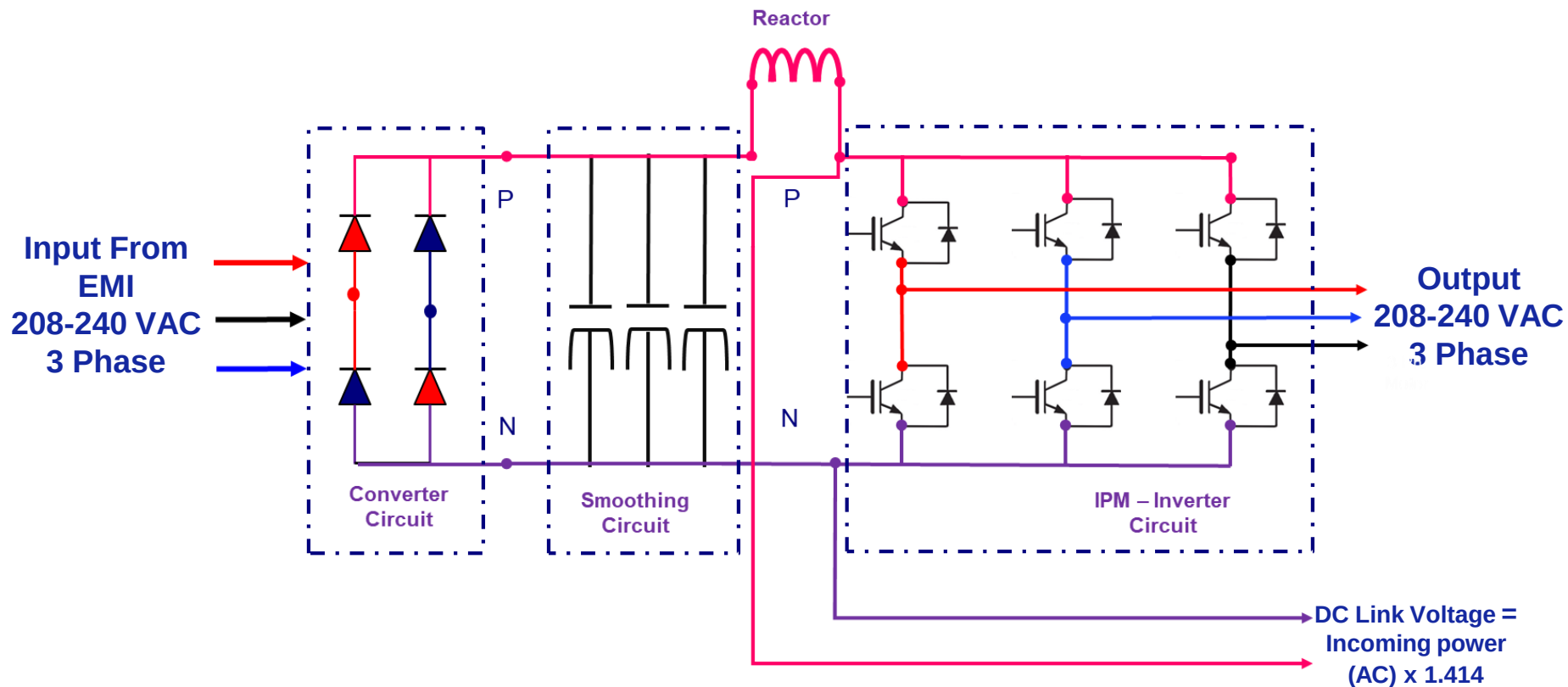
- Purpose:
 - To operate the compressor.
 - Supply DC link Voltage to the fan IPM-IGBT
- How:
 - Convert the power being supplied from the EMI into a High DC Link Voltage.
 - The DC link voltage is supplied to the condenser fan PCB.
 - The DC link voltage is supplied the IPM-IGBT.
 - The IPM-IGBT supplies 3 phase AC power to the compressor.





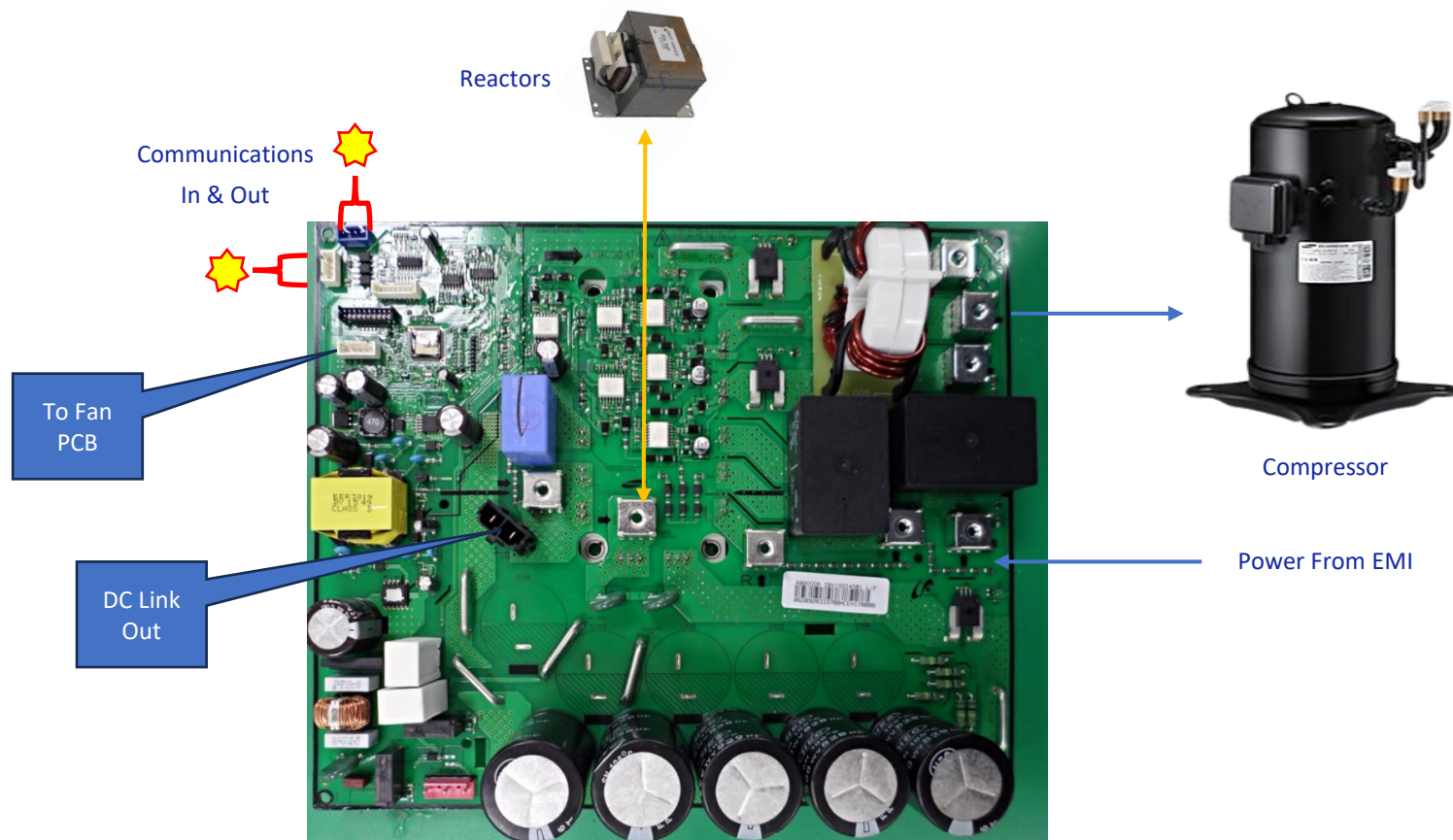
Inverter Circuit Components

Inverter PCB





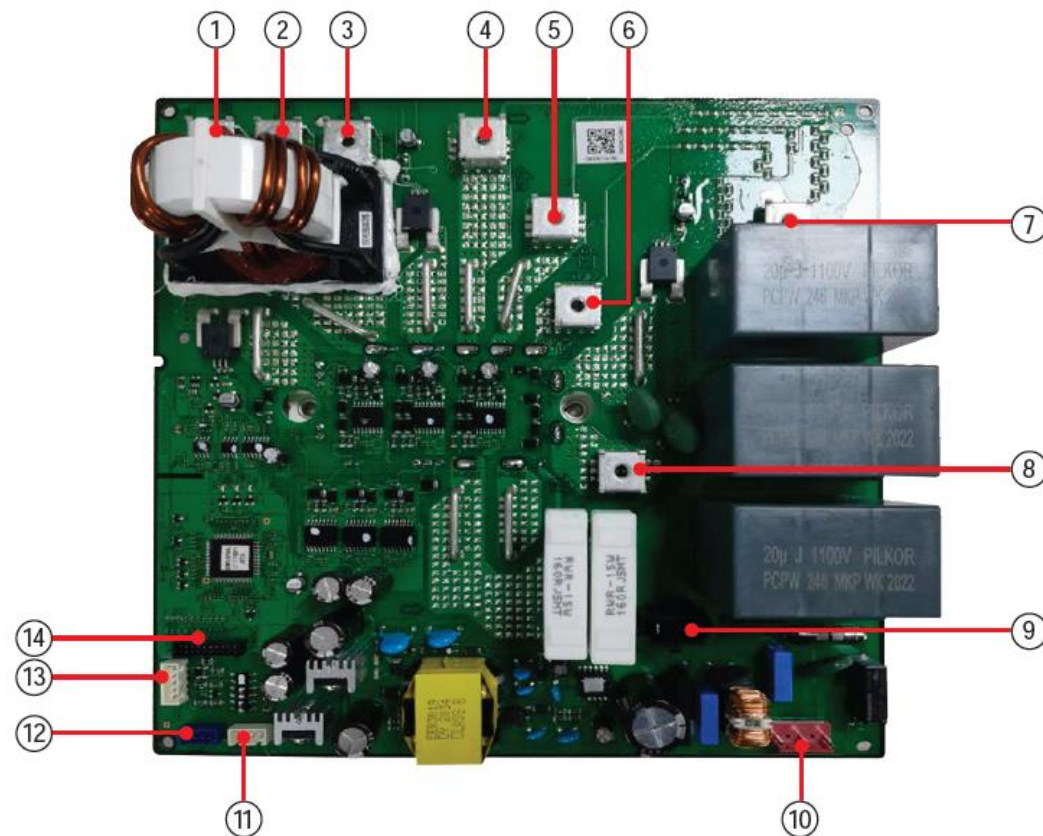
Inverter PCB





Inverter PBA Power In and Out

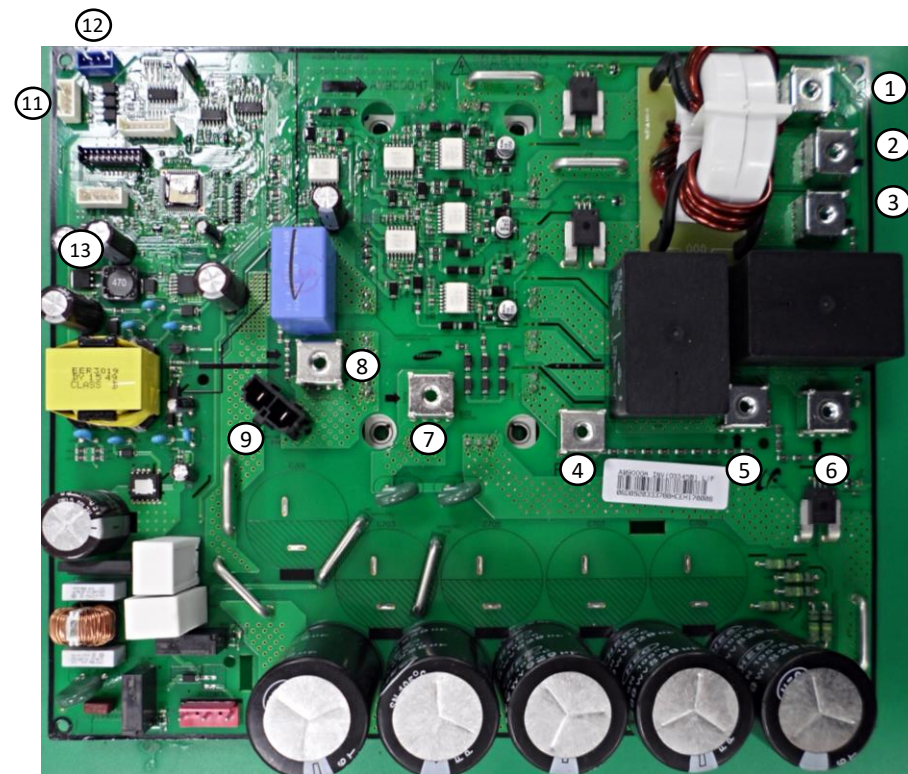
① U-COMPU #1: COMPU	② V-COMP V #1: COMPV	③ W-COMP W #1: COMPW	④ T-INPUTTOP #1: T-IN
⑤ S-INPUTTOP #1: S-IN	⑥ R-INPUTTOP #1: R-IN	⑦ CN705-REACTOR1 #1: REACTOR1	⑧ CN704-REACTOR 2 #1: REACTOR 2
⑨ CN15-FAN POWER #1: DC_LINK #3: GND	⑩ CNP13-AC POWER #1: AC LIVE #3: AC NEUTRAL	⑪ CN31-MAIN COMM 2 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN	⑫ CN32-MAIN COMM1 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN
⑬ CN91-FAN DC #1: 18V #2: GND #3: 5V-FAN #4: AD-SELECT	⑭ CNS22-INV MICOM DOWNLOAD #1: RX_DEBUG #2: TX_DEBUG #3: BOOT #4: JTAG_TDO #5: JTAG_TCLK #6: JTAG_TDI #7: JTAG_TMS #8: JTAG RESET #9: GND #10: VCC #11: VCC #17: GND #18: DAC_CLK #19: DAC_CS #20: DAC_DATA		

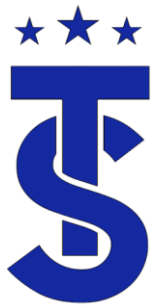




Inverter PBA Power In and Out

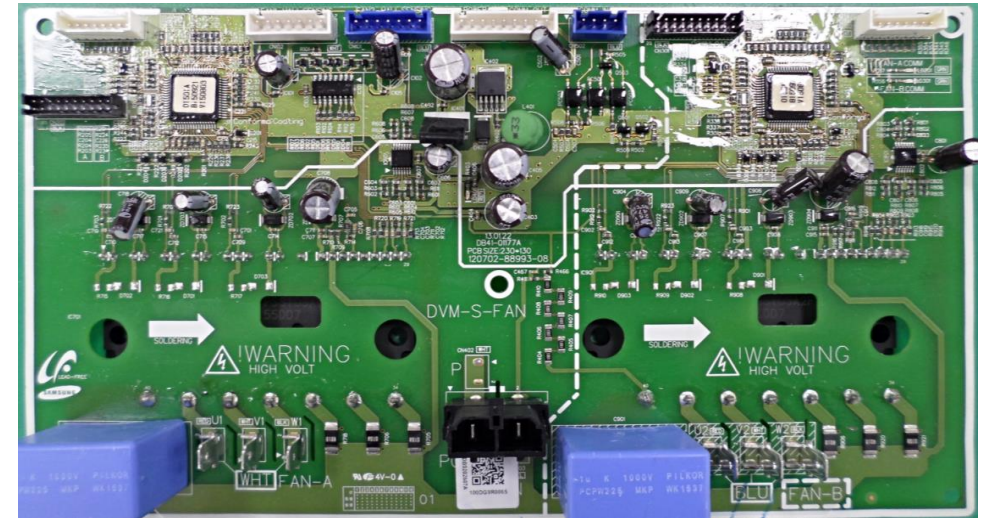
① U-COMPU #1: COMPU	② V-COMPV #1: COMPV	③ W-COMPW #1: COMPW	④ T-INPUTTOP #1: T-IN
⑤ S-INPUTTOP #1: S-IN	⑥ R-INPUTTOP #1: R-IN	⑦ CN705-REACTOR1 #1: REACTOR1	⑧ CN704-REACTOR 2 #1: REACTOR 2
⑨ CN15-FAN POWER #1: DC_LINK #3: GND	⑩ CNP13-AC POWER #1: AC LIVE #3: AC NEUTRAL	⑪ CN31-MAIN COMM 2 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN	⑫ CN32-MAIN COMM1 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN
⑬ CN91-FAN DC #1: 18V #2: GND #3: 5V-FAN #4: AD-SELECT	⑭ CNS22-INV MICOM DOWNLOAD #1: RX_DEBUG #2: TX_DEBUG #3: BOOT #4: JTAG_TDO #5: JTAG_TCLK #6: JTAG_TDI #7: JTAG_TMS #8: JTAG RESET #9: GND #10: VCC #11: VCC #17: GND #18: DAC_CLK #19: DAC_CS #20: DAC_DATA		





Fan PCB

- Purpose:
 - Operate the condenser fans.
- How:
 - The DC link voltage that is supplied by Inverter # 1 PCB is supplied the IPM-IGBT circuit.
 - The IPM-IGBT supplies 3 phase AC power to the fan.
 - If there are 2 fans, then there are 2 separate IPM-IGBT circuits.

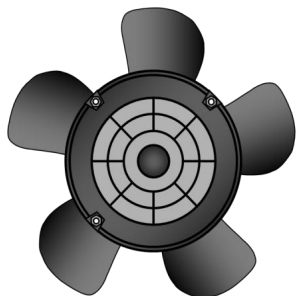
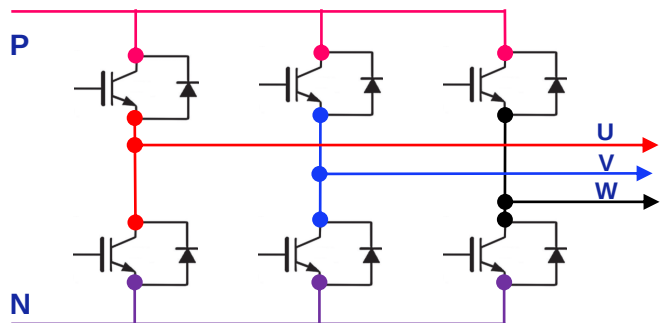




Inverter Circuit Components

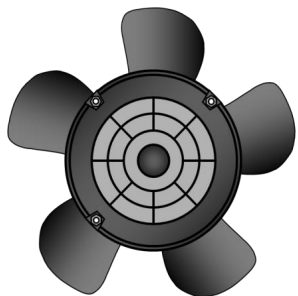
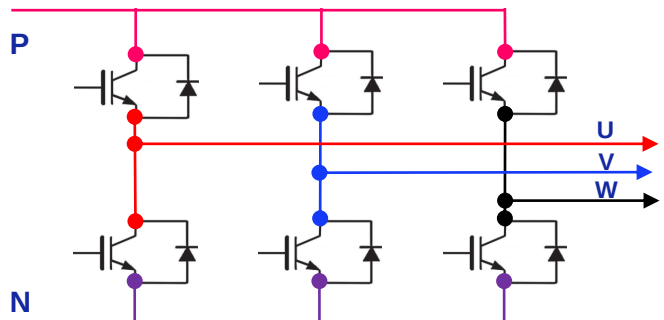
Fan PCB

Input
DC Link
Voltage
From Inverter
PCB

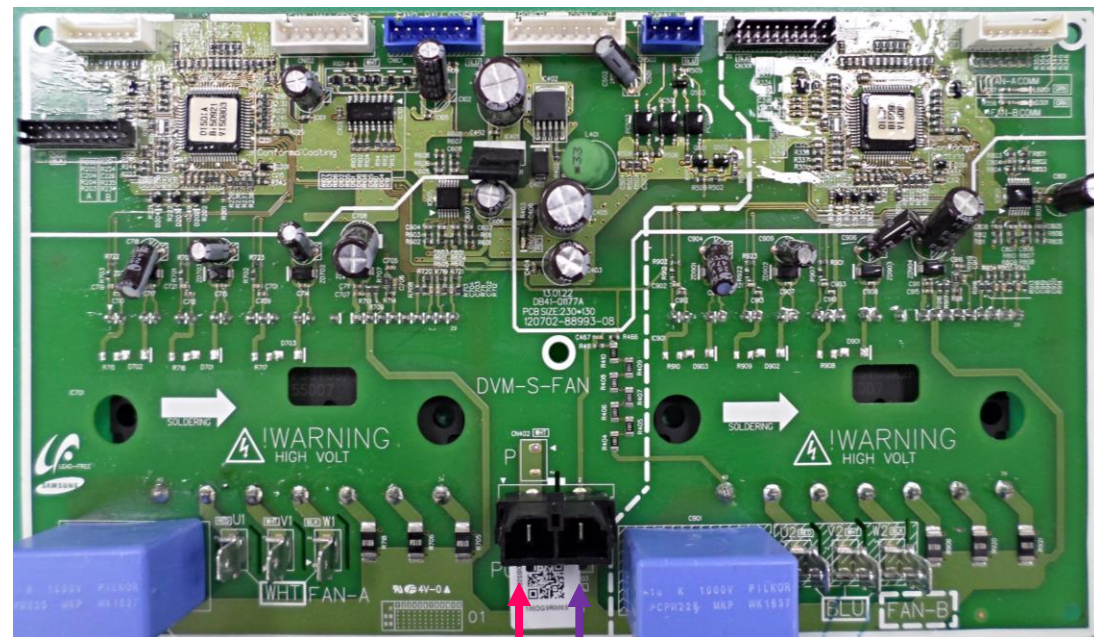


3 Ph VAC Motor

Input
DC Link
Voltage
From Inverter PCB



3 Ph VAC Motor



Output 1
208-240 VAC
3 Phase

Input
DC Link
Voltage
From Inverter
PCB

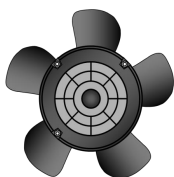
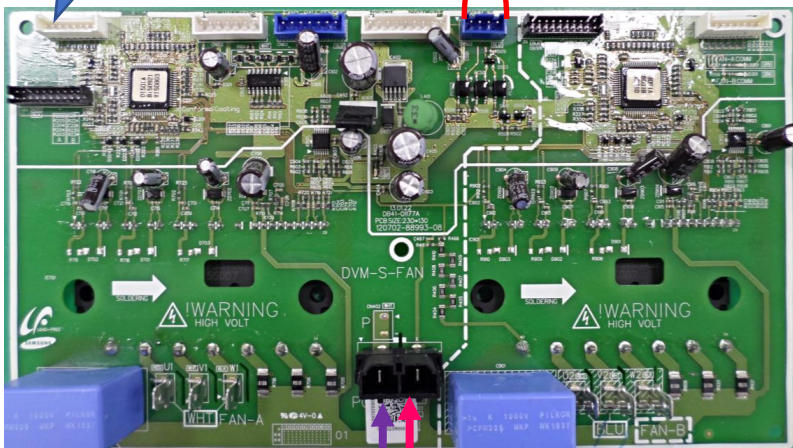
Output 2
208-240 VAC
3 Phase



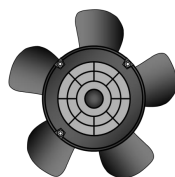
Fan PBA Input and Outputs

Hall Sensor Connectors

Power In and Comm Out



Input
DC Link
Voltage
From Inverter
PCB



① CN102-FA1 HALL SENSING #1 : HALL-U #2 : 5V #3 : HALL-V #4 : GND #5 : HALL-W #6 : MOTOR-TEMP #7 : GND	② CN202-DOWNLOAD1 #1 : RX-DEBUG #2 : TX-DEBUG #3 : BOOT #4 : TDO #5 : TCK #6 : TDI #7 : TMS #9 : GND #10 : 5V	③ CN502-COMMUNICATION #1 : 12V-MAIN #2 : INV SMPS RELAY-MAIN #3 : COMM-MAIN #4 : GND-MAIN	④ CN501-COMMUNICATION #1 : 18V-INV #2 : GND-INV #4 : GND-INV #6 : 12V-MAIN #7 : INV SMPS RELAY-INV #8 : COMM-INV #9 : GND-INV
⑤ CN101-FAN2 HALL SENSING #1 : HALL-U #2 : 5V #3 : HALL-V #4 : GND #5 : HALL-W #6 : MOTOR-TEMP #7 : GND	⑥ CN301-DOWNLOAD2 #1 : RX-DEBUG #2 : TX-DEBUG #3 : BOOT #4 : TDO #5 : TCK #6 : TDI #7 : TMS #9 : GND #10 : 5V	⑦ U1-V1-W1 #1 : FAN1-U #2 : FAN1-V #3 : FAN1-W	⑧ CN401-POWER #1 : DC Link #2 : GND
⑨ U2-V2-W2 #1 : FAN2-U #2 : FAN2-V #3 : FAN2-W			



The Hub PCB

- Purpose:
 - Operate high and low voltage loads.
 - High voltage loads include solenoids and the reversing valve.
 - Low voltage loads include thermistors, transducers and EEV's.
 - Supply low voltage power the main board.
- How:
 - Power is supplied from the EMI through the high-pressure switches.
 - The SMPS creates 5 Vdc & 12 Vdc low voltage supplies.
 - The main PBA determines when each load is energized.





The Hub PCB

- Power In:
 - 208 – 240 VAC
- Power Out:
 - 5 VDC
 - Thermistors
 - Pressure Transducers
 - 12 VDC
 - EEV Coils
 - Power to the Main Board



Input
208-240 VAC

Output
12 VDC to Main
PBA

Output
5 VDC Devices

Output
12 VDC Devices

Output
208-240 VAC



Hub PBA Inputs & Outputs

CN 96

Main PBA connection
(uART comm., DC5, 12V
supply to Main PBA)

 C Chassis
Communications

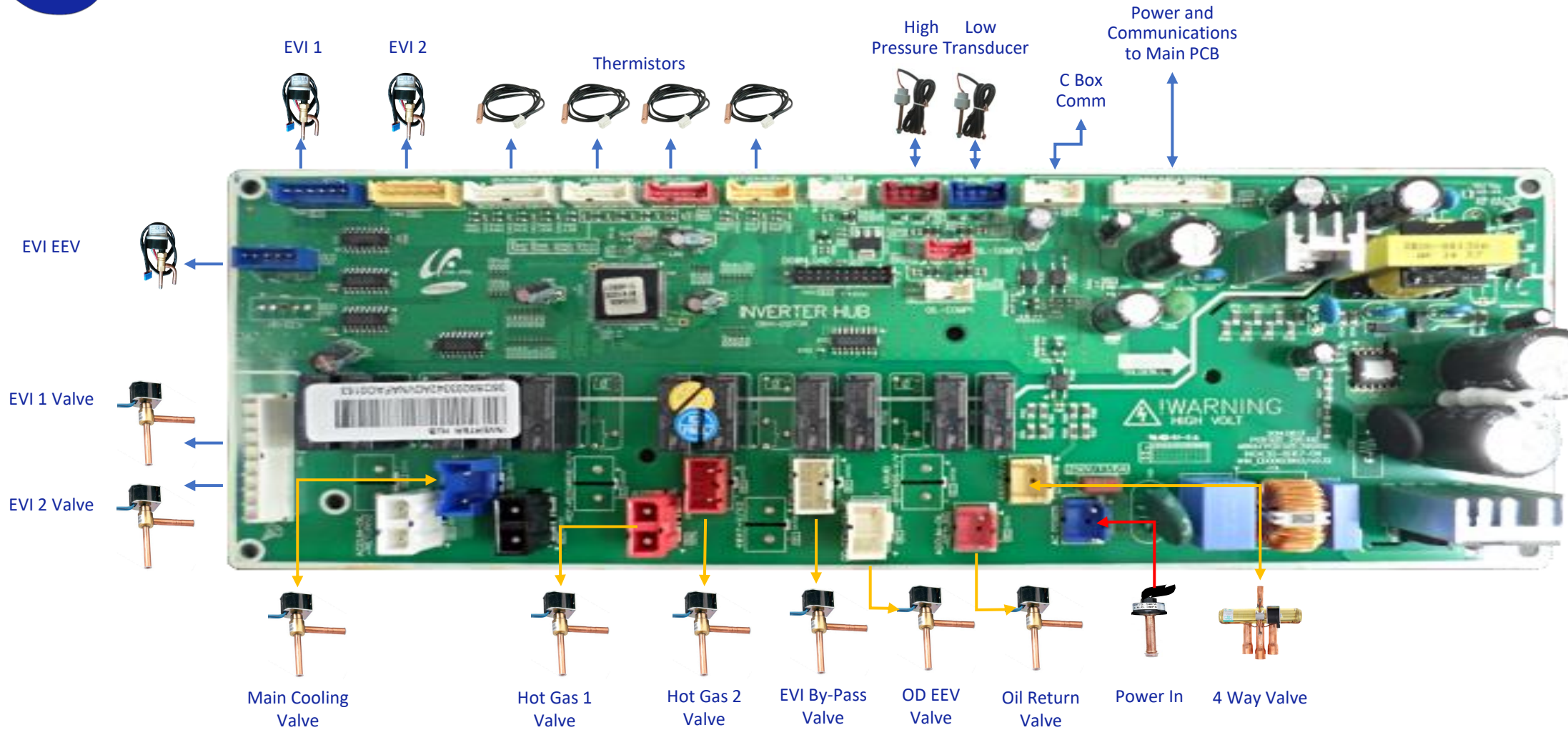


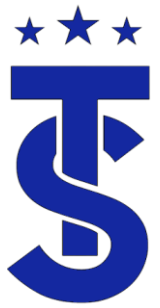
① CN83-EV1 EEV #1: EEV3_A_OUT #2: EEV3_B_OUT #3: EEV3_A'_OUT #4: EEV3_B'_OUT #5: 12V	② CN81-EEV1 #1: EEV1_B'_OUT #2: EEV1_A'_OUT #3: EEV1_B_OUT #4: EEV1_A_OUT #5: 12V #6: 12V	③ CN82-EEV2 #1: EEV2_B'_OUT #2: EEV2_A'_OUT #3: EEV2_B_OUT #4: EEV2_A_OUT #5: 12V #6: 12V	④ CN43-TEMP.SENSOR #1: COMP1 DISACHRGE #2: COMP1 DISCHARGE #3: COMP1 TOP #4: COMP1 TOP1 #5: COND OUT #6: COND OUT #7: OUTDOOR TEMP. #8: OUTDOOR TEMP.
⑤ CN45-TEMP.SENSOR #1: LIQUID #2: LIQUID #3: COMP2 DISCHARGE #4: COMP2 DISCHARGE #5: COMP2 TOP #6: COMP2 TOP	⑥ CN46-SUCT #1: SUCTION 2 #2: SUCTION 2 #3: GND #4: GND #6: GND	⑦ CN44-TEMP.SENSOR #1: SUCTION 1 #2: SUCTION 1 #3: EVI INLET #4: ENI INLET #5: ENI OUT #6: EVI OUT	⑧ CN906-SNOW SENSOR #1: 12V #3: GND #4: SNOW_SENSOR #5: PSD_POWER
⑨ CN42-HIGH PRESSURE SENSOR #1: HIGH PRESSURE SENSOR #3: GND #4: VCC	⑩ CN41-LOW PRESSURE SENSOR #2: LOW PRESSURE SENSOR #3: GND #4: VCC	⑪ CN97-INV COMM #1: 12V #2: INV_SMPS_RELAY #3: COMM OUT #4: GND	⑫ CN96-MAIN-HUB COMM. #1: 12V #2: INV_SMPS_RELAY #3: COMM-MAIN #4: GND #5: HIGH-PRESSURE-SENSOR #6: LOW-PRESSURE-SENSOR #7: ZERO-CROSSING #8: GND #9: VCC

■ DC

① CN714-CCH1 #1: CCH1 #2: CCH1	② CN715-MAIN-COOLING #1: MAIN-COOLING #2: MAIN-COOLING	③ CN713-CCH2 #1: CCH2 #2: CCH2	④ CN704-HOTGAS-VALVE1 #1: HOTGAS BYPASS1 #2: HOTGAS BYPASS1
⑤ CN705-HOTGAS-BYPASS2 #1: HOTGAS BYPASS2 #2: HOTGAS BYPASS2	⑥ CN703-EVI-BYPASS #1: EVI BYPASS1 #2: EVI BYPASS1	⑦ CN716-OD-EEV-VALVE #1: OD EEV VALVE #2: OD EEV VALVE	⑧ CN711-OIL-RETURN-VALVE #1: ACCUM OIL RETURN VALVE #2: ACCUM OIL RETURN VALVE
⑨ CN708-4-WAY-VALVE #1: 4-WAY VALVE #2: 4-WAY VALVE	⑩ CN70-AC POWER INPUT #1: AC LIVE #2: AC NEUTRAL	⑪ CN701 EVI VALVE 1,2 #1: EVI VALVE 1 #3: EVI VALVE 2 #7: EVI VALVE 1 #8: EVI VALVE 2 #9: AC NEUTRAL	

Hub PBA





The Main PCB

- Purpose:
 - Is the brains of the system.
 - Is the user interface.
 - Is where communications start.
- How:
 - Contains the EPROM and Micom.
 - The SMPS creates 12 & 5 VDC for communications
 - Creates the communication signals for indoor units, outdoor units and upper-level controls.





Main PBA Inputs & Outputs



C Chassis
Communications

① CN22-DOWNLOAD #1 : RX-DOWN #2 : TX-DOWN #3 : N-TRST #4 : TDO #5 : TCK #6 : TDI #7 : TMS #8 : #9 : GND #10 : VCC	② CN43-COMMTEST #1 : 12V #2 : INVERTER-INRUSH-OUT #3 : INVERTER-COMM #4 : GND	③ CN301-EEPROM #1 : GND #2 : #3 : VCC #4 : EEPROM-SELECT #5 : EEPROM-SO #6 : EEPROM-SI #7 : EEPROM-CLOCK	④ CN42 - HUB COMM #1 : 12V #2 : INVERTER-INRUSH-OUT #3 : INVERTER-COMM #4 : GND #5 : HIGH-PRESSURE-SENSOR #6 : LOW-PRESSURE-SENSOR #7 : ZERO-CROSSING #8 : GND #9 : VCC
⑤ CN901-DRED #1 : KEY3 #2 : GRID #3 : KEY4 #4 : GND #5 : VCC	⑥ OPT1 -MODE SELECTOR #1 : KEY3 #2 : GRID #3 : KEY4	⑦ CN85-CONDITION CHECK #1 : 12V #2 : ERROR-CHECK-OUT #3 : 12V #4 : COMP-CHECK-OUT	⑧ CN86-EXTERNAL CONTROL #1 : CONTROL #2 : GND
⑨ CN12 - 12V POWER #1 : 12V #2 : GND	⑩ CN45 - OUTDOOR COMM #1 : COM-C #2 : COM-D #3 : #4 : 12V #5 : GND	⑪ CN44 - INDOOR COMM #1 : COM-A #2 : COM-B #3 : 5V #4 : AGND	⑫ CN34- UNUSED COMM #1 : COM-E #2 : COM-F
⑬ CN33-INDOOR COMM (EXTRA) #2 : COM-B	⑭ CN13-5V POWER #1 : COM-A #2 : COM-B	⑮ CN88-GAS LEAK #1 : VCC #2 : COM-B	⑯ CN87-PUMP DOWN #1 : 12V #2 : PUMP DOWN START OUT #3 : 12V #4 : PUMP DOWN END OUT





Sub Communications Board

- Purpose:
 - To communicate with the indoor units.
 - F1 – F2 Indoor unit communications .5+/-VDC
 - Communicate with other outdoor units in a modular systems.
 - OF1 –OF2 Outdoor unit communications
 - Communicate with upper-level controls.
 - R1 – R2 Upper-level controls communications
- How:



No.	Function
1	Earth connection
2	Main PBA connection (OF1/OF2)
3	Main PBA connection (F1/F2)
4	Main PBA connection (R1/R2)
5	Indoor-Outdoor communication line connection (F1/F2)
6	Outdoor-Outdoor communication line connection (OF1/OF2)
7	Solution-Outdoor communication line connection (R1/R2)
8	N/A (Manufacture only)



3 Phase Main PBA

C Chassis
Communication Connection
(uART comm., DC5, 12V)

Communications
Check



CN 43
Comm Check



Main PBA

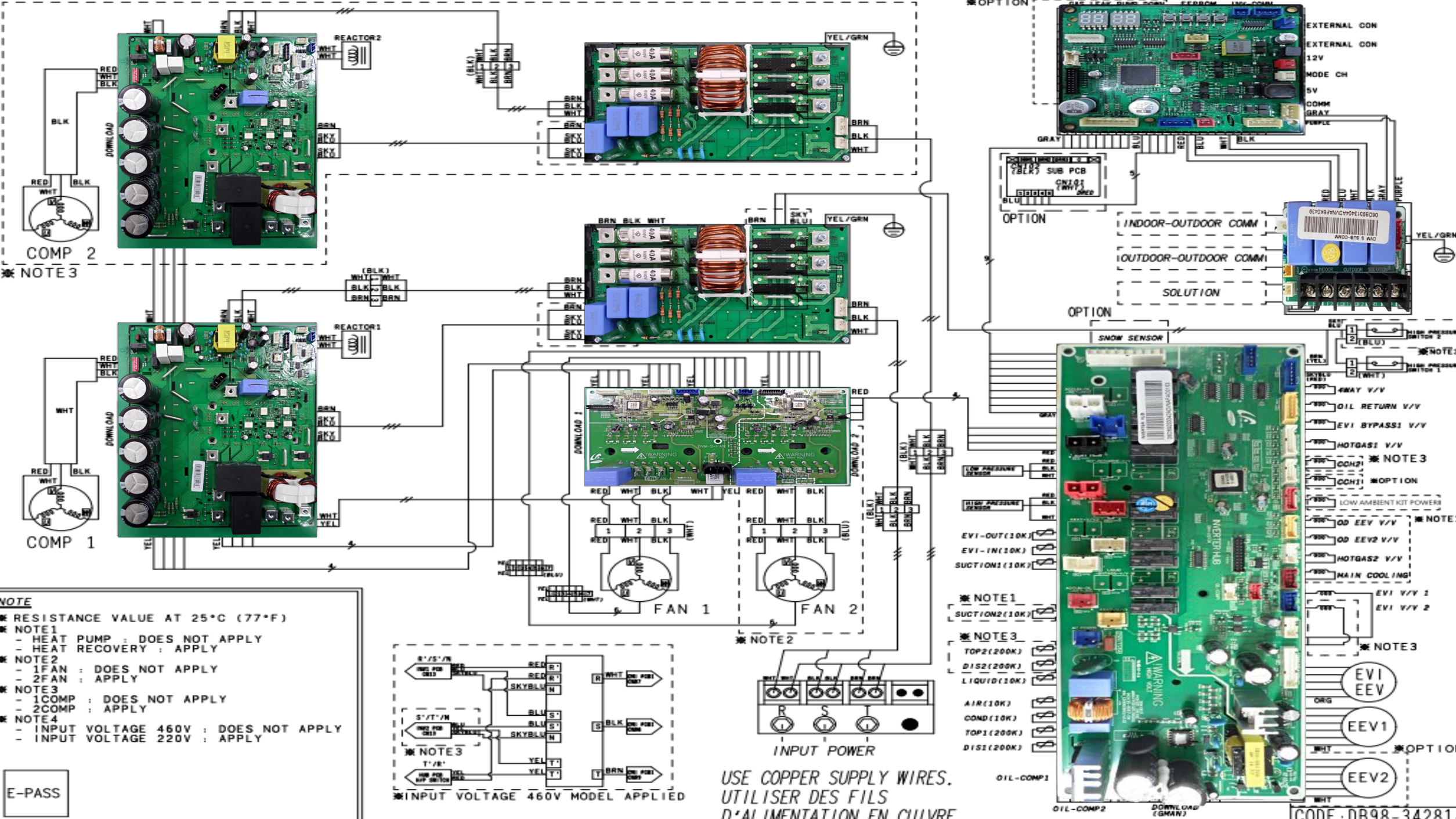
Power From and
Communications
to Hub PCB



Comm. PBA

Indoor Comm

Outdoor Comm

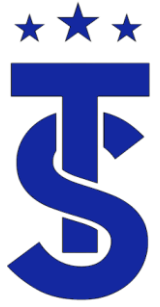


NOTE
* RESISTANCE VALUE AT 25°C (77°F)
* NOTE1
- HEAT PUMP : DOES NOT APPLY
- HEAT RECOVERY : APPLY
* NOTE2
- 1FAN : DOES NOT APPLY
- 2FAN : APPLY
* NOTE3
- 1COMP : DOES NOT APPLY
- 2COMP : APPLY
* NOTE4
- INPUT VOLTAGE 460V : DOES NOT APPLY
- INPUT VOLTAGE 220V : APPLY

E-PASS

USE COPPER SUPPLY WIRES.
UTILISER DES FILS
D'ALIMENTATION EN CUIVRE.

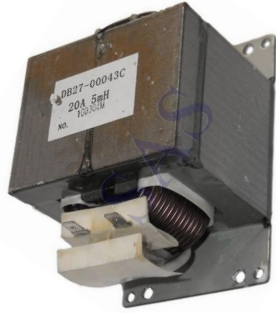
CODE:DB98-34281A



Electrical Components Basic Overview



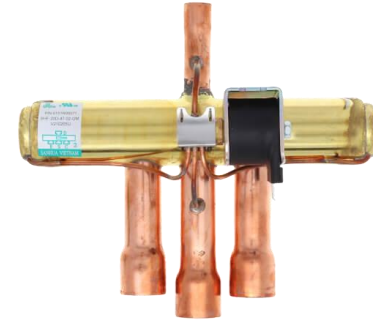
EEV



Reactor



High Pressure Sw.



Reversing Valve



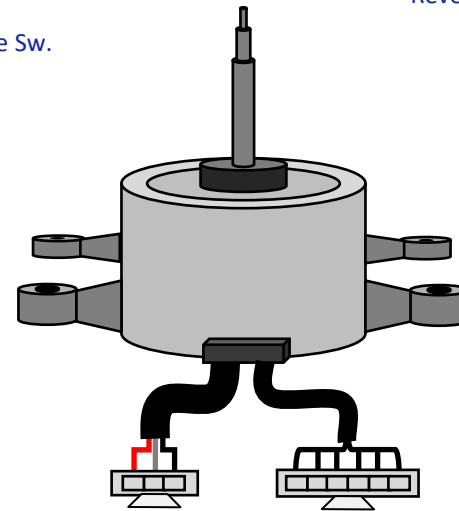
Thermistors



Solenoid Valves



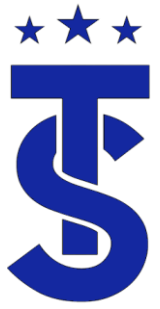
Pressure Transducer



Condenser fan



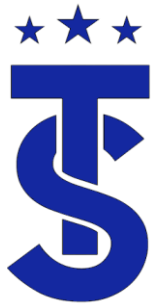
Compressor



The Compressor

- Purpose:
 - Circulate refrigerant through the system.
- How:
 - The compressor speed can be varied depending on the voltage and frequency being supplied from the Inverter PCB.
 - The compressor **MUST** be supplied with a balanced voltage between all 3 phases.

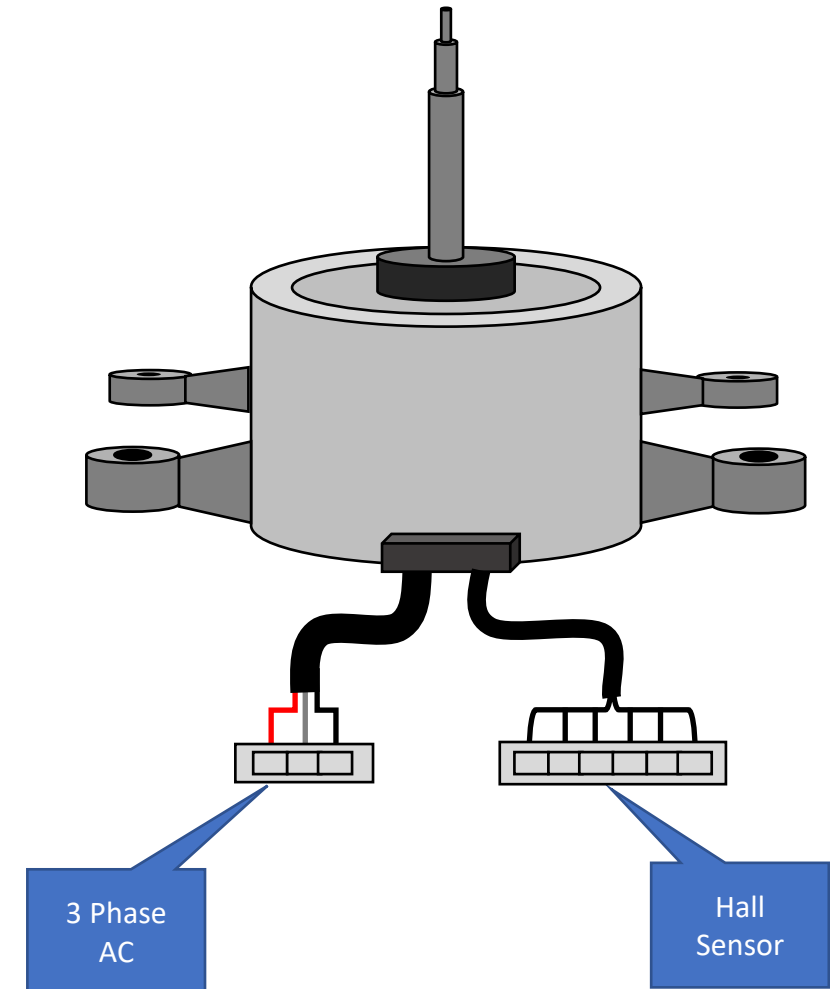


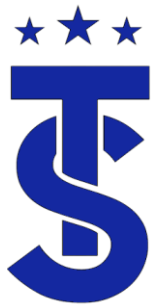


Outdoor Fan Motor

Conventional DVM S

- Purpose:
 - Move air across the outdoor coil.
- How:
 - The motor is supplied with 3 phase AC voltage from the Fan IPM.
 - The hall sensor circuit provides feed back to the fan PCB.

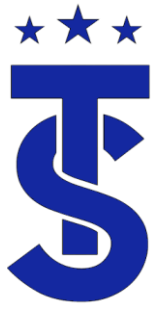




Reactor

- Purpose:
 - Increase the power factor of the system in conjunction with the power factor correction circuit.
- How:
 - The reactor is a coil that when an electrical current is passed through it, it creates a magnetic field.
 - It is placed in series with the Positive power supply lines of the DC Bus voltage

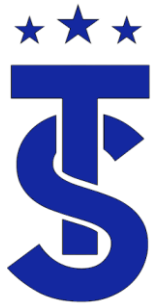




Solenoid Valve

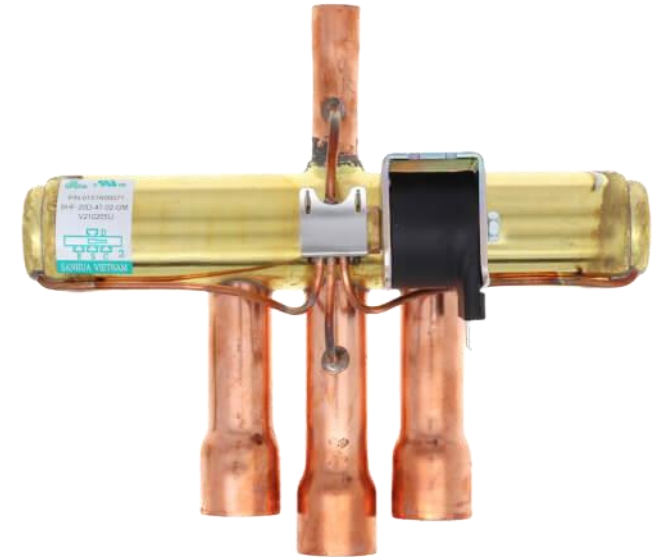
- Purpose:
 - Control the flow or refrigerant through the system.
- How.:
 - When the coil is supplied with 208-240 VAC the valve opens and allows flow through the device.





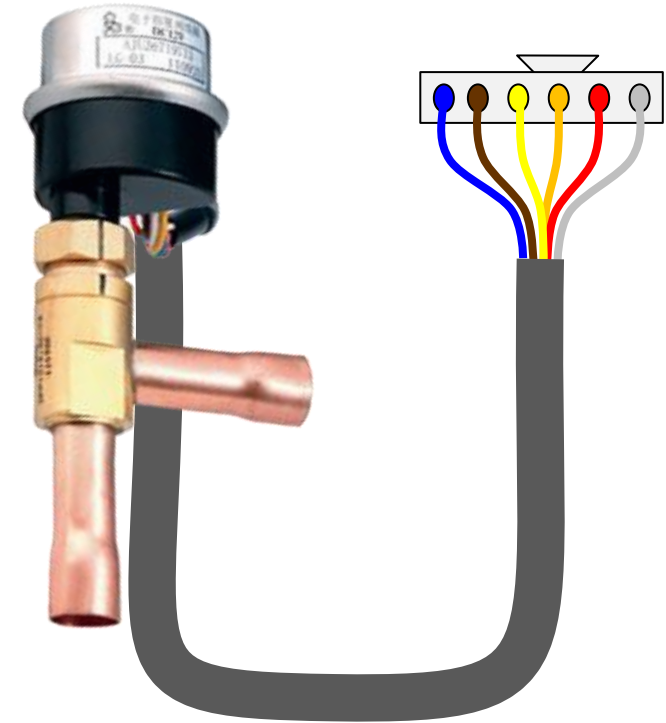
Reversing Valve

- Purpose:
 - The reversing valve directs hot gas to either the outdoor coil in cooling mode or the indoor units in the heating mode.
- How:
 - The solenoid valve is supplied with 208-240 VAC the heating mode from the Main PCB.
 - In order for the valve to switch there needs to be a pressure differential between the high and low side of the refrigeration system.





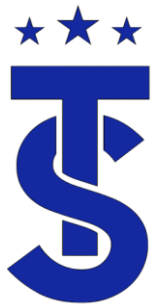
- Purpose:
 - Maintain a targeted temperature differential across a component.
- How:
 - The EEV coil is supplied with a rotational 12VDC pulsating signal to open and close the valve
 - The Main PCB changes the direction of the signal to open and close the valve based on system operation
 - If the temperature differential is high the valve opens
 - If the temperature differential is low the valve closes



Thermistors

- Purpose:
 - To measure temperatures through out the system.
 - The system uses 10 K Ohm and 200 K Ohm NTC thermistors.
- How:
 - Thermistor are supplied with 5 DVC.
 - The resistance of the thermistor changes as its temperature changed.
 - As temperature goes up its resistance goes down.



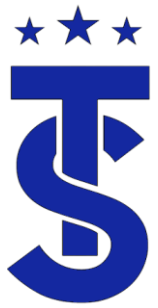


Pressure Transducers

- Purpose:
 - To measure High Side and Low Side pressures.
- How:
 - They are supplied with 5 VDC between Red and Black wires.
 - The Voltage out on the white wire varies as the pressure changes.
 - The system converts the voltage out of the white wire into a pressure.
 - As Pressure increases so does the voltage output.



For reference
purposes only!

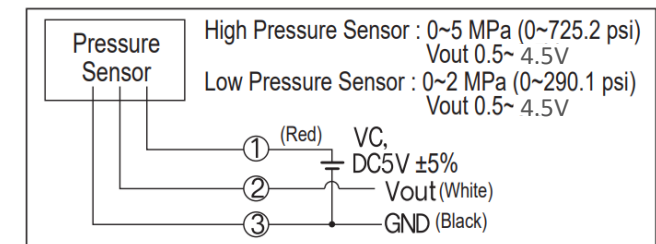
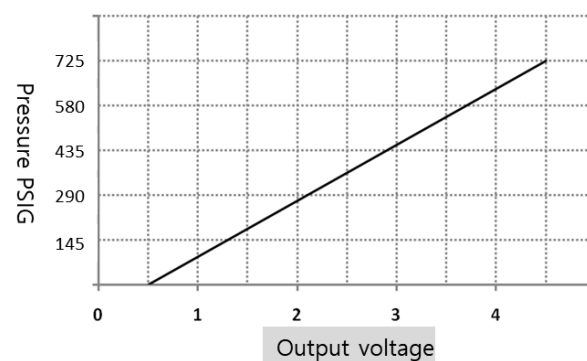
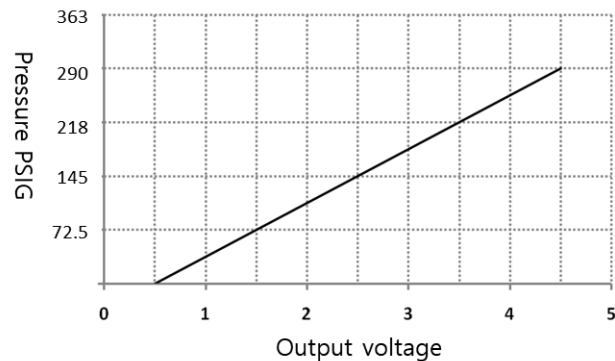


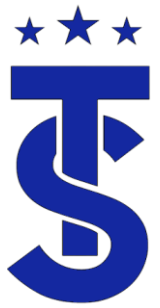
Pressure Transducers

- Purpose:
 - To measure High and Low Side pressures.
- How:
 - They are supplied with 5 VDC Between Red and Black
 - The Voltage out on the white wire varies as the pressure changes.
 - The system converts the voltage out of the white wire into a pressure.
 - As Pressure increases so does the voltage output



For reference purposes only!





High Pressure Switch

- Purpose:
 - Shut the system down when there is a high-pressure condition in the refrigerant circuit.
- How:
 - The switch or switches are wired in series with the power supply of the Hub PCB.
 - When the pressure gets to its limit the switch opens removing power from the Hub PCB.
 - This causes the Main PCB to lose power and shuts the system down.



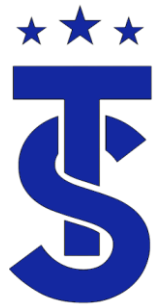
For reference
purposes only!



Looking Deeper



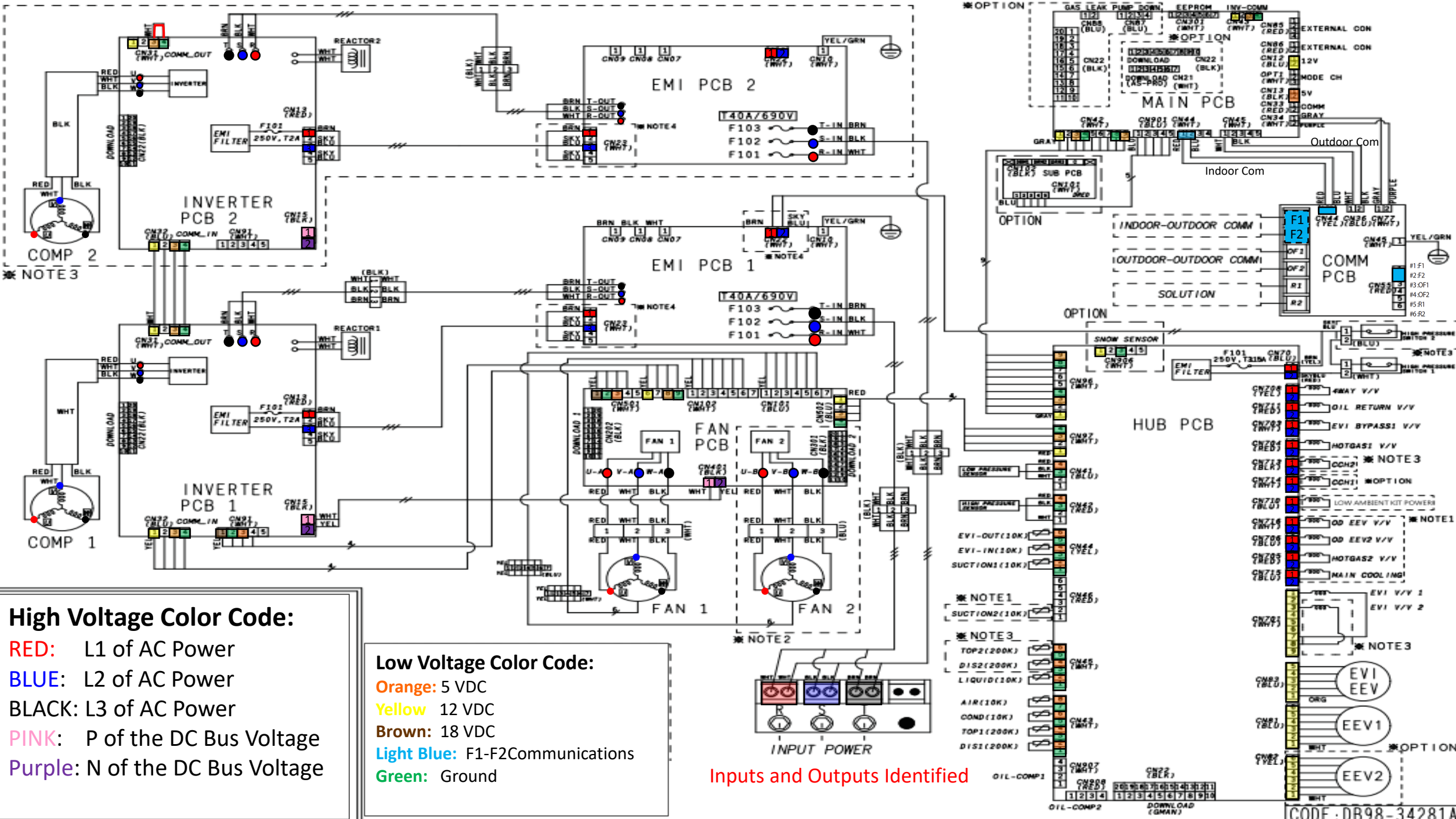
- Now that we have a basic understanding of the components within the electrical circuit let's take a more detailed look at each of the circuits board to determine where voltage enters and leaves the boards.
- We can start to look at how the boards communicate with each other.
- With this information we can get a better understanding of what happens within the electrical circuit.
- This information can be found in the service manuals.



Putting It All Together

- Now that we know all the system components functions, inter-relationship to other components, their input and out put voltages we can put the pieces together to make a whole system.
- We will map the electrical flow and communication through out the system.





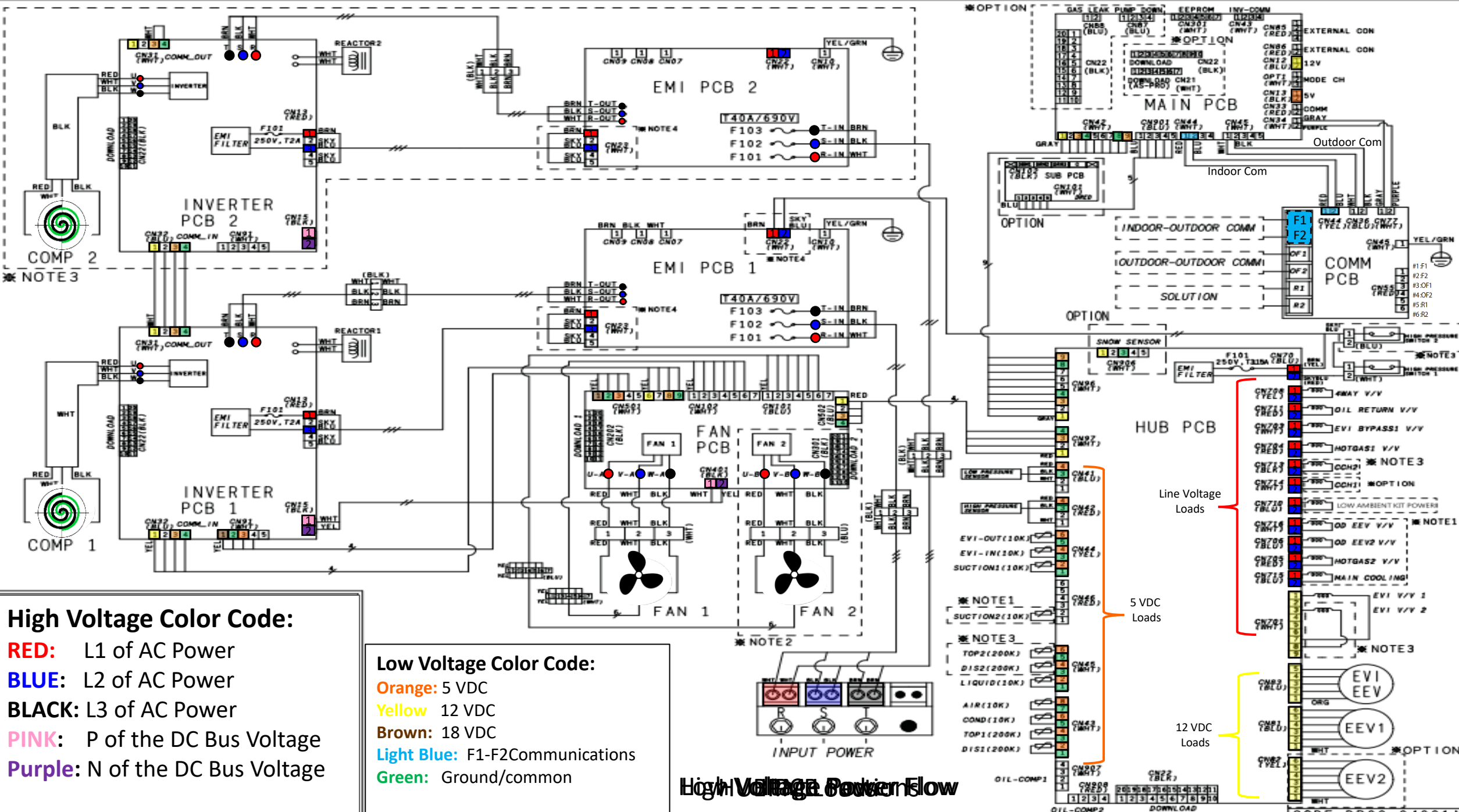
High Voltage Color Code:

- RED: L1 of AC Power
- BLUE: L2 of AC Power
- BLACK: L3 of AC Power
- PINK: P of the DC Bus Voltage
- Purple: N of the DC Bus Voltage

Low Voltage Color Code:

- Orange: 5 VDC
- Yellow: 12 VDC
- Brown: 18 VDC
- Light Blue: F1-F2 Communications
- Green: Ground

Inputs and Outputs Identified



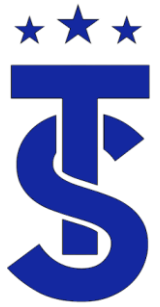
High Voltage Color Code:

- RED:** L1 of AC Power
- BLUE:** L2 of AC Power
- BLACK:** L3 of AC Power
- PINK:** P of the DC Bus Voltage
- Purple:** N of the DC Bus Voltage

Low Voltage Color Code:

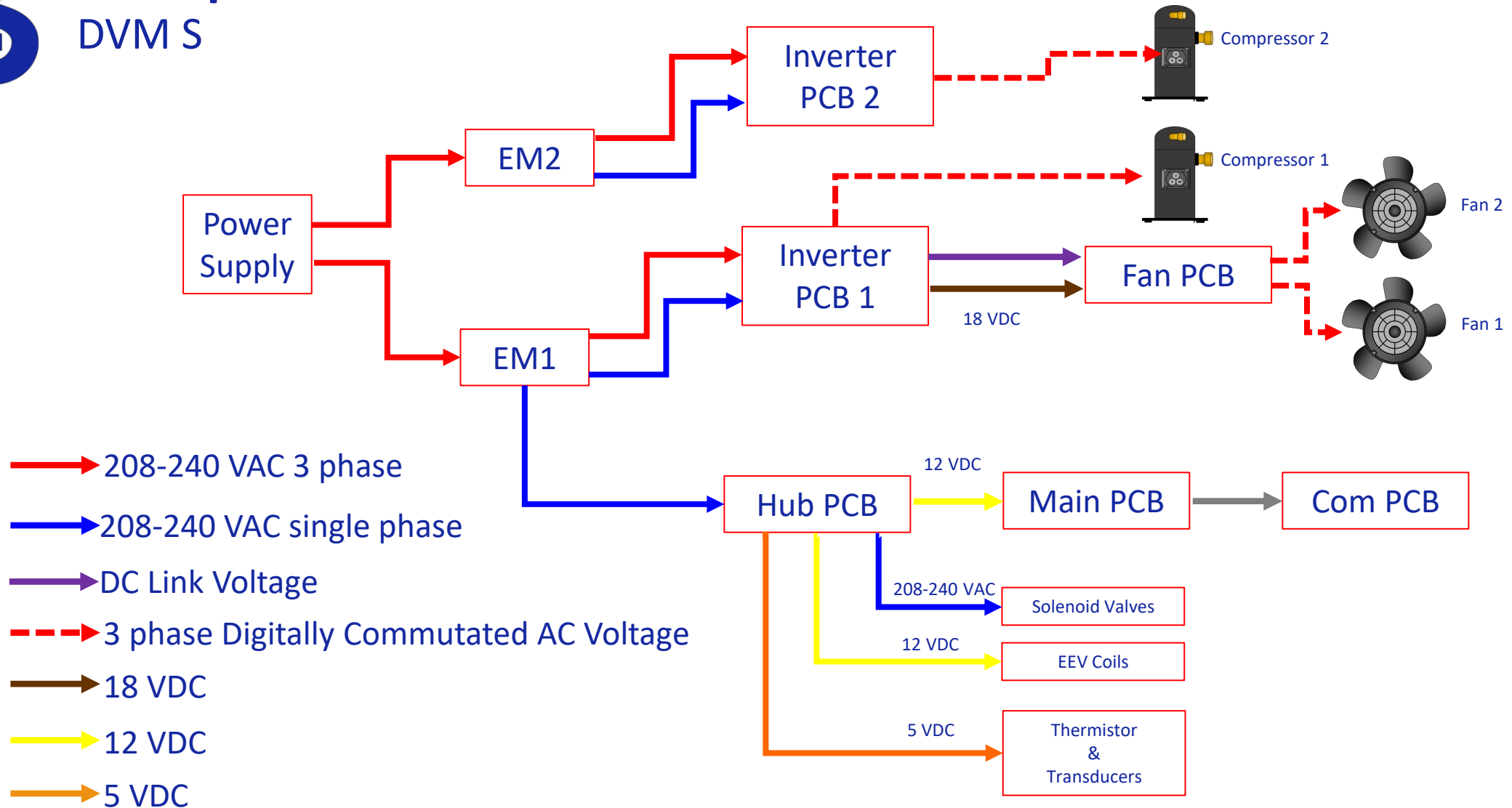
- Orange:** 5 VDC
- Yellow:** 12 VDC
- Brown:** 18 VDC
- Light Blue:** F1-F2 Communications
- Green:** Ground/common

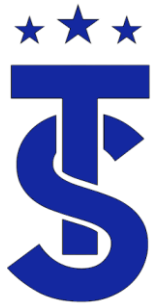
High Voltage Power Flow



Simplified Power Flow

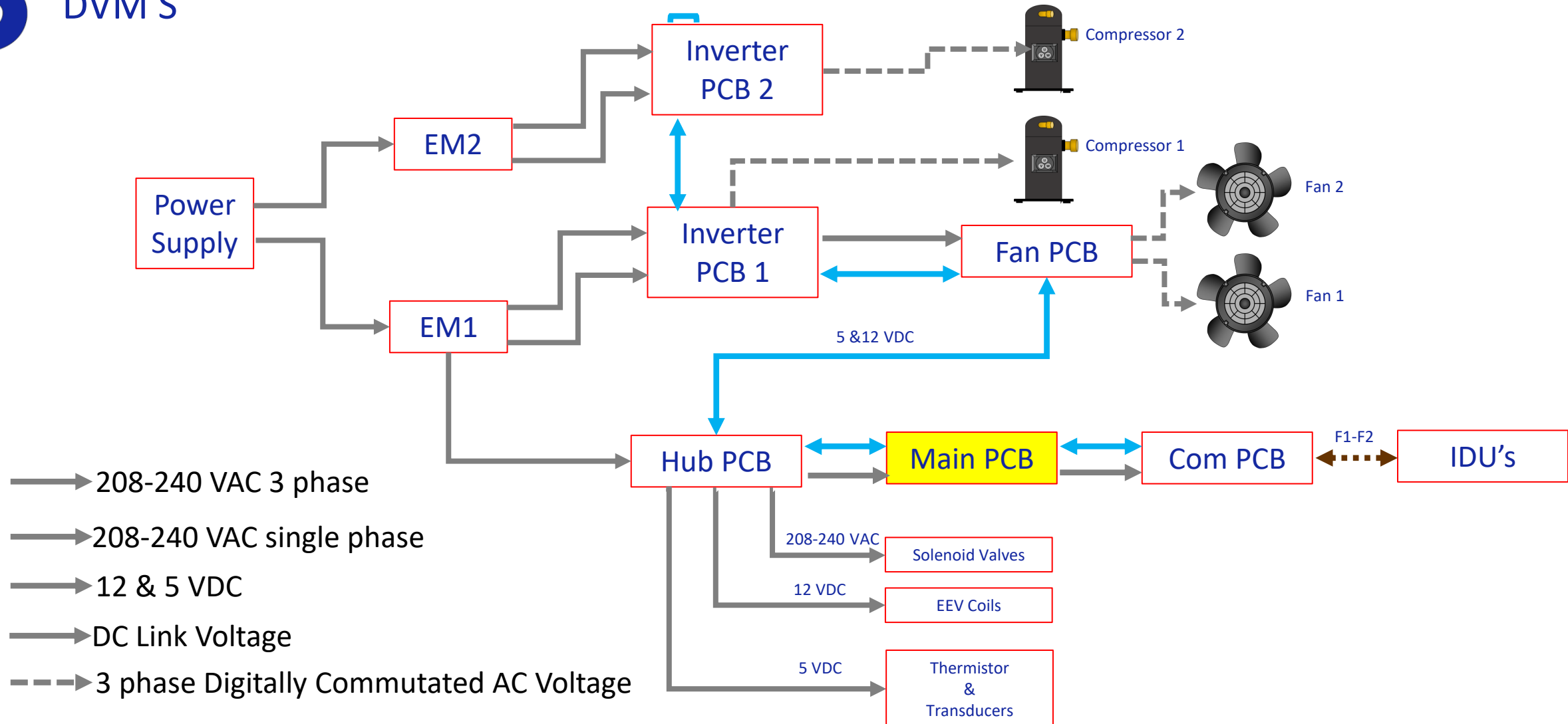
DVM S





Simplified Communication Flow

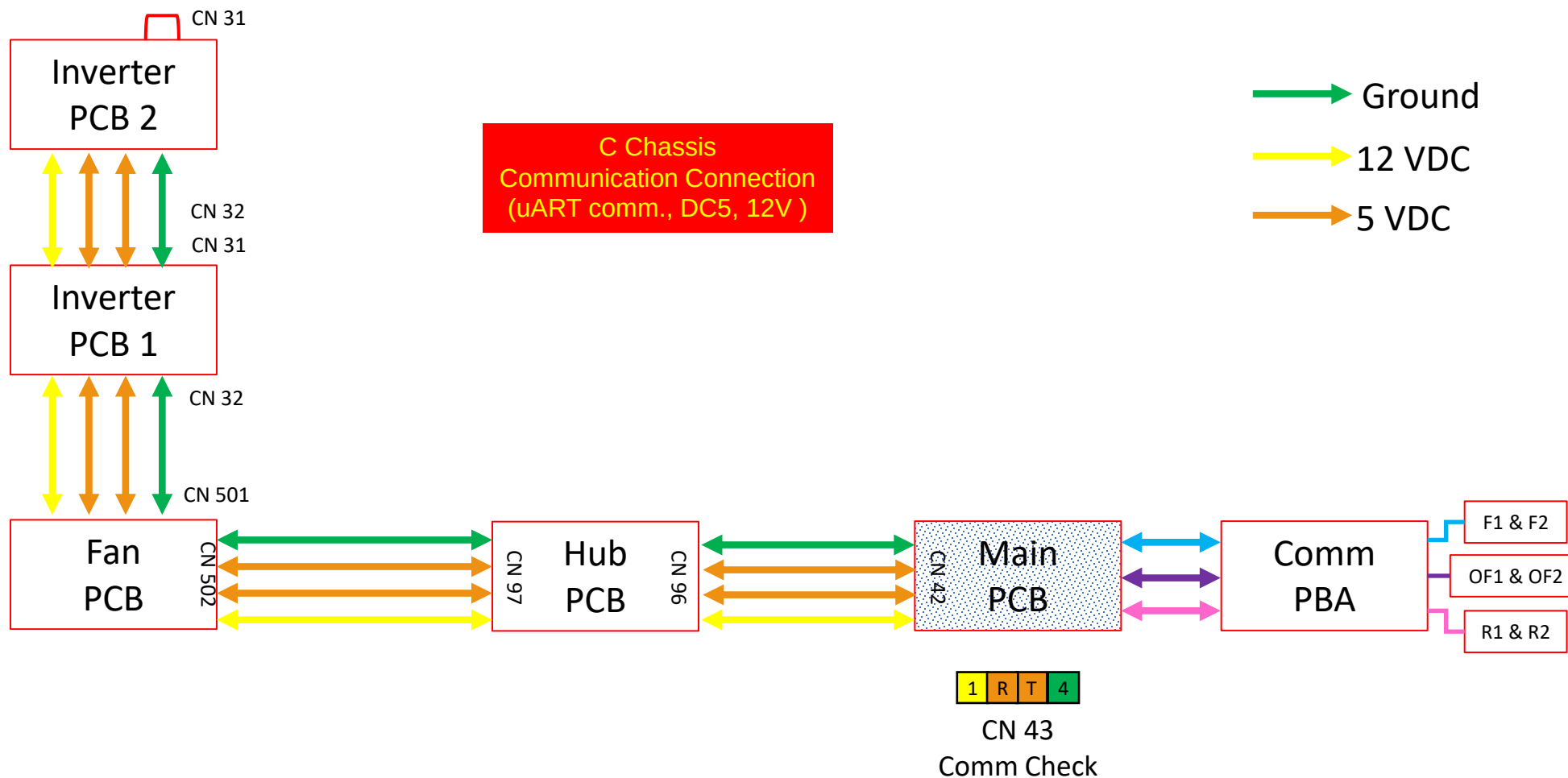
DVM S





Communications Flow

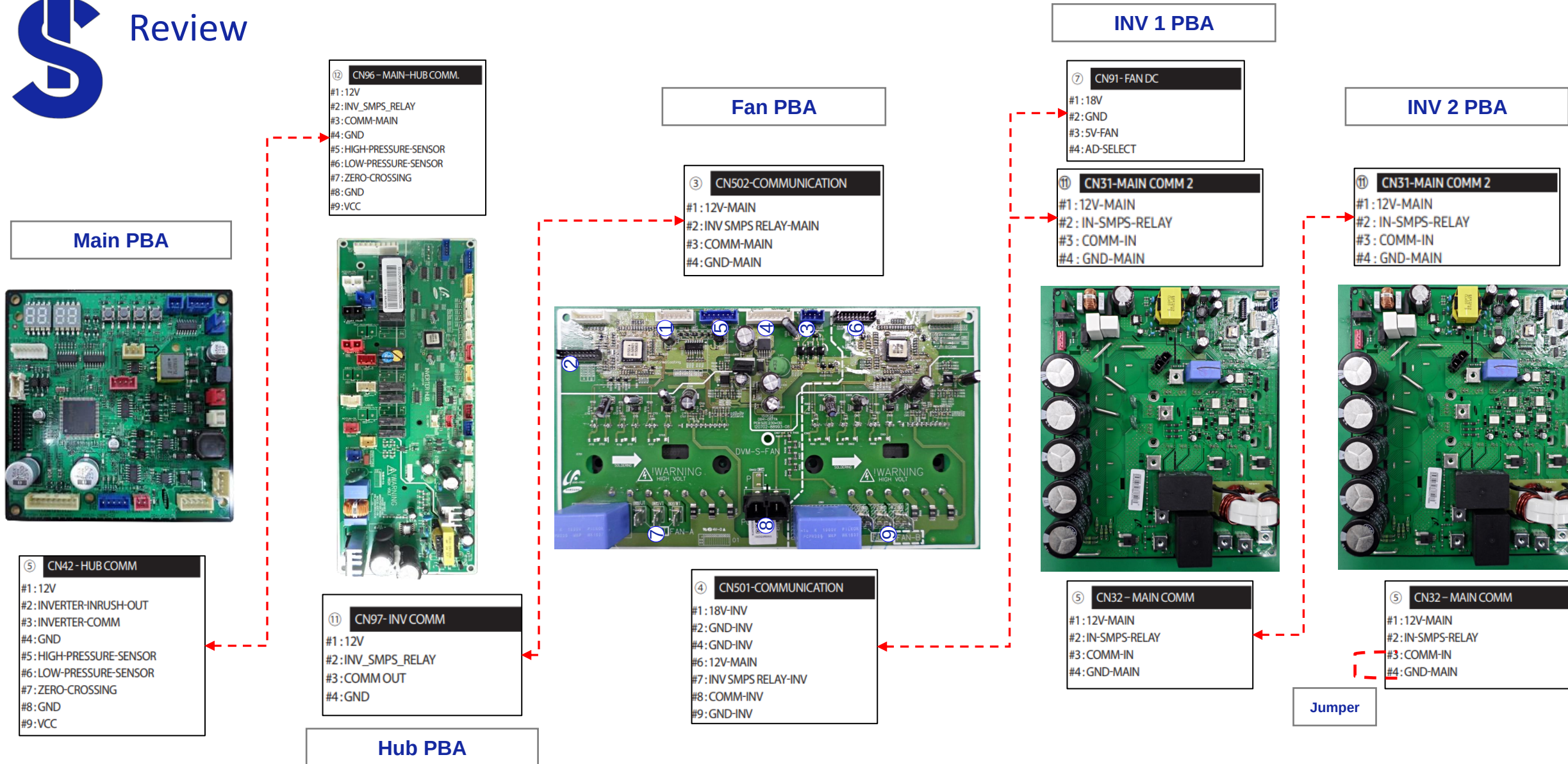
DVM S

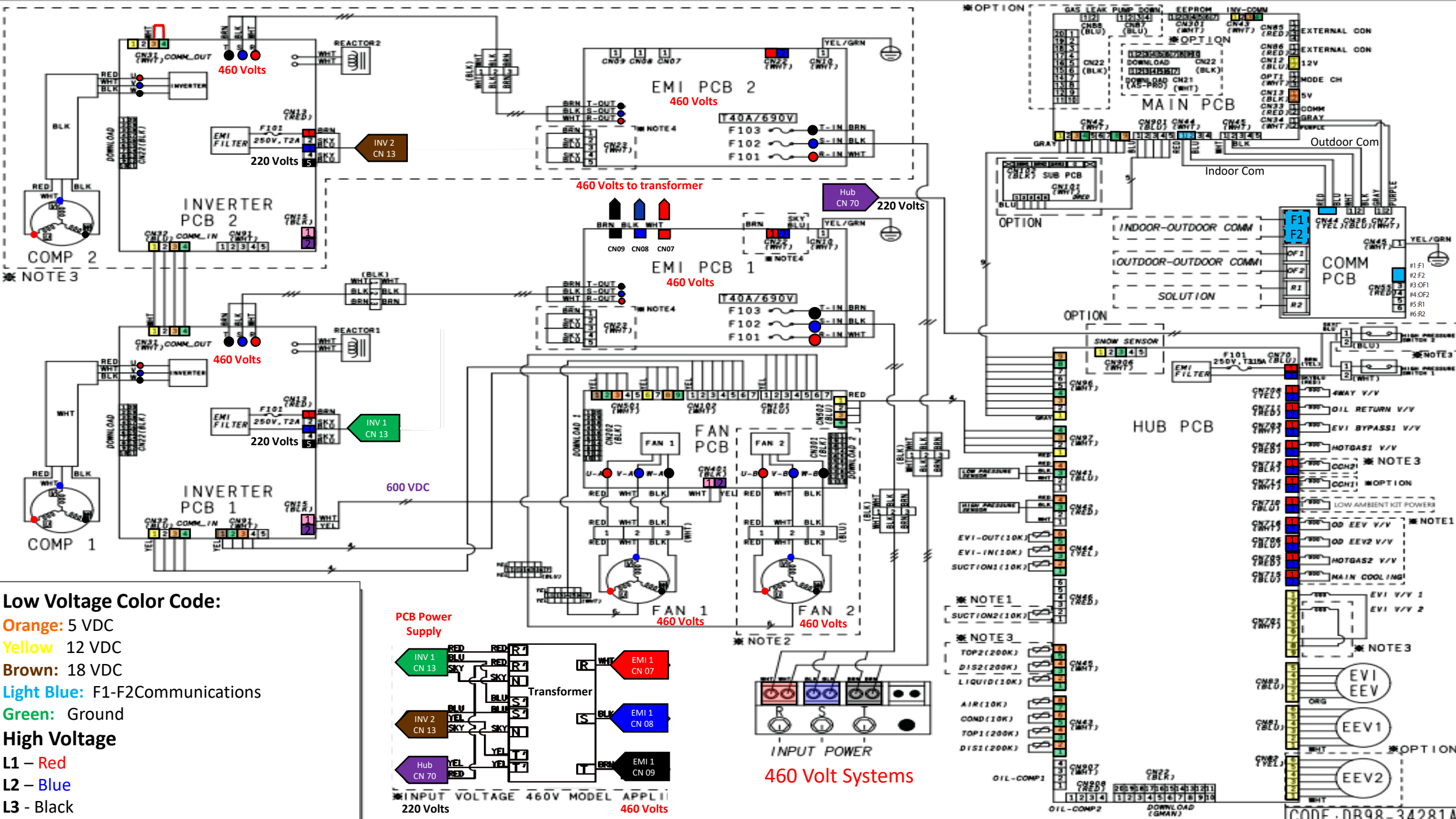




DVM S Communication Network

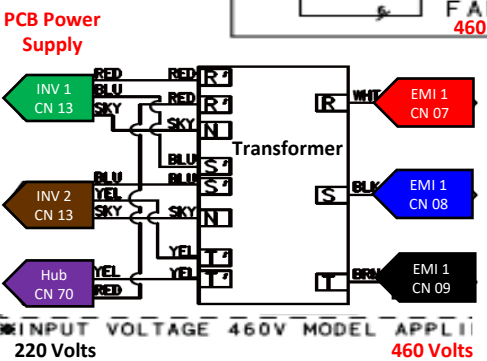
Review



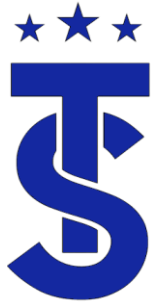


Low Voltage Color Code:

- Orange: 5 VDC
- Yellow: 12 VDC
- Brown: 18 VDC
- Light Blue: F1-F2 Communications
- Green: Ground
- High Voltage**
- L1 – Red
- L2 – Blue
- L3 - Black



460 Volt Systems

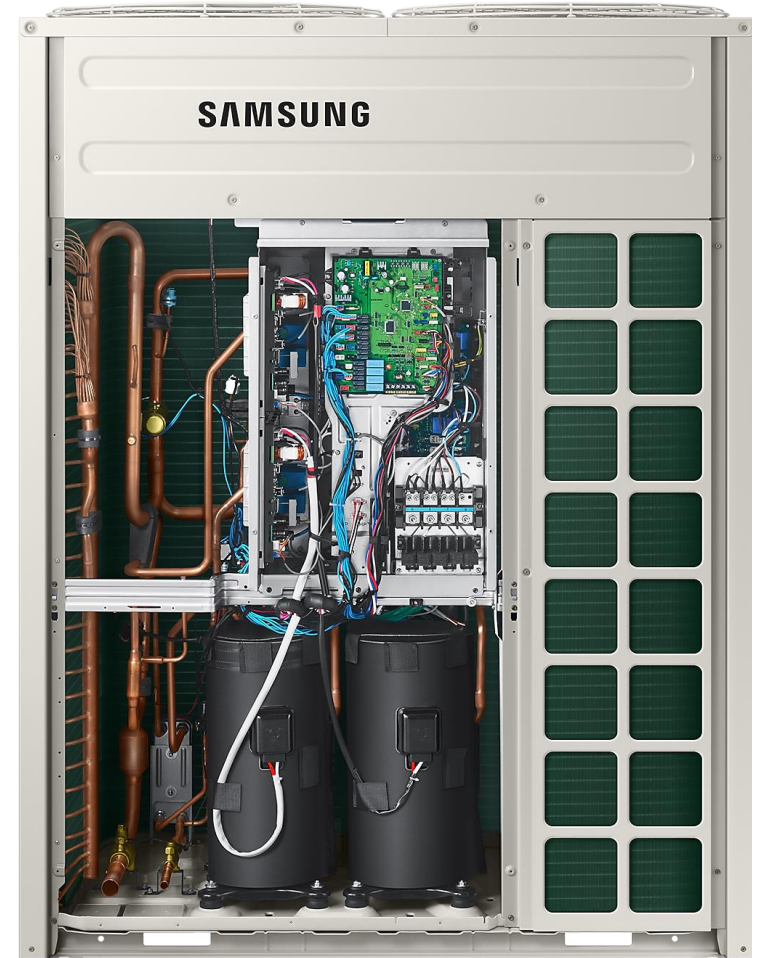


As much as things change, they stay the same

These systems electrically are more alike than they are different

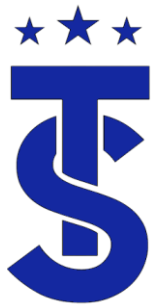


DVM S



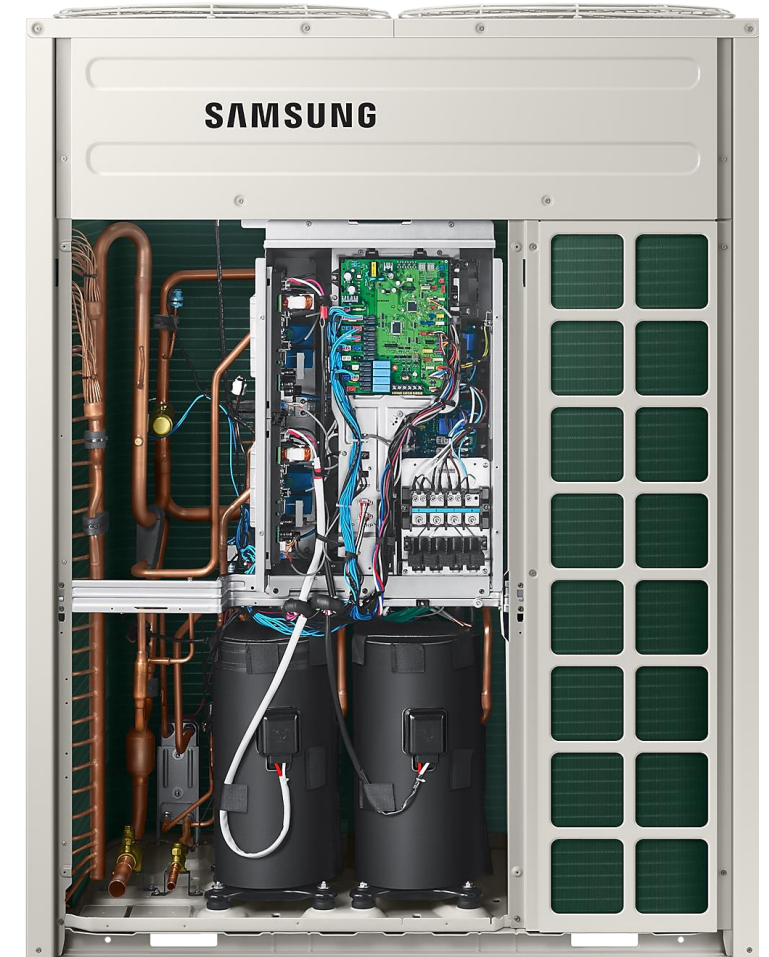
DVM S 2

SAMSUNG



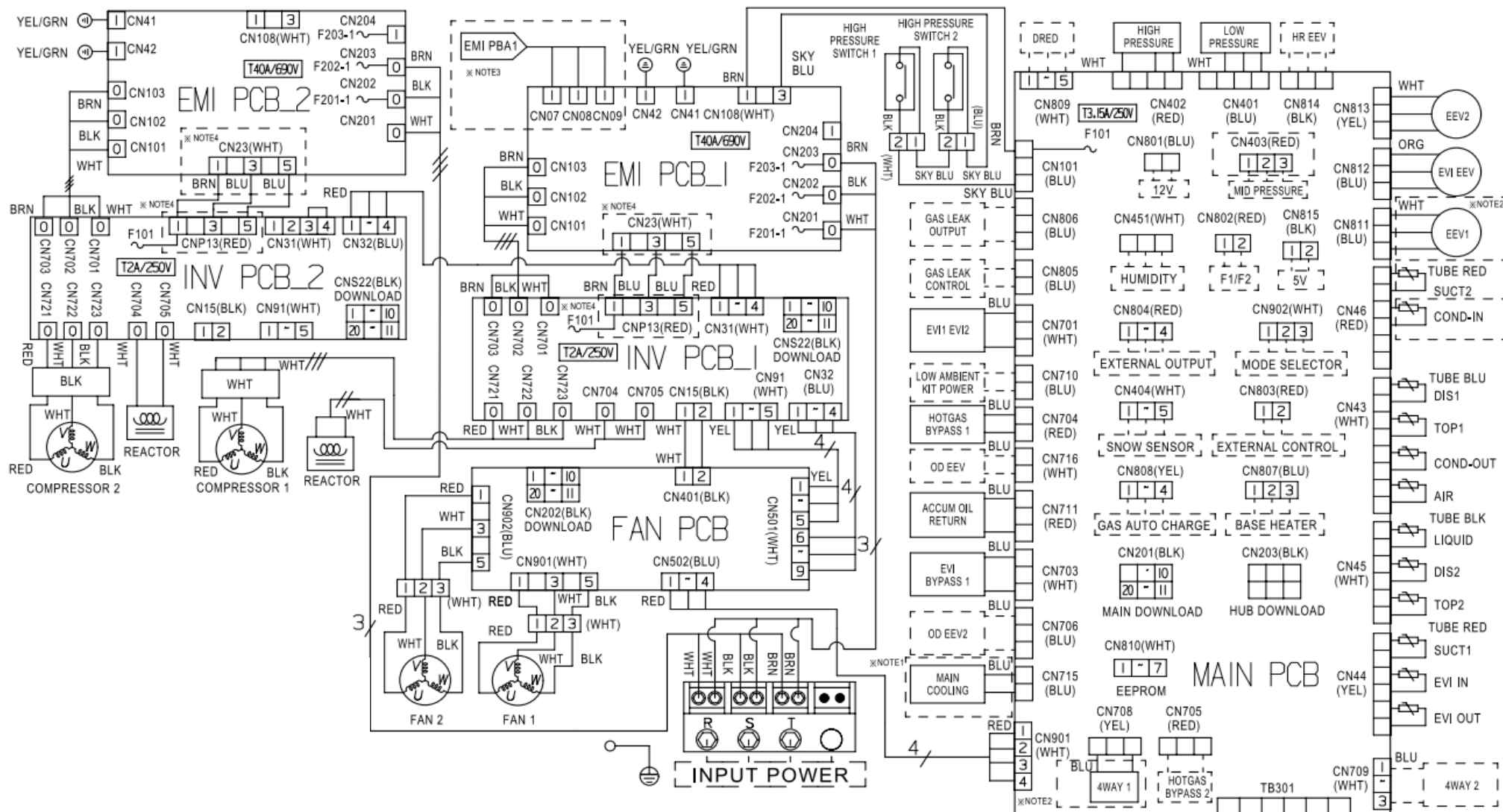
DVM S 2 Electrical

- Let's follow the same process as previously to:
 - Identify system components.
 - Determine their relationship to other components within the system.
 - Map out electrical and communications flow through the system.
 - Identify the differences between DVM S & DVM 2 circuits.





Identify the components of the system

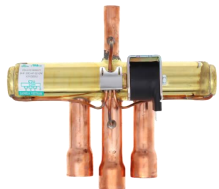




DVM S 2 System Components



Solenoid Valves



Reversing Valves



EEV



High Pressure Sw.



Pressure Transducer



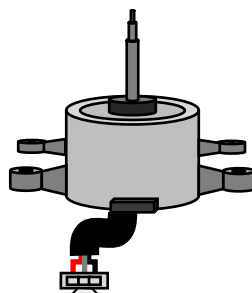
Thermistors



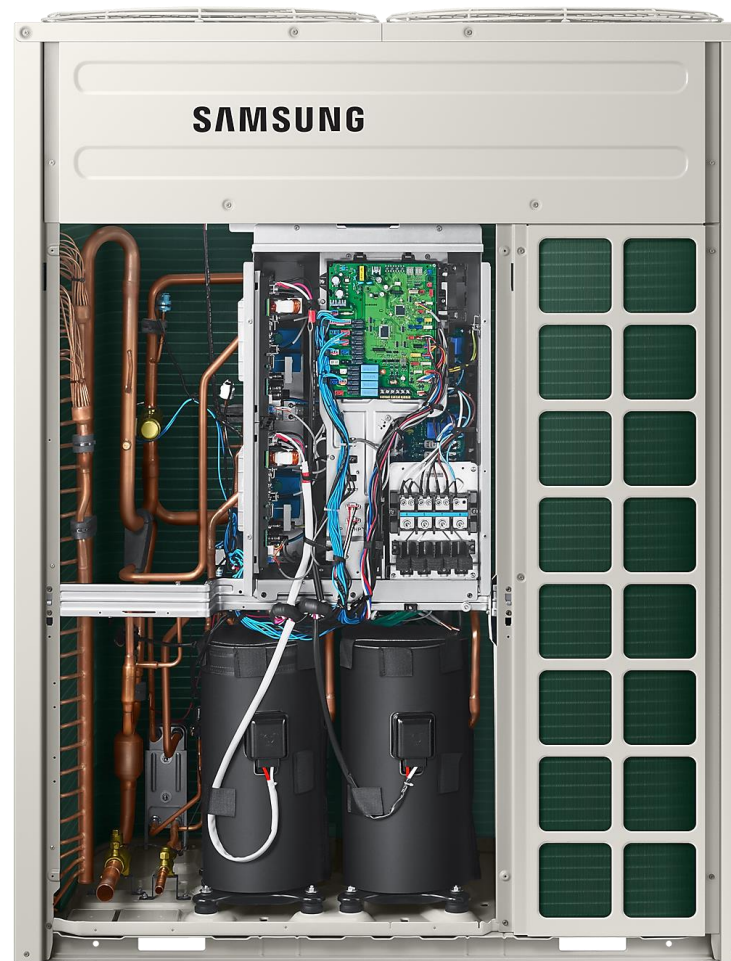
Compressor



Reactor



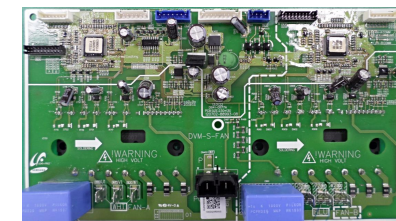
Condenser fan



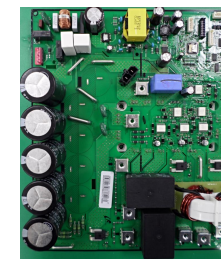
Component pictures are for demonstration purposes only.
Actual components may look slightly different



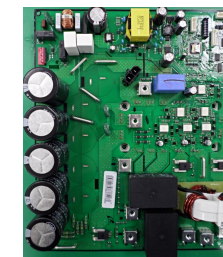
Main, Hub &
Communications
PBA



Fan PBA



Inverter 1 PBA



Inverter 2 PBA



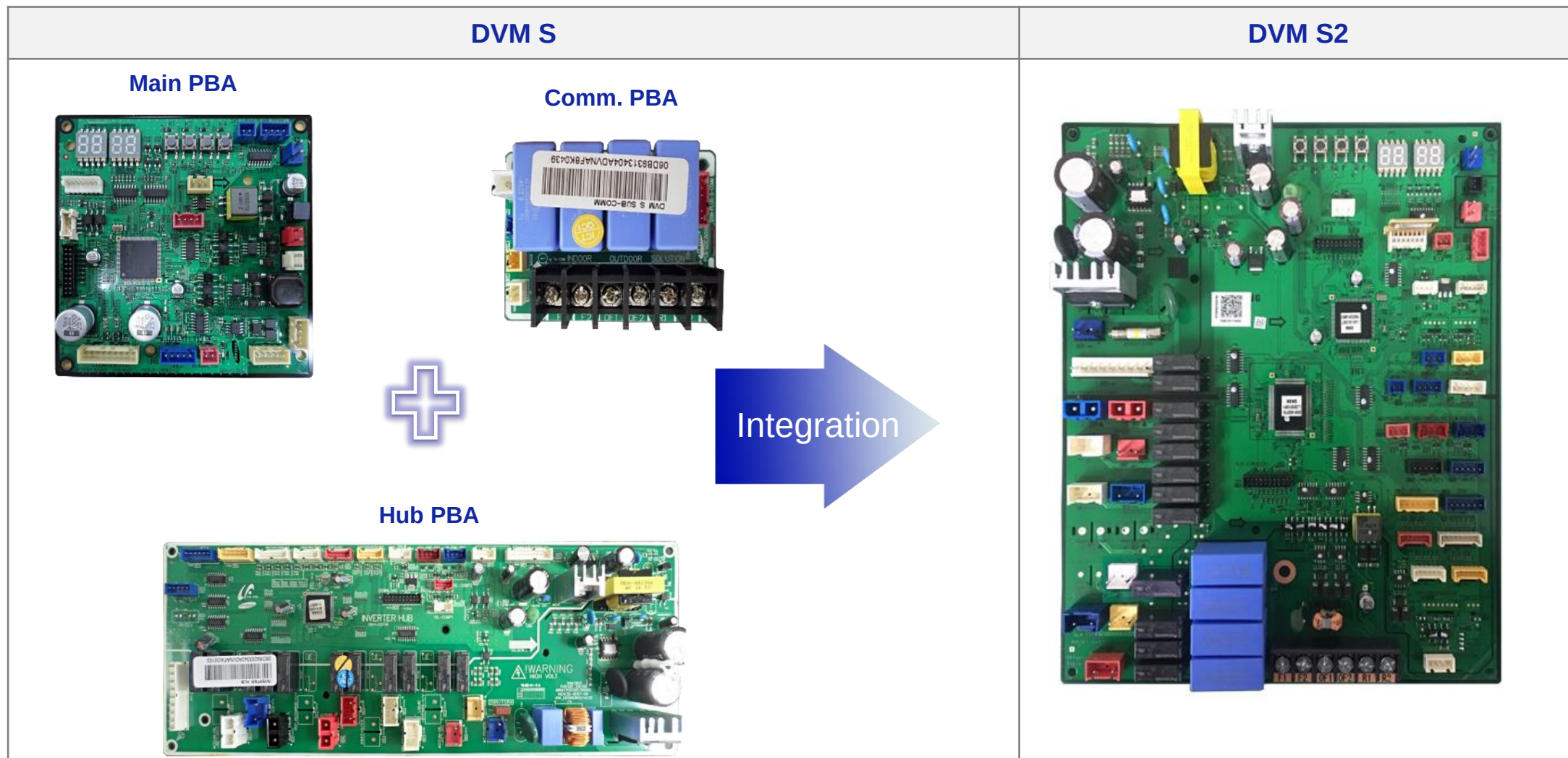
EMI 1

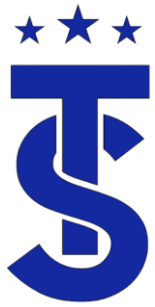


EMI 2

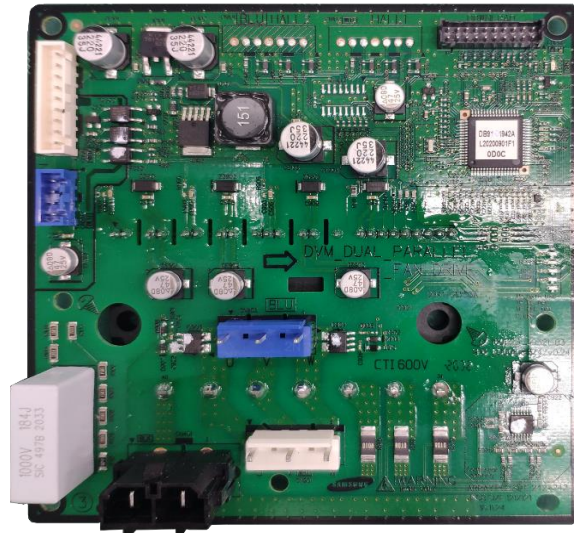


DVM S 2 Improvements



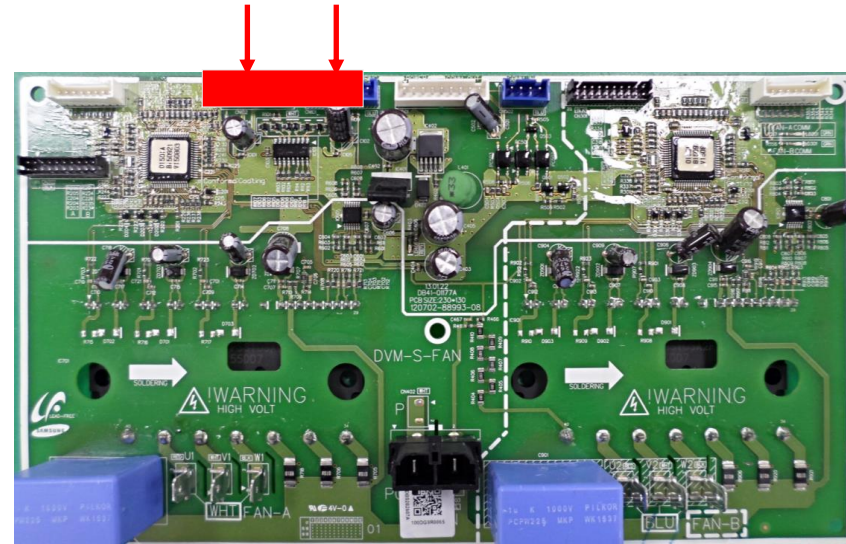


DVM S 2 Fan PBA



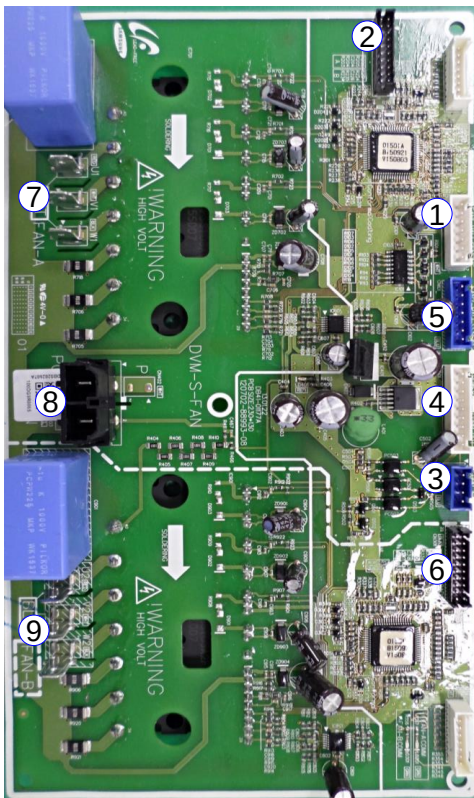
DVM S2
S1&S2 Platform

No Hall Sensor Connections



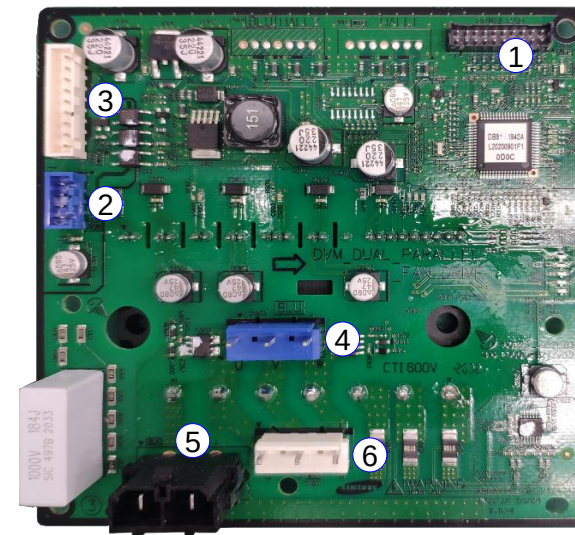
DVM S2
S3 Platform

Fan PBA



Conventional
&
DVM S2 (S3 platform)

Function	Conventional fan PBA		New fan PBA
	DVM S	DVM S2 (S3 platform)	DVM S2 (S1, S2 platform)
Fan motor #1 hall IC detector	1	Unused	-
S/W download for Fan PBA (micom #1)	2	2	1
uART communication	3 (With hub PBA)	3 (With main PBA)	2 (With main PBA) (CN502)
Inverter PBA connection (uART comm., DC 5,12,18V supply)	4	4	3 (CN501)
Fan motor #2 hall IC detector	5	Unused	-
S/W download for Fan PBA (micom #2)	6	6	-
Fan motor #1 input U,V,W connection	7	7	6
Inverter PBA connection (DC Power supply from Inverter PBA)	8	8	5 (CN401)
Fan motor #2 input U,V,W connection	9	9	4

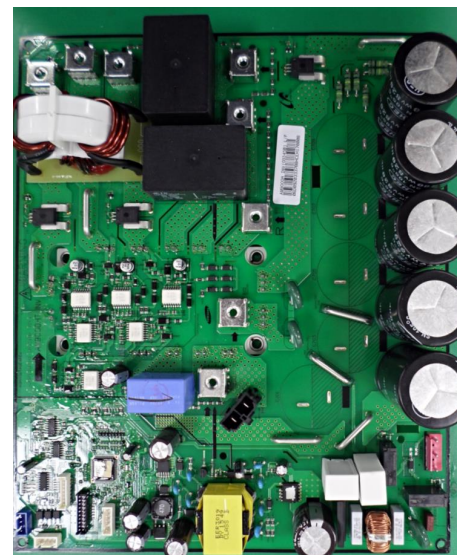


DVM S2
(S1, S2 platform)

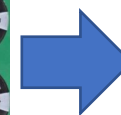


DVM S 2 Inverter PBA

- Film capacitor
 - (Inverter PCB Downsize 17 %)
 - (Increase 10 times Capacitor Lifetime)
- PWM Carrier Frequency
 - 5.5k ~ 7kHz Compressor operation
 - (Increased inverter efficiency about 1.5%)



DVM S

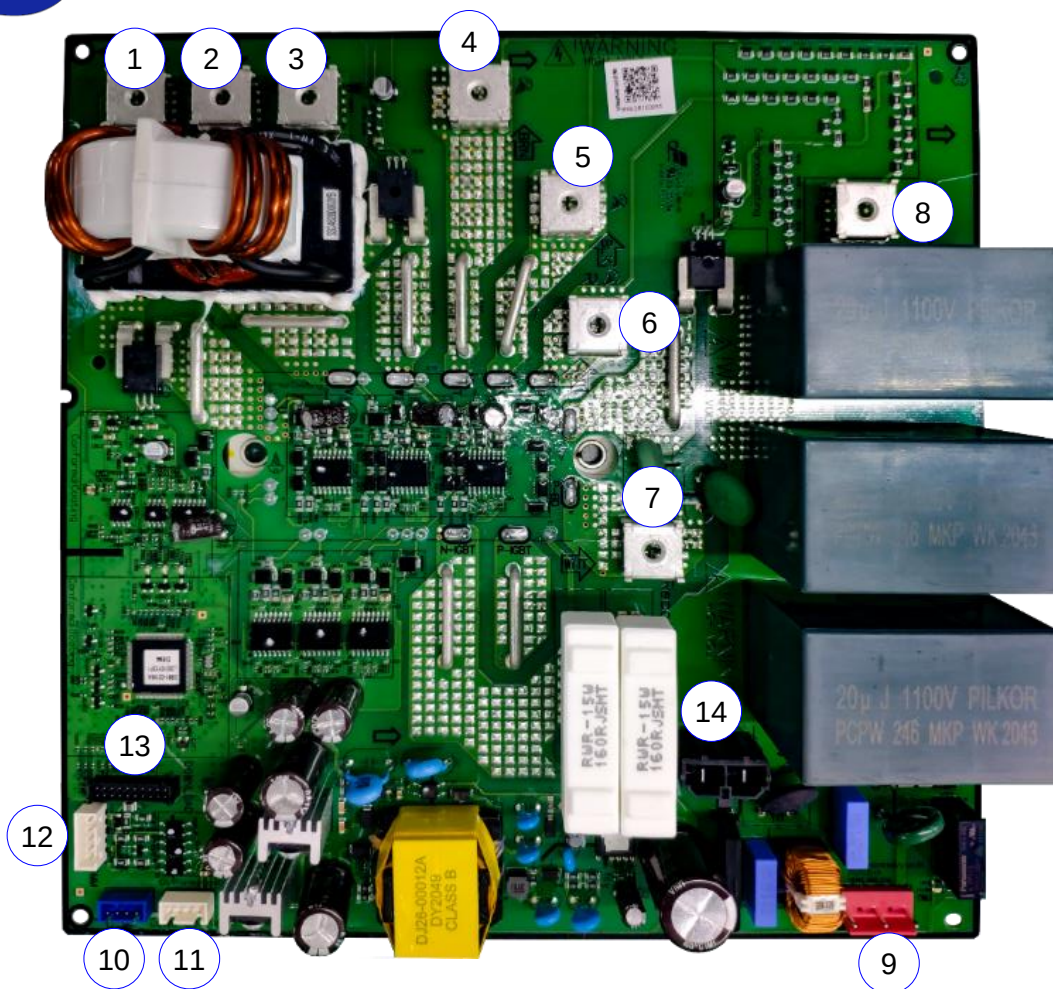


DVM S 2



#8 Inverter PBA (Not integrated with the) FAN PBA

- Function : Inverter compressor control



※ AM(096~192)BXVGJ(H/R)/AA & AM(096~144)BXVGF(H/R)/AA

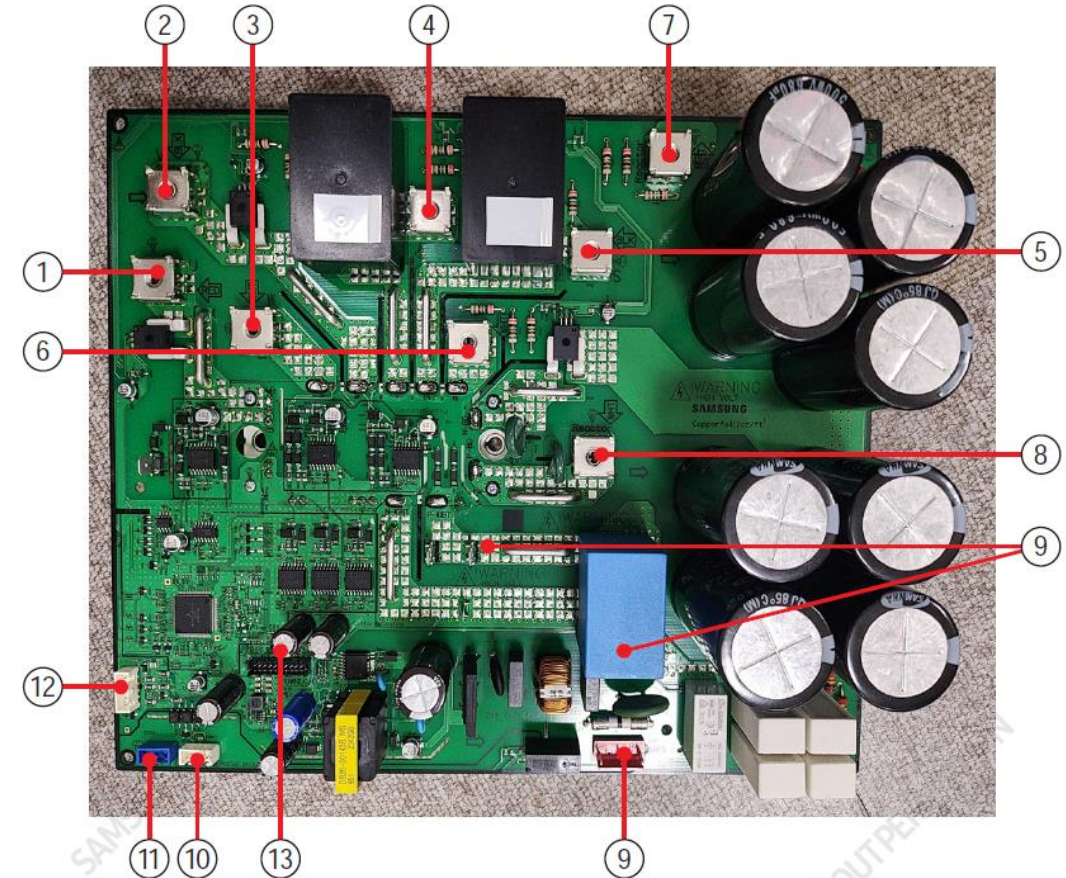
No.	CN.	Function
1		Compressor input U connection
2		Compressor input V connection
3		Compressor input W connection
4		T IN (It supplies *AC power from the EMI PBA)
5		S IN (It supplies *AC power from the EMI PBA)
6		R IN (It supplies *AC power from the EMI PBA)
7,8		DC Reactor connection
9		EMI PBA connection (F model : 1Φ AC power supply)
		Down trans. connection (J model : 1Φ AC power supply)
10	CN32	Comm. #1 (↔ Fan PBA or INV #1 PBA)
11	CN31	Comm. #1 (↔ INV #2 PBA or Comm. jumper)
12	CN901	Fan PBA connection (DC 5, 12, 18 V supply)
13		S/W download for INV PBA
14	CN15	Fan PBA connection (F model : DC 294 ~ 325 V, J model : DC 650 V supply)

* F model : 208 ~ 230 V
J model : 460 V



575 Volt Inverter

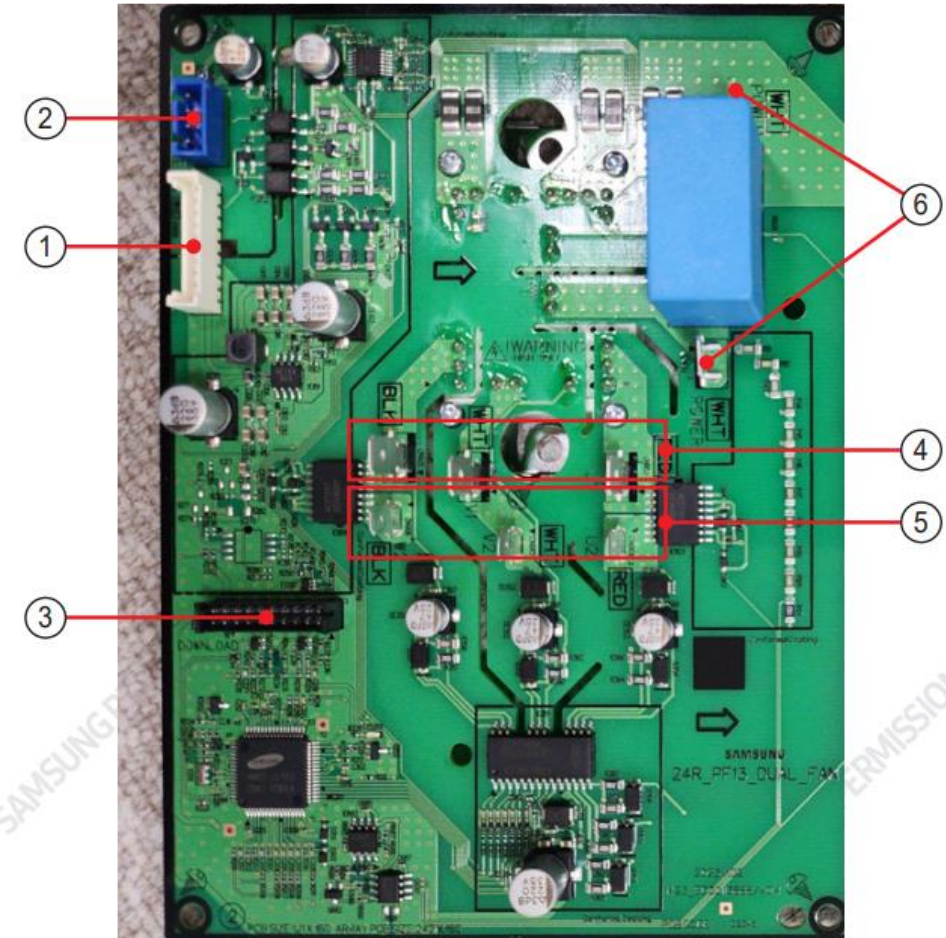
① U-COMPU #1: COMPU	② V-COMP V #1: COMP V	③ W-COMP W #1: COMP W	④ T-INPUT TOP #1: T-IN
⑤ S-INPUT TOP #1: S-IN	⑥ R-INPUT TOP #1: R-IN	⑦ CN705-REACTOR 1 #1: REACTOR 1	⑧ CN704-REACTOR 2 #1: REACTOR 2
⑨ CN15-FAN POWER #1: DC_LINK #3: GND	⑩ CNP13-AC POWER #1: AC LIVE #3: AC NEUTRAL	⑪ CN31-MAIN COMM 2 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN	⑫ CN32-MAIN COMM1 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN
⑬ CN91-FAN DC #1: 18V #2: GND #3: 5V-FAN #4: AD-SELECT	⑭ CNS22-INV MICOM DOWNLOAD #1: RX_DEBUG #2: TX_DEBUG #3: BOOT #4: JTAG_TDO #5: JTAG_TCLK #6: JTAG_TDI #7: JTAG_TMS #8: JTAG RESET #9: GND #10: VCC #11: VCC #17: GND #18: DAC_CLK #19: DAC_CS #20: DAC_DATA		

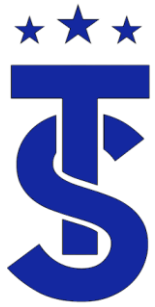




575 Volt Fan Board

① INVERTER to FAN #1 INVERTER DC18V #2 INVERTER GND #3 - #4 INVERTER GND #5 - #6 MAIN DC12V to INV #7 INVERTER SMPS RELAY to INV #8 COMM to INV #9 GND to INV	② Main to FAN #1 DC12V #2 INV.SMPS RELAY #3 COMM #4 GND	③ Download Connector #1 RXD #2 TXD #3 BOOT #4 TDO #5 TCK #6 TDI #7 TMS #8 NTRST #9 P_GND #10 DC 5V	④ IPM to Motor1 #1 U1 #2 - #3 V1 #4 - #5 W1
⑤ IPM to Motor2 #1 U2 #2 - #3 V2 #4 - #5 W2	⑥ DC LINK VOLTAGE SENSING #1 540Vdc from INV #2 P_GND from INV		





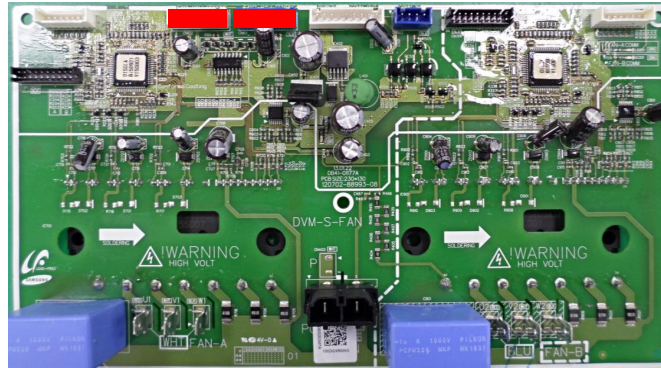
DVM S 2 Circuit Boards

Main, Comm & Hub
PBA

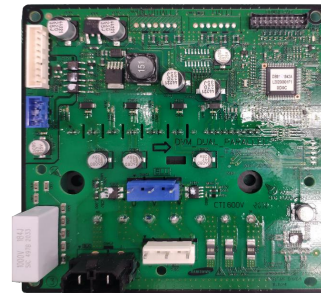


Fan PBA

No Hall Sensing Circuits

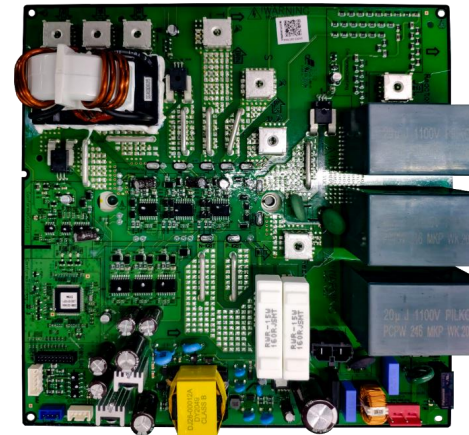


S3 Platforms

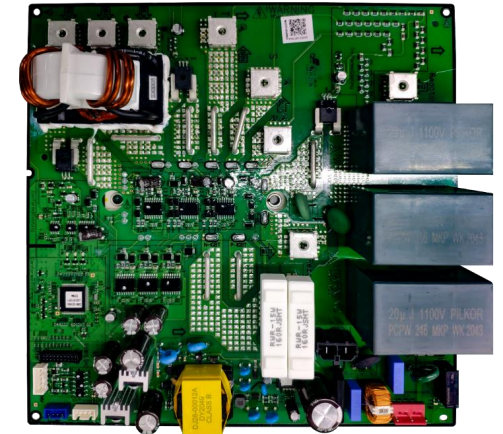


S1 & S2 Platforms

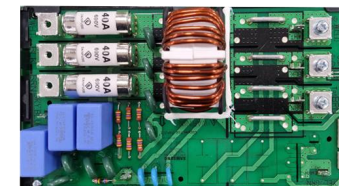
Inverter # 1 PBA



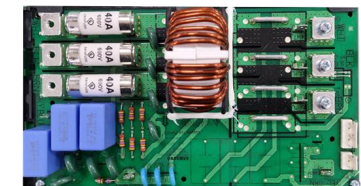
Inverter # 2 PBA



EMI # 1

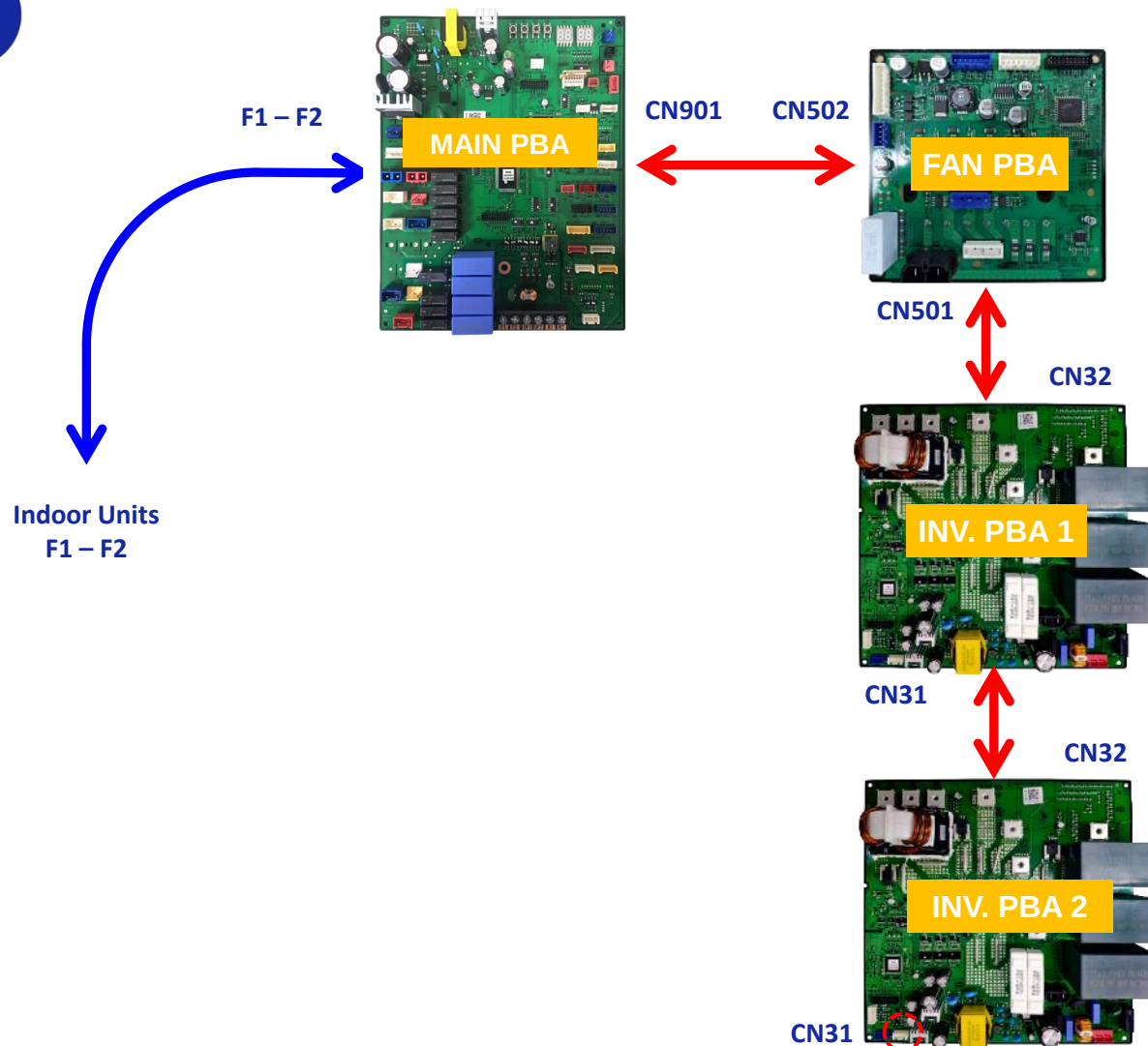


EMI # 2



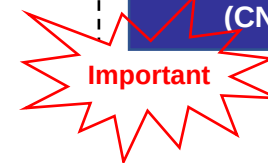
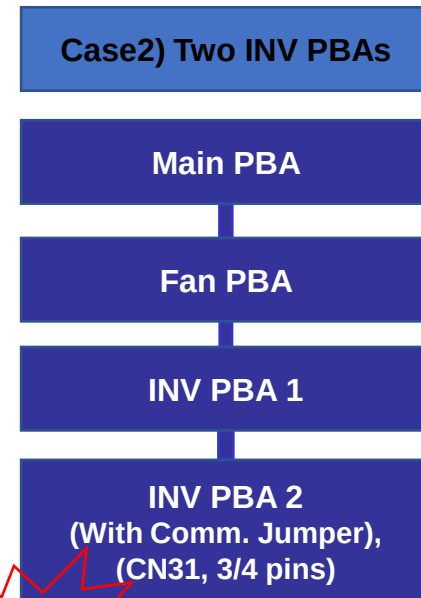
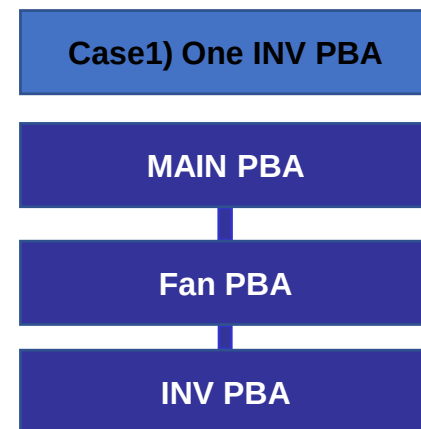


DVM S 2 Communications Flow



— uART Comm.
— RS485 Comm.

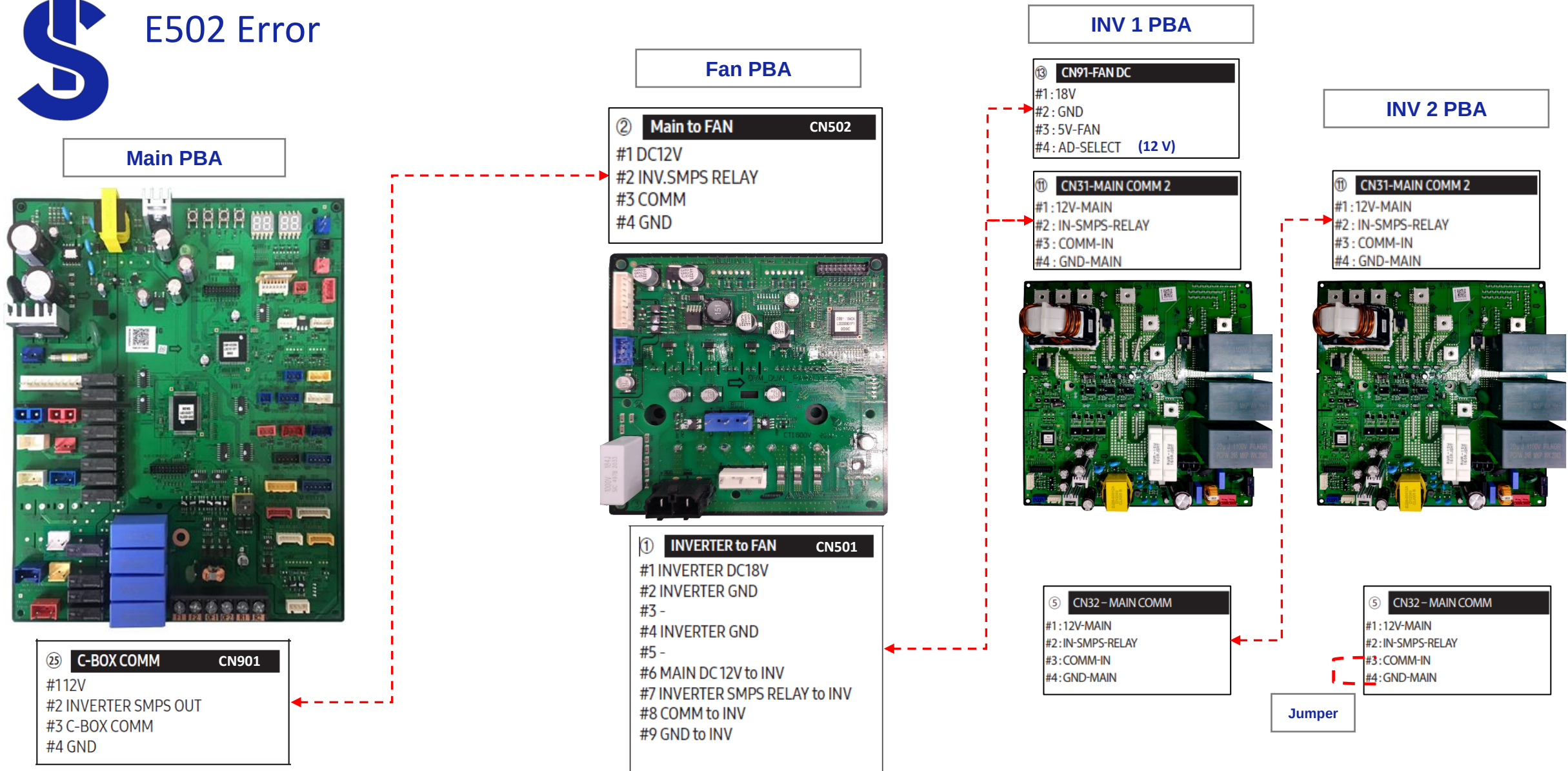
PBA Com. connection order

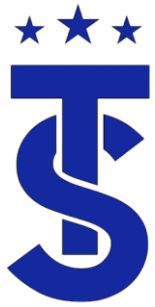




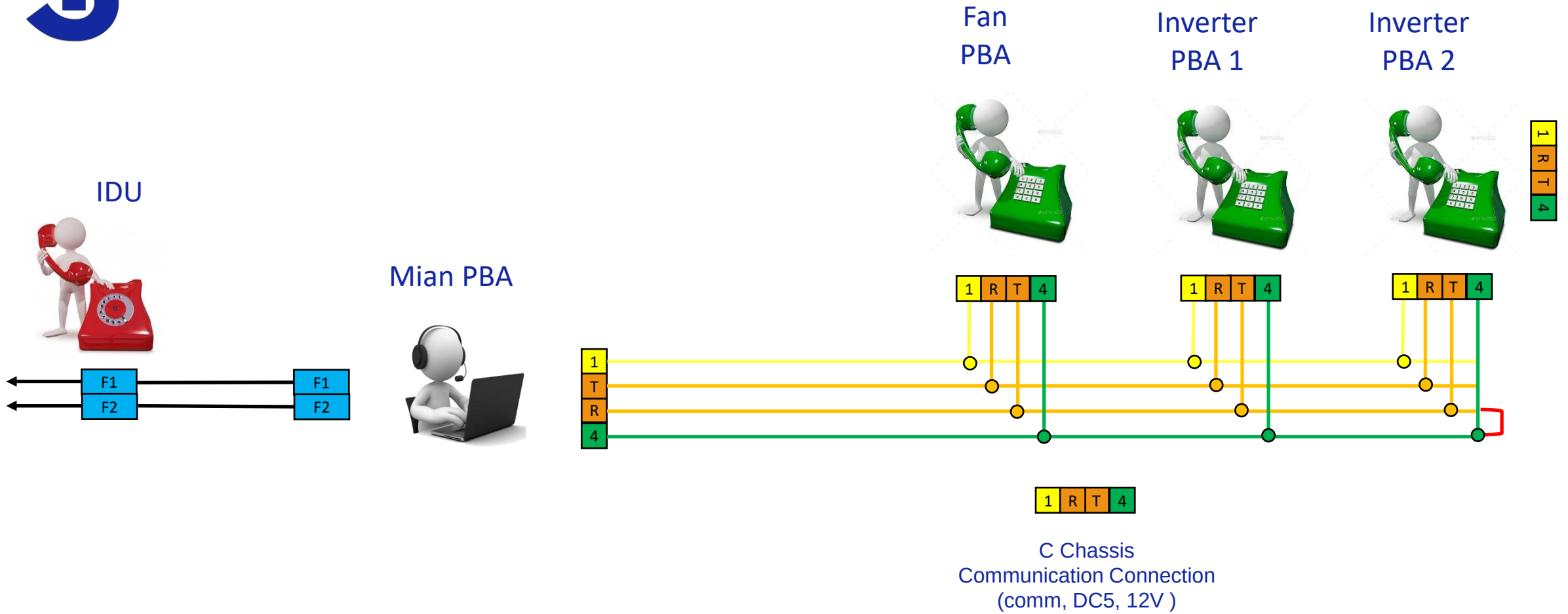
DVM S 2 Communication Network

E502 Error





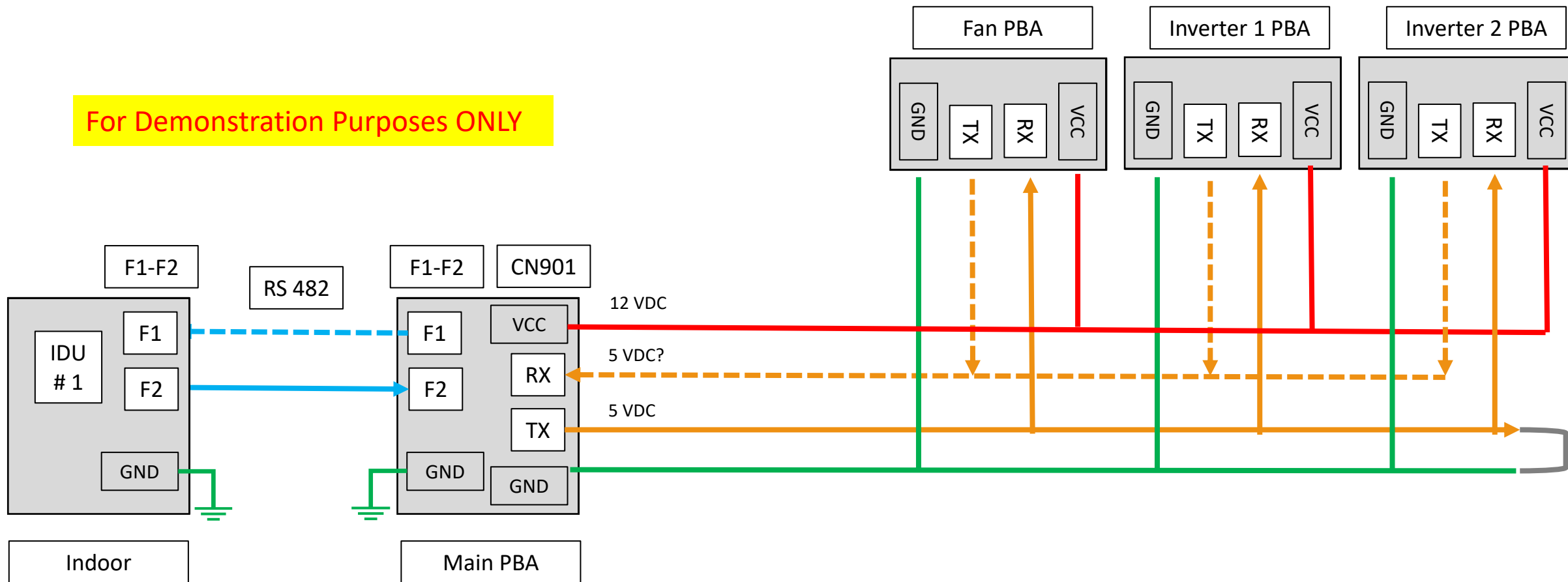
DVM S2 Communications





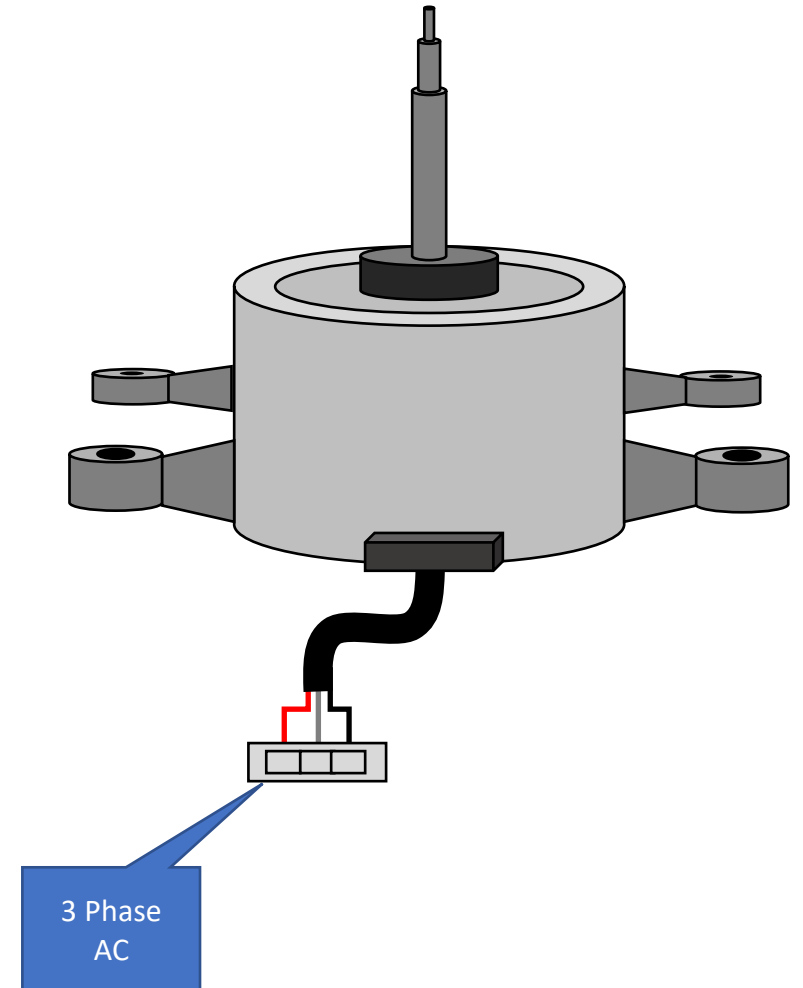
DVM S2 C Chassis Communication

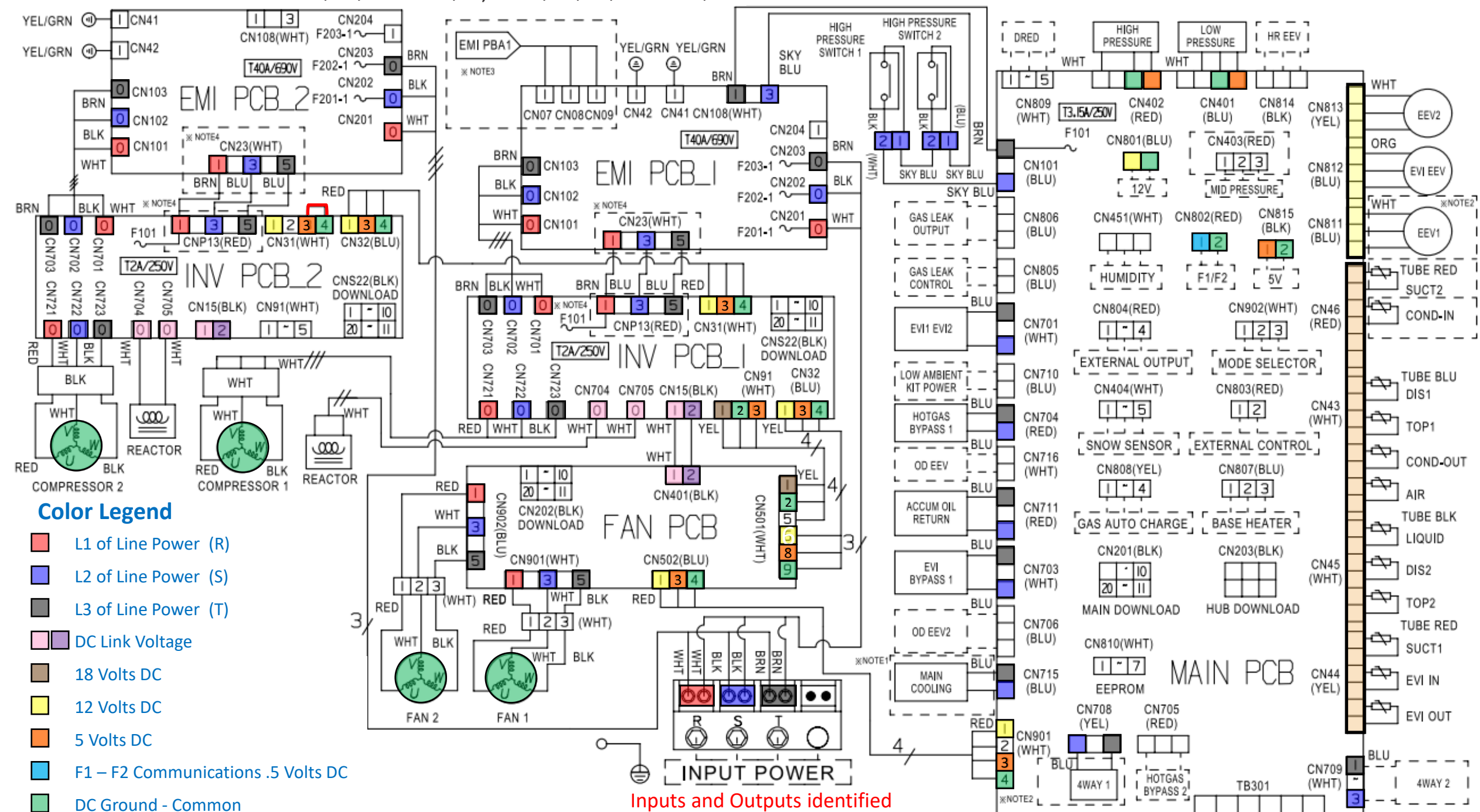
For Demonstration Purposes ONLY

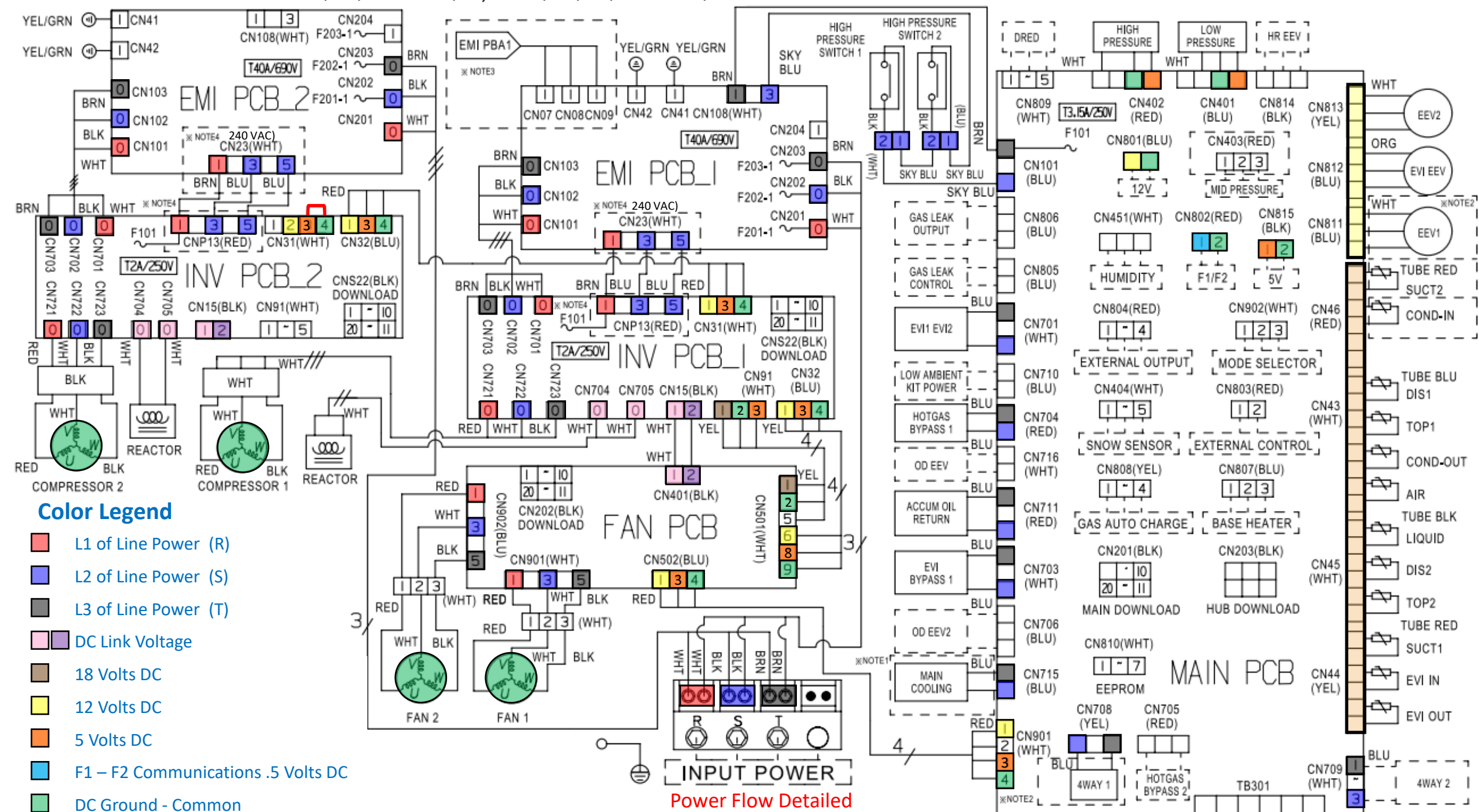


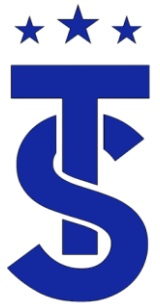
DVM S 2 Outdoor Fan Motor

- Purpose:
 - Move air across the outdoor coil.
- How:
 - The motor is supplied with 3 phase AC voltage from the Fan IPM.
 - There is no hall sensor circuit.



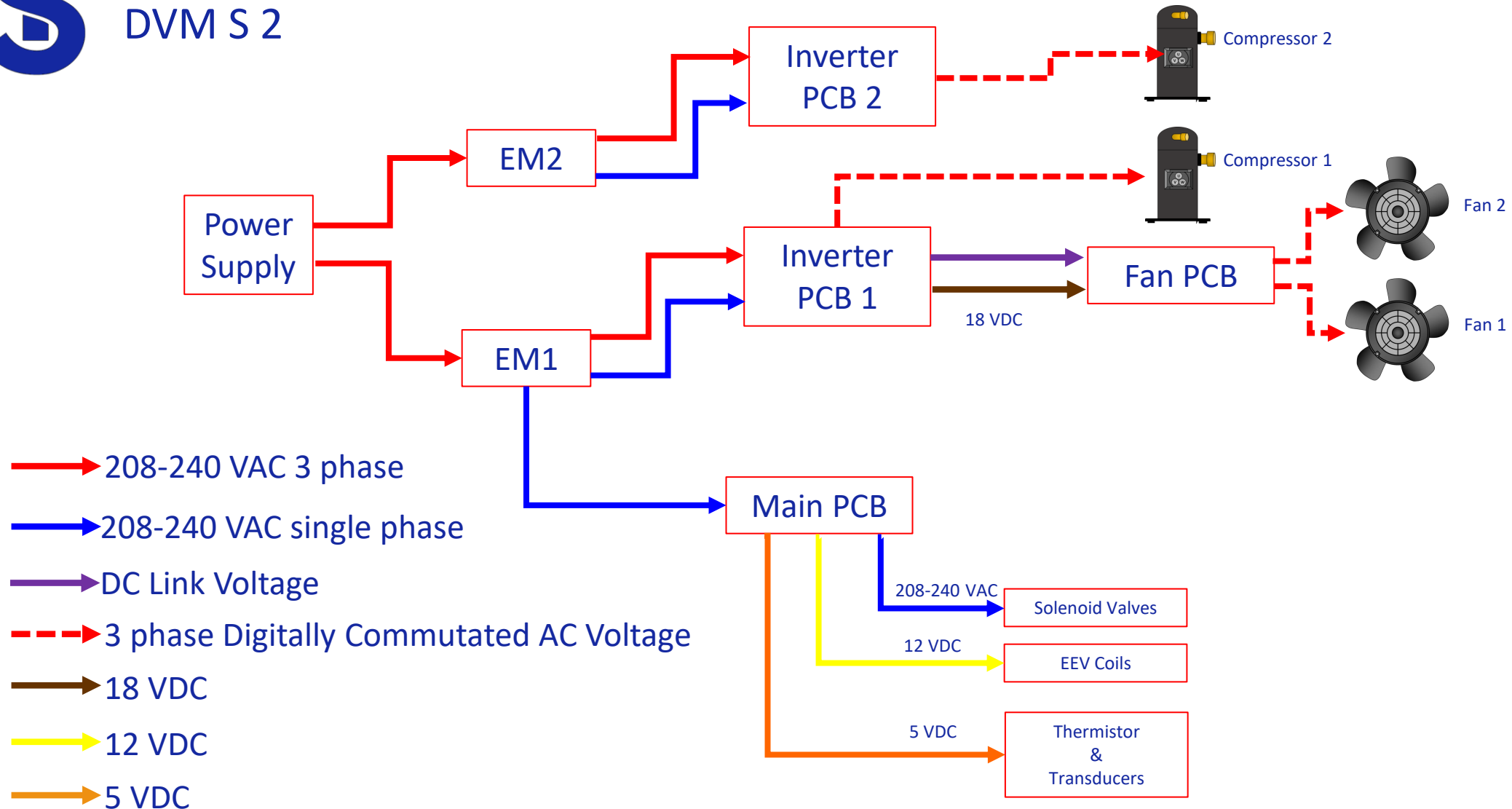


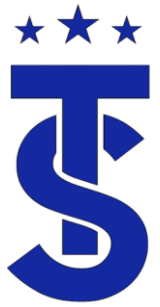




Simplified Power Flow

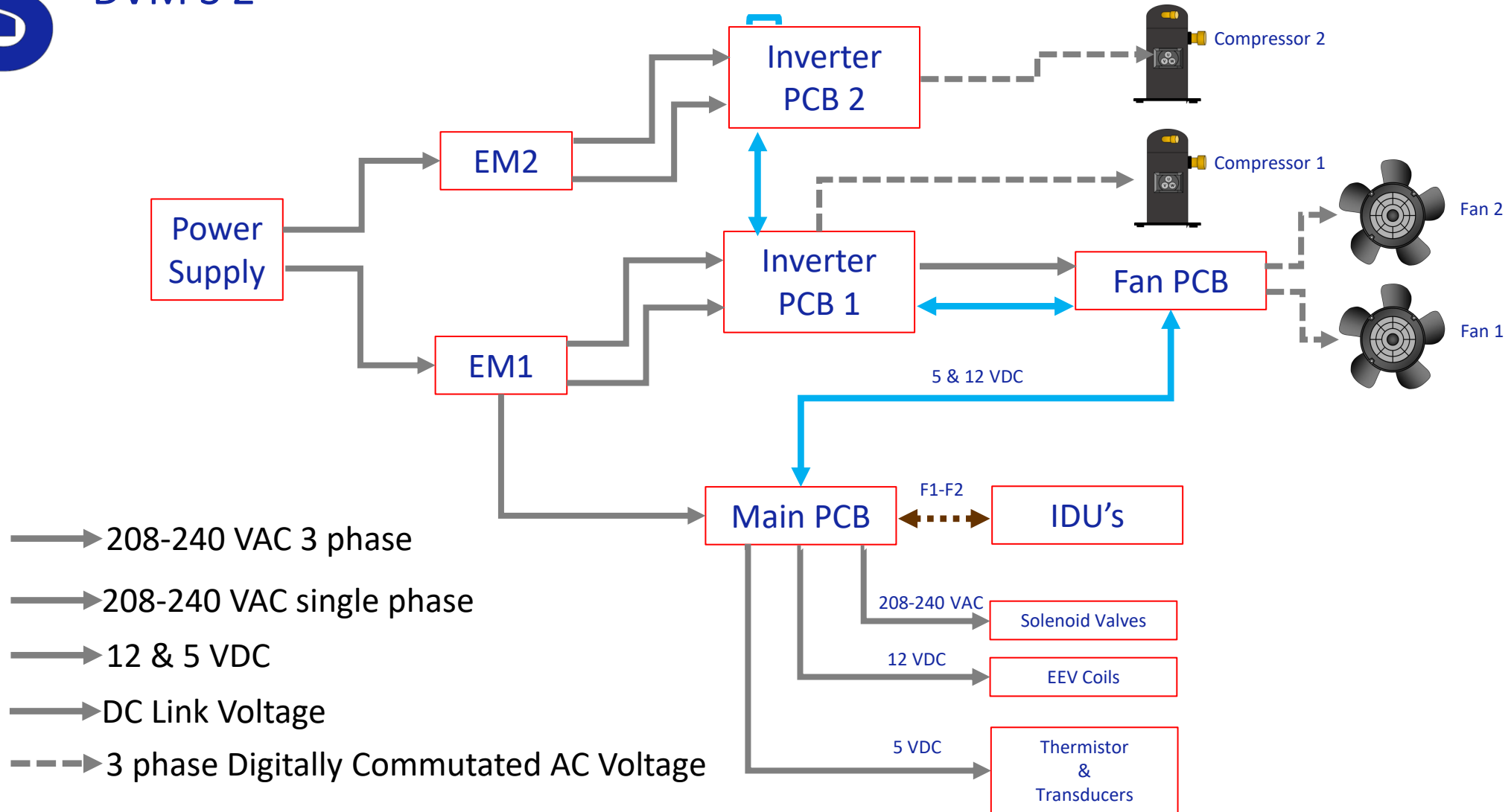
DVM S 2

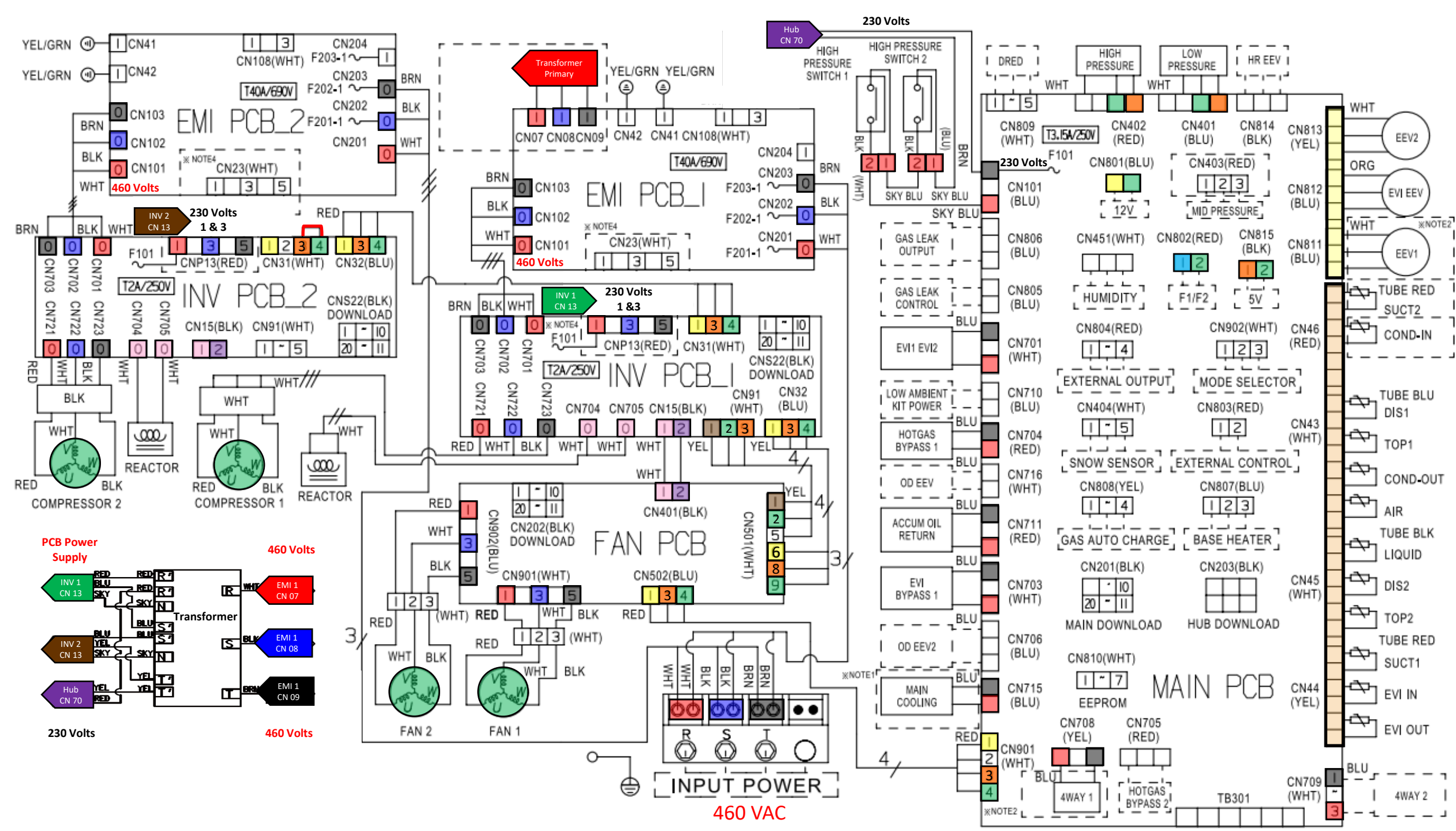


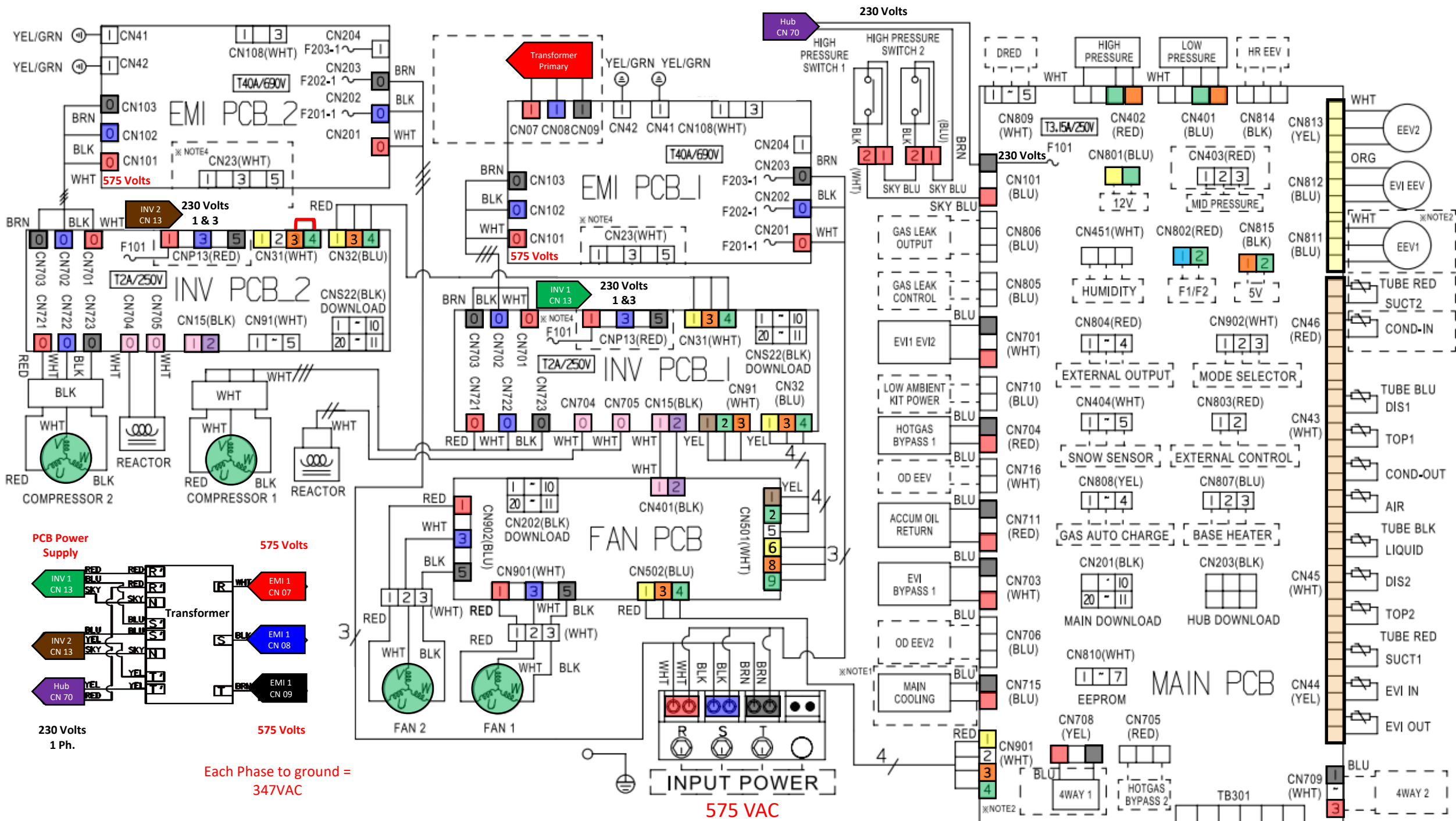


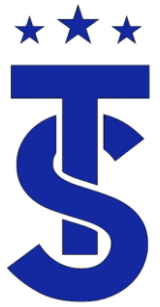
Simplified Communication Flow

DVM S 2









FJM Vs DVM S Eco

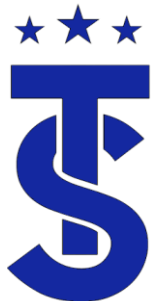
- What Do they have in common?
- What's Different?



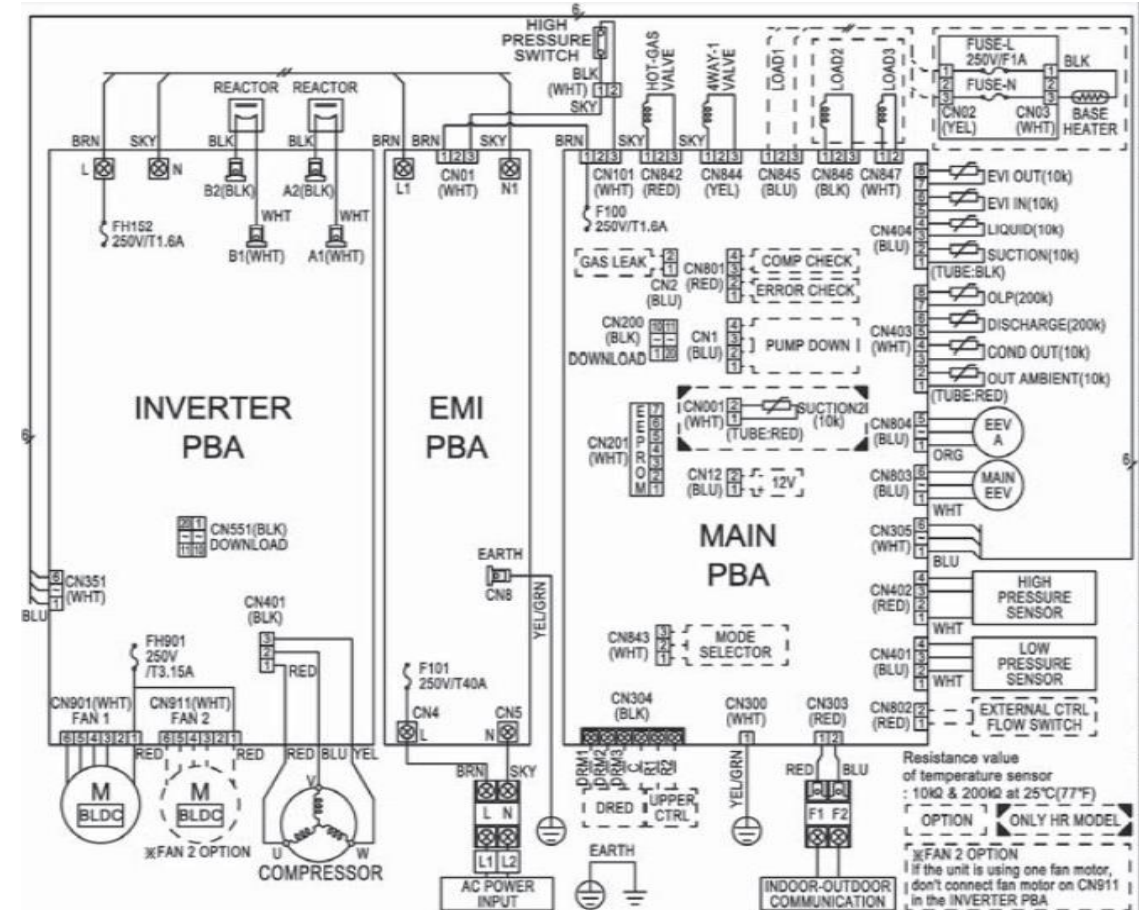
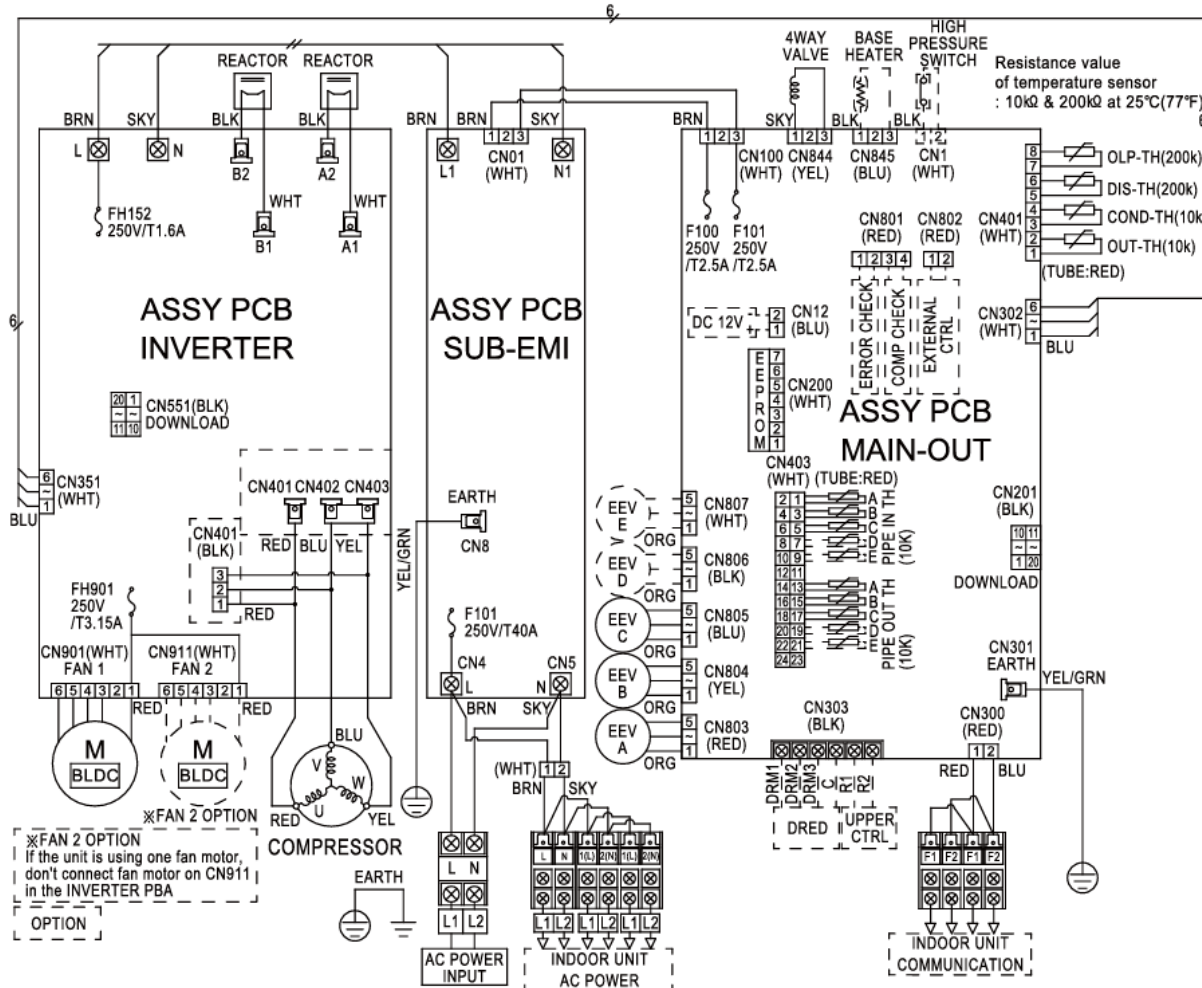
FJM ODU



Eco ODU



FJM Electrical Vs DVM S Eco Electrical



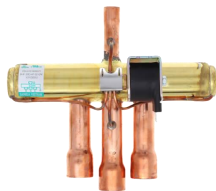


Eco System Components

Component pictures are for demonstration purposes only.
Actual components may look slightly different



Solenoid Valves



Reversing Valve



EEV



High Pressure Sw.



Pressure Transducer



Thermistors



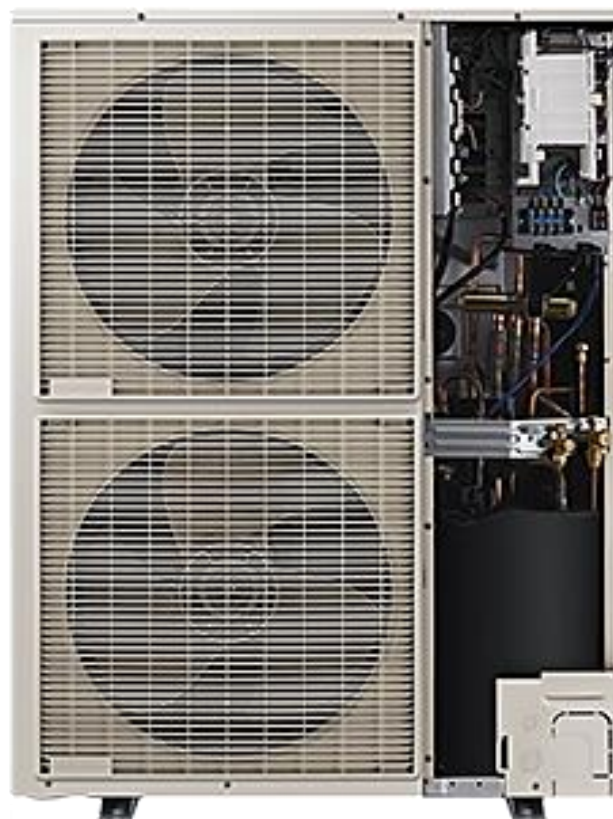
Reactor



BLDC fan Motor



Compressor



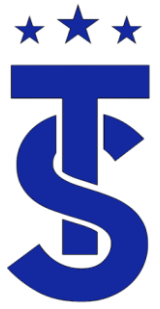
EMI



Inverter PCB



Main PCB



EMI Board

- The EMI board is used to eliminate noise and harmonic from the 208 -240 VAC electrical supply
- The EMI board has an internal fuse
- The EMI supplies 208-240VAC power to the Inverter PCB and Main PCB

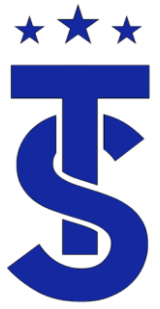




EMI Input and Outputs

No.	Local	Function	Description
1	L1	AC POWER	OT-048
2	EARTH	EARTH	YEL/GRN WIRE
3	L	AC POWER	BRN WIRE
4	N	AC POWER	SKY/BLU WIRE
5	N1	AC POWER	OT-048
6	CN01	AC POWER	YW396-03AV WHT





The Main PCB

- Is the control center of the system
- Is supplied with 208-240 VAC from the EMI
- Supplies 208-240 VAC to the SMPS and line voltage devices.
- The SMPS creates 5,12 & 15 DCV for communication. EEV coils and thermistors
- Communicates with the indoor units via the F1 & F2 controls layer

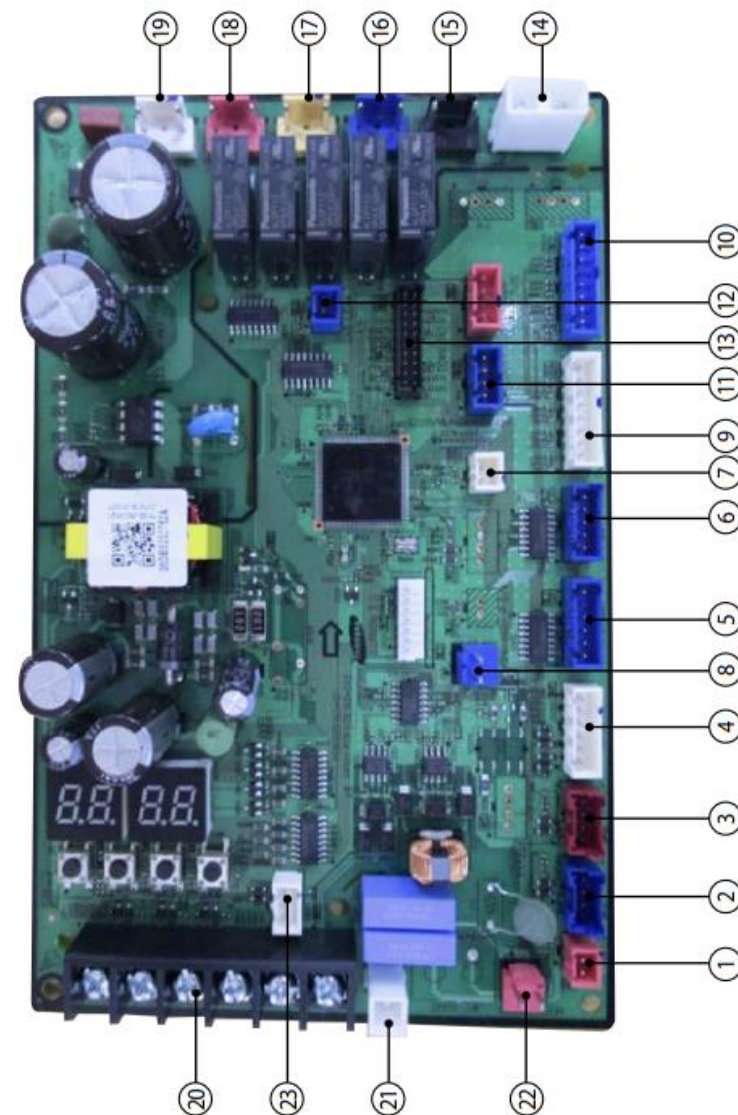


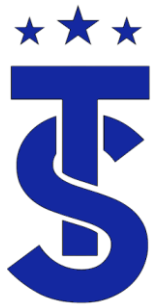


Main PBA Inputs and Outputs

Main PCB (cont.)

No.	Local	Function	Description
1	CN800	EXTERNAL CTRL	SMW250-02 RED
2	CN401	LOW PRESSURE SENSOR	B04B-XARK-1 BLU
3	CN402	HIGH PRESSURE SENSOR	B04B-XARK-1 RED
4	CN305	COMM INV	SMW250-06 WHT
5	CN803	MAIN EEV	SMW250-06 BLU
6	CN804	EEV A	SMW250-05 BLU
7	CN001	TH3	SMW250-02 WHT
8	CN12	DC12V	YW396-02V BLU
9	CN403	OUT/COND/DIS/OLP	SMW250-08 WHT
10	CN404	SUCTION/D_TUBE_TH1_TH2	SMW250-08 BLU
11	CN1	GAS LEAK	SMW250-02 BLU
12	CN2	GAS LEAK	SMW250-04 BLU
13	CN200	DOWN LOAD	YDW200-20 BLK
14	CN847	LOAD1	YDW236-02 WHT
15	CN846	LOAD2	YW396-03AV BLK
16	CN845	LOAD4	YW396-03AV BLU
17	CN844	4WAY	YW396-03AV YEL
18	CN842	HOTGAS	YW396-03AV RED
19	CN101	A/C POWER	YW396-03AV WHT
20	CN304	DRED	DAPC-2009-6P BLK
21	CN300	EARTH	YDW236-01 WHT
22	CN303	COMM-INDOOR	YW396-02V RED
23	CN843	MODE SELECTOR	SMW250-03 WHT





Inverter PCB

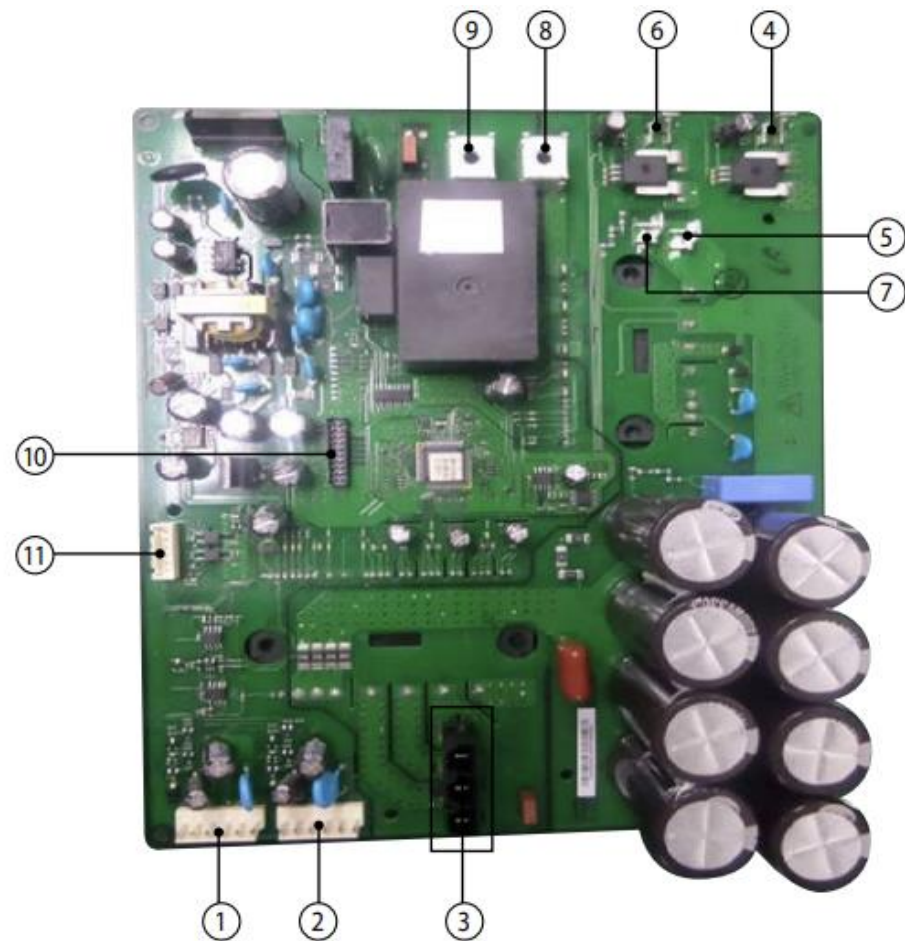
- Receives 208-240 VAC from power from the EMI board.
- Its converts AC power to DC Bus Voltage
- DC bus voltage is supplied to the condenser fan, SMPS and compressor IGBT
- The compressor IGBT/IPM supplies a 3 phase digitally commutated AC voltage to the compressor
- The SMPS is creates 5, 12 and 15 VDC





Inverter Input and Outputs

No.	Local	Function	Description
1	CN901	FAN1	YW396-06V WHT
2	CN911	FAN2	YW396-06V WHT
3	CN401	COMP	42819-3213 BLK
4	REACTOR_A2	REACTOR_A2	YTR250
5	REACTOR_A1	REACTOR_A1	YTR250
6	REACTOR_B2	REACTOR_B2	YTR250
7	REACTOR_B1	REACTOR_B1	YTR250
8	N_	AC POWER	OT-048
9	L_	AC POWER	OT-048
10	CN551	DOWNLOAD	YDW200-20 BLK
11	CN351	COMM-MAIN	SMW250-06 WHT





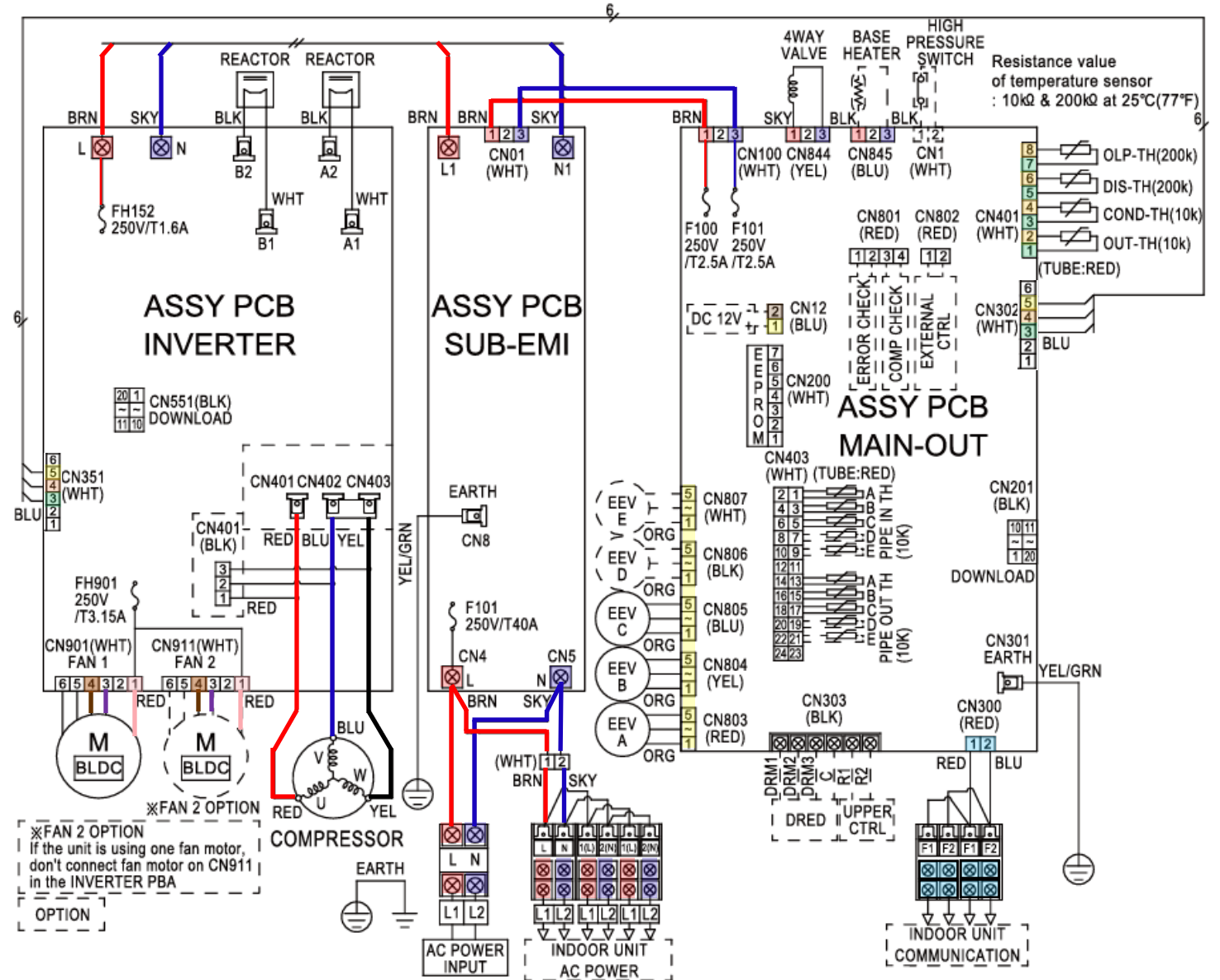
BLDC Outdoor Fan Motor

- The motor is constantly supplied with DC Bus Voltage in the range of 290+ VDC
- It requires a control voltage input of 15 DVC
- Its speed is adjusted by a varying DC voltage supplied to the motor in the rang of 2-5 VDC
- It send a rotational feedback signal to the to PCB



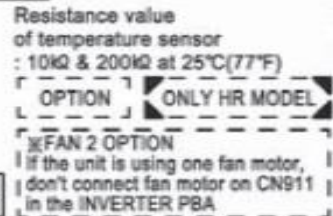
FJM ODU Power Flow

- 5 VDC (communications)
- 12VDC (control)
- 15VCD (control Fan)
- GND
- DC Link 305+ VDC





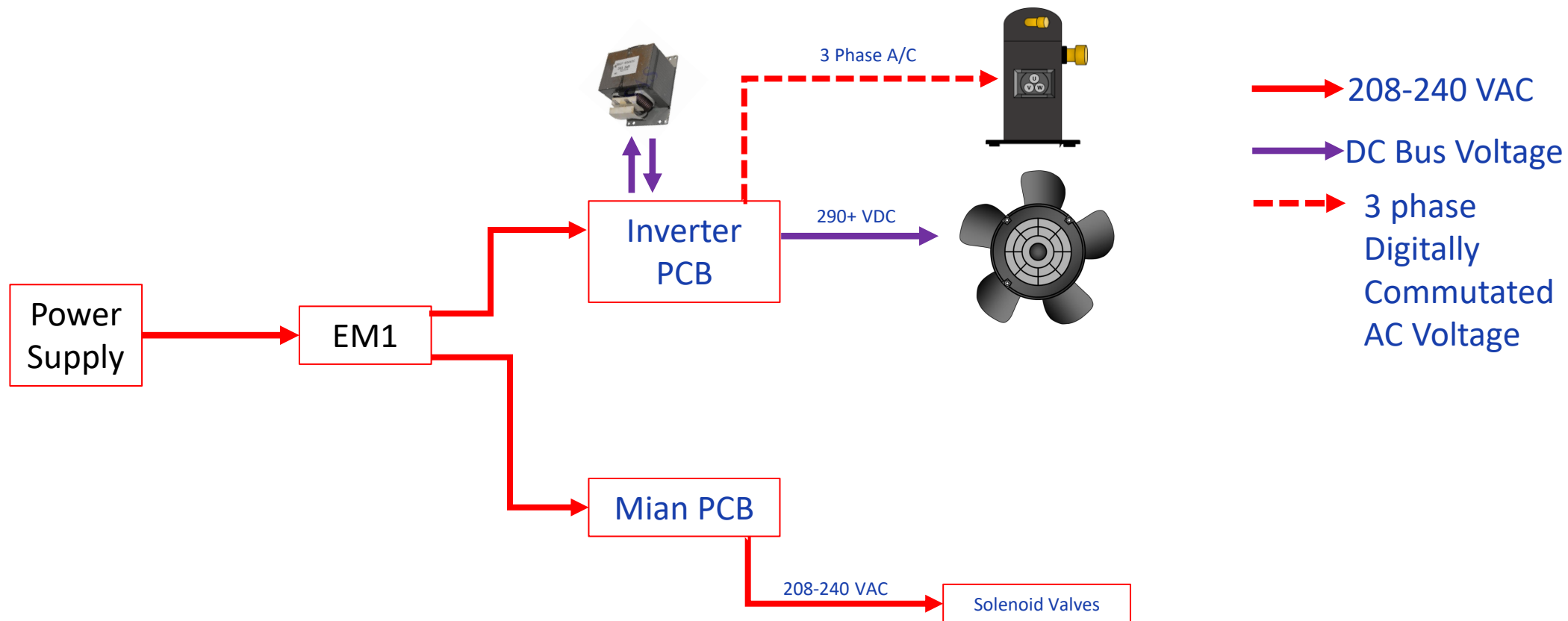
- 





Simplified Eco ODU Power Flow

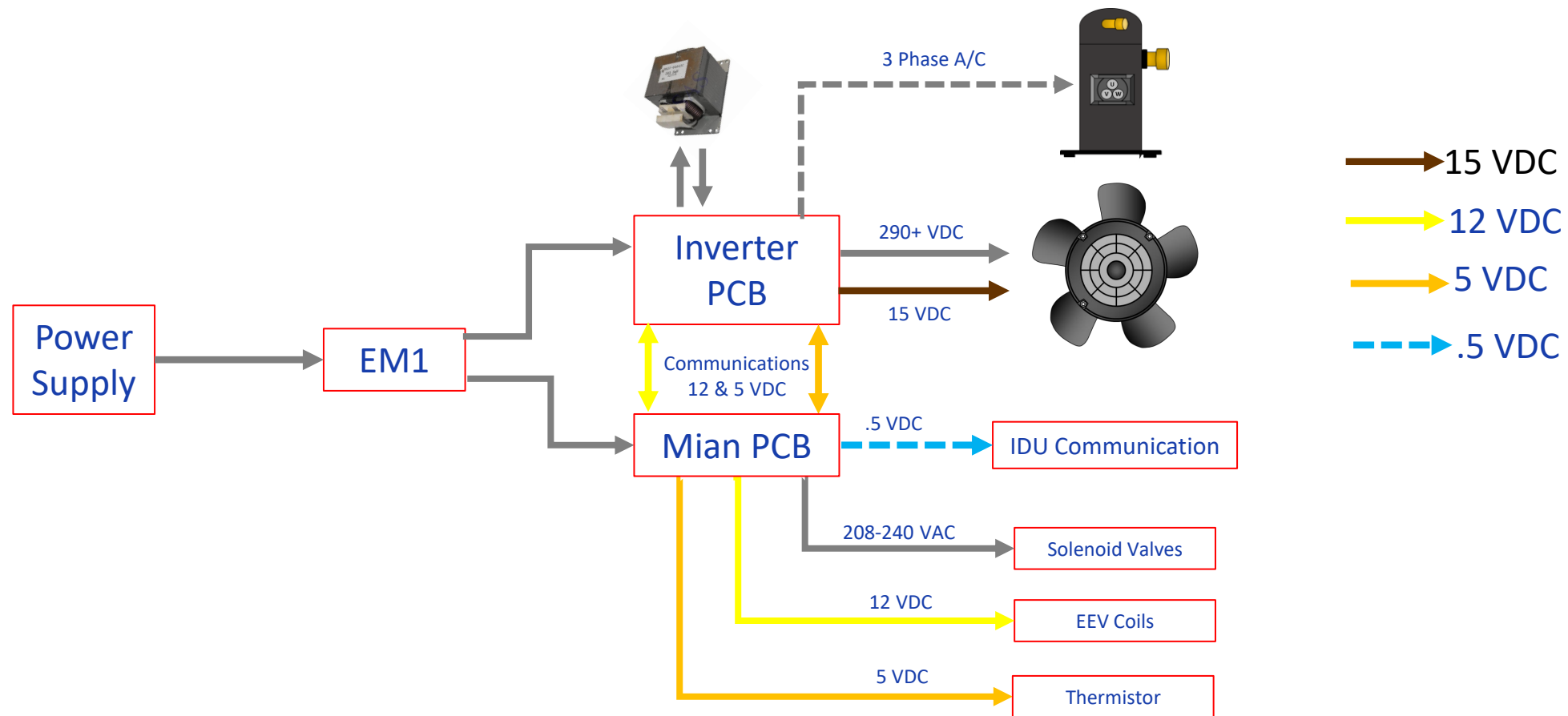
High Voltage





Simplified Eco ODU Power Flow

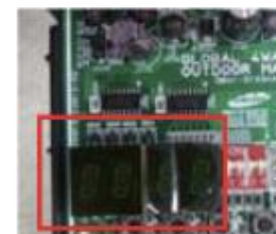
Control Voltages





Change the outdoor unit inverter or main PBA.

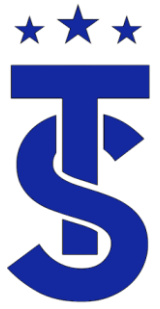
1. If the 7-segment on the outdoor unit main PBA shows E203 error, change the PBA → outdoor unit main PBA is normal.
2. When the LED on the outdoor unit inverter is shown as following; Red-On, Green-On, Orange-Blinking and if the 7-segment on the outdoor unit main PBA does not display anything, change the main PBA → Outdoor unit inverter PBA is normal.



[7-segment on
Main PBA (Common)]

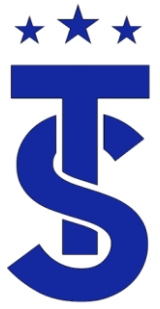


[AM****XMDC* series
inverter PBA LED]

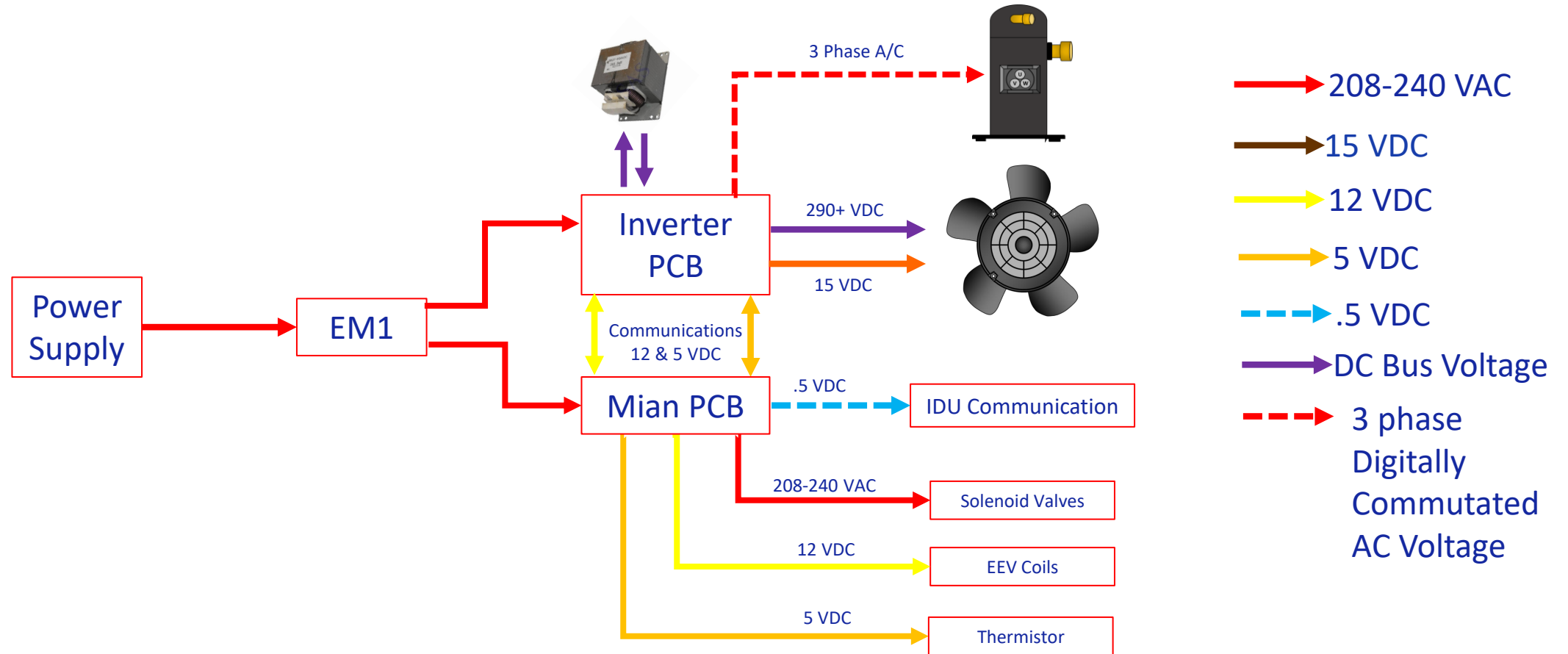


Inverter PCB LED's

- Inverter PCB LED's
- There are 3 LEDs on the PCB that can be used for service
- The LEDs are powered by the main PCB
- Under normal circumstances Red & Green LED's are lit
- The Orange LED blinks when the inverter PCB and Main PCB are communicating.



Simplified Eco ODU Power Flow



Indoor Unit Line Up



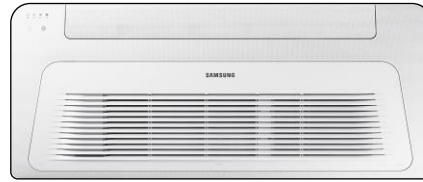
360 Cassette
9K - 48K



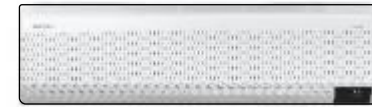
WindFree™* 4-Way Cassette
6K - 48K



WindFree™* Mini 4-Way Cassette
5K - 20K



WindFree™* 1-Way Cassette
7K-12K



WindFree™* 2.0
5K - 28K



Max
32K



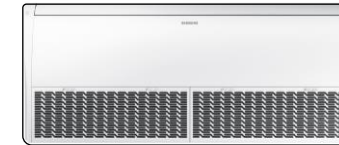
Duct S
6K - 48K



Slim Duct
7K - 48K



Floor Standing (Concealed)
6K - 24K



Big Ceiling
36K - 48K



HSP Duct
54K



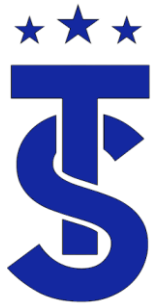
Floor Standing
(Exposed)
6K - 24K



Under Ceiling
18K - 24K



Multi-Position Air Handler
12K - 60K



IDU System Components

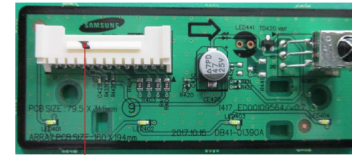
Click on any component to see a description of the component



Fan Motor



EEV



PCB



Controller



Louver Motor



Drain Pump



PCB



5 VDC Thermistor

IDU PCB

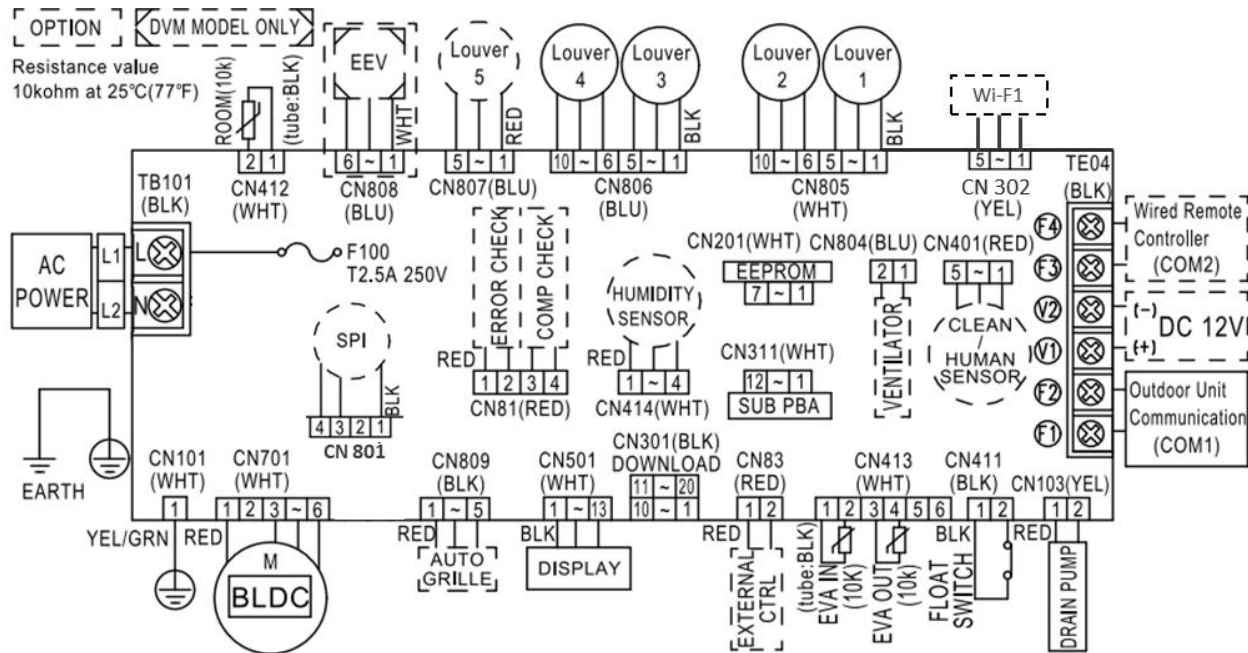
- The indoor PCB is the heart of the indoor unit.
- It is supplied with power from the outdoor unit.
- It contains an inverter circuit that produces the DC Link Voltage for the fan and a SMPS for that creates the communication and low voltage supply.





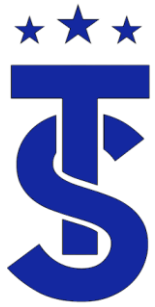
4 Way Cassette

- Electrical Test point locations

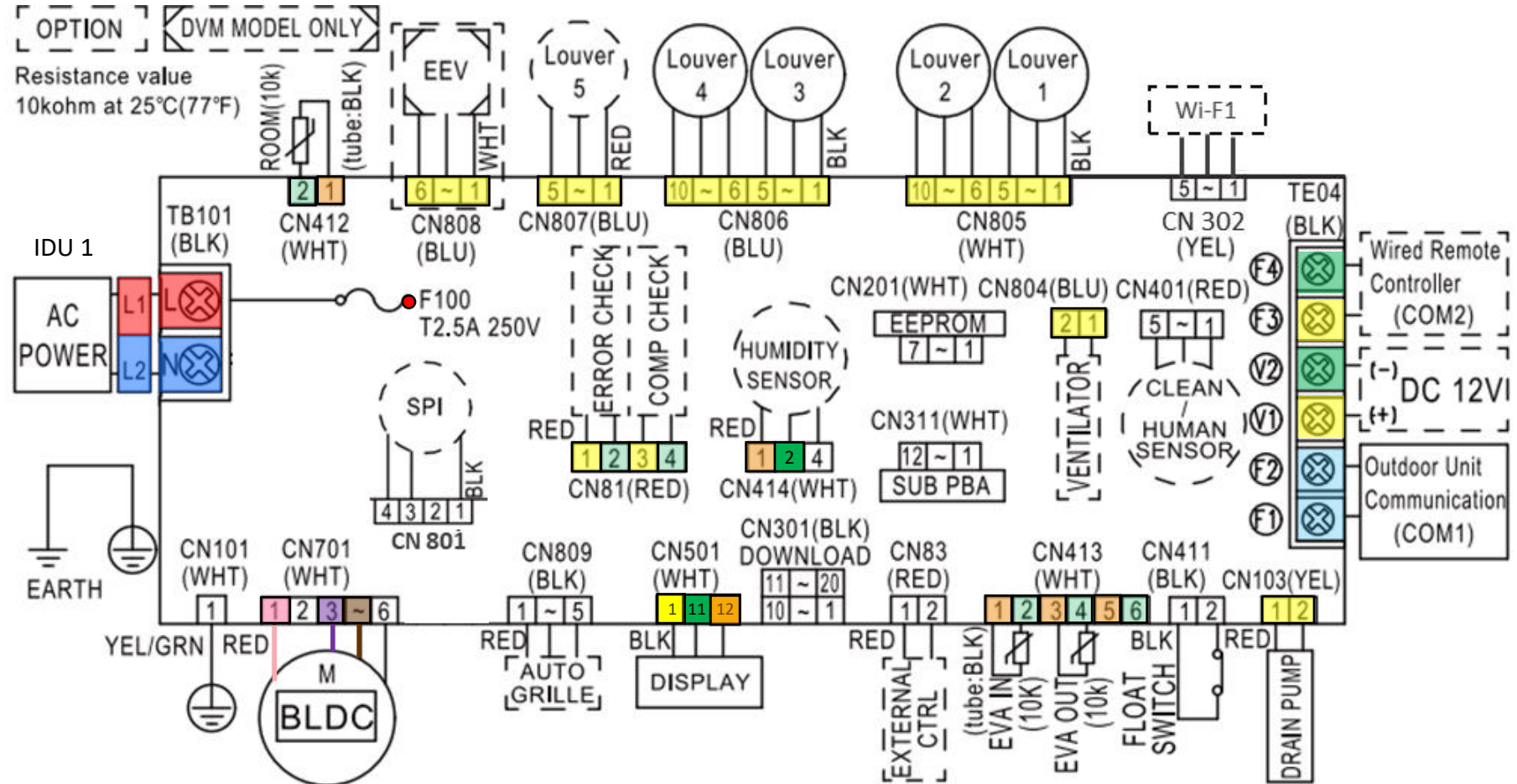


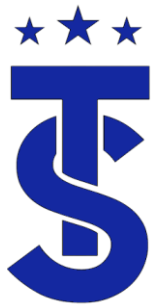
4way cassette , mini 4way cassette type

① CN101-GND #1: GND	② CN701-BLDC MOTOR #1: DC310V #3: GND #4: DC15V #5: FAN RPM #6: RPM FEEDBACK	③ CN140-FUSE CHECK #1: FUSE CHECK SIGNAL #2: GND	④ CN809-AUTO GRILL #1: DC12V #4: REMOCON SIGNAL #5: GND
⑤ CN501-DISPLAY #1: DC12V #2: LED_0 #3: LED_1 #4: LED_2 #5: LED_3 #6: LED_4 #7: LED_5 #8: REMOCON_OUTPUT_SIGNAL #9: AUTO SWITCH #10: REMOCON_INPUT_SIGNAL #11: GND #12: DC5V #13: GND	⑥ CN301-DOWNLOAD	⑦ CN83-EXT CTRL #1: GND #2: EXT-CTRL SIGNAL	⑧ CN413:THERMISTOR #1: EVA-IN THERMISTOR #2: GND #3: EVA-OUT THERMISTOR #4: GND #5: DISCHARGE THERMISTOR #6: GND
⑨ CN411-FLOAT SWITCH #1: F/S SIGNAL #2: GND	⑩ CN103-DRAIN PUMP #1: D/P POWER(DC12V) #2: GND	⑪ CN81-ERROR/COMP CHECK #1: DC12V #2: ERROR SIGNAL OUTPUT(GND) #3: DC12V #4: COMP/OPER. SIGNAL OUTPUT(GND)	⑫ CN201-EEPROM #1: GND #3: DC5V #4: EEPROM_SELECT #5: EEPROM_SO #6: EEPROM_SI #7: EEPROM_CLK
⑬ CN311-2WIRED REMOCON	⑭ CN804-VENTILATOR #1: DC12V #2: VENT SIGNAL OUTPUT(GND)	⑮ CN401-HUMAN SENSING #1: DC12V #2: HUMAN SENSOR COMM(TXD) #3: HUMAN SENSOR COMM(RXD) #4: GND	⑯ CN412-ROOM THERMISTOR #1: ROOM THERMISTOR #2: GND
⑰ CN808-EEV #1~#4: EEV SIGNAL OUTPUT #5: DC12V #6: DC12V	⑱ CN807-LOUVER5 #1: DC12V #2~#5: LOUVER SIGNAL OUTPUT	⑲ CN806-LOUVER3/4 #1: DC12V #2~#5: LOUVER SIGNAL OUTPUT #6: DC12V #7~#10: LOUVER SIGNAL OUTPUT	⑳ CN805-LOUVER1/2 #1: DC12V #2~#5: LOUVER SIGNAL OUTPUT
㉑ CN801-SPI #1: GND #2: GND #3: SPI POWER OUTPUT(DC12V)	㉒ TB101-AC POWER #1: POWER(L) #2: POWER(N)	㉓ TE04-COMMUNICATION #1: COM1(F1) #2: COM1(F2) #3: V1(DC12V) #4: V2(GND) #5: COM2(F3) #6: COM2(F4)	



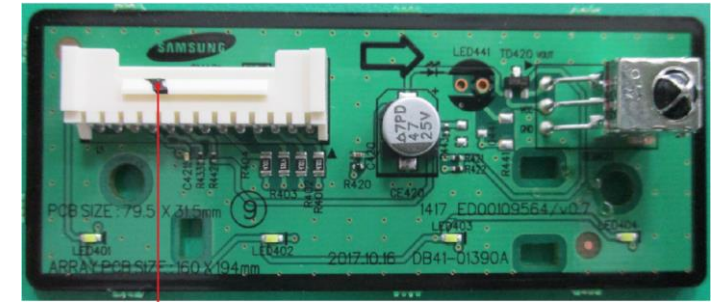
IDU Electrical





Display Board

- The display board receives the signal from the handheld controller and communicates that information with the main board.

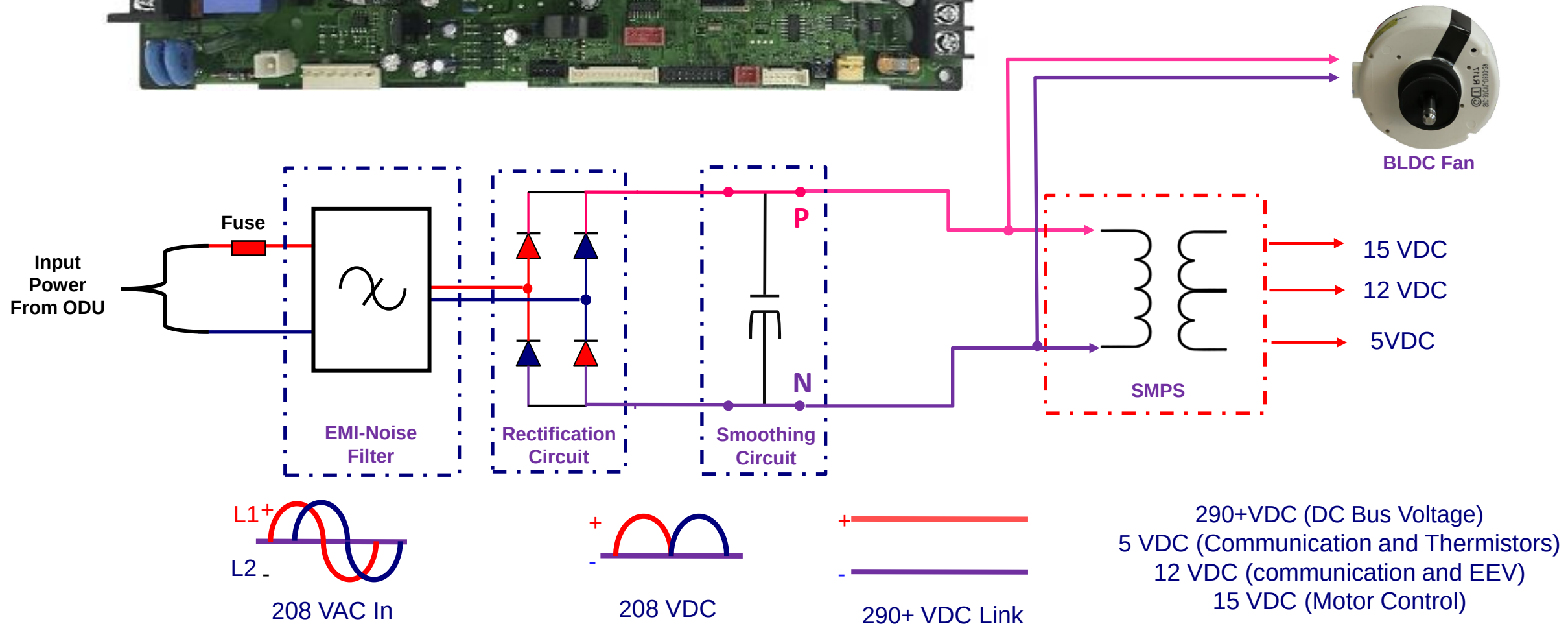


1. CN01-DISPLAY

- #1 : DC12V
- #2 : LED_Operation
- #3 : LED_Defrost
- #4 : LED_Timer
- #5 : -
- #6 : LED_Filter
- #7 : -
- #8 : Remocon Signal Out
- #9 : Panel Select
- #10 : Remocon Signal In
- #11 : GND
- #12 : DC5V
- #13 : -



IDU Power Flow





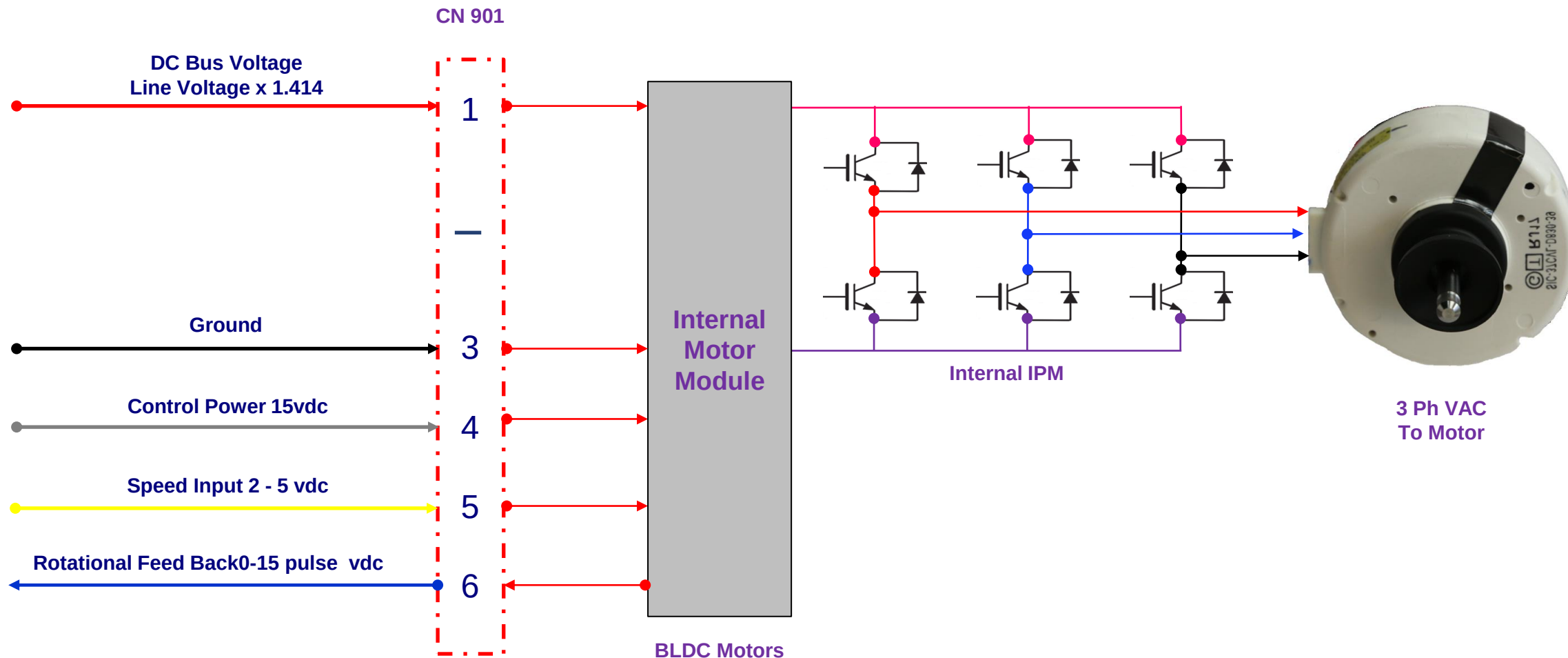
Indoor Motor

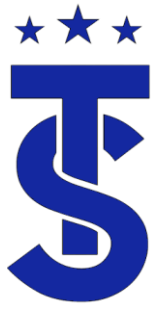
- The motor is constantly supplied with DC Bus Voltage 290+ VDC
- It requires a control voltage input of 15 DVC
- Its speed is adjusted by a varying DC voltage supplied the rang of 2-5 VDC
- It send a rotational feedback signal to the to PCB





Indoor Fan Motor Voltages





Drain Pump

- The Drain pump motor operates on 12 VDC
- There is a N.O float switch connected to the PCB
- The pump must be activated through the option code setting
 - 020000-1X0000-200000-300000
 - Section 2
 - Page 1
 - Segments 8



Thermistors

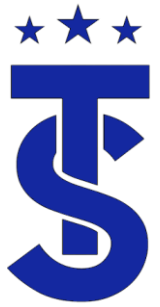
- Thermistors use 5 Vdc.
- They are a NTC device where resistance decreases as temperature increases.
- Indoor units use 10K ohm thermistors to measure:
 - Pipe in Temperature (Liquid Pipe)
 - Pipe Out Temperature (gas Pipe)
 - Ambient -Return Air



Louver Motor

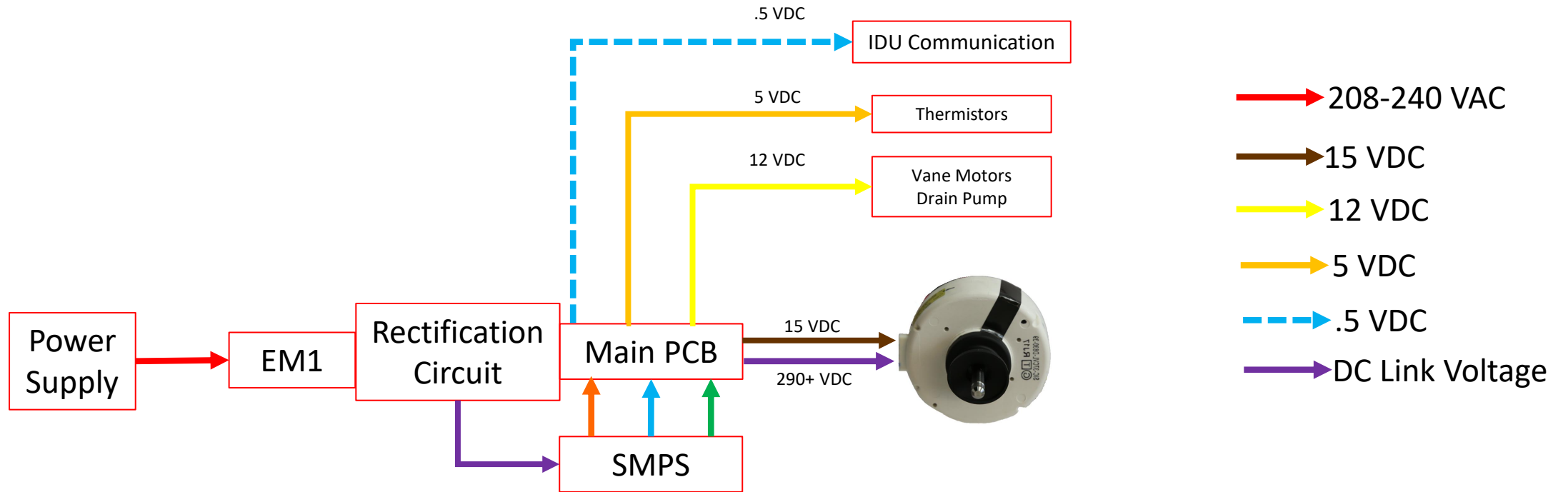
- Louver motors are powered with 12 VDC and controlled by the main PCB





Simplified IDU Power Flow

Integrated SMPS





The End

- ☐ DO WE WANT TO PUT A SECTION IN HERE ON HOW TO BENCH TEST EACH COMPONENT?



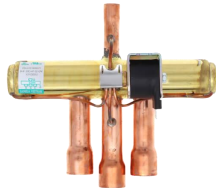
Component Troubleshoot



Component Troubleshooting



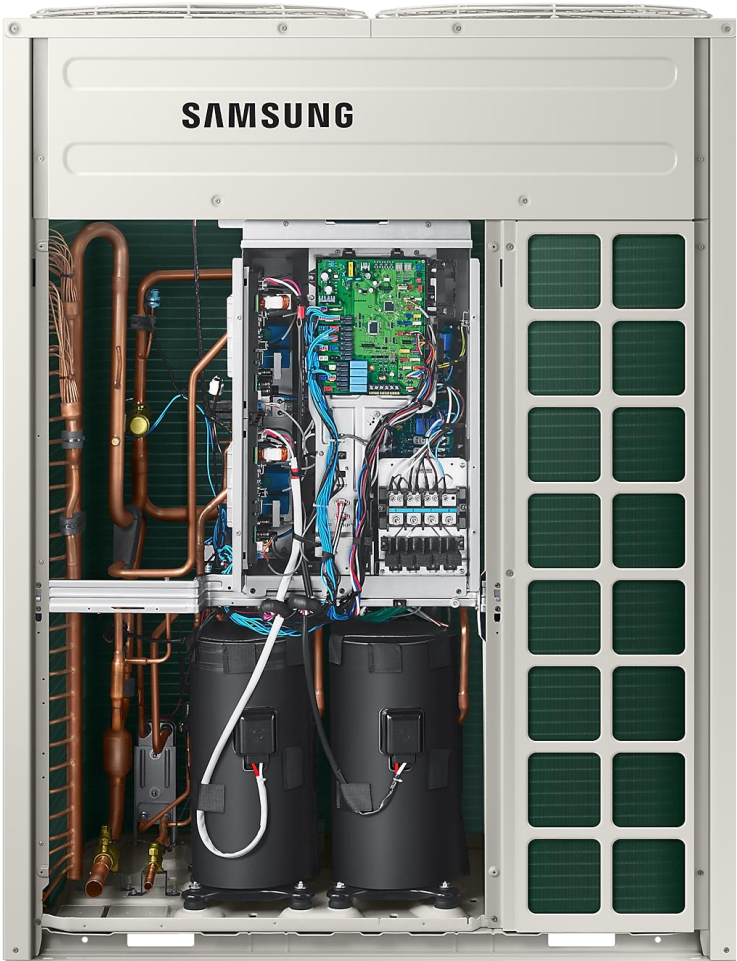
Solenoid Valves



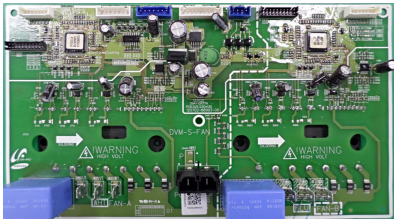
Reversing Valves



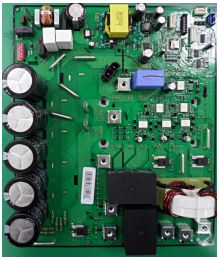
EEV



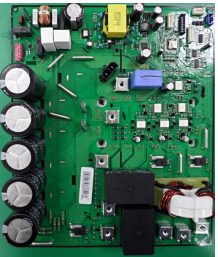
Mian, Hub & Communications PBA



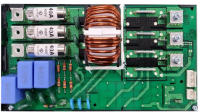
Fan PBA



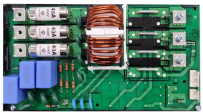
Inverter 1 PBA



Inverter 2 PBA



EMI 1



EMI 2



High Pressure Sw.



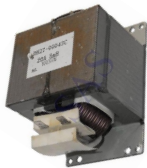
Pressure Transducer



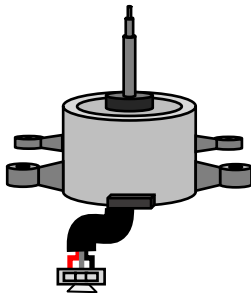
Thermistors



Compressor



Reactor



Condenser fan

Component pictures are for demonstration purposes only.
Actual components may look slightly different



Error Codes

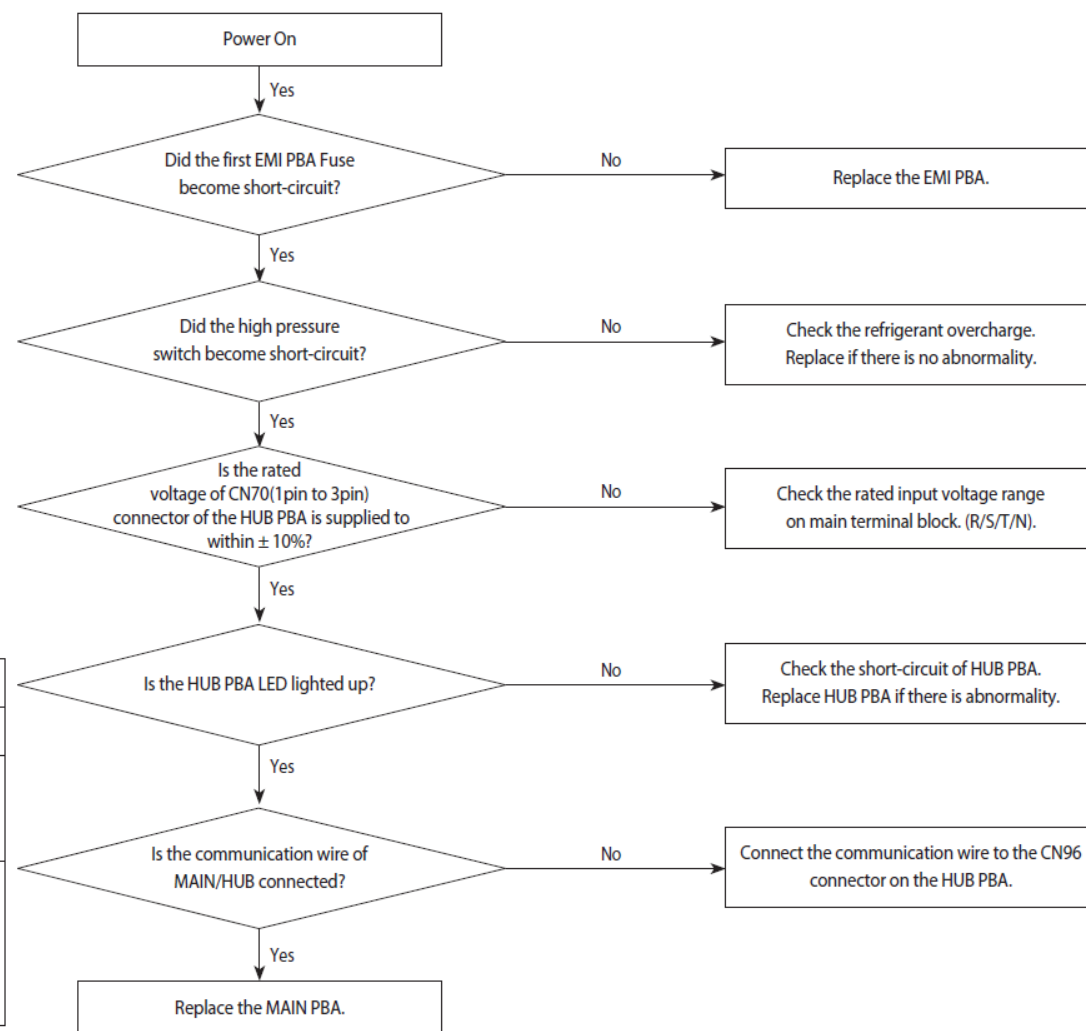
Displayed alphabet	Explanation	
<i>E</i>	When displaying Error 101~700	
<i>P</i>	When displaying Error 701~800	
<i>L</i>	When E206 error occurs	Displays address of subordinate within the set C001 : HUB, C002: FAN, C003: INV1, C004: INV2
	When MCU error occurs	Displays address of MCU Ex) C100: MCU address 0, C101: MCU address 1, C102: MCU address 2
<i>P</i>	When displaying outdoor unit address Ex) U200~203 main, Sub 1, 2, 3	
<i>U</i>	When displaying indoor unit address Ex) A000: Indoor unit adress 0, A001: Indoor unit address 1, A002: Indoor unit address 2	



Fault Main PBA had no power

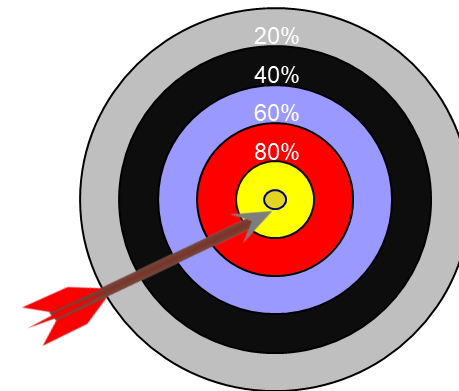
Type of Power Source	AC Input Voltage	Standard AC Voltage(Vac)		Standard DC Voltage(Vdc)	
		Min.	Max.	Min.	Max.
F	208~230V	177	265	250	374
J	460V	391	552	553	781

Outdoor unit display	Main PBA has no power phenomenon (7-seg does not blink)
Judgment Method	Hub PBA power and connection wire to detect.
Connector check Method	CN96 on HUB PBA - 1pin to 4pin : DC 12V - 9pin to 4pin : DC 5V
Cause of problem	· HUB PBA connector wire defects and the connection is not. · Main PBA defective. · Hub PBA defective. · High pressure switch operation · Water hub PBA defective. CN96 on HUB PBA



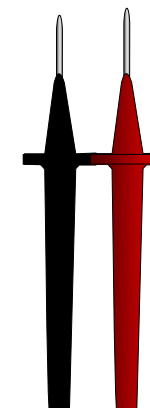
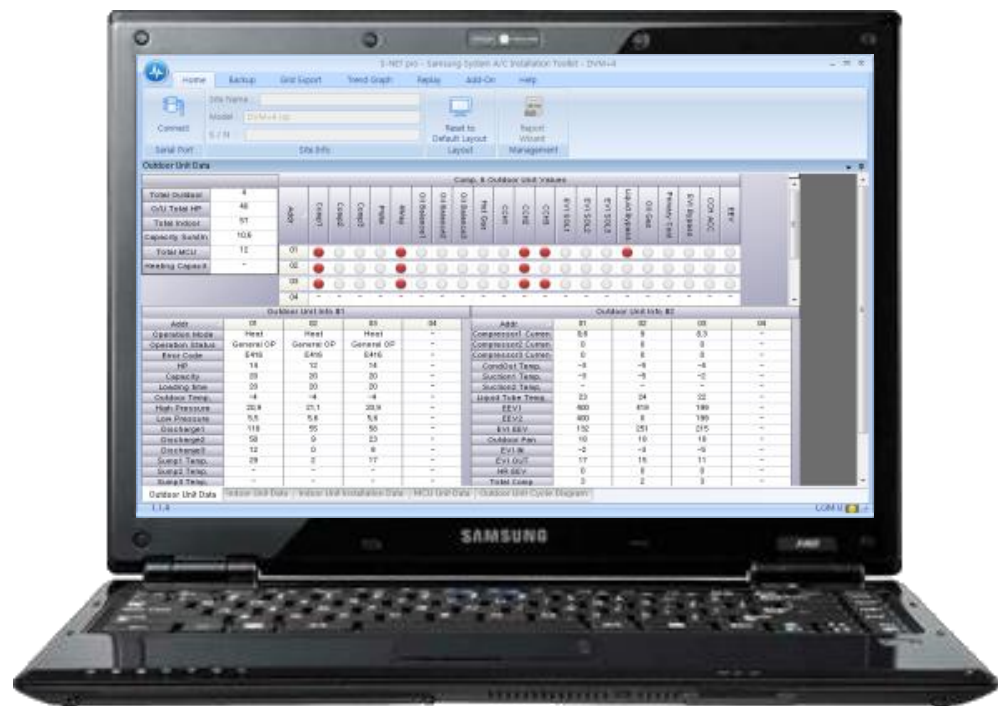


Troubleshooting Process



- ☐ Process:
 - ☐ A series of actions or steps taken in order to achieve a particular end.
- ☐ Is an Error Code being Displayed?
- ☐ What code is being displayed?
- ☐ What does the code mean?
- ☐ Is there a troubleshooting flow chart or video?
- ☐ If not you have to start with basic voltage in and out test.

Gathering Information.



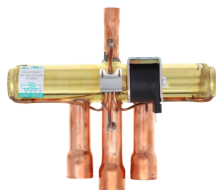
Which one can you do with out?



Component Troubleshooting



Solenoid Valves
Water Hub
Hub PBA



Reversing Valves
Hub PBA



EEV
Hub PBA



High Pressure Sw.
Water Hub



Pressure
Transducer
Hub PBA



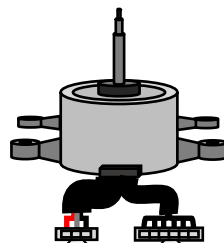
Thermistors



Compressor
Inverter PBA



Reactor
Inverter PBA



ODU Fan
Motor

❑ Troubleshooting Components are straight forward, their test procedures do not change

❑ Troubleshooting boards is more challenging but if the manufactures procedures are followed the process becomes easier



Main PBA



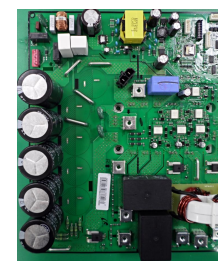
Comm. PBA



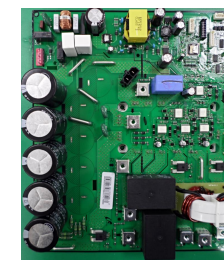
Hub PBA



Fan PBA



Inverter # 1 PBA



Inverter # 2 PBA



EMI # 1



EMI # 2

SAMSUNG



PBA Troubleshooting

- ❑ Troubleshooting boards is more challenging but if the manufactures procedures are followed the process becomes easier.
- ❑ Keep it simple. Verify inputs and outputs.
- ❑ Use the service manuals to identify the Voltages at each connector



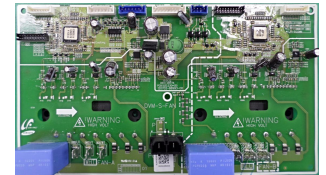
Main PBA



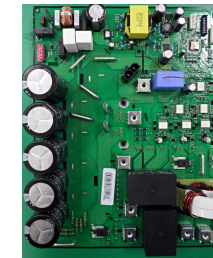
Comm. PBA



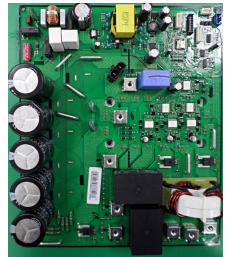
Hub PBA



Fan PBA



Inverter # 1 PBA



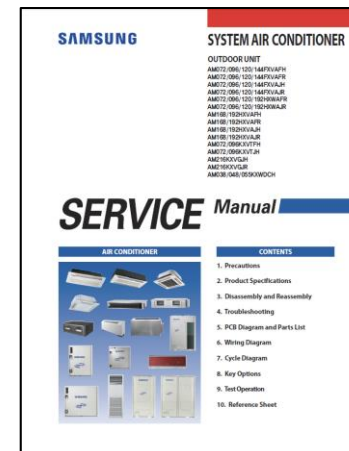
Inverter # 2 PBA



EMI # 1



EMI # 2



SAMSUNG



Troubleshooting the Hub PCB

- ☐ Power In:
 - ☐ 208 – 240 VAC
- ☐ Power Out:
 - ☐ 5 VDC
 - ☐ Thermistors
 - ☐ Pressure Transducers
 - ☐ 12 VDC
 - ☐ EEV Coils
 - ☐ Power to the Main Board

Input
208-240 VAC

★ C Chassis
Communications
input



Output
5 & 12 VDC to
Main PBA

Output
5 VDC Devices

Output
12 VDC Devices

★ C Chassis
Communications
output

Output
208-240 VAC



Hub PBA Inputs & Outputs

★ C Chassis
Communications

CN 96

Main PBA connection
(uART comm., DC5, 12V
supply to Main PBA)



① CN83-EV1 EEV #1: EEV3_A_OUT #2: EEV3_B_OUT #3: EEV3_A'_OUT #4: EEV3_B'_OUT #5: 12V	② CN81-EEV1 #1: EEV1_B'_OUT #2: EEV1_A'_OUT #3: EEV1_B_OUT #4: EEV1_A_OUT #5: 12V #6: 12V	③ CN82-EEV2 #1: EEV2_B'_OUT #2: EEV2_A'_OUT #3: EEV2_B_OUT #4: EEV2_A_OUT #5: 12V #6: 12V	④ CN43-TEMP.SENSOR #1: COMP1 DISACHRGE #2: COMP1 DISCHARGE #3: COMP1 TOP #4: COMP1 TOP1 #5: COND OUT #6: COND OUT #7: OUTDOOR TEMP. #8: OUTDOOR TEMP.
⑤ CN45-TEMP.SENSOR #1: LIQUID #2: LIQUID #3: COMP2 DISCHARGE #4: COMP2 DISCHARGE #5: COMP2 TOP #6: COMP2 TOP	⑥ CN46-SUCT #1: SUCTION 2 #2: SUCTION 2 #3: GND #4: GND #6: GND	⑦ CN44-TEMP.SENSOR #1: SUCTION 1 #2: SUCTION 1 #3: EVI INLET #4: ENI INLET #5: ENI OUT #6: EVI OUT	⑧ CN906-SNOW SENSOR #1: 12V #3: GND #4: SNOW_SENSOR #5: PSD_POWER
⑨ CN42-HIGH PRESSURE SENSOR #1: HIGH PRESSURE SENSOR #3: GND #4: VCC	⑩ CN41-LOW PRESSURE SENSOR #2: LOW PRESSURE SENSOR #3: GND #4: VCC	⑪ CN97-INV COMM #1: 12V #2: INV_SMPS_RELAY #3: COMM OUT #4: GND	⑫ CN96-MAIN-HUB COMM. #1: 12V #2: INV_SMPS_RELAY #3: COMM-MAIN #4: GND #5: HIGH-PRESSURE-SENSOR #6: LOW-PRESSURE-SENSOR #7: ZERO-CROSSING #8: GND #9: VCC

■ DC

① CN714-CCH1 #1: CCH1 #2: CCH1	② CN715-MAIN-COOLING #1: MAIN-COOLING #2: MAIN-COOLING	③ CN713-CCH2 #1: CCH2 #2: CCH2	④ CN704-HOTGAS-VALVE1 #1: HOTGAS BYPASS1 #2: HOTGAS BYPASS1
⑤ CN705-HOTGAS-BYPASS2 #1: HOTGAS BYPASS2 #2: HOTGAS BYPASS2	⑥ CN703-EVI-BYPASS #1: EVI BYPASS1 #2: EVI BYPASS1	⑦ CN716-OD-EEV-VALVE #1: OD EEV VALVE #2: OD EEV VALVE	⑧ CN711-OIL-RETURN-VALVE #1: ACCUM OIL RETURN VALVE #2: ACCUM OIL RETURN VALVE
⑨ CN708-4-WAY-VALVE #1: 4-WAY VALVE #2: 4-WAY VALVE	⑩ CN70-AC POWER INPUT #1: AC LIVE #2: AC NEUTRAL	⑪ CN701 EVI VALVE 1,2 #1: EVI VALVE 1 #3: EVI VALVE 2 #7: EVI VALVE 1 #8: EVI VALVE 2 #9: AC NEUTRAL	



Troubleshooting the Main PBA

Inputs & Outputs



C Chassis
Communications

① CN22-DOWNLOAD #1 : RX-DOWN #2 : TX-DOWN #3 : N-TRST #4 : TDO #5 : TCK #6 : TDI #7 : TMS #8 : #9 : GND #10 : VCC	② CN43-COMM TEST #1 : 12V #2 : INVERTER-INRUSH-OUT #3 : INVERTER-COMM #4 : GND	③ CN301-EEPROM #1 : GND #2 : #3 : VCC #4 : EEPROM-SELECT #5 : EEPROM-SO #6 : EEPROM-SI #7 : EEPROM-CLOCK	④ CN42 - HUB COMM #1 : 12V #2 : INVERTER-INRUSH-OUT #3 : INVERTER-COMM #4 : GND #5 : HIGH-PRESSURE-SENSOR #6 : LOW-PRESSURE-SENSOR #7 : ZERO-CROSSING #8 : GND #9 : VCC
⑤ CN901-DRED #1 : KEY3 #2 : GRID #3 : KEY4 #4 : GND #5 : VCC	⑥ OPT1 -MODE SELECTOR #1 : KEY3 #2 : GRID #3 : KEY4	⑦ CN85-CONDITION CHECK #1 : 12V #2 : ERROR-CHECK-OUT #3 : 12V #4 : COMP-CHECK-OUT	⑧ CN86-EXTERNAL CONTROL #1 : CONTROL #2 : GND
⑨ CN12 - 12V POWER #1 : 12V #2 : GND	⑩ CN45 -OUTDOOR COMM #1 : COM-C #2 : COM-D #3 : #4 : 12V #5 : GND	⑪ CN44 - INDOOR COMM #1 : COM-A #2 : COM-B #3 : 5V #4 : AGND	⑫ CN34- UNUSED COMM #1 : COM-E #2 : COM-F
⑬ CN33-INDOOR COMM (EXTRA) #2 : COM-B	⑭ CN13-5V POWER #1 : COM-A #2 : COM-B	⑮ CN88-GAS LEAK #1 : VCC #2 : COM-B	⑯ CN87-PUMP DOWN #1 : 12V #2 : PUMP DOWN START OUT #3 : 12V #4 : PUMP DOWN END OUT

Troubleshooting the Inverter PBA Power In and Out

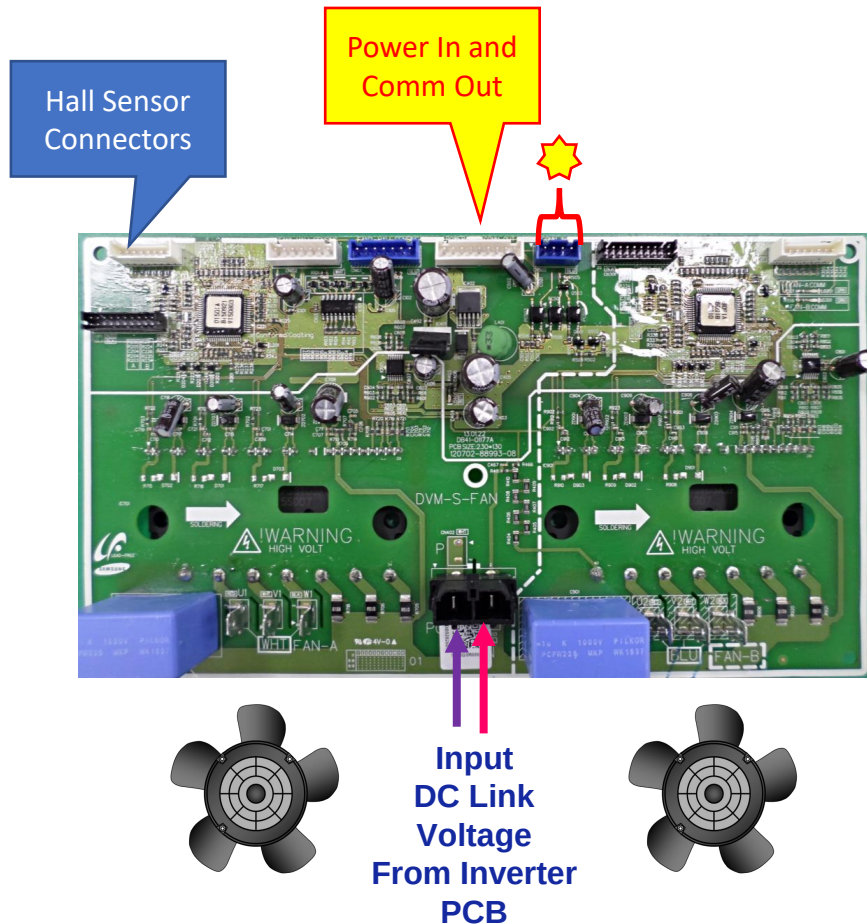
① U-COMPU #1: COMPU	② V-COMPV #1: COMPV	③ W-COMPW #1: COMPW	④ T-INPUTTOP #1: T-IN
⑤ S-INPUTTOP #1: S-IN	⑥ R-INPUTTOP #1: R-IN	⑦ CN705-REACTOR1 #1: REACTOR1	⑧ CN704-REACTOR 2 #1: REACTOR 2
⑨ CN15-FAN POWER #1: DC_LINK #3: GND	⑩ CNP13-AC POWER #1: AC LIVE #3: AC NEUTRAL	⑪ CN31-MAIN COMM 2 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN	⑫ CN32-MAIN COMM1 #1: 12V-MAIN #2: IN-SMPS-RELAY #3: COMM-IN #4: GND-MAIN
⑬ CN91-FAN DC #1: 18V #2: GND #3: 5V-FAN #4: AD-SELECT	⑭ CNS22-INV MICOM DOWNLOAD #1: RX_DEBUG #2: TX_DEBUG #3: BOOT #4: JTAG_TDO #5: JTAG_TCLK #6: JTAG_TDI #7: JTAG_TMS #8: JTAG RESET #9: GND #10: VCC #11: VCC #17: GND #18: DAC_CLK #19: DAC_CS #20: DAC_DATA		



 C Chassis Communications



Fan PBA Input and Outputs

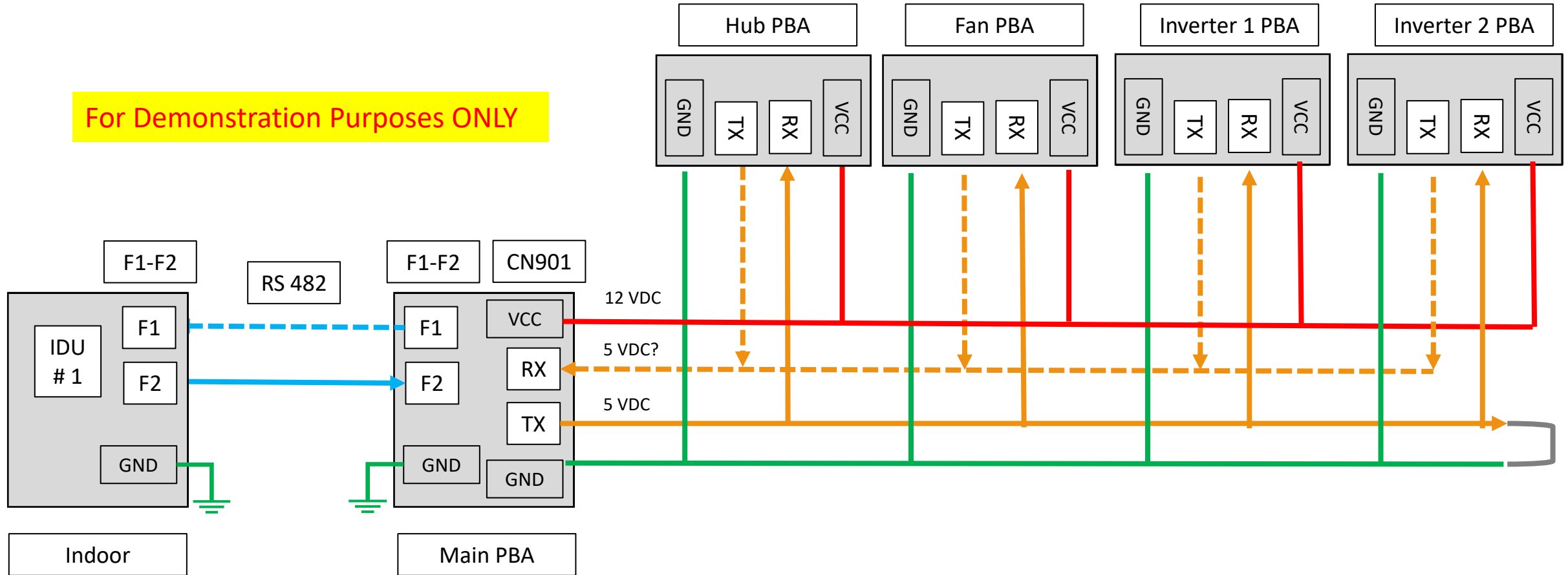


① CN102-FA1 HALL SENSING #1 : HALL-U #2 : 5V #3 : HALL-V #4 : GND #5 : HALL-W #6 : MOTOR-TEMP #7 : GND	② CN202-DOWNLOAD1 #1 : RX-DEBUG #2 : TX-DEBUG #3 : BOOT #4 : TDO #5 : TCK #6 : TDI #7 : TMS #9 : GND #10 : 5V	③ CN502-COMMUNICATION #1 : 12V-MAIN #2 : INV SMPS RELAY-MAIN #3 : COMM-MAIN #4 : GND-MAIN	④ CN501-COMMUNICATION #1 : 18V-INV #2 : GND-INV #4 : GND-INV #6 : 12V-MAIN #7 : INV SMPS RELAY-INV #8 : COMM-INV #9 : GND-INV
⑤ CN101-FAN2 HALL SENSING #1 : HALL-U #2 : 5V #3 : HALL-V #4 : GND #5 : HALL-W #6 : MOTOR-TEMP #7 : GND	⑥ CN301-DOWNLOAD2 #1 : RX-DEBUG #2 : TX-DEBUG #3 : BOOT #4 : TDO #5 : TCK #6 : TDI #7 : TMS #9 : GND #10 : 5V	⑦ U1-V1-W1 #1 : FAN1-U #2 : FAN1-V #3 : FAN1-W	⑧ CN401-POWER #1 : DC Link #2 : GND
⑨ U2-V2-W2 #1 : FAN2-U #2 : FAN2-V #3 : FAN2-W			

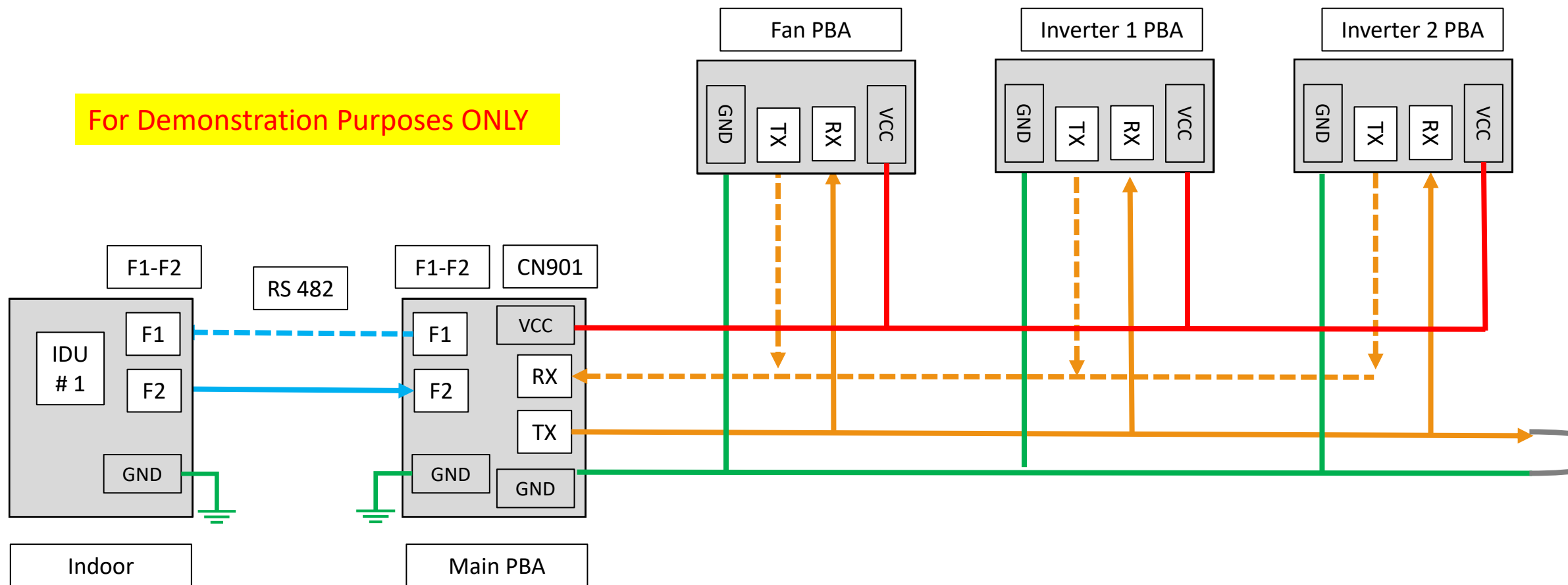


UART Communication??

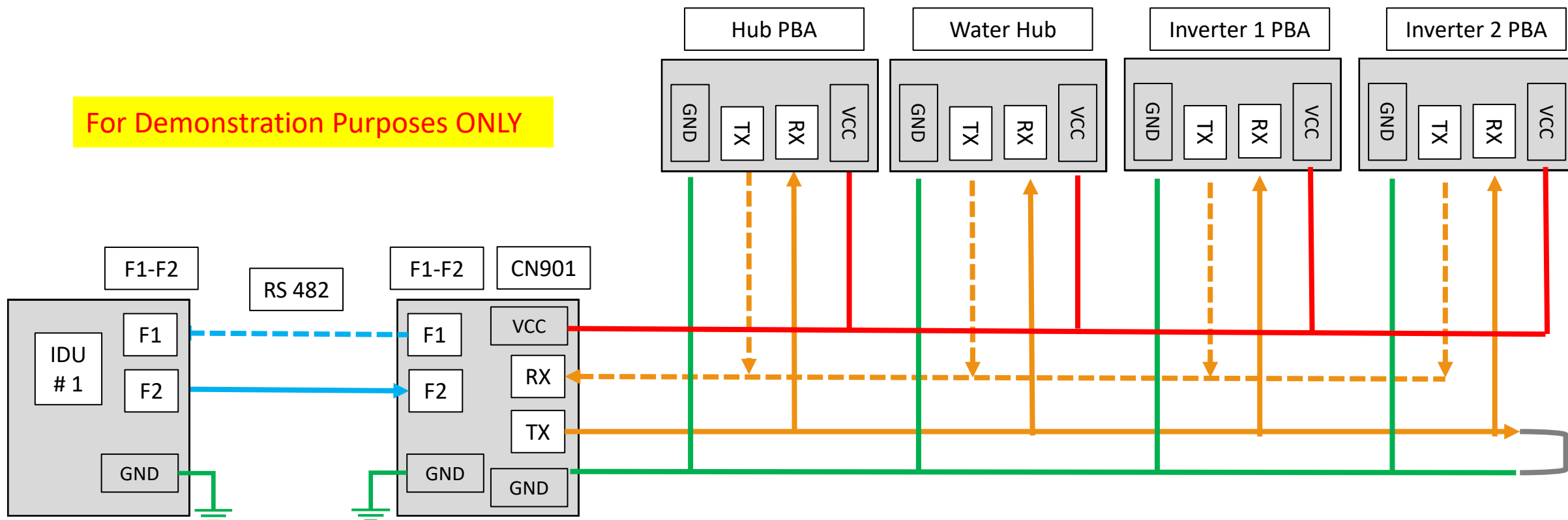
For Demonstration Purposes ONLY



For Demonstration Purposes ONLY



For Demonstration Purposes ONLY

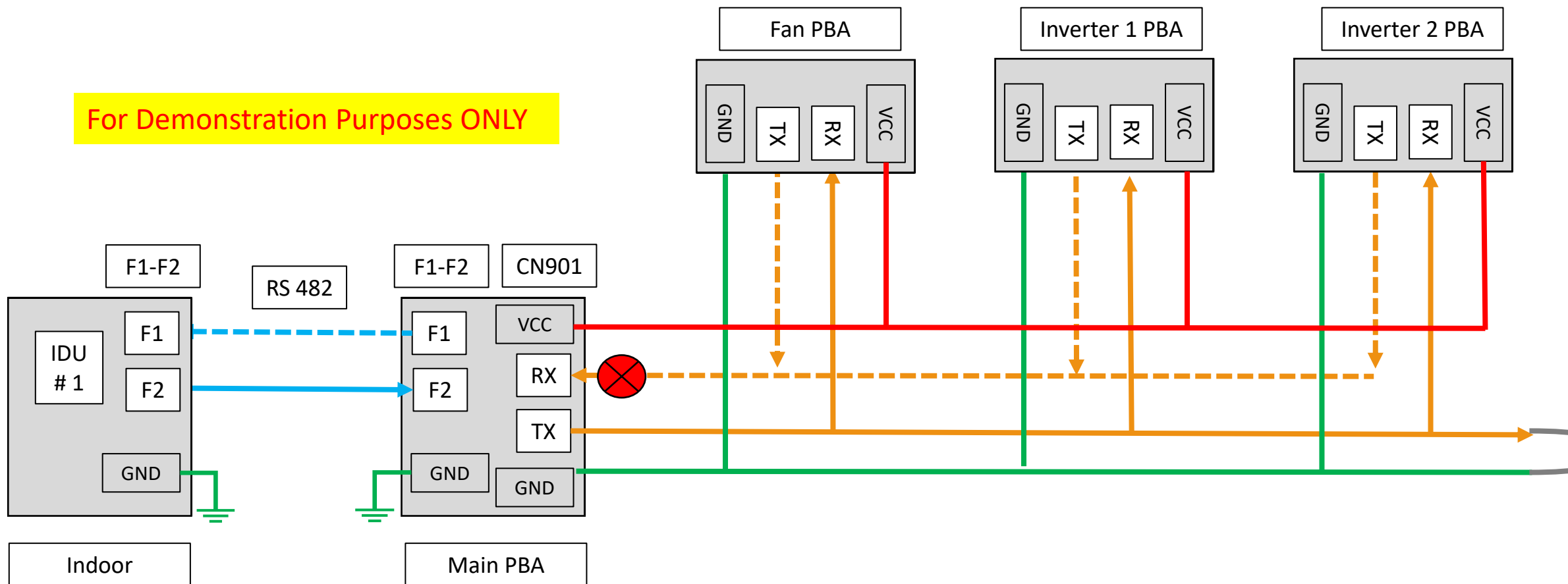




E205

Occurs when the Main micom does not receive any data.

For Demonstration Purposes ONLY

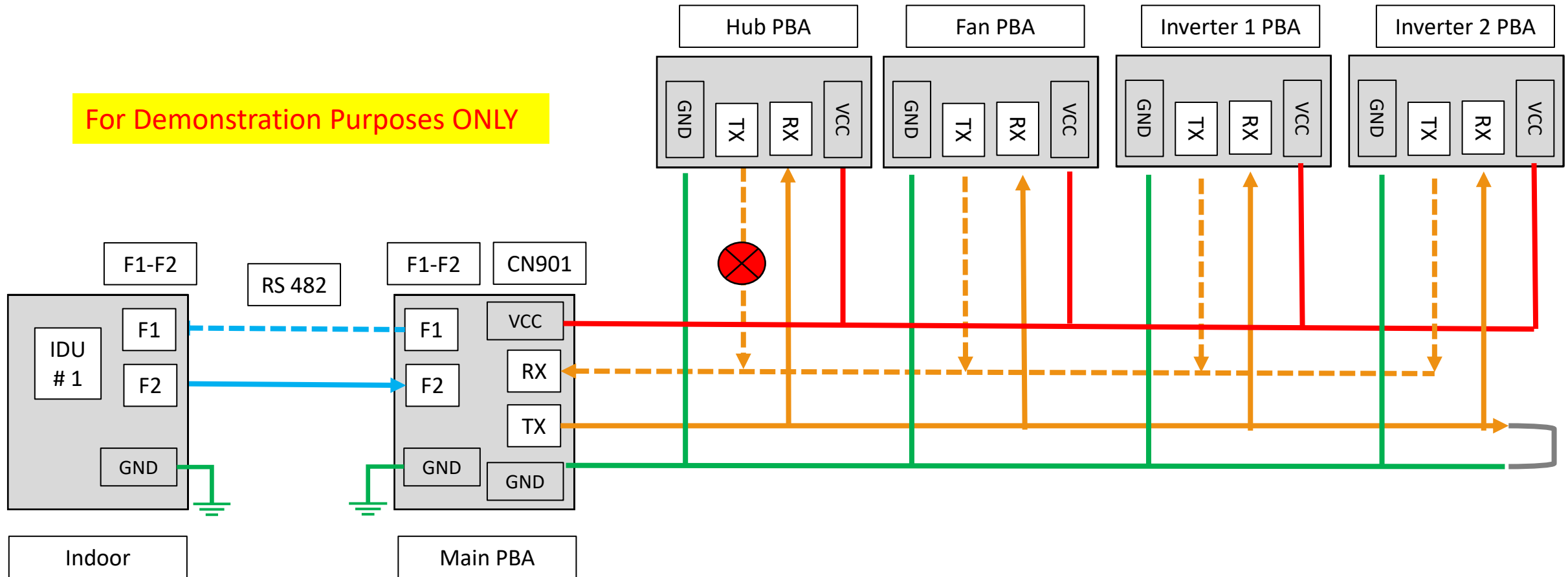




E206 C001

Occurs when the Main micom does not receive any data from the Hub micom (DVM S)

For Demonstration Purposes ONLY

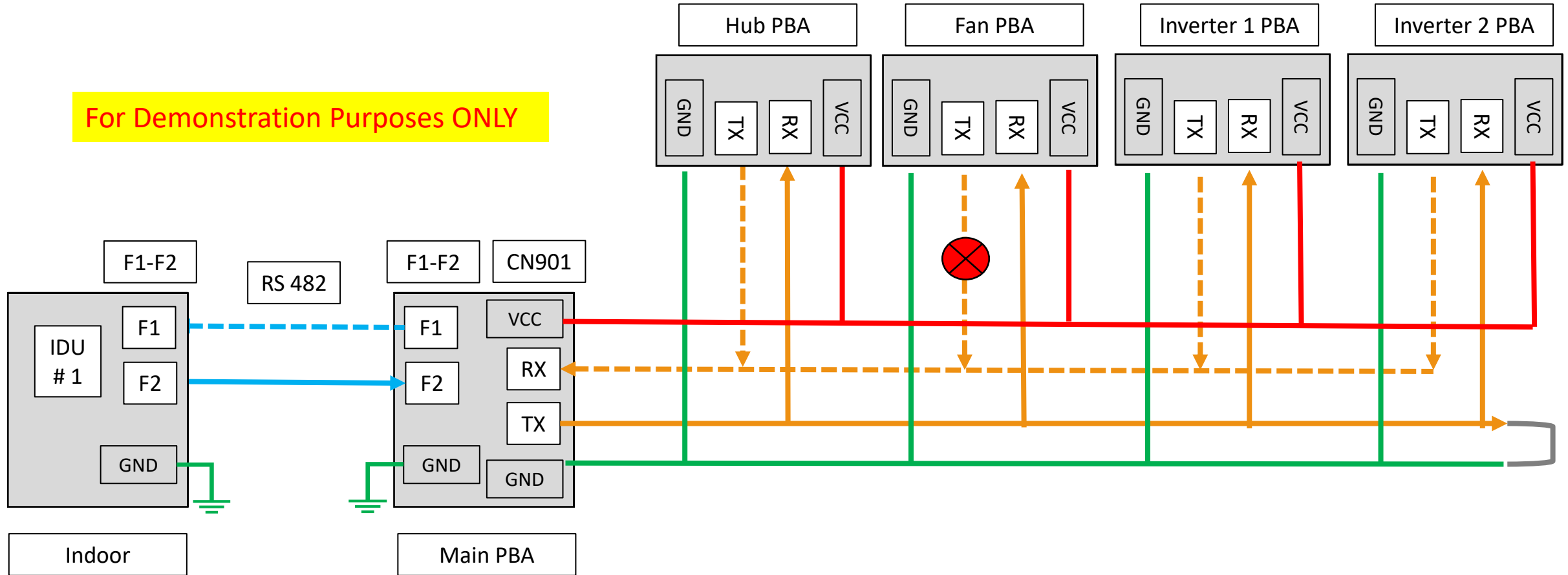




E206 C002

Occurs when the Main micom does not receive any data from the Fan Micom

For Demonstration Purposes ONLY

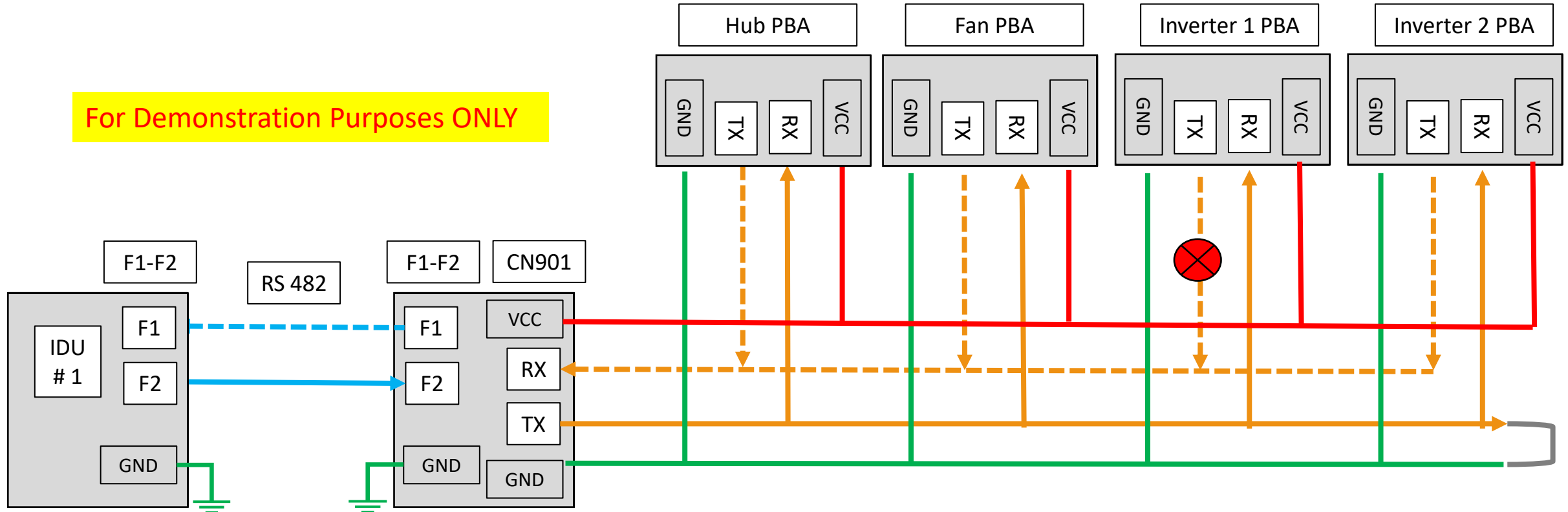


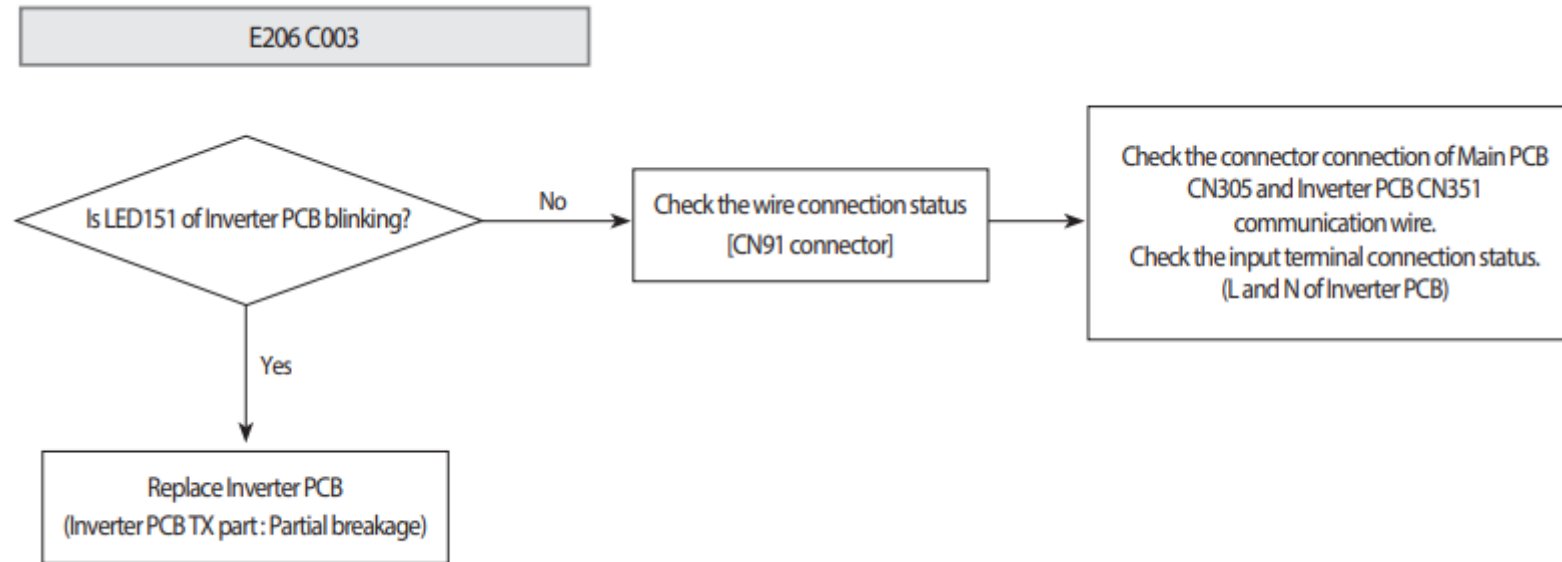


E206 C003

Occurs when the Main micom does not receive any data from the Inverter 1

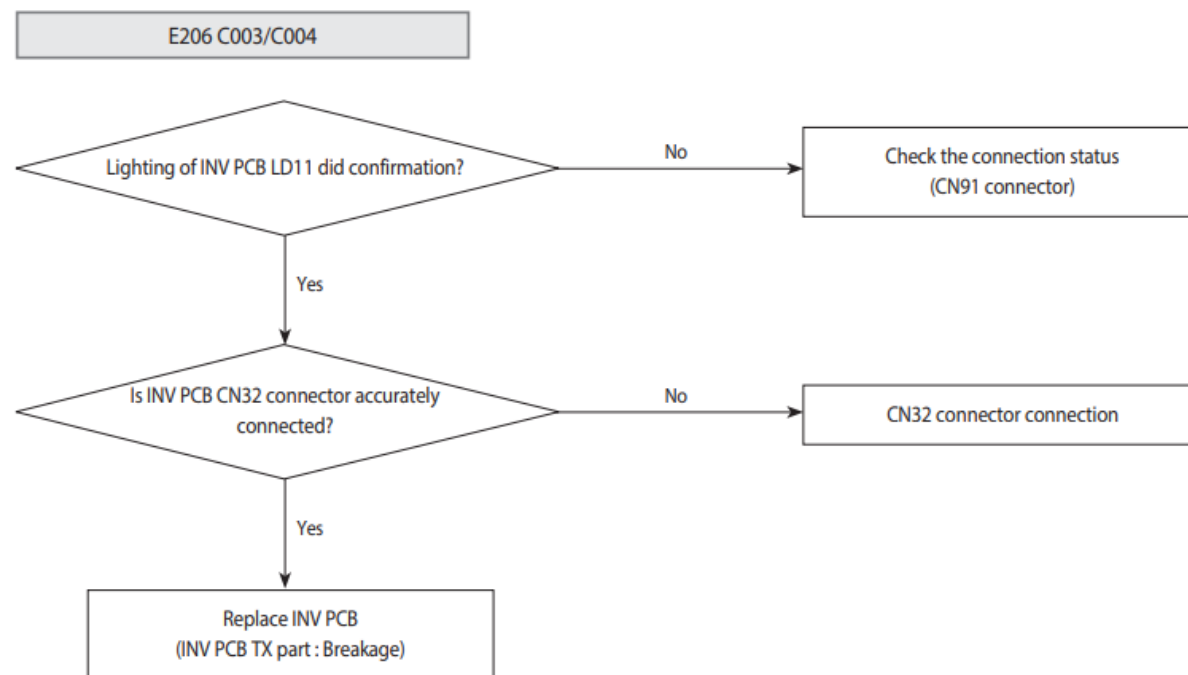
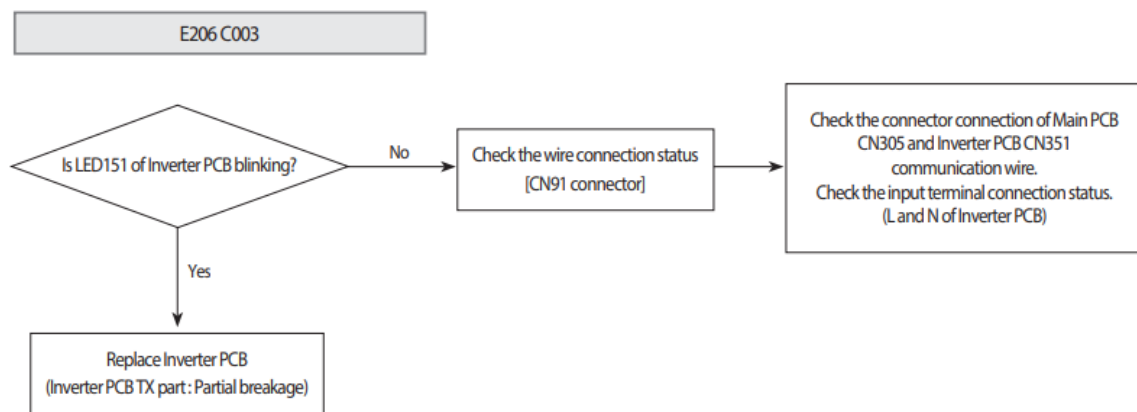
For Demonstration Purposes ONLY







E206 C003/C004

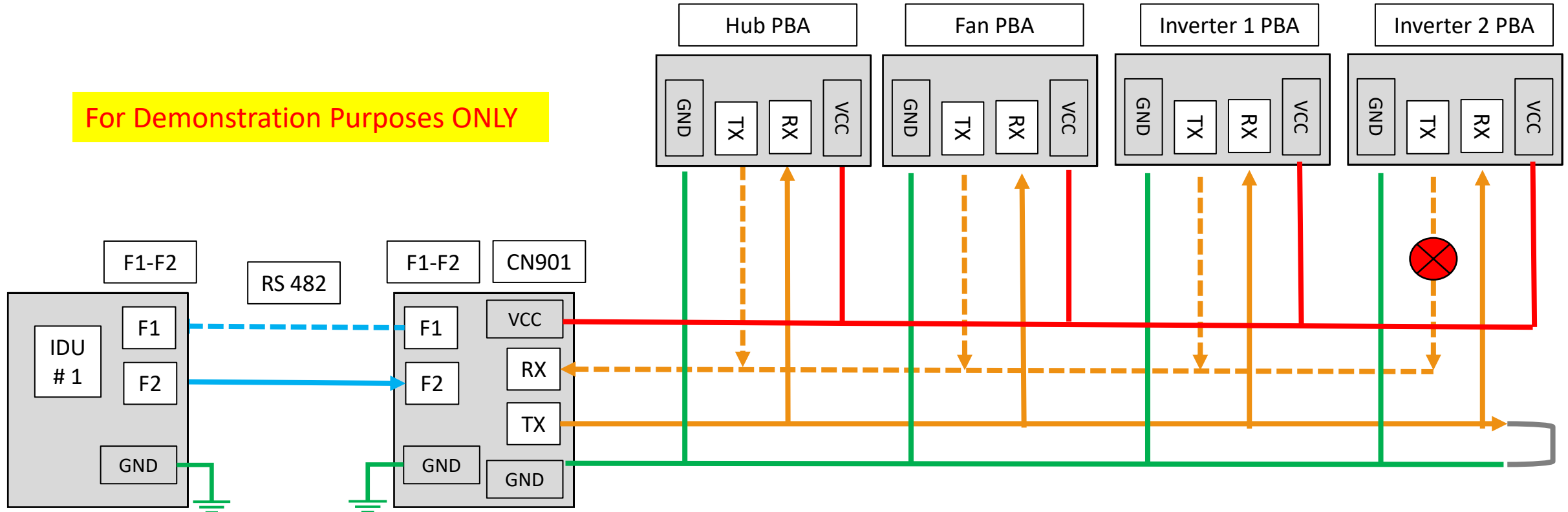




E206 C004

Occurs when the Main micom does not receive any data from the Inverter 2

For Demonstration Purposes ONLY

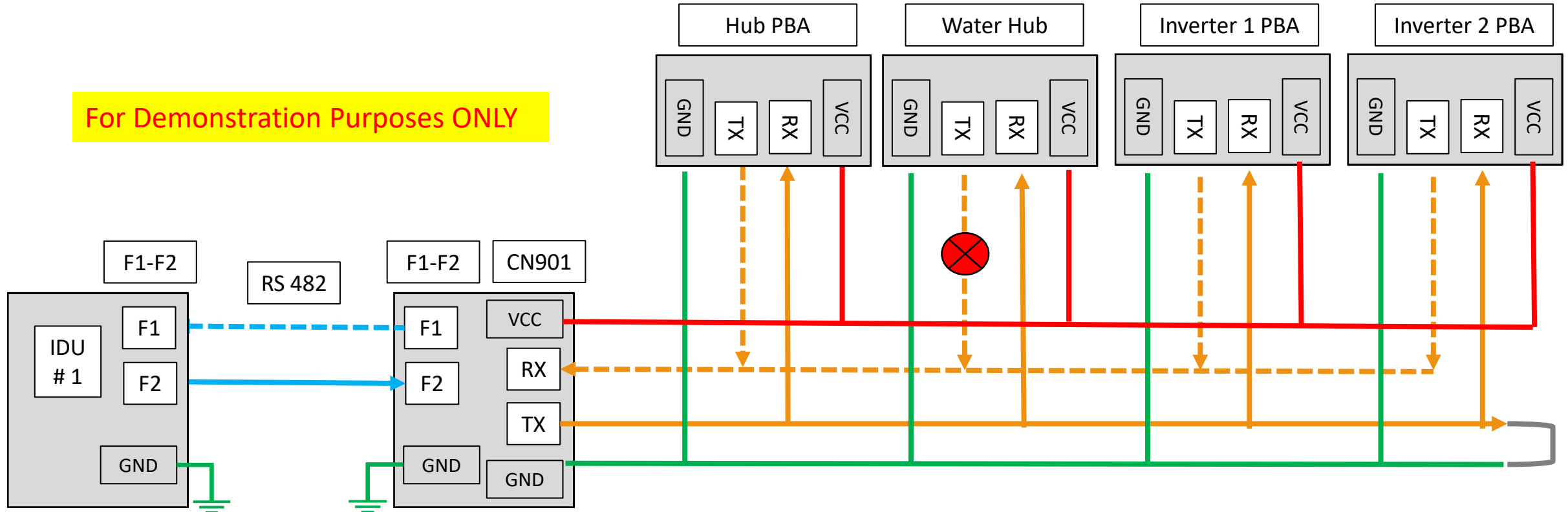




E206 C005

Occurs when the Main micom does not receive any data from the water hub

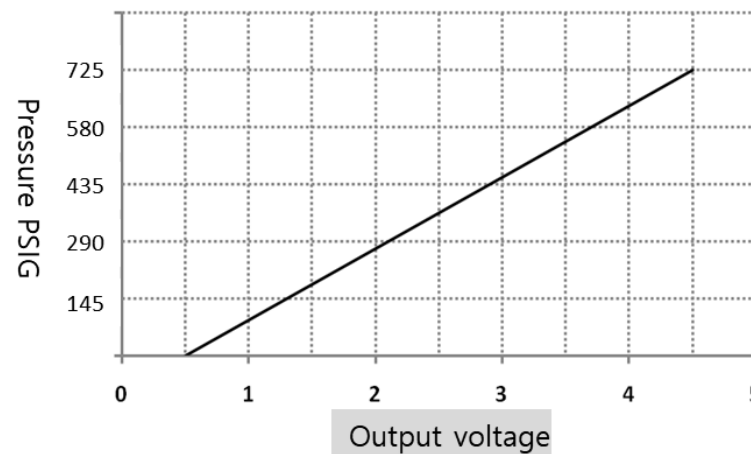
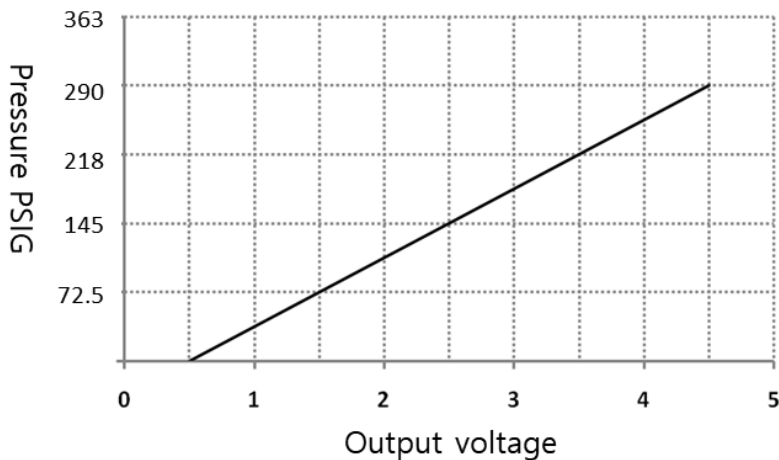
For Demonstration Purposes ONLY



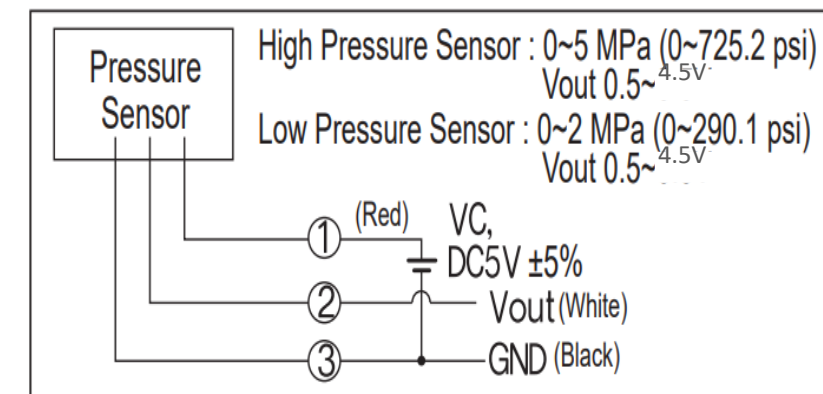


Pressure Transducers

- ❑ Are used to measure High and Low Side pressures.
- ❑ They are supplied with 5 VDC Between Red and Black
- ❑ The Voltage out on the white wire varies as the pressure changes.
- ❑ The system converts the voltage out of the white wire into a pressure.
- ❑ Measure system pressure and voltages at the transducer.
- ❑ S-Net makes this process easier



For reference purposes only!





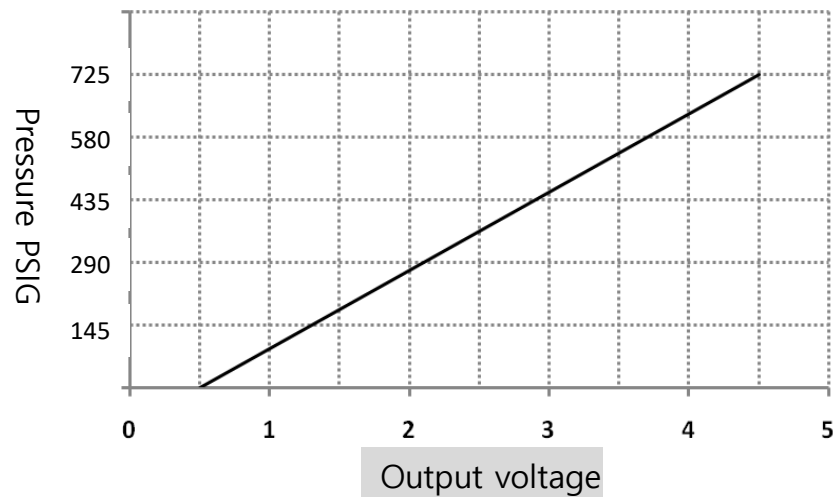
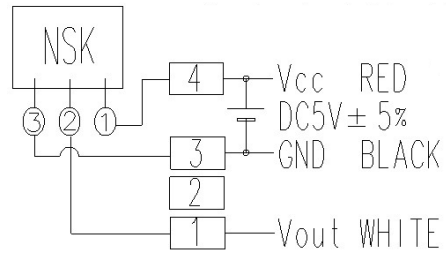
Pressure Transducer Testing



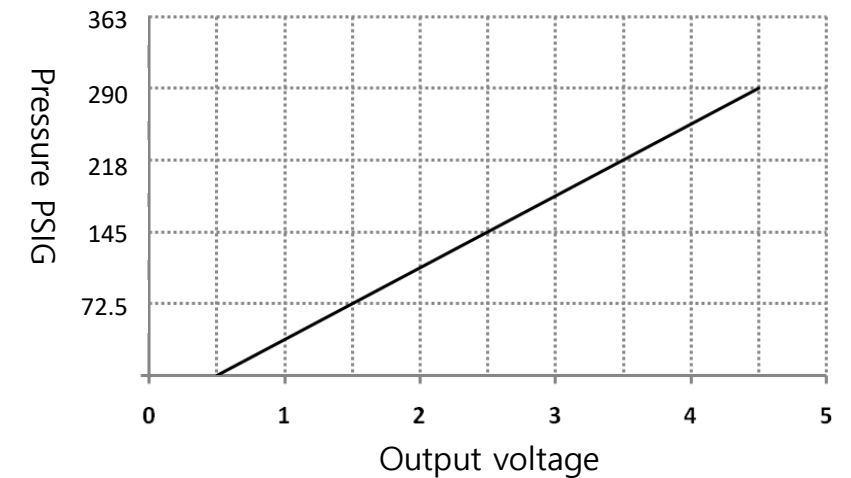
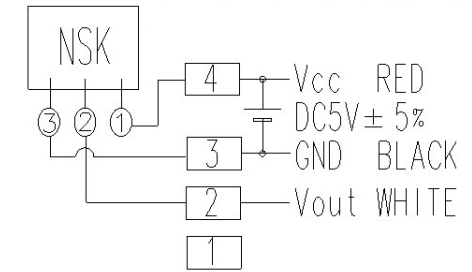


Pressure Transducers

◆ High pressure sensor error. (E 291)



◆ Low pressure sensor error. (E 296)





Solenoid Valve Testing



- ☐ Verify voltage at the coil
- ☐ If the voltage is correct and the valve is not working, check the resistance of the coil.
- ☐ If the resistance is within the allowable range, then the valve mechanism is stuck.



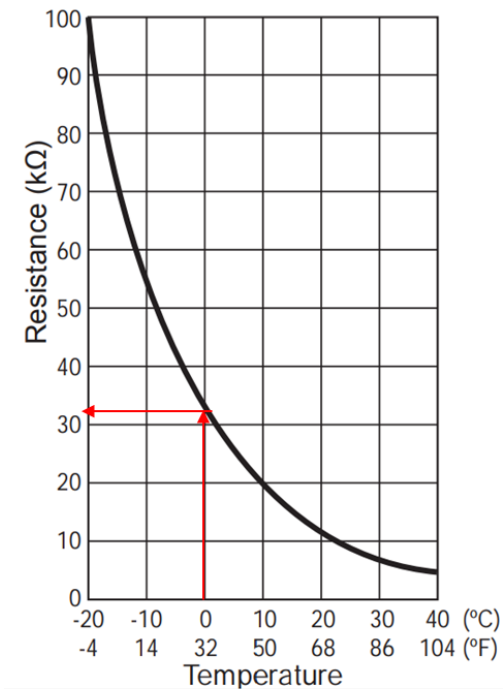
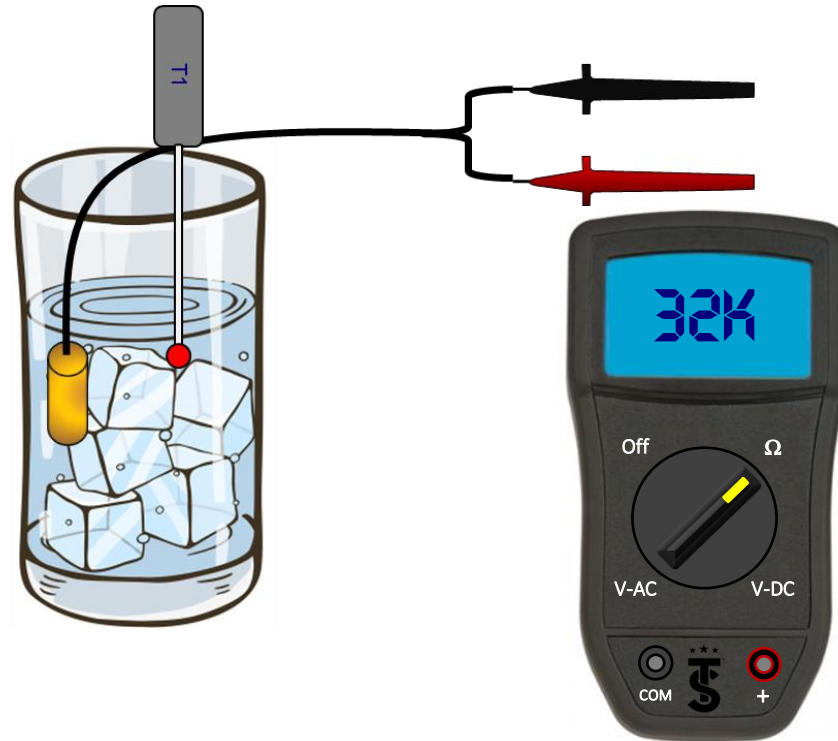
Thermistor Test

Refer to the specific unit's service manual for exact values

- Is the thermistor connector seated on the PCB?
- Is there 5 VDC available at the PCB with the thermistor removed?
- Does the resistance value of the thermistor equal its temperature



TEMP. (°F)	MAX (kΩ)	CENTER (kΩ)	Min (kΩ)
32	563.1	553.50	515.2
41	476.1	446.20	417.1
50	385.1	362.40	340.2
59	312.6	295.40	278.5
68	256.6	242.50	229.5
77	210	200.00	190.0
86	174.6	165.70	156.8
95	145.8	137.80	130.0
104	122.5	115.40	108.4
113	103.3	96.95	90.8
122	87.87	81.92	76.5
131	74.47	69.44	64.6
140	63.65	59.16	54.9
149	54.55	50.54	46.7





Compressor Testing

- ☐ Is the compressor getting a balanced 3 phase voltage.
- ☐ Test the resistance of the windings all 3 winding should be equal and match specification listed in the service manual.
A motor can fail in 1 of 4 ways, it can be
 - ☐ Shorted
 - ☐ Grounded
 - ☐ Open
 - ☐ Mechanically locked up.





Compressor Errors

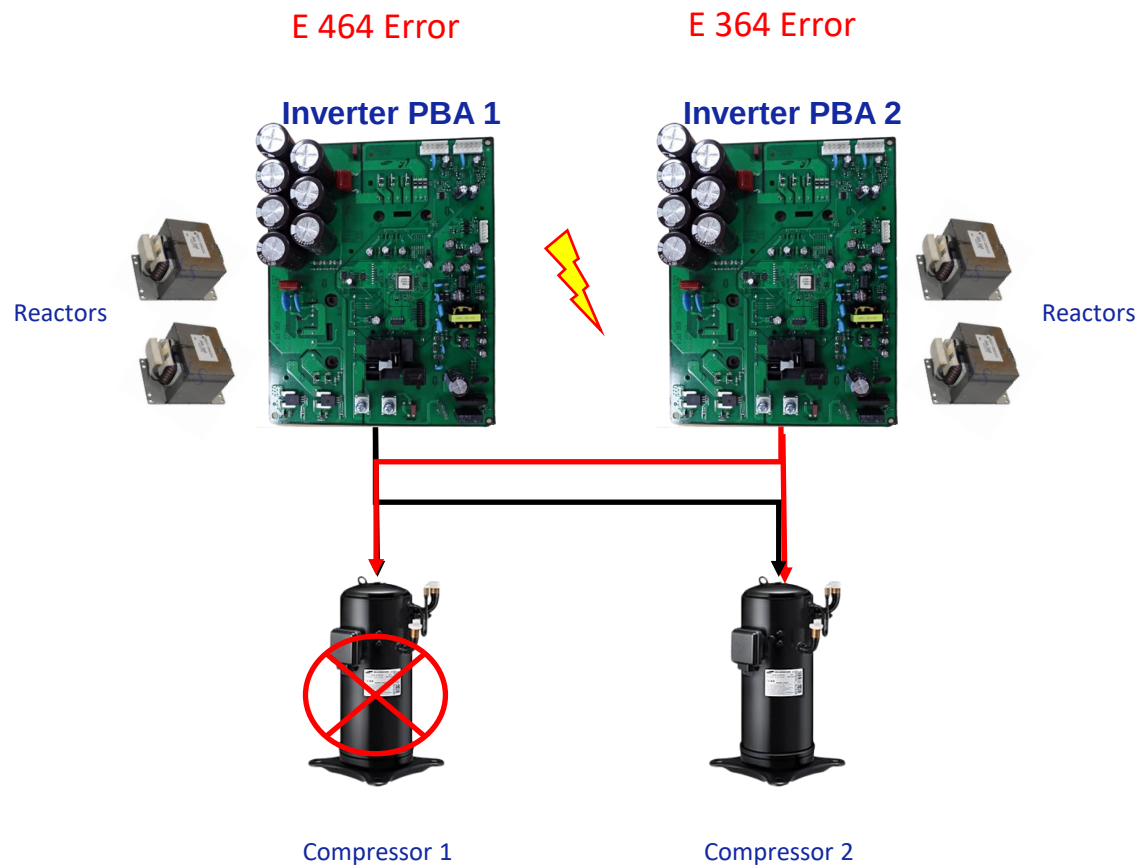
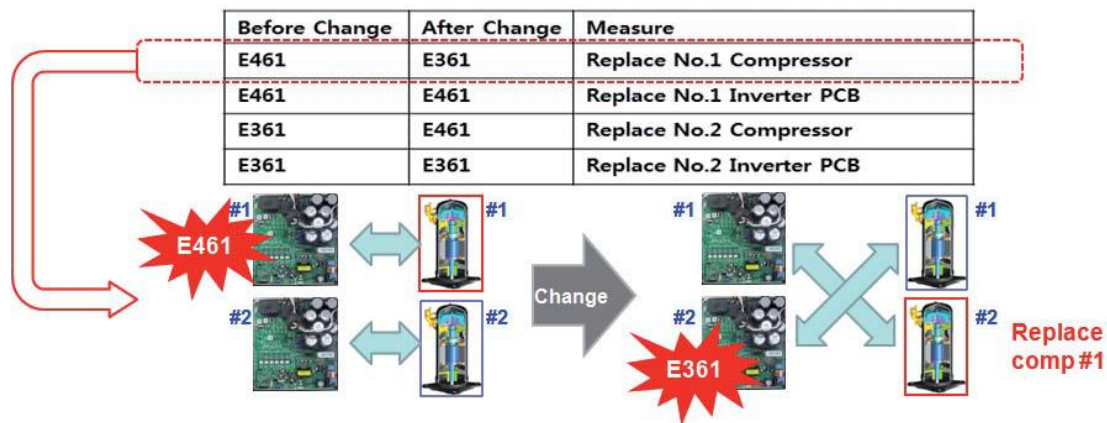
- ☐ If there is a compressor error it could be associated with the:
 - ☐ Main Power Supply
 - ☐ Reactor
 - ☐ Inverter PCB
 - ☐ Or compressor
- ☐ A common test is to switch the compressor leads and see if the problem follows the compressor or stays with the inverter PBA.
- ☐ The Compressor, reactor and compressor all can be tested.





Compressor Errors

- ❑ After Checking Supply and DC link voltage swap the compressor leads
- ❑ Check fault code
 - ❑ Did the compressor stay with the inverter or follow the compressor?



Compressor and Inverter PBA Test

E 464 Error

Before Change

Inverter PBA 1



Inverter PBA 2



Before Change	After Change	Diagnosis
E461	E 361	Replace # 1 Compressor
E 461	E461	Replace # 1 Inverter PBA
E 361	E461	Replace # 2 Compressor
E 361	E 361	Replace # 2 Inverter PBA

E 364 Error

After Change

Inverter PBA 1



Inverter PBA 2



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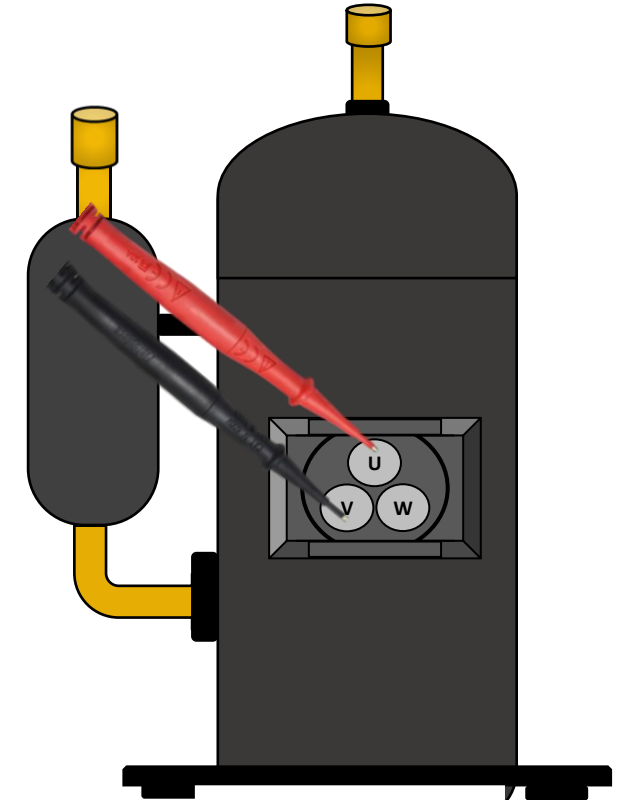
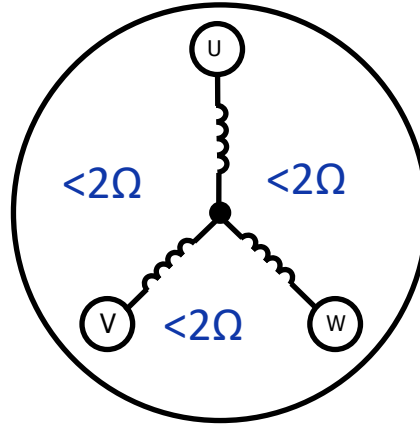


Compressor Winding Test

Resistance test	Normal range
Resistance value of (U↔V,V↔W,W↔U) on compressor	less than 2Ω
Resistance value between the body of compressor and chassis	MΩ

❑ The following resistance were measured between:

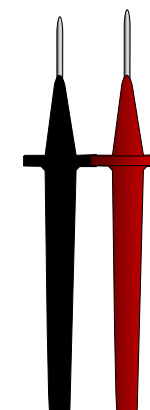
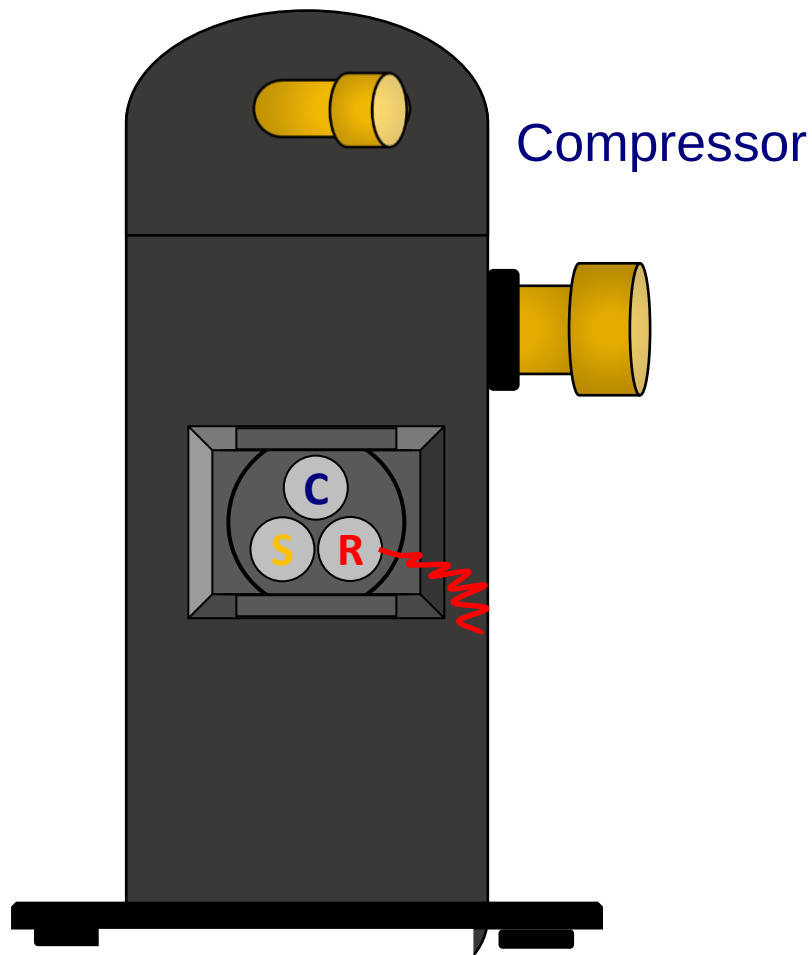
- ❑ U-V - .68 Ω
- ❑ V-W- .68 Ω
- ❑ W-U - .68 Ω
- ❑ U-Gnd - OL
- ❑ V - Gnd - OL
- ❑ W - Gnd - OL



No	Comp Model	Resistance (20°C)								
		C-R or U-V (Ω)			C-S or U-W (Ω)			V-W (Ω)		
		Spec.	Min	Max	Spec.	Min	Max	Spec.	Min	Max
1	DS-GB052FA++	0.21	0.20	0.22	0.21	0.20	0.22	0.21	0.20	0.22
2	DS-GB052FB++	0.13	0.12	0.13	0.13	0.12	0.13	0.13	0.12	0.13
3	DS-GB066FA++	0.14	0.13	0.15	0.14	0.13	0.15	0.14	0.13	0.15
4	DS-GB070FA++	0.11	0.11	0.12	0.11	0.11	0.12	0.11	0.11	0.12



Resistance Check to Ground



SAMSUNG



Inverter Checking

Section 4-3

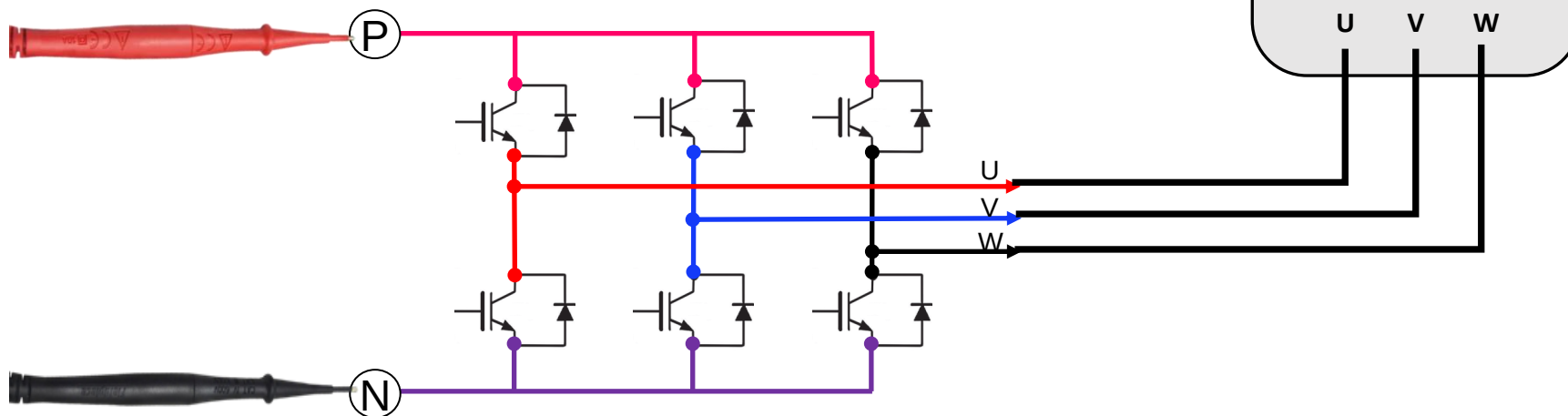
- ☐ How to use inverter checker
- ☐ Check between MOTOR and FAN PBA
- ☐ 1) After cut off, connect inverter checker with U,V,W of Motor
- ☐ 2) After turn on, enter Comp. check mode by pushing K2 in main PBA
- ☐ 3) Judgment
 - ☐ 6 LEDs of inverter checker are lightning successively (MOTOR PBA OK, MOTOR NG)
 - ☐ If one of 6 LEDs in inverter checker is not lightning (MOTOR PBA NG, MOTOR OK)

Type	DVM S	
Model	Air Cooled	Water Cooled
COMP 1	9times (KD__)	8times (KD__)
COMP 2	10times (KE__)	9times (KE__)
MOTOR 1	11times (KF__)	
MOTOR 2	12times (KG__)	



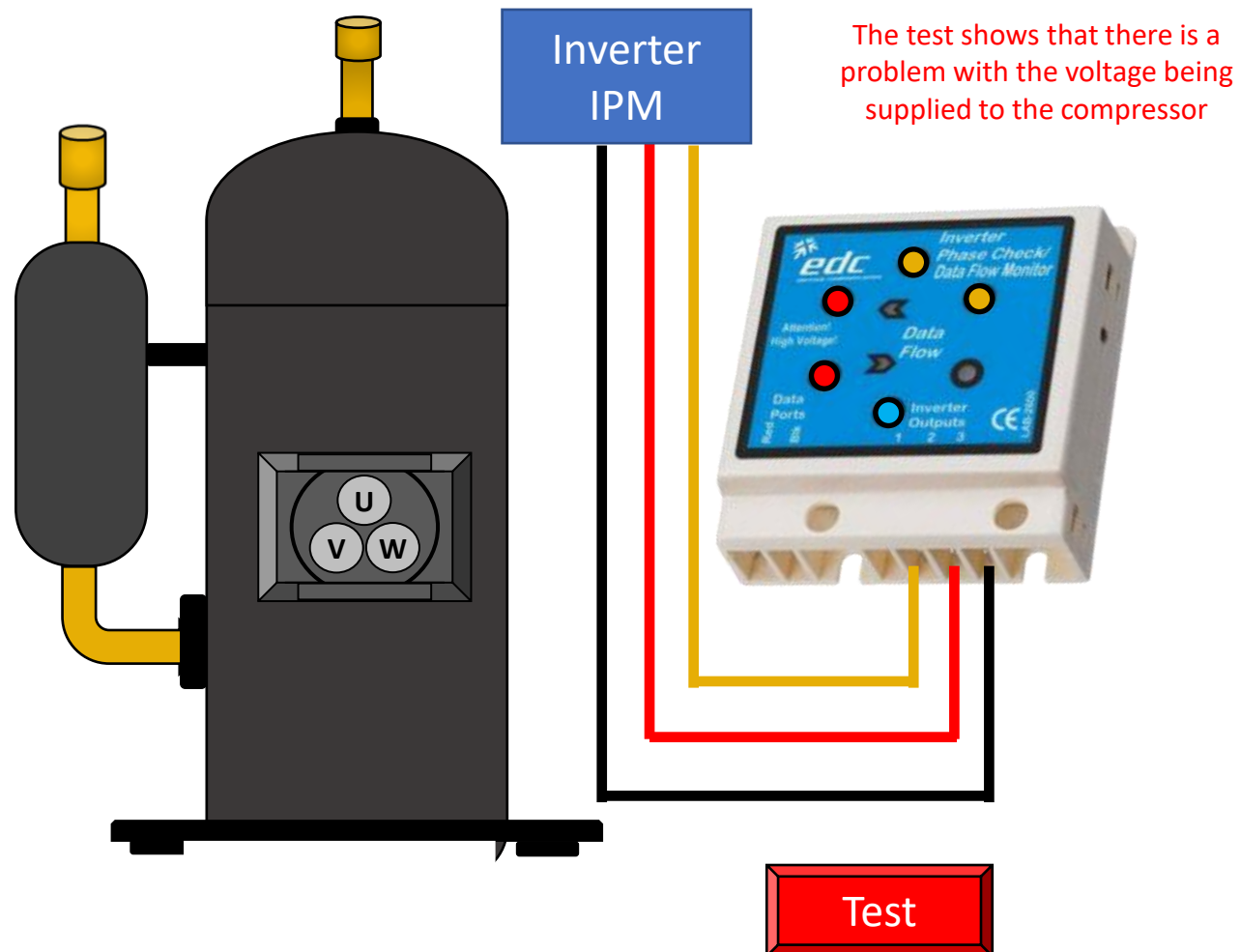
Inverter Testing

- ☐ Verify the DC Link Voltage
- ☐ Properly connect the tester to the Inverter Board
- ☐ Apply power.
- ☐ Check the light sequence.



Testing the compressor IPM Voltage

- ❑ Use an inverter tester to test the voltage to the compressor.
- ❑ There must be a balanced 3 phase A/C voltage supplied to the compressor.
- ❑ Refer to the manufacturer's user manual for assistance.
- ❑ Click test to see test the inverter power supply.

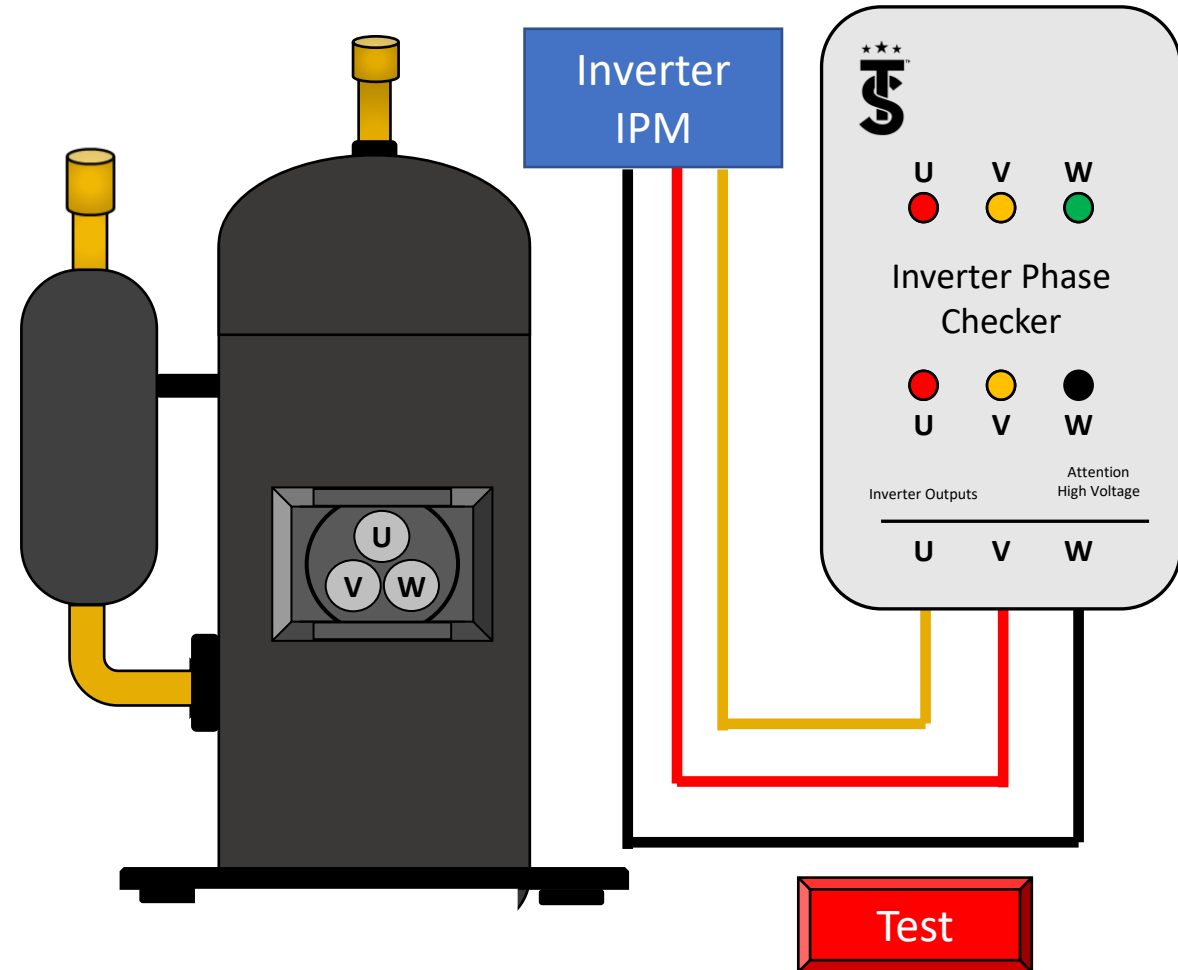




Testing the compressor IPM Voltage 1

- ❑ Use an inverter tester to test the voltage to the compressor.
- ❑ There must be a balanced 3 phase A/C voltage supplied to the compressor.
- ❑ Refer to the manufacturer's user manual for assistance.
- ❑ Click test to see test the inverter power supply

The test shows that there is a problem with the voltage being supplied to the compressor

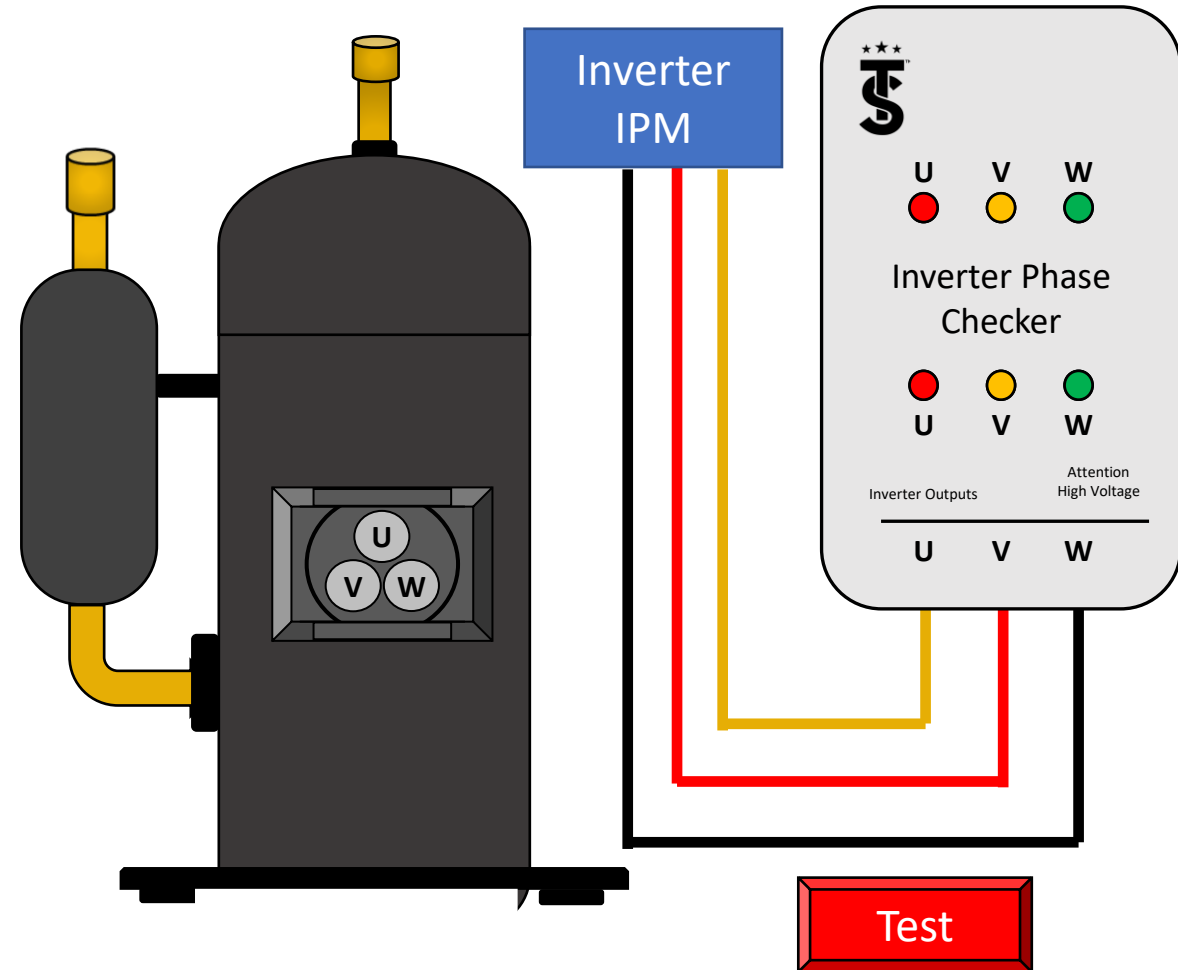




Testing the compressor IPM Voltage 2

- ❑ Use an inverter tester to test the voltage to the compressor.
- ❑ There must be a balanced 3 phase A/C voltage supplied to the compressor.
- ❑ Refer to the manufacturer's user manual for assistance.
- ❑ Click test to see test the inverter power supply

The test shows that there is a problem with the voltage being supplied to the compressor



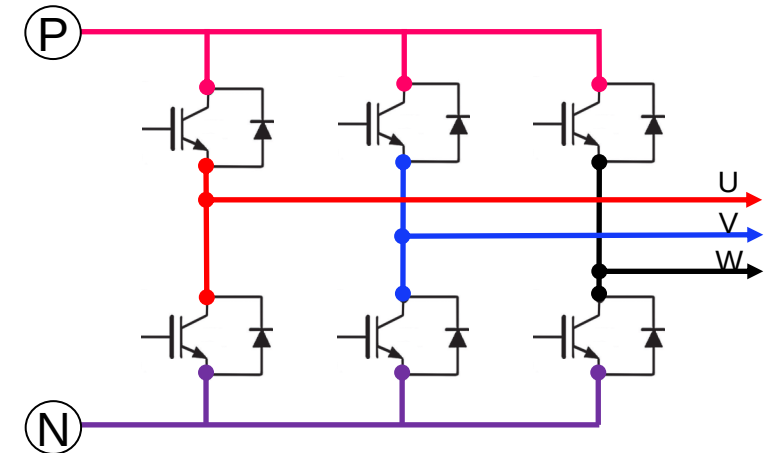


Inverter IPM Check

- ❑ You can check the Diode (switch) and Load (coil) of each IGBT with a meter.



Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500 kΩ	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times.
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
	W	N-IGBT		
Measure the diode voltage values	U	P-IGBT	0.3~0.7V	
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		

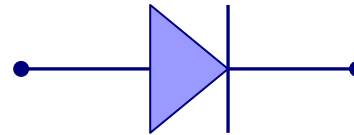




Diode Testing



- Using the diode testing feature of you meter you can measure a voltage drop across the diode in one and direction

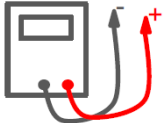




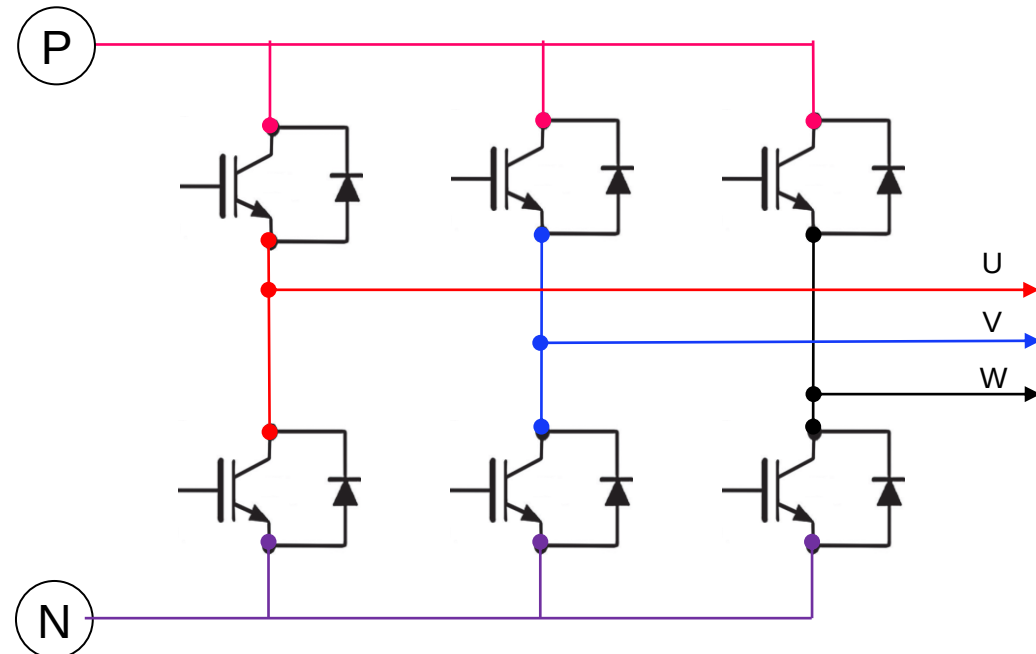
IGBT 1 Testing



Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500KΩ	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times.
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
	W	N-IGBT		
Measure the diode voltage values	U	P-IGBT	0.3~0.7V	
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		



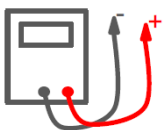
Directions: Click on the test you want to perform



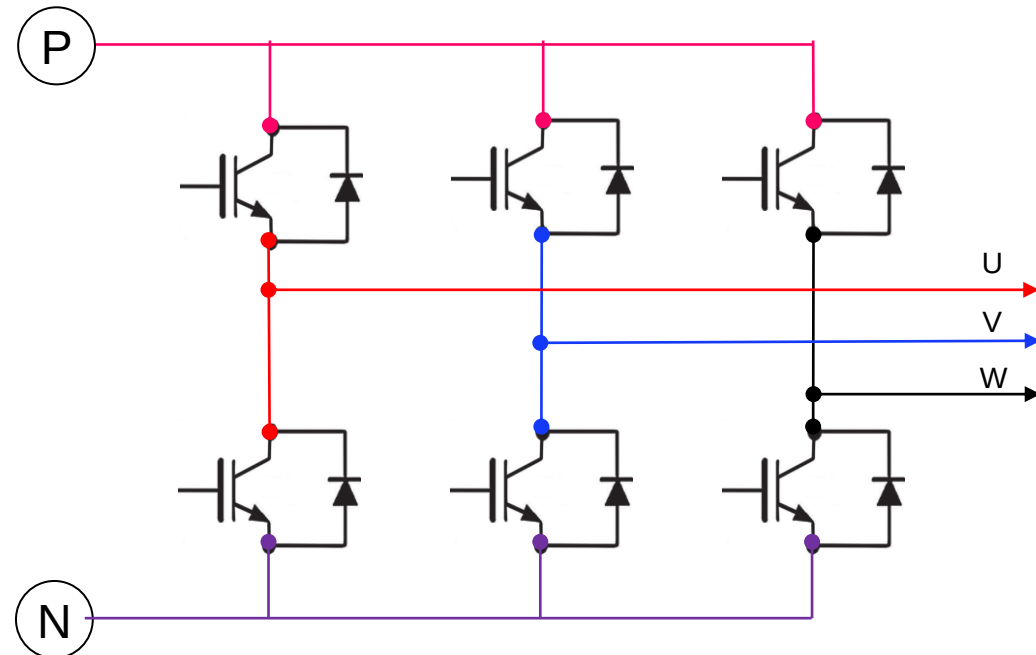


IGPT Resistance Check Red Leads on P



Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500KΩ	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times. 
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
	W	N-IGBT		
Measure the diode voltage values	U	P-IGBT	0.3~0.7V	
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		

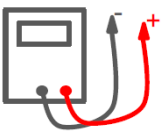
Directions: Click on the test you want to perform



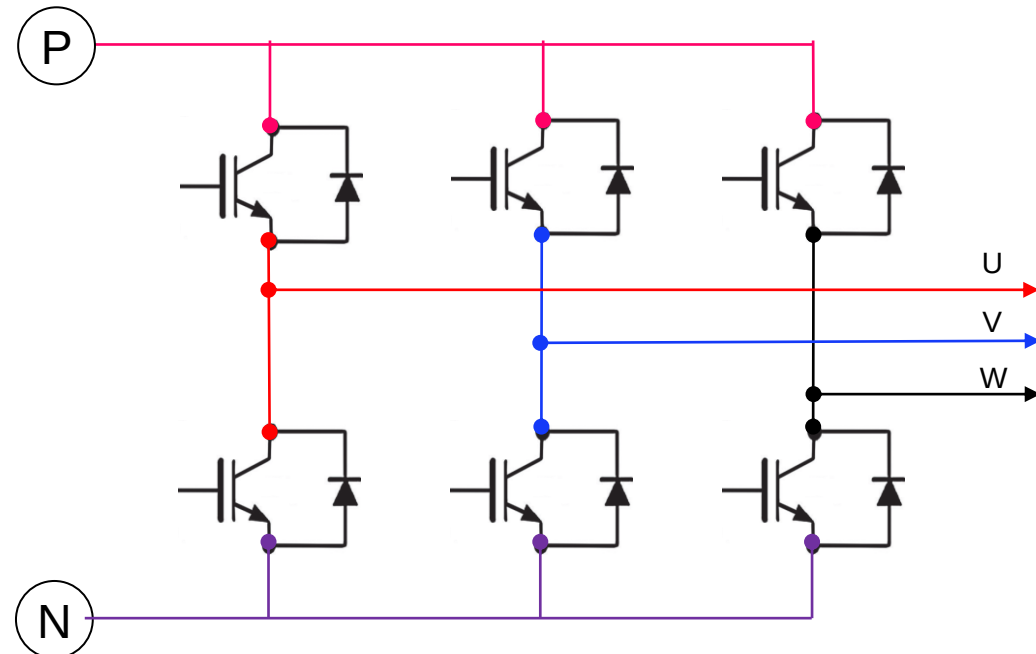


IGBT Resistance Check Black Lead on N



Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500KΩ	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times.
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
	W	N-IGBT		
Measure the diode voltage values	U	P-IGBT	0.3~0.7V	
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		

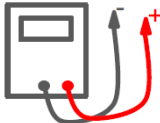
Directions: Click on the test you want to perform





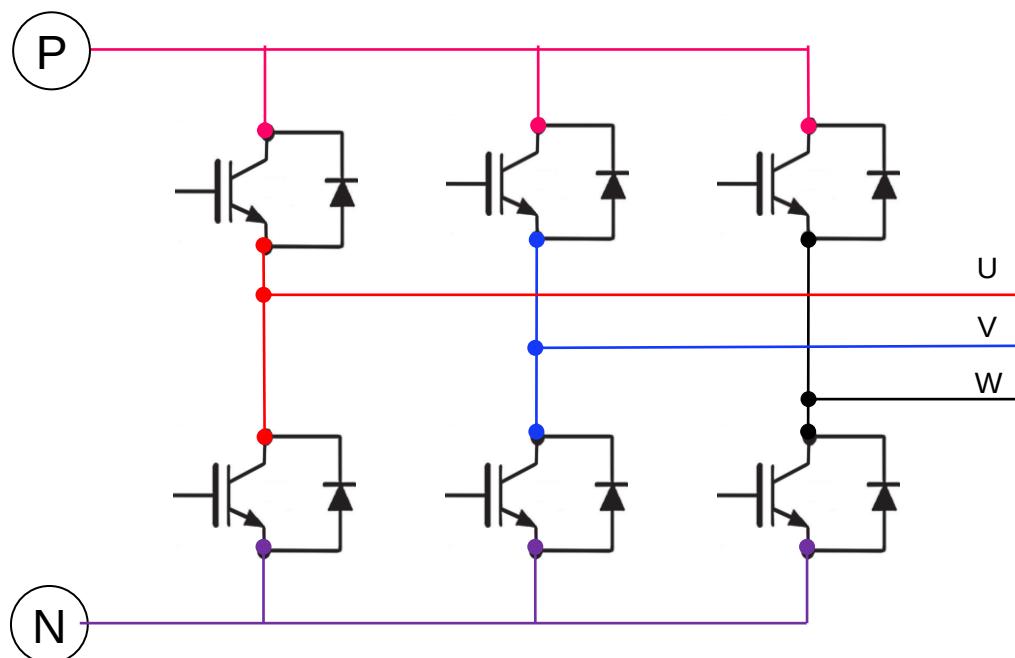
IGBT 1 Diode Test Black Lead on P

Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500KΩ	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times.
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
Measure the diode voltage values	W	N-IGBT	0.3~0.7V	
	U	P-IGBT		
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		



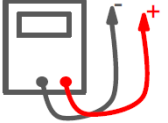


Directions: Click on the test you want to perform

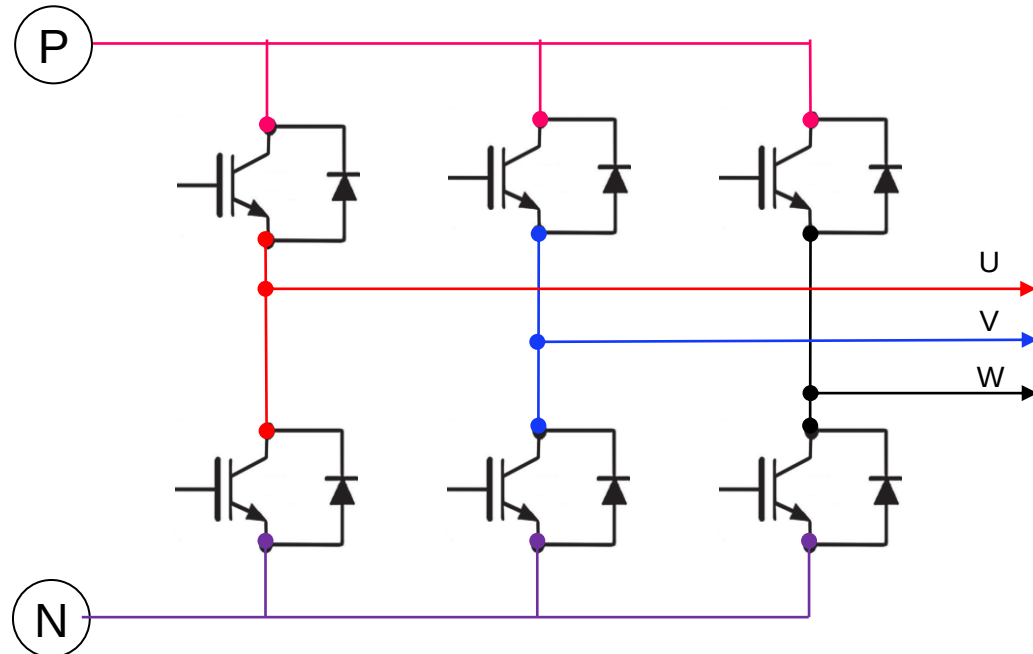




IGPT 1 Diode Check Red Leads on N

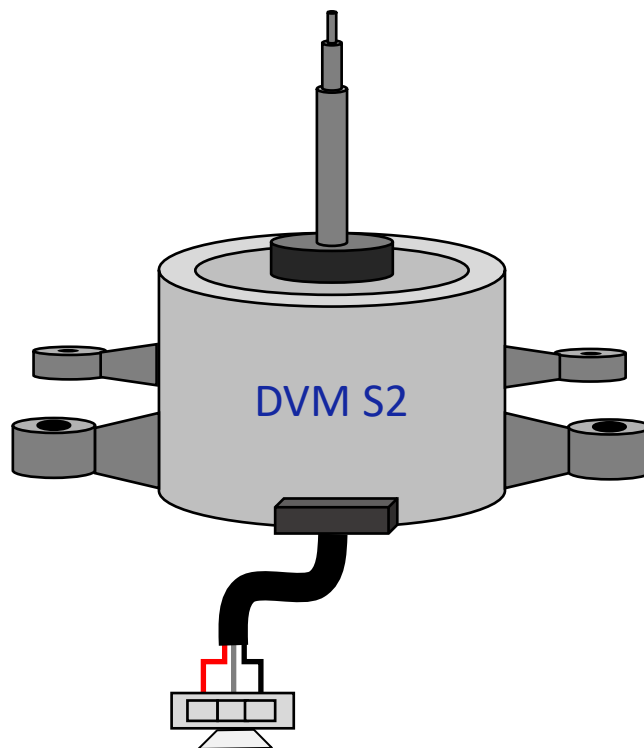
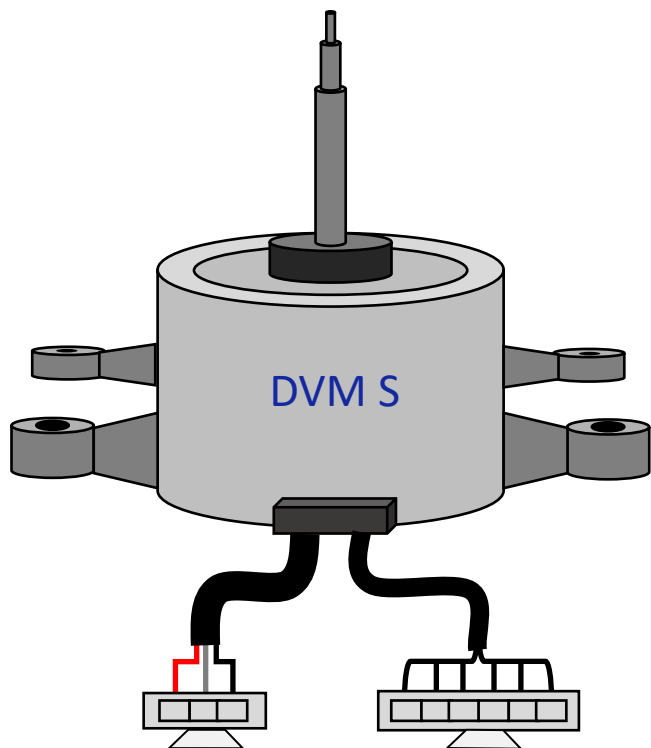
Division	Measured Point		Criterion	Remark
	+	-		
Measure the resistance values	P-IGBT	U	More than 500K Ω	Measurement error can occur for reasons such as the initial measurement condenser discharge. Measured over at least three times.
	P-IGBT	V		
	P-IGBT	W		
	U	N-IGBT		
	V	N-IGBT		
	W	N-IGBT		
Measure the diode voltage values	U	P-IGBT	0.3~0.7V	
	V	P-IGBT		
	W	P-IGBT		
	N-IGBT	U		
	N-IGBT	V		
	N-IGBT	W		

Directions: Click on the test you want to perform





ODU Fan Motors

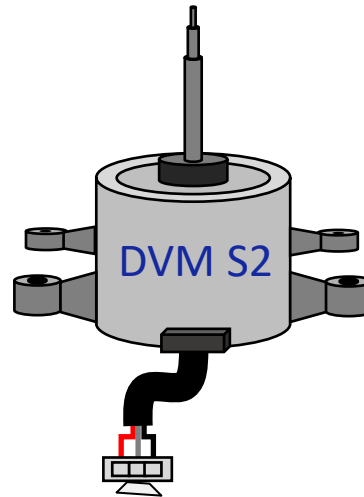
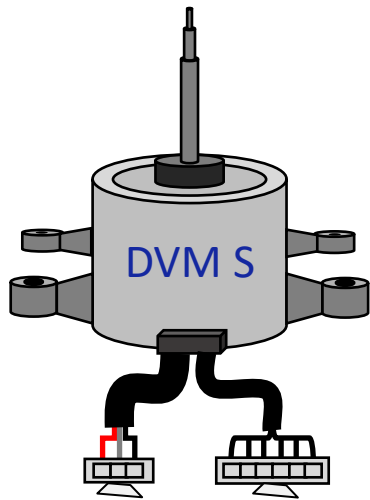


Eco BLDC



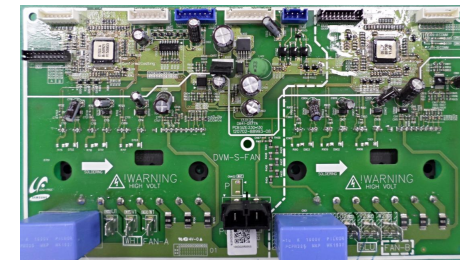
ODU Fan Motor Trouble Shooting

PBA or Fan Motor

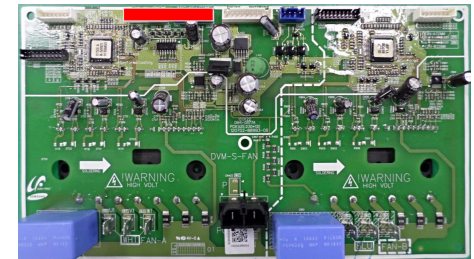


Eco BLDC

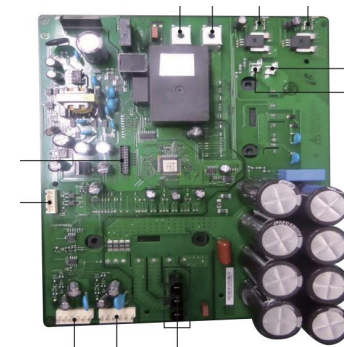
DVM S



DVM S2



Eco



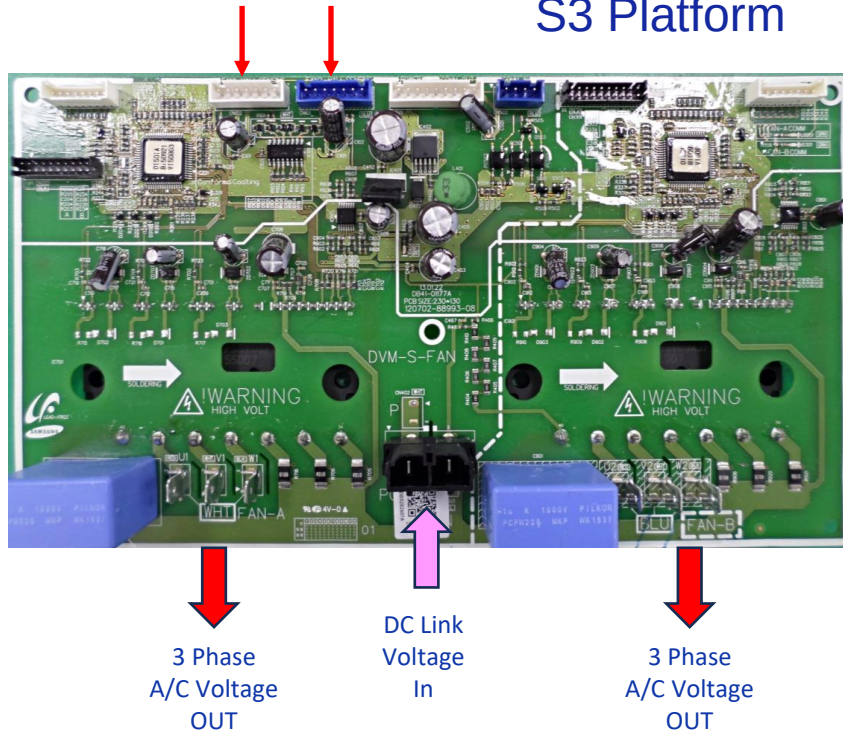


3 Phase ODU Fan PCB

Inputs and Output Voltages

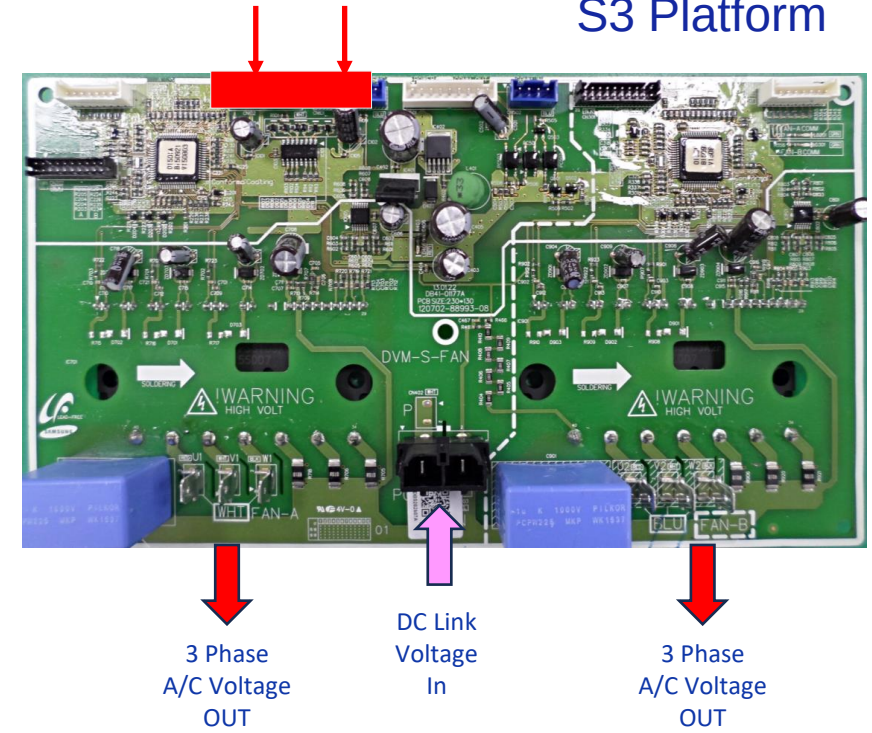
Hall Sensor Connections

DVM S
S3 Platform



No Hall Sensor Connections

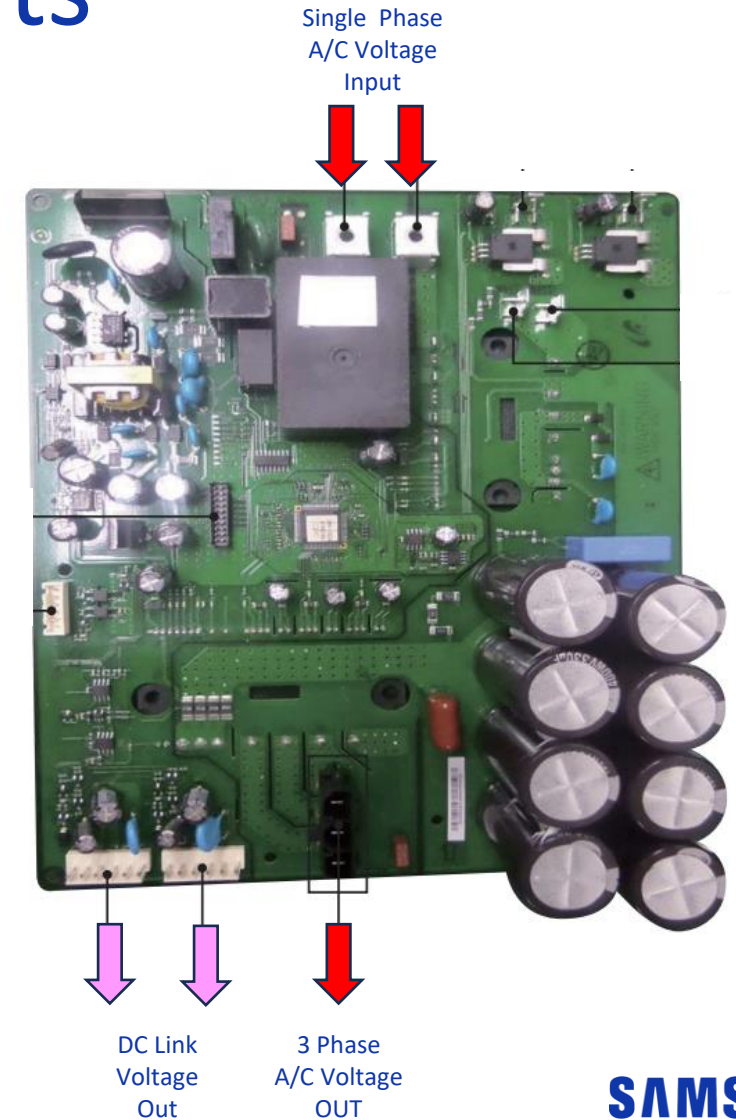
DVM S2
S3 Platform



Single Phase ODU Inverter/Fan PBA

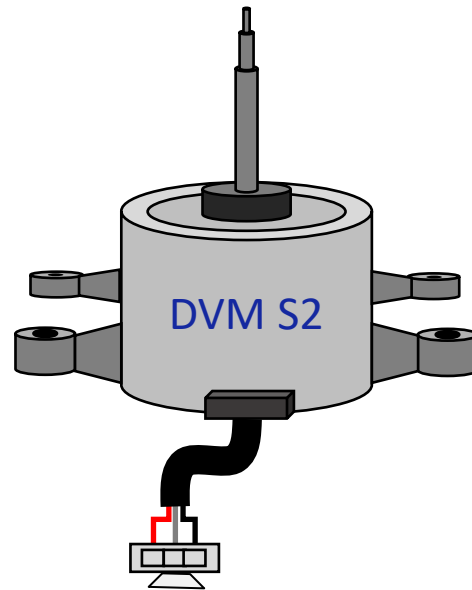
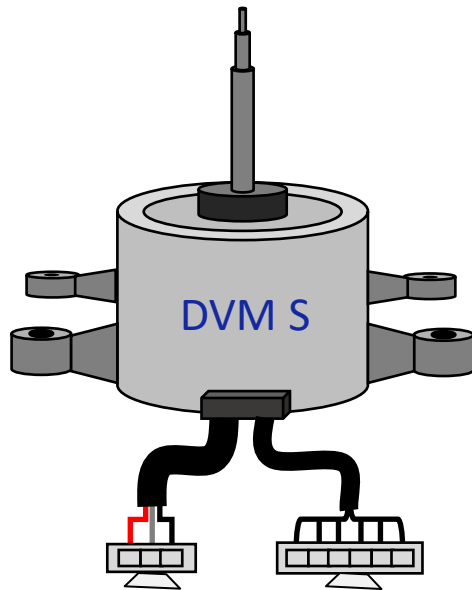
High Voltage Input and Outputs

- ❑ The single-phase ODU Inverter PBA is supplied with a single-phase A/C voltage.
- ❑ It supplies a 3 phase A/C Voltage to the compressor.
- ❑ It supplies DC Link voltage to the BLDC fan motor
- ❑ It also supplies the 15 VDC control voltage to the BLDC motor



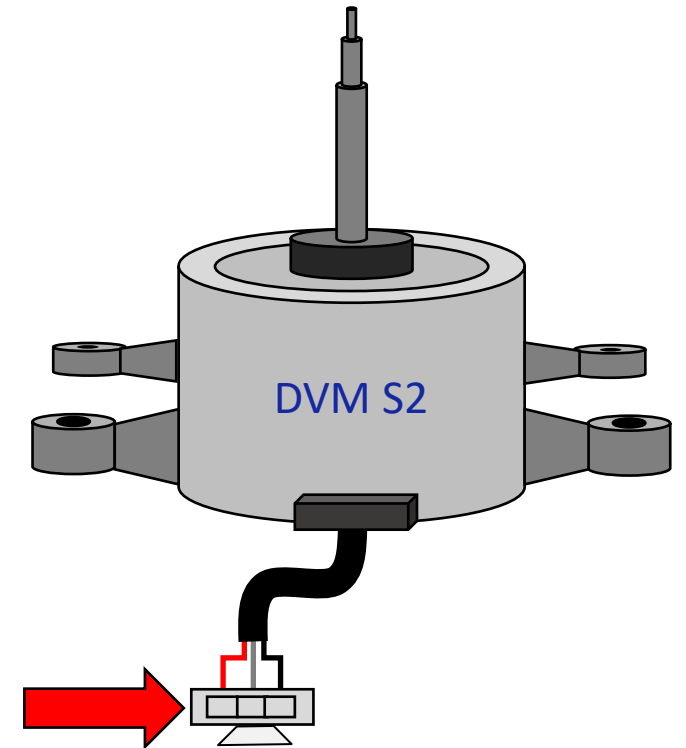
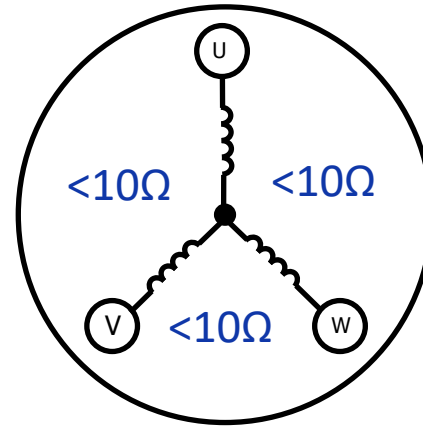
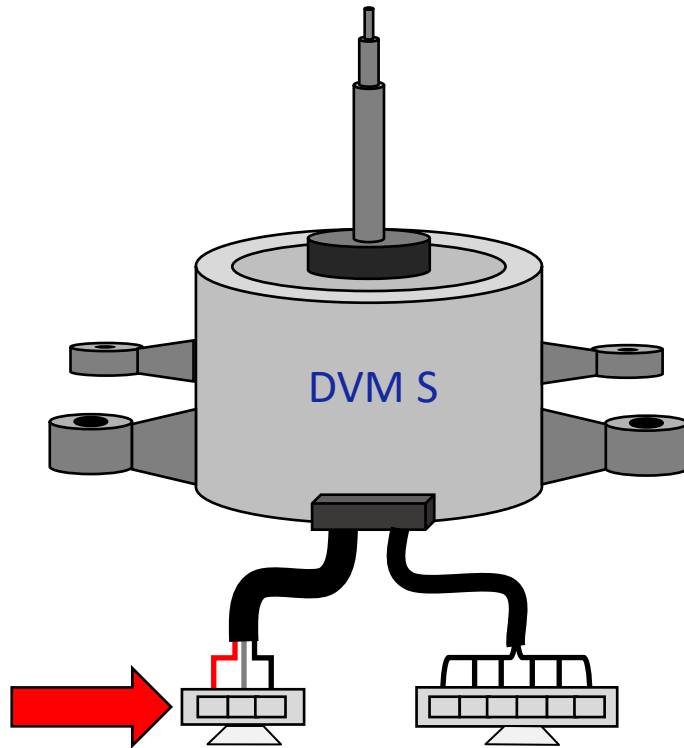


3 Phase ODU Fan Motor Testing





3 Phase ODU Fan Motor Resistance Check



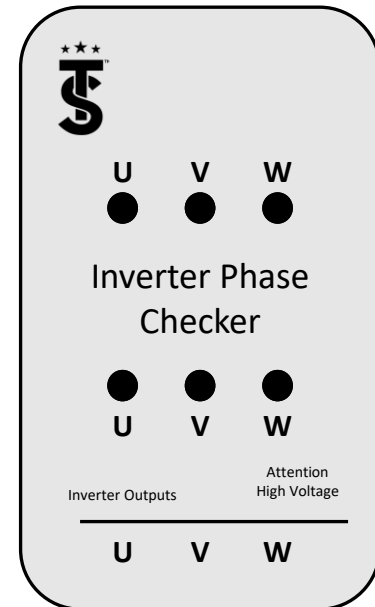
Is the resistance value between different phases
(U - V, V - W, W - U) less than 10Ω and equal between each phase



Checking for a balanced 3 phase A/C voltage



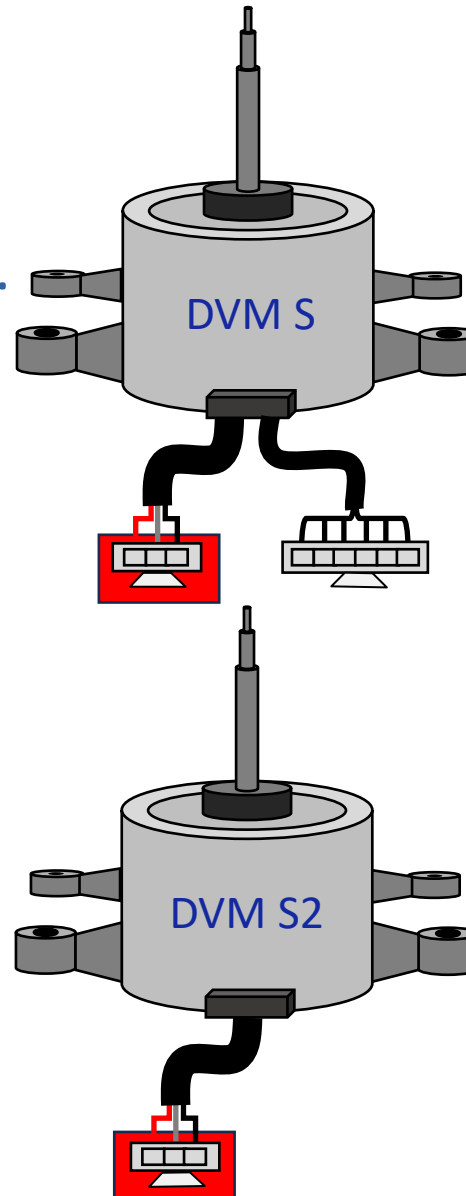
- ❑ In order for the motor to operate correctly it must be supplied with a balanced 3 phase A/C voltage,
- ❑ A meter or inverter checker can be used to check for a balanced voltage



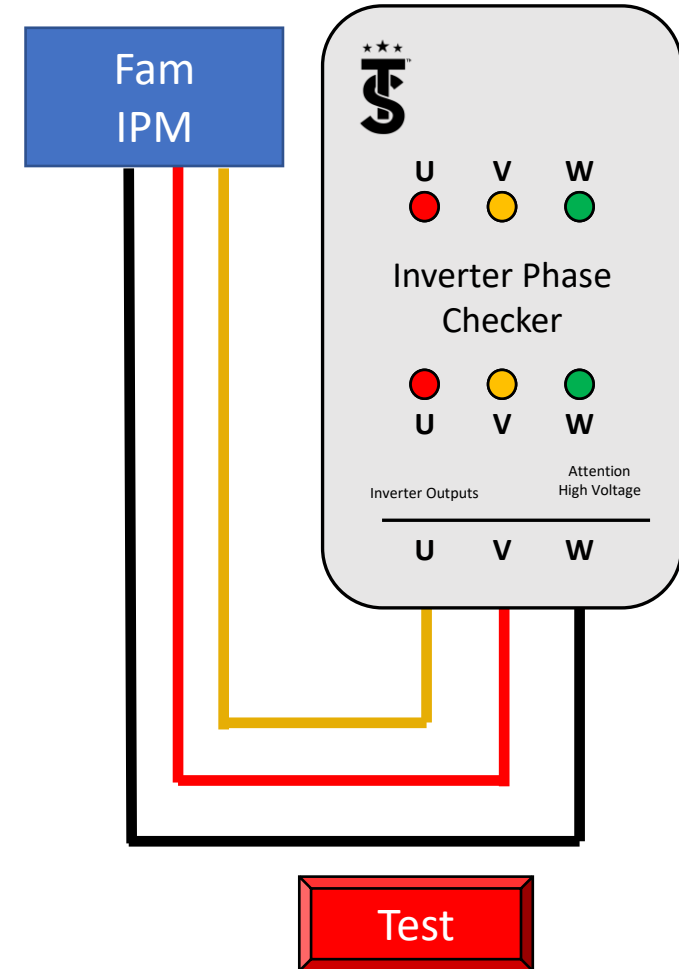


Testing the compressor IPM Voltage 2

- ❑ Use an inverter tester to test the voltage to the compressor.
- ❑ There must be a balanced 3 phase A/C voltage supplied to the fan motor.
- ❑ Refer to the manufacturer's user manual for assistance.
- ❑ Click test to see test the inverter power supply



The test shows that there is a problem with the voltage being supplied to the compressor





Single Phase ODU BLDC Fan Motor Testing

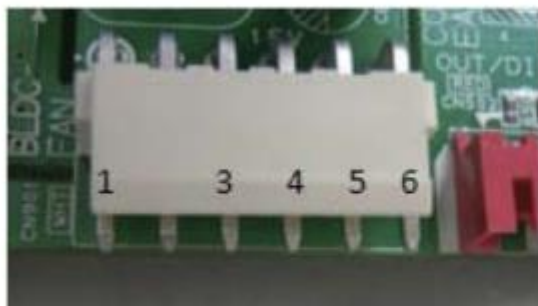


Eco BLDC





Eco BLDC Motor Testing



#1-#3 DC_Link voltage (Between DC_link – GND)	280Vdc ~ 360Vdc
#4-#3 DC voltage (Between DC15V-GND)	14Vdc ~ 16Vdc
#5-#3 Fan command voltage (Between Fan control – GND)	2Vdc ~ 5Vdc

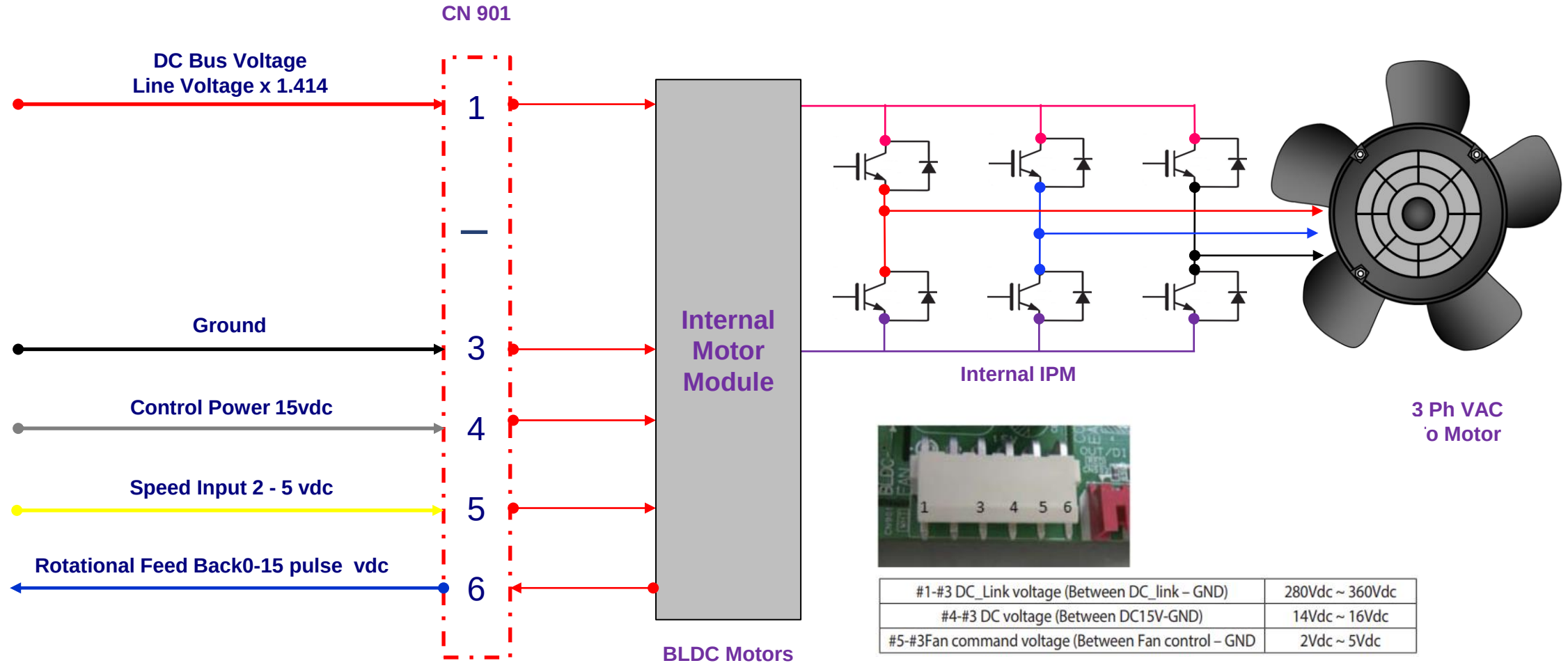
Voltage Test

Measuring part	Inspection part	Value
 Motor connector	#1(RED)-#3(BLK) Resistance (Between DC_link-GND)	Open (Infinite resistance value)
	#4(WHT)-#3(BLK) Resistance (Between DC_15V-GND)	About 1.0kΩ ~ 3kΩ
	#5(YEL)-#3(BLK) Resistance (Between Fan control-GND)	About 230kΩ ~ 280kΩ
	#6(BLU)-#3(BLK) Resistance (Between Feedback-GND)	Over 1MΩ or open Infinite resistance value)

Resistance Test



BLDC ODU Fan Motor Voltages

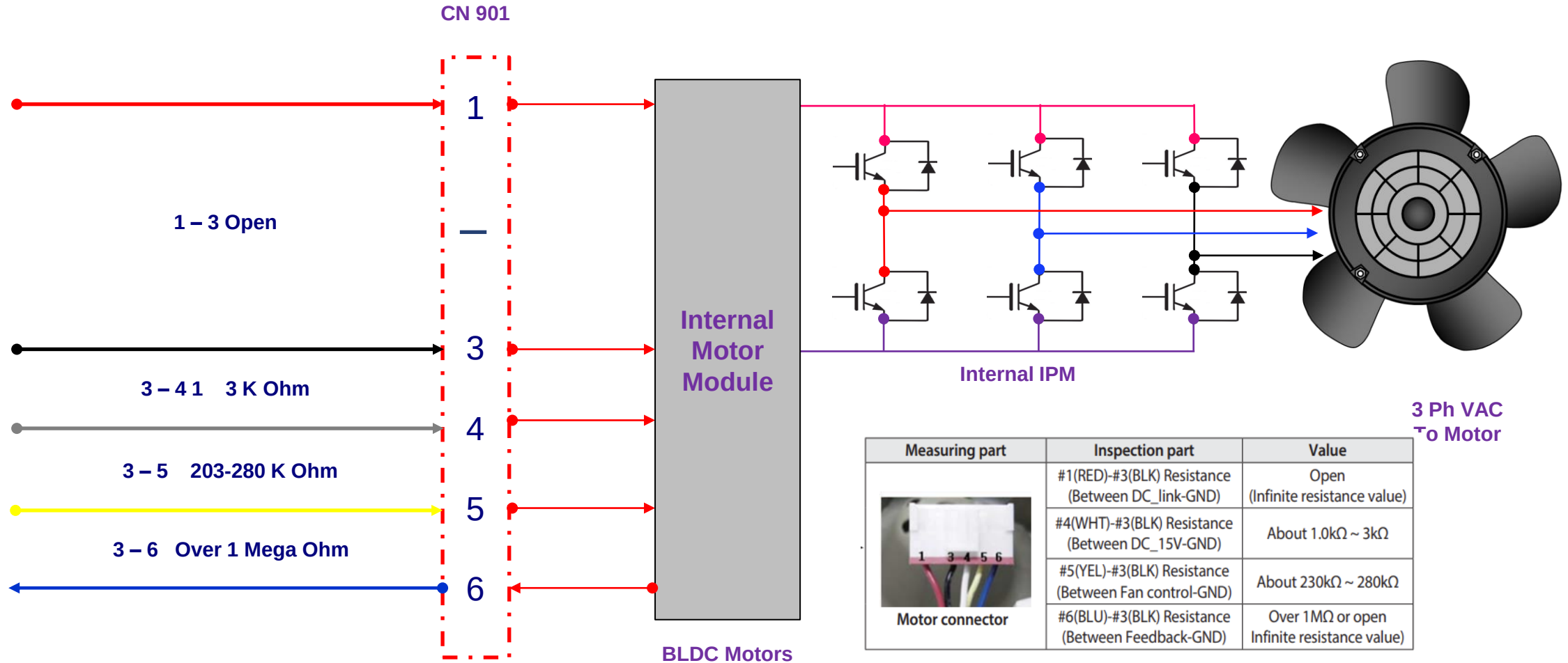


Voltage Test

SAMSUNG



BLDC ODU Fan Motor Resistance Measurements

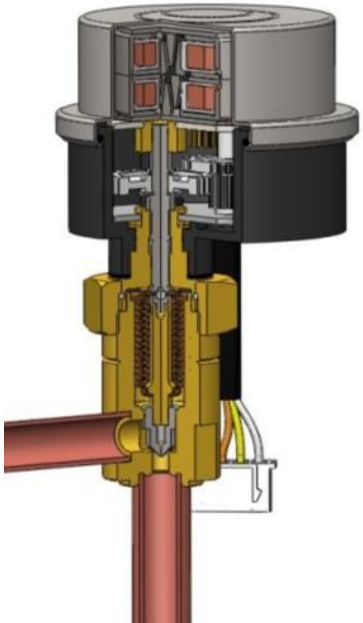


Resistance Test

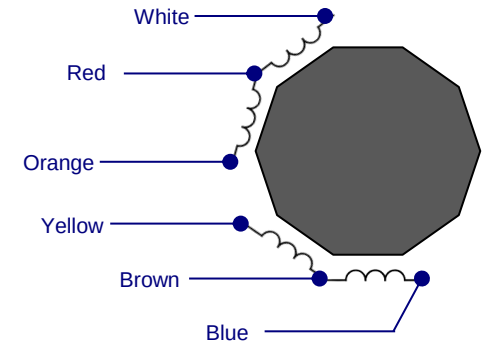
SAMSUNG



EEV Resistance Testing



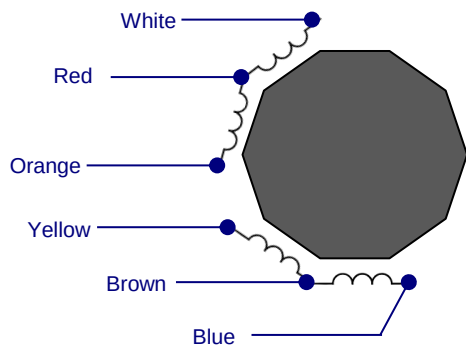
- ❑ Test the resistance of the driver motor.
- ❑ If motor windings test to be good check for flow.
- ❑ Carefully back off the



RED – WHITE	150±15Ω
RED – ORANGE	//
BROWN-YELLOW	//
BRWON-BLU	//
COM : RED, BROWN	

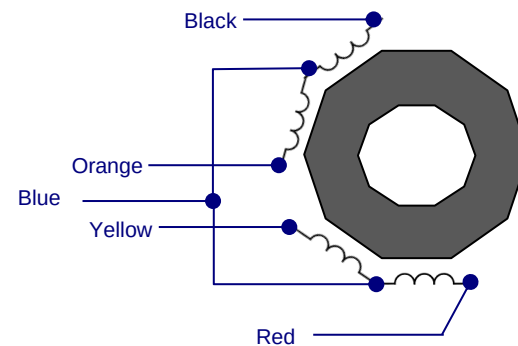
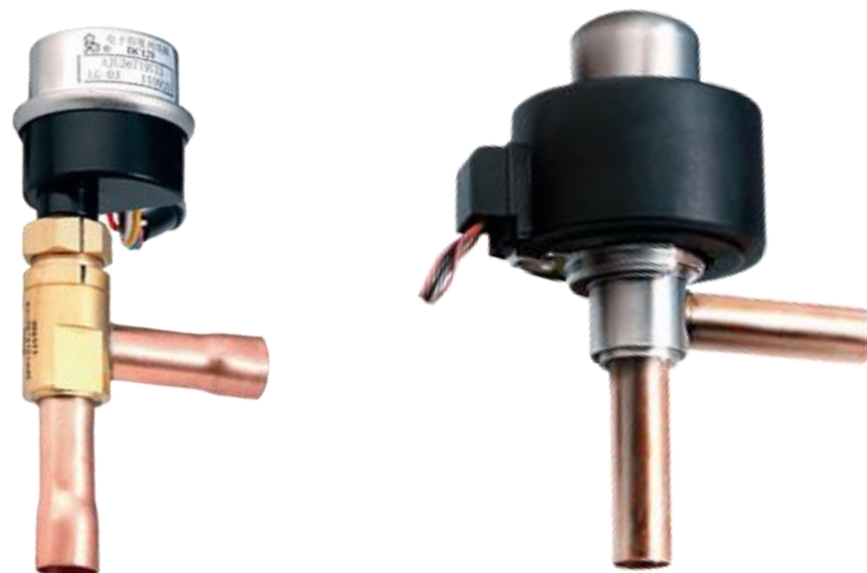


EEV Coil Resistance



RED – WHITE	$150 \pm 15\Omega$
RED – ORANGE	"
BROWN-YELLOW	"
BRWON-BLU	"

COM : RED, BROWN



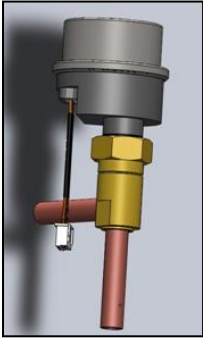
BLUE-BLACK	$46 \pm 2\Omega$
BLUE-YELLOW	"
BLUE-RED	"
BLUE-ORANGE	"

COM : Blue
RAC CAC
FJM

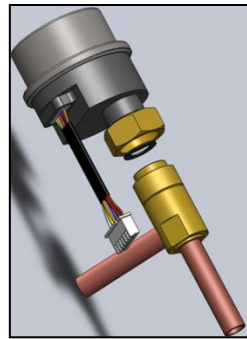
BLUE-BLACK	$92\Omega \pm 2\Omega$
BLUE-YELLOW	$92\Omega \pm 2\Omega$
BLUE-RED	$92\Omega \pm 2\Omega$
BLUE-ORANGE	$92\Omega \pm 2\Omega$
COM : Blue	DVMS EVI EEV



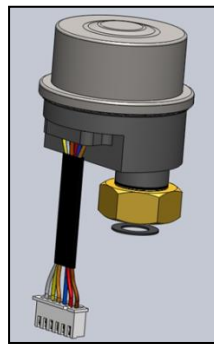
EEV Troubleshooting



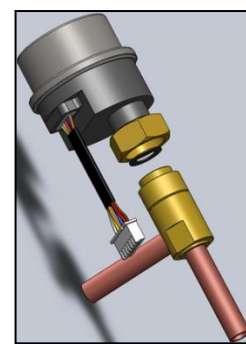
Disassemble EEV body with a wrench



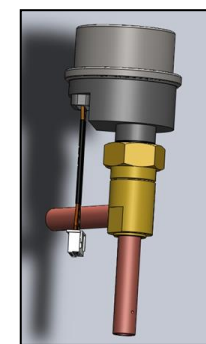
Remove motor



Replace motor with new washer



Assemble ① with a torque wrench (11 ft.lbs.), and a backup wrench



Finished

- When removing the motor, it must be in the full open position (New motor is provided at the 2000 pulse position)
- Motor should only be replaced when the pressure on the needle shaft is less than 145 psi.



145 psi.



203 psi.



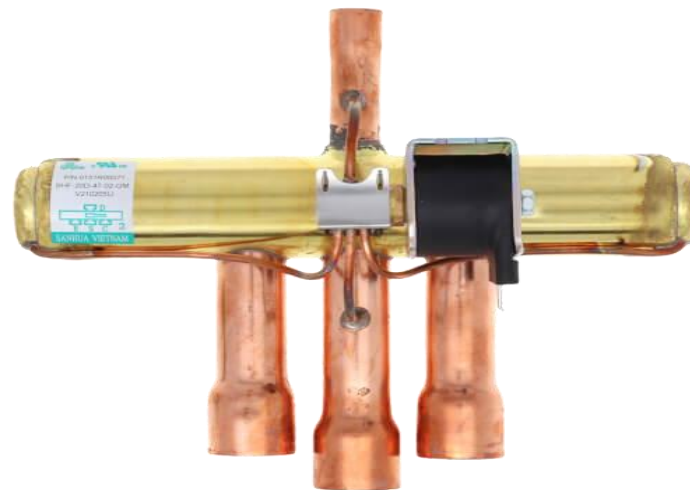
290 psi.

Shaft which is connected with Bellows extends out based on pressure behind the bellows
Deformation of Bellows
→ Opening level may be changed



The Reversing Valve Testing

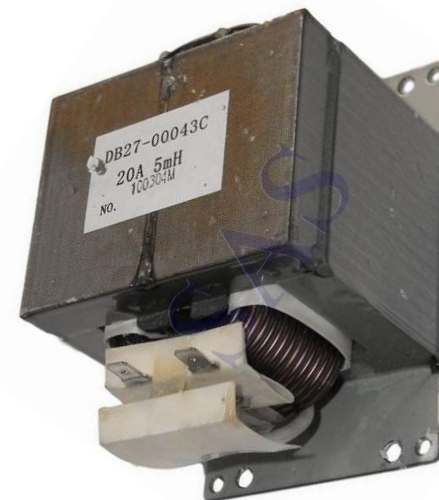
- ☐ It is energized in the heating mode
- ☐ Measure the voltage
- ☐ Is there a pressure differential between the high and low side of the system,
- ☐ Its the difference between the high and low side pressure that causes the reversing valve to shift.
- ☐ Use magnets to find the location of the slider
- ☐ Is there a temperature difference between the suction pipes?





Reactor Testing

- ☐ All this device does is creates a magnetic field.
- ☐ Check that it has continuity through the coil
- ☐ If it is open it is bad
- ☐ If there is rodent damage replace if.





Nom. Ton	Model	Units		Module #1					
				RLA		FLA		Power Supply	
		Hz	Volts	Comp1	Comp2	FAN1	FAN2	MCA	MOP
6Ton	AM072BXVGF*	60	208~230	18.7	-	4.6	-	28	35
8Ton	AM096BXVGF*	60	208~230	12.2	12.2	4.2	4.2	36	40
10Ton	AM120BXVGF*	60	208~230	14.4	14.4	4.2	4.2	40.8	45
12Ton	AM144BXVGF*	60	208~230	19.6	19.6	4.2	4.2	52.6	60
14Ton	AM168BXVGF*	60	208~230	20.4	20.4	4.2	4.2	54.4	60
16Ton	AM192BXVGF*	60	208~230	22.5	22.5	4.6	4.6	60	70
18Ton	AM216BXVGF*	60	208~230	24.3	24.3	4.6	4.6	64	80
20Ton	AM240BXVGF*	60	208~230	26.1	26.1	4.6	4.6	68	80

Nom. Ton	Model	Units		Module #1					
				RLA		FLA		Power Supply	
		Hz	Volts	Comp1	Comp2	FAN1	FAN2	MCA	MOP
6Ton	AM072BXVGJ*	60	460	10	-	2.3	-	15	20
8Ton	AM096BXVGJ*	60	460	6	6	2.1	2.1	18	20
10Ton	AM120BXVGJ*	60	460	6.6	6.6	2.1	2.1	19.4	25
12Ton	AM144BXVGJ*	60	460	9.6	9.6	2.1	2.1	26.2	35
14Ton	AM168BXVGJ*	60	460	11	11	2.1	2.1	29	35
16Ton	AM192BXVGJ*	60	460	13	13	2.3	2.3	34	40
18Ton	AM216BXVGJ*	60	460	14.8	14.8	2.3	2.3	38	50
20Ton	AM240BXVGJ*	60	460	15.6	15.6	2.3	2.3	40	50

Non. Ton	Model	Units		Module #1					
				RLA		FLA		PowerSupply	
		Hz	Volts	Comp1	Comp2	FAN1	FAN2	MCA	MOP
6Ton	AM072DXVGQ*	60	575	8.5	-	2.3	-	13.4	20
8Ton	AM096DXVGQ*	60	575	5	5	2.3	2.3	16.1	20
10Ton	AM120DXVGQ*	60	575	5.5	5.5	2.3	2.3	17.4	20
12Ton	AM144DXVGQ*	60	575	7.6	7.6	2.3	2.3	22.2	25
14Ton	AM168DXVGQ*	60	575	8.8	8.8	2.3	2.3	24.8	30
16Ton	AM192DXVGQ*	60	575	11.2	11.2	2.3	2.3	30.5	40
18Ton	AM216DXVGQ*	60	575	12.8	12.8	2.3	2.3	34	45
20Ton	AM240DXVGQ*	60	575	13.5	13.5	2.3	2.3	35.8	45



Questions?